

Metric-measure limits of subcritical FK and spanning-tree planar maps

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Abstract

We resolve the finite spherical cases of Gwynne and Miller’s graph-metric conjecture for critical Fortuin–Kasteleyn maps at each fixed $q \in (0, 4)$ and for uniform spanning-tree-decorated maps. After deterministic rescaling of graph distances, these maps, equipped with the vertex probability measure proportional to degree, converge in Gromov–Hausdorff–Prokhorov law to their ordinary unit-area Liouville quantum gravity spheres. Distances use every primal edge, and convergence holds through all positive integer edge counts.

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1 Introduction

Random planar maps model random two-dimensional geometry. A statistical-mechanical weight changes the law of the map and, in the continuum description, the parameter of Liouville quantum gravity (LQG). Contour encodings and mating of trees identify the corresponding quantum surfaces. The metric problem asks whether shortest-path distances in the maps converge to the intrinsic distances on those surfaces.

We consider two finite spherical ensembles: critical Fortuin–Kasteleyn (FK) maps and maps decorated by a spanning tree. The FK proof applies to every fixed $q \in (0, 4)$. The spanning-tree endpoint has a separate proof using independent walk increments. Both conclusions retain quantum area through a natural discrete measure. Here *subcritical* refers to the LQG range $\gamma < 2$, or equivalently $q < 4$; the FK model itself is at its self-dual critical point.

1.1 The ensembles and the limit

A rooted planar map is a finite connected multigraph embedded in the oriented sphere, considered up to orientation-preserving homeomorphism, with one distinguished oriented edge. Loops and multiple edges are allowed. Fix a positive integer n .

For the FK ensemble, let $\mathcal{F}_n$ consist of rooted pairs (M, A) , where M has n edges and $A \subseteq E(M)$. The spanning subgraph $(V(M), A)$ includes isolated vertices. In the dual map, let A^* contain the edges crossing $E(M) \setminus A$. Writing k for the number of connected components, set

$$\ell(M, A) = k(A) + k(A^*) - 1 = 2k(A) + |A| - |V(M)|.$$

This counts the interfaces between primal and dual FK components. For fixed $q \in (0, 4)$, sample according to

$$\mathbb{P}[(M_n, A_n) = (M, A)] = \frac{q^{\ell(M, A)/2}}{Z_n(q)}, \quad Z_n(q) = \sum_{(M, A) \in \mathcal{F}_n} q^{\ell(M, A)/2}. \quad (1.1)$$

For the spanning-tree ensemble, sample (M_n, T_n) uniformly from rooted map–spanning-tree pairs with n edges. Equivalently, its map marginal is weighted by the number of spanning trees.

In either ensemble, d_n denotes graph distance using every primal map edge, each of length one. Equip the vertex set with the probability measure

$$\mu_n(v) = \frac{\deg_{M_n}(v)}{2n}, \quad (1.2)$$

where a loop contributes twice to the degree. This is the law of the vertex incident to a uniformly sampled corner of the map.

For FK maps, choose the unique $\gamma \in (\sqrt{2}, 2)$ satisfying

$$q = 2 + 2 \cos(\pi \gamma^2 / 2). \quad (1.3)$$

For spanning-tree maps, set $\gamma = \sqrt{2}$. The FK inventory parameter and Brownian correlation are

$$b_F = \frac{\sqrt{q}}{2 + \sqrt{q}}, \quad \rho = \frac{b_F}{1 - b_F} = \frac{\sqrt{q}}{2} = -\cos(\pi\gamma^2/4).$$

We reserve $p = 1/d_\gamma$ for the reciprocal metric dimension; it is not the inventory parameter. These conventions are used throughout. Let (S, h) be the ordinary unit-area γ -quantum sphere: the quantum-sphere measure of [13, Definition 4.21(ii)], disintegrated at area one, with its distinguished points forgotten. Write μ_h for its quantum-area probability measure and D_h for its intrinsic LQG metric, with one fixed deterministic multiplicative normalization [23, 22].

Theorem 1.1 (Metric and measure convergence). *For each fixed $q \in (0, 4)$ in the FK ensemble, and for the spanning-tree ensemble, there are deterministic positive numbers $a_n \rightarrow 0$ such that*

$$(V(M_n), a_n d_n, \mu_n) \xrightarrow[n \rightarrow \infty]{\text{law}} (S, D_h, \mu_h)$$

in Gromov–Hausdorff–Prokhorov topology on compact metric probability spaces modulo measure-preserving isometry. The limit has the parameter specified above. Convergence holds through all positive integers n ; the root and decoration are forgotten.

Gromov–Hausdorff–Prokhorov distance simultaneously compares the metric spaces and their measures after isometric embedding in a common space; we give the precise definition in Section 2. In particular, the theorem implies convergence in the ordinary Gromov–Hausdorff topology. The normalization may depend on the chosen ensemble and on the fixed FK parameter. No uniformity in q , endpoint FK statement, normalization power law, or convergence rate is asserted. The measure in Equation (1.2) is part of the theorem; a conclusion for the uniform measure on vertices would require a separate argument.

1.2 History and significance

The FK weights originate in the random-cluster model of Fortuin and Kasteleyn [15]. Mullin’s bijection [36] encodes maps with a spanning tree by walks in a quadrant; Bernardi gives a parenthesis-shuffle formulation [4], and the precise finite convention used here appears in [12, Section 2.3, Theorem 1]. Sheffield’s inventory bijection gives the FK encoding and its bilateral Brownian limit [38]; Gwynne and Sun established the finite empty-word limit [27]. The quantum-area measure was developed by Duplantier and Sheffield [14]. Duplantier, Miller, and Sheffield identified Brownian contours with curve-decorated quantum surfaces [13], and Miller and Sheffield proved the corresponding fixed-area sphere identification [34]. These results determine the encoding and surface limits. The graph metric needs additional control of local passages and of vertices which typical samples might miss.

The construction of the subcritical LQG metric rests on the LFPP tightness theorem of Ding, Dubédat, Dunlap, and Falconet [9], the weak-metric theory of Dubédat, Falconet, Gwynne, Pfeffer, and Sun [11], and the uniqueness theorem of Gwynne and Miller [23]. Our comparison argument draws on annular independence and covering arguments for GFF-local metrics [21, Lemma 3.1 and Section 4], and on the local-field-perturbation and geodesic-saving strategy of the uniqueness proof. Here the additional task is to obtain and compare conditional passage laws from discrete maps.

Gwynne and Miller’s graph-metric conjecture explicitly includes FK weights for $\gamma \in (\sqrt{2}, 2)$ and spanning-tree weights at $\gamma = \sqrt{2}$ [23, Conjecture 1.7]. Theorem 1.1 resolves its specified finite spherical FK family and its finite spherical spanning-tree case, with the additional measured

convergence stated above. It does not address every ensemble or version of that conjecture. For several models with independent-increment encodings, including infinite spanning-tree-weighted maps, Gwynne, Holden, and Sun coupled finite contour portions with portions of mated-CRT maps, obtaining rough isometries with polylogarithmic distortion [17, Theorem 1.5]. Ding and Gwynne’s volume-growth estimates [10, Theorem 1.6, model 2] enter our spanning-tree proof. For the infinite spanning-tree map, Gwynne and Pfeffer identified the internal-diameter exponent [26, Theorem 3.1]; their volume-counting argument [26, Lemma 3.2] is adapted here to endpoint moments.

At $q = 1$, all 2^n edge subsets of a rooted n -edge map have the same weight, so the map marginal is uniform. Following the Brownian-map scaling-limit and uniqueness theorems of Le Gall and Miermont [29, 31], its metric limit is known from the uniform-map theorem [5, Corollary 1.2] and the identification of the Brownian sphere with $\sqrt{8/3}$ -LQG [35, Theorem 1.4] [23, Corollary 1.4]. At $q = 2$, Equation (1.3) gives $\gamma = \sqrt{3}$, so the FK–Ising ensemble is included in the same theorem. The spanning-tree clause is proved independently and is not obtained by taking a limit of the FK parameter.

1.3 Proof overview

The common objective is to identify local graph distances, control every vertex, and then retain the measure under the passage to the limit. The two encodings require different estimates for the first two steps.

Local information from words. A contour records a decorated map by a finite word or walk. Buffered compact regions admit finite chart descriptions containing retained word intervals, exact boundary heights, and gluing instructions. Refining these charts recovers every confined passage and its finite marked subpaths. For FK words, flexible orders can depend on earlier letters outside these intervals. We retain the finite supplier-comparison instructions and the order of the relevant traversals, then smooth relative heights on the exact lattice before passing to the limit. This proves conditional laws for local passage observations, including discontinuous shortest-path costs (Sections 5–7). The tree proof supplies its own reconstruction and conditional laws from independent increments in Section 15.

Identification by local field changes. We prove a common comparison and rigidity theorem for local passage laws. FK maps supply its normalization through surrounding circuits before arbitrary endpoints are controlled; spanning-tree maps supply it through internal moment bounds. Annuli at prescribed quantum scales compare these laws with D_h . If the optimal lower and upper comparison constants differed, a local change of the real field would create a strict shortcut near a saturated geodesic segment. A weighted count of the possible output passages gives a contradiction. The established Weyl rule controls D_h ; exact local conditional laws transport the graph-passage observations. This establishes a *deterministic* multiplier at the level of the whole limiting law. Each model verifies the local-kernel, noncollapse, and joining hypotheses separately.

Uniform control of endpoints. For FK maps, the Brownian two-wall exponent satisfies $2/\gamma^2 > 1/2$. This strict margin pays for the choices of locations in finite contact diagrams. The resulting entropy estimate and protected planar sewing produce small surrounding circuits; a recursive covering argument then controls arbitrary discrete endpoints (Sections 10–12). For

spanning trees, internal moment bounds at record scales and the SLE_8 interval topology give the endpoint modulus. A first-failure argument extends these bounds to every scale (Sections 14–17).

Finite spheres and their area measures. The FK local laws transfer to ordinary sphere charts with their actual field normalization. Two conditionally independent uniform corner roots of the same map, together with one common sequence of corner samples, exclude vertices escaping all finite sample nets. For spanning trees, explicit excursion densities control the two ends of the contour and give uniform convergence of the metric pulled back to contour time. The encodings identify the discrete corner measure with the pushforward of uniform time, and the continuum curve is parameterized by quantum area. The sampling and common-parameter lemmas of Section 2 therefore yield measured convergence in Sections 18 and 19. A single elementary normalization lemma then removes the deterministic subsequential multiplier using the unconditional law of log diameter.

Several ingredients are useful independently of the final ensembles: the transfer of discontinuous local observations by exact lattice smoothing, comparison by circuits before endpoint tightness, the balance between contact rank and the entropy of chosen locations, and the two-root completion argument. The measured convergence criterion separates finite distance arrays from the additional covering condition needed to control all points of a compact space.

2 Sampling and deterministic normalization

We use two elementary ways to retain the sampling measure when a metric limit is identified. The first uses finite distance arrays and covering radii; the second uses a common time parametrization. We then give one deterministic normalization argument for both models.

2.1 From sampled distances to measured convergence

A compact metric probability space is a triple (X, d, μ) , where (X, d) is a nonempty compact metric space and μ is a Borel probability measure. We identify triples by measure-preserving isometries. Its Gromov–Hausdorff–Prokhorov distance from (Y, e, ν) is the infimum, over isometric embeddings in a common metric space, of the maximum of the Hausdorff distance between their images and the Prokhorov distance between their measures. The latter distance is the infimum of $r > 0$ such that $\mu(A) \leq \nu(A^r) + r$ and $\nu(A) \leq \mu(A^r) + r$ for every Borel set A , where A^r is its open r -neighborhood. The space of these isometry classes is separable and complete in this metric [30, Theorem 6 and Proposition 8(i)].

Sampled distance arrays underlie the Gromov-weak framework of Greven, Pfaffelhuber, and Winter [16]. Controlling the full support requires an additional condition, as in the lower-mass criterion of Athreya, Löhner, and Winter [2]. We use the following covering-radius version and give its proof. The covering assumption rules out points which carry little mass but remain far from all typical samples.

Lemma 2.1 (Sample arrays and covering radii). *Let (X_n, d_n, μ_n) and (X, d, μ) be random compact metric probability spaces, with μ almost surely of full support. Conditional on each space, sample a sequence $(x_{n,i})_{i \geq 1}$, respectively $(x_i)_{i \geq 1}$, independently with its probability measure. Suppose that, for every fixed k ,*

$$(d_n(x_{n,i}, x_{n,j}))_{i,j \leq k} \implies (d(x_i, x_j))_{i,j \leq k},$$

and that, for every $r > 0$,

$$\lim_{k \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P}(R_{n,k} > r) = 0, \quad R_{n,k} := \sup_{z \in X_n} \min_{i \leq k} d_n(z, x_{n,i}). \quad (2.1)$$

Then (X_n, d_n, μ_n) converges in distribution to (X, d, μ) in Gromov–Hausdorff–Prokhorov topology. Only the indicated conditional sampling laws are assumed; no joint coupling of the spaces is needed.

Proof. Write $\hat{\mu}_{n,m} = m^{-1} \sum_{i=1}^m \delta_{x_{n,i}}$. We first prove the uniform empirical approximation

$$\lim_{m \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P}(d_{\mathbb{P}}^{X_n}(\mu_n, \hat{\mu}_{n,m}) > \varepsilon) = 0 \quad (\varepsilon > 0). \quad (2.2)$$

Use the next k samples as auxiliary centers $y_{n,j} = x_{n,m+j}$. They are independent of the first m samples conditional on the space. Partition X_n into the Borel cells $C_1, \dots, C_k$ of the nearest center, breaking ties by the smallest index. Conditional on the space and the centers, put

$$p_j = \mu_n(C_j), \quad \hat{p}_j = \frac{1}{m} \sum_{i=1}^m \mathbf{1}_{\{x_{n,i} \in C_j\}}, \quad T = \frac{1}{2} \sum_{j=1}^k |\hat{p}_j - p_j|.$$

The conditional multinomial law and Cauchy–Schwarz give

$$\mathbb{E}[T \mid X_n, d_n, \mu_n, (y_{n,j})_{j \leq k}] \leq \frac{1}{2\sqrt{m}} \sum_{j=1}^k \sqrt{p_j(1-p_j)} \leq \frac{1}{2} \sqrt{\frac{k}{m}}. \quad (2.3)$$

If these centers form an r -net, moving every point to its cell's center moves both μ_n and $\hat{\mu}_{n,m}$ by distance at most r . The two resulting measures on the centers have total variation distance at most T , and hence Prokhorov distance at most T . Indeed their common mass can be coupled at identical centers and the remaining mass is at most T . Thus

$$d_{\mathbb{P}}^{X_n}(\mu_n, \hat{\mu}_{n,m}) \leq 2r + T.$$

The auxiliary covering radius has the same law as $R_{n,k}$. Markov's inequality therefore yields, for $r, t > 0$,

$$\mathbb{P}(d_{\mathbb{P}}^{X_n}(\mu_n, \hat{\mu}_{n,m}) > 2r + t) \leq \mathbb{P}(R_{n,k} > r) + \frac{1}{2t} \sqrt{\frac{k}{m}}. \quad (2.4)$$

Choose r, t with $2r + t < \varepsilon$, then k using Equation (2.1), and finally let m tend to infinity. This proves Equation (2.2). The independence of the auxiliary centers is essential: the cells were fixed before the sample counts in Equation (2.3) were drawn.

Let $F_{n,m}$ be the finite metric space consisting of the first m sampled points, equipped with $\hat{\mu}_{n,m}$, with repeated points identified and their masses added. Inside X_n its Hausdorff distance from X_n is exactly $R_{n,m}$. Consequently

$$\lim_{m \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P}(d_{\text{GHP}}((X_n, d_n, \mu_n), F_{n,m}) > \varepsilon) = 0. \quad (2.5)$$

For fixed m , the distance-array assumption gives $F_{n,m} \Rightarrow F_m$, where F_m is the analogous empirical space from $x_1, \dots, x_m$. Here is the needed continuity, including possible repeated points. Two m -label distance arrays differing entrywise by at most a give a correspondence pairing equal labels with distortion at most a . For each $b > a/2$, join paired points in the disjoint union by edges of length b and take the induced path metric. The distortion inequality ensures that no new

path shortens either original metric. Paired points are at distance at most b , so the Hausdorff distance is at most b ; the coupling which gives mass $1/m$ to each label pair makes the Prokhorov distance at most b as well. Letting $b \downarrow a/2$ proves the claimed continuity. Zero distances simply identify labels.

Almost surely the target samples are dense: every nonempty ball in a countable base has positive measure and is eventually sampled. Compactness then implies $R_m := \sup_{z \in X} \min_{i \leq m} d(z, x_i) \rightarrow 0$. The same auxiliary-center argument, now for this single random space, shows that its empirical Prokhorov distances tend to zero in probability. Therefore $F_m \rightarrow (X, d, \mu)$ in probability in GHP distance.

For completeness, let f be bounded by one and Lipschitz with constant one for $1 \wedge d_{\text{GHP}}$. Insert $F_{n,m}$ and F_m between the two arguments of f . For fixed m the middle expectation difference tends to zero. The other differences tend to zero after $m \rightarrow \infty$ by Equation (2.5) and the target approximation. Bounded Lipschitz functions determine weak convergence on the space of compact metric probability spaces. This proves the result. $\square$

Lemma 2.2 (The measure in a common sample completion). *Suppose two random compact metric probability spaces with full-support measures have sample sequences which are conditionally iid in each marginal. If their labeled distance arrays agree almost surely, their unique label-preserving completion isometry preserves their measures. No conditional independence given both spaces is required.*

Proof. In each marginal, almost surely the samples are dense and their empirical measures converge weakly to the sampling measure. For the latter assertion, condition on a fixed compact space and apply the strong law to a countable uniformly dense subset of its continuous functions. The two probability-one events intersect under any coupling. The equal arrays define an isometry of the dense sample sets, hence a unique isometry of the compact completions. It maps every empirical measure in one space to the corresponding empirical measure in the other. Passing to weak limits proves the assertion. $\square$

Lemma 2.3 (A common parameter and its measure). *Let $f_n : [0, 1] \rightarrow X_n$ and $f : [0, 1] \rightarrow X$ be measurable surjections onto compact metric spaces, and let their pushforwards of Lebesgue measure be μ_n and μ . If*

$$e_n := \sup_{s, t \in [0, 1]} |d_n(f_n(s), f_n(t)) - d(f(s), f(t))| \rightarrow 0,$$

then $d_{\text{GHP}}((X_n, d_n, \mu_n), (X, d, \mu)) \leq e_n/2$.

Proof. The correspondence $\{(f_n(t), f(t)) : t \in [0, 1]\}$ covers both spaces and has distortion at most e_n . Use the common metric constructed in the proof of Lemma 2.1, with bridges of length $b > e_n/2$. The pushforward of Lebesgue measure under $t \mapsto (f_n(t), f(t))$ is a coupling of the two measures, supported on pairs at distance at most b . Both Hausdorff and Prokhorov distances are at most b . Let $b \downarrow e_n/2$. $\square$

2.2 Removing a deterministic subsequential multiplier

Lemma 2.4 (A continuous translation-equivariant center). *For every finite real random variable Y , there is a unique $m(Y) \in \mathbb{R}$ satisfying*

$$\mathbb{E} \arctan(Y - m(Y)) = 0. \tag{2.6}$$

It depends only on the law of Y , satisfies $m(Y + c) = m(Y) + c$ for every deterministic $c \in \mathbb{R}$, and is continuous under convergence in distribution of finite real random variables.

Proof. For fixed Y , the function $t \mapsto \mathbb{E} \arctan(Y - t)$ is continuous, strictly decreasing, and has limits $\pi/2$ and $-\pi/2$. Thus its zero exists uniquely. Translation equivariance follows by substitution. If $Y_j \Rightarrow Y$, expectations converge at every fixed t , since the integrand is bounded and continuous. Their strict opposite signs at $m(Y) \pm \varepsilon$ trap the zeros, so $m(Y_j) \rightarrow m(Y)$. No moment assumption is used. $\square$

Lemma 2.5 (Deterministic normalization of a measured limit). *Let (V_n, d_n, μ_n) be random compact metric probability spaces, let $s_n > 0$ be deterministic with $s_n \rightarrow \infty$, and let (S, D, μ) be a random compact metric probability space whose diameter H is finite and strictly positive almost surely. Suppose every subsequence has a further subsequence on which*

$$(V_n, d_n/s_n, \mu_n) \Longrightarrow (S, cD, \mu)$$

in GHP topology, for some deterministic $c \in (0, \infty)$ which may depend on that further subsequence. With m as in Lemma 2.4, write $H_n = \text{diam}(V_n, d_n)$ and put

$$b_n = \exp\{m(\log(H_n \vee 1)) - m(\log H)\}, \quad a_n = b_n^{-1}.$$

Then $a_n > 0$, $a_n \rightarrow 0$, and

$$(V_n, a_n d_n, \mu_n) \Longrightarrow (S, D, \mu)$$

along the full sequence.

Proof. Take any subsequence and a further subsequence from the hypothesis. Diameter is continuous for GH, and GHP convergence implies GH convergence, so $H_n/s_n \Rightarrow cH$. Since $s_n \rightarrow \infty$ and $cH > 0$ almost surely,

$$\log(H_n \vee 1) - \log s_n \Rightarrow \log c + \log H.$$

Continuity and translation equivariance of m give $b_n/s_n \rightarrow c$. Multiplying the metrics by the positive deterministic ratios $s_n/b_n \rightarrow c^{-1}$ gives the stated normalized GHP limit on this further subsequence, with the measures unchanged. Positive scaling is jointly continuous in GHP: scale a common embedding for fixed factors; for convergent factors, the identity correspondence has distortion bounded by their difference times the diameter, and the identity coupling controls the measures in the same embedding. Every subsequence has such a further subsequence, which proves the full-sequence assertion. If b_n failed to tend to infinity, a bounded subsequence of b_n would have a further subsequence with $b_n/s_n \rightarrow c > 0$, a contradiction.

All centers here use unconditional diameter laws, so the normalization is deterministic. The truncation $H_n \vee 1$ includes one-vertex maps. The hypothesis that c is deterministic is indispensable: this argument does not remove a random multiplier. $\square$

Part I

Critical FK maps

3 Encoding and continuum geometry

In the FK proof $q \in (0, 4)$ is fixed. Let d_γ be the almost sure Hausdorff dimension of the γ -LQG metric, a deterministic constant [25, Corollary 1.7]. Put

$$b_F = \frac{\sqrt{q}}{2 + \sqrt{q}}, \quad \rho = \frac{b_F}{1 - b_F}, \quad \theta = \frac{\pi\gamma^2}{4}, \quad p = \frac{1}{d_\gamma}, \quad \xi = \gamma p, \quad Q = \frac{2}{\gamma} + \frac{\gamma}{2}. \quad (3.1)$$

Thus $\rho = -\cos \theta$, $\pi/2 < \theta < \pi$, and

$$Q - 2 = \frac{(\gamma - 2)^2}{2\gamma} > 0, \quad \frac{\pi}{2\theta} - \frac{1}{2} = \frac{2}{\gamma^2} - \frac{1}{2} > 0. \quad (3.2)$$

All constants may depend on q . A constant depending on q is fixed before any scale tends to infinity.

We first identify the word law and its continuum surface, then record the local field and metric properties used in the proof. These inputs identify the limiting geometry of the encoding; they do not assert convergence of discrete graph distances.

3.1 The discrete and continuum encodings

An inventory letter is a burger of type 1 or 2, a rigid order of one of these types, or a flexible order. Their respective probabilities are

$$\frac{1}{4}, \quad \frac{1}{4}, \quad \frac{1 - b_F}{4}, \quad \frac{1 - b_F}{4}, \quad \frac{b_F}{2}.$$

An order takes the freshest available burger of an allowed type. Reduction commutes orders past burgers of the other type and cancels a burger with its matching order. We use both a bilateral iid word and a word of length $2n$ conditioned to reduce to the empty word. After the flexible orders are resolved, the two net burger counts give the contour $Z = (L, R)$.

We use the inventory bijection in the convention of [38]. It produces the primal all-edge map, its tree/cotree sewing, its FK decoration, and its oriented root. The primal tour vertex at integer time j is denoted by $v_n(j)$. Consecutive tour vertices are equal or joined by a primal tree edge. A match in the other contour supplies a remaining primal edge through its quadrangle. Consequently every graph edge used below is an edge of the primal map, whether or not it belongs to the FK subgraph.

Lemma 3.1 (The exact finite law). *The inventory word of length $2n$ conditioned to reduce to the empty word induces the joint law proportional to $q^{\ell(M,A)/2}$ in the statement of [Theorem 1.1](#). Conditional on the unrooted decorated map, its root corner is uniform.*

Proof. An empty word has n burger letters and n order letters. If f of its orders are flexible, its unconditioned probability is

$$4^{-n} \left(\frac{1 - b_F}{4} \right)^{n-f} \left(\frac{b_F}{2} \right)^f = c_{n,q} \left(\frac{2b_F}{1 - b_F} \right)^f.$$

The bijection has $\ell = f + 1$. Since $2b_F/(1 - b_F) = \sqrt{q}$, the displayed weight is a factor independent of (M, A) times $q^{\ell/2}$. Rooted isomorphism classes correspond under the same bijection. Re-rooting does not change the weight. Equivalently, on an unrooted class one may choose a uniform dart; the automorphism group acts freely on darts, so this gives the asserted rooted law. A corner and its outgoing dart are equivalent root conventions. $\square$

The public continuum inputs will be used in the following precise forms.

Theorem 3.2 (Contour and surface inputs). *After fixed deterministic changes of time and height units:*

- (i) *The bilateral resolved contour, viewed at time scale m and height scale $\sqrt{m}$, converges locally uniformly in law to a two-dimensional Brownian motion with equal coordinate variances and correlation ρ . The empty-word contour at total time $2n$, with total rescaled time one, converges to the corresponding quadrant excursion from zero to zero.*
- (ii) *The bilateral Brownian contour determines a curve-decorated γ -quantum cone. Its curve $\eta : \mathbb{R} \rightarrow \mathbb{C}$ is continuous, transient, space filling, and parameterized by quantum area. The excursion determines the ordinary unit-area DMS quantum sphere with its area-parameterized space-filling curve. The latter sphere is the quantum-sphere measure disintegrated at area one.*
- (iii) *The spatial equivalence relation of the sewing is generated by finite chains of scalar horizontal chords. For a scalar coordinate X , a horizontal chord joins $s < t$ if*

$$X(s) = X(t) = \inf_{[s,t]} X.$$

At the deterministic or independently sampled cuts used below, the two boundary strands are contour-tree lineages parameterized by quantum boundary length. In the cone coupling, the intrinsic decorated surface of each fixed bounded time interval is determined by its contour increments. This assertion may be used simultaneously for a countable collection of deterministic cuts or cuts sampled independently of the surface, and for later selection among that collection.

- (iv) *On a buffered spatial patch, the imaginary field determines its directed local traversals and their frontier flow lines. Together with the real field and local Poisson area marks, these determine the clocked local pieces, their incoming and outgoing sides, and the quantum lengths of the local frontier arcs. This statement does not determine the global order of separate traversals.*

References and conventions. Part (i) is the bilateral inventory invariance principle of [38] and the finite-volume result [27, Theorem 1.8]. Parts (ii)–(iii) are the mating-of-trees results of [13] and [34, Theorem 1.1]; cone interval-surface determination in the beaded regime is stated explicitly in [19, Lemma 3.13]. Independent random cuts follow by conditioning on their values, and countable intersections allow the stated later selections. This is intrinsic determination, modulo quantum-surface equivalence. The finite-sphere proof below instead transfers spatial observations using actual local fields; it does not require a sphere version of the isolated-interval reconstruction statement.

The finite-chain convention in (iii) is the strict-chord relation of [13, Lemma 8.9], with its generated classes identified in [13, Lemmas 8.14–8.15]. A weak scalar chord splits into at most two strict chords: otherwise two distinct interior local minima would have the same height, excluded by Theorem 4.9. The same frontier-tree relation applies to the sphere’s whole-plane space-filling curve. Quantum-area reparametrization and its quantum boundary-length coordinates preserve the tree identifications; the initial and final times are identified at the exploration root.

For (iv), [24, Lemma 2.4] reconstructs directed parameterized local traversals with a buffer; its quantum-area clock is described in the proof of [24, Lemma 4.3]. The frontier arcs are the labelled local flow-line boundaries in that construction. Their quantum lengths use the boundary Gaussian multiplicative chaos construction and its conformal coordinate rule [37, Sections 1.2

and 5.1]. More explicitly, choose the specified side of a compact frontier subarc and a sided domain inside the buffer, then straighten that domain conformally and construct boundary length from the restriction of the actual normalized real field. The boundary coordinate rule identifies this measure with the same side's global quantum boundary length and makes it independent of the chosen smaller domain. Exhaust the arc by compact subarcs to include its ends. The sided domains and prime-end labels come from the local imaginary field, so this construction uses only the indicated field restrictions. Contained Poisson-cell locality is also given in [8, Lemma 3.7]. We always shrink the testing region before using these locality statements.

The imaginary whole-plane field is taken modulo its angular period, with its stationary phase, rather than circle-average-pinned with a prescribed phase. Local decompositions consist of Dirichlet GFF parts and harmonic parts modulo that period. The real and imaginary fields are independent before the curve is clocked by real-field area. $\square$

3.2 Ordinary patches of the sphere

The finite-volume argument will transfer local observations from a quantum cone to the ordinary sphere. Such a comparison must retain the field's additive constant: that constant fixes both area and intrinsic distance. The next lemma proves the required one-way domination with the actual unit-area normalization in place.

Lemma 3.3 (Actual-field absolute continuity on a sphere patch). *Let $(\mathbb{C}, h, \infty, 0, 1)$ be the ordinary unit-area quantum sphere with three conditionally independent quantum-area marks in the indicated embedding. If U is bounded and $\bar{U} \subset \mathbb{C} \setminus \{0, 1\}$, and R is large enough that a neighborhood of $\bar{U}$ lies in $B_R(0)$, then*

$$\text{Law}(h|_U) \ll \text{Law}(G_R|_U),$$

where G_R is the whole-plane GFF pinned to have zero average on $\partial B_R(0)$. This is a statement about actual distributions, with their additive constants retained. The right-hand law is mutually absolutely continuous with the restriction of a deterministically dilated circle-average embedding of a γ -quantum cone whose normalization circle is $\partial B_R(0)$. Both comparisons remain valid jointly with the complete independent stationary-phase imaginary field and its associated unparameterized whole-plane space-filling curve.

Proof. We use [7, Definition 2.2], whose three-marked construction agrees with the ordinary DMS sphere by [1, Theorem 1.1 and Remark 1.2]. Write G_1 for the whole-plane GFF pinned on the unit circle, and let d be the deterministic Liouville drift in that definition. The sphere field is obtained by a positive finite integrable weighting of

$$N(G_1 + d), \quad N(H) = H - \gamma^{-1} \log \mu_H(\mathbb{C}).$$

The drift has logarithmic singularities only at $0, 1, \infty$; on compact subsets avoiding $0, 1$ it has finite Dirichlet energy. In particular, the possible crease of $\log(|z| \vee 1)$ at the unit circle causes no difficulty for Cameron–Martin comparison.

Couple the pinned GFFs by $G_1 = G_R - (G_R)_1(0)$. Since $N(H + c) = N(H)$, including for a field-dependent scalar c , the normalized field is also $N(G_R + d)$. The original weighting is still a positive finite integrable density on this Gaussian probability space, although its formula after changing the pin need not be the original one. It therefore suffices first to prove the domination for the unweighted field $N(G_R + d)$.

Choose $g \in C_c^\infty(B_R(0))$ with $0 \leq g \leq 1$, equal to one on a neighborhood of $\bar{U}$. Gaussian orthogonal decomposition in the Cameron–Martin direction g gives $G_R = Xg + G^\perp$, where X is a nondegenerate real Gaussian independent of $G^\perp$. Conditional on $G^\perp$, set $\nu = \mu_{G^\perp+d}$ and

$$A(x) = \int_{\mathbb{C}} e^{\gamma x g(z)} \nu(dz), \quad T(x) = x - \gamma^{-1} \log A(x).$$

Almost surely ν is finite, nonzero, and assigns positive mass to an open set outside $\text{supp } g$. These assertions follow from the corresponding properties of μ_{G_R+d} , since removing the bounded function Xg changes that measure by a bounded positive factor. The same comparison applies to every finite value of x . Differentiation under the integral on compact x -intervals yields

$$T'(x) = \frac{\int_{\mathbb{C}} (1 - g(z)) e^{\gamma x g(z)} \nu(dz)}{\int_{\mathbb{C}} e^{\gamma x g(z)} \nu(dz)} > 0.$$

Thus T is a strictly increasing C^1 diffeomorphism onto its image. Conditional on $G^\perp$, the law of $T(X)$ has a Lebesgue density and hence is absolutely continuous with respect to the nondegenerate Gaussian law of X . On U ,

$$N(G_R + d)|_U = G^\perp|_U + d|_U + T(X).$$

Integrating the preceding conditional domination proves

$$\text{Law}(N(G_R + d)|_U) \ll \text{Law}((G_R + d)|_U).$$

A finite-energy cutoff equal to d near $\bar{U}$, supported away from $0, 1$ and the pinning circle, is a Cameron–Martin shift of G_R . Its restriction removes $d|_U$ by mutual absolute continuity. Restoring the positive weighting proves the first assertion.

For the comparison with the cone, [13, Definition 4.10 and the following discussion] identifies its field inside the unit disk with $G_1 - \gamma \log |z|$. Under the deterministic coordinate map $z \mapsto z/R$, its representation on $B_R(0)$ is

$$G_R - \gamma \log |z| + (\gamma - Q) \log R.$$

On $\bar{U}$, the added deterministic function has a finite-energy cutoff away from 0 and $\partial B_R(0)$. Cameron–Martin gives mutual absolute continuity with $G_R|_U$. In particular the direction needed for transferring a cone probability-one event is

$$\text{Law}(h|_U) \ll \text{Law}(G_R|_U) \ll \text{Law}(h^{\text{cone}, R}|_U).$$

No comparison across the cone’s pinning circle is required.

Finally, the complete imaginary field is independent of the real field and has the same stationary-phase law in both constructions. Taking the product with this common law, and then taking measurable images to obtain its curve, preserves each asserted domination. $\square$

Remark 3.4 (Auxiliary marks and affine charts). In a singly marked sphere with exploration root $p_* = \infty$, sample two auxiliary points Q_1, Q_2 independently from quantum area, conditionally on the surface and independently of the imaginary field. Sending them to $0, 1$ by the unique complex affine map gives the three-marked embedding above, by [7, Lemma 2.3] and [13, Proposition A.13]. Explicitly, if $\psi(z) = Q_1 + (Q_2 - Q_1)z$ maps the new coordinate to the old one, then $h_{\text{new}} = h \circ \psi + Q \log |\psi'|$. The affine map is independent of the imaginary field; conditional on the real surface and marks, affine invariance of the whole-plane space-filling SLE and the stationary imaginary-field convention preserve its complete law. Thus the transformed imaginary field remains independent of the transformed real field, before quantum-area parametrization.

A second independent auxiliary pair Q_3, Q_4 gives another such chart with the same exploration root. The two pairs are disjoint almost surely. Every point other than p_* is ordinary in at least one chart, so countably many compactly buffered patches in the two charts cover the punctured sphere. This is a local covering assertion; it does not assert that every large curve interval lies in one ordinary patch.

3.3 The reference intrinsic metric

The metric below is the already constructed LQG metric. Later we compare an extracted discrete passage rule with it. At present that passage rule has neither an established metric property nor a Weyl-scaling law.

Theorem 3.5 (Metric inputs). *For the fixed subcritical parameter, the LQG metric D_h is a continuous length metric inducing the Euclidean topology. In whole-plane GFF coordinates it is proper and geodesic, and geodesics between each fixed pair of distinct points are almost surely unique. Internal metrics are local in the field. Conformal covariance and Weyl scaling hold: adding a continuous function f multiplies path length locally by $e^{\xi f}$.*

For a compact set K in an ordinary field chart and $0 < \chi < \xi(Q - 2)$, there are almost surely finite random constants controlling connections between nearby points of K inside Euclidean balls of a fixed multiple of their separation by $C|z - w|^\chi$. The corresponding internal diameters of small dyadic squares, with a bounded enlargement still in the chart, have the same type of bound. Fixed boxes of bounded aspect ratio admit access to their flat faces by completion of the internal metric.

In whole-plane GFF coordinates, strong confluence holds simultaneously for all geodesics: an interior segment of any geodesic persists in all geodesics whose endpoints sufficiently closely approximate slightly longer interior subsegments. The same local assertion holds for short geodesic subarcs in buffered ordinary surface charts. In particular, every nonempty open geodesic subarc contains a segment of a geodesic between points of a deterministic countable dense set.

References and use. Metric construction, locality, and Weyl scaling are supplied by [23]; properness is [11, Lemma 3.8], and general conformal coordinate changes are [22, Theorem 1.3]. Fixed-pair geodesic uniqueness is [32, Theorem 1.2]. The internal connection and dyadic-square estimates are [11, Lemma 3.20, Equations (3.66)–(3.67)]. Strong confluence and its interior-endpoint form are [6, Theorem 2 and Proposition 6]. Local statements transfer to fixed compact subsets of ordinary cone or sphere charts by local absolute continuity. For confluence, first take a shorter geodesic subarc in a buffered chart. Its sufficiently short neighboring connections cannot reach the chart boundary, whose internal distance from a compact inner neighborhood is positive. They are consequently determined by the same internal metric, so the whole-plane confluence event applies locally. This does not require absolute continuity of an entire surface’s global geodesics. Singular marked points will be treated separately.

For clarity, the open-square internal bound in [11, Proposition 3.10] does not by itself include boundary points. To access a specified flat-face or corner point of a rectangle of size δ , choose inward points at Euclidean distances comparable to $2^{-j}\delta$. Consecutive inward points can be joined inside the rectangle by a bounded chain of balls or dyadic squares of that scale. The internal connection bound gives total cost at most

$$C \sum_{j \geq 0} (2^{-j}\delta)^\chi = \frac{C\delta^\chi}{1 - 2^{-\chi}}.$$

The chain is Cauchy for the internal metric. Its Euclidean limit is the prescribed boundary point, and the same inward-chain construction between two approximations shows uniqueness in the internal completion. A bounded number of these connections also crosses any fixed-shape polygonal collar while staying inside that collar. No bound uniform over degenerating aspect ratios is claimed.

These properties are asserted only for D_h . No analogous Weyl rule, metric property, or endpoint modulus is assumed for a subsequential discrete passage rule. $\square$

4 Fresh inventory blocks and Brownian contacts

We first establish the contour estimates used in local graph reconstruction and in the later control of discrete endpoints. A fresh inventory block has the same Brownian limit uniformly over its incoming stack. Brownian survival estimates then rule out specific contact configurations, including contacts whose endpoints are chosen after the contour is observed. All parameters are fixed as in Equation (3.1); no passage metric or graph-distance tightness is used in this section.

4.1 Bounds determined by a fresh raw word

Write the inventory alphabet as $\{h, c, H, C, F\}$, with h, c producing the two burger types, H, C ordering the corresponding type, and F ordering the freshest available burger. The symbol probabilities are $1/4, 1/4, (1-b_F)/4, (1-b_F)/4, b_F/2$, respectively. For a finite word $w = w_1 \cdots w_m$, let $R(w)$ be its reduction from an empty burger stack, retaining unsatisfied orders permanently. Set

$$L_k(w) = |R(w_1 \cdots w_k)|, \quad M_m(w) = \max_{0 \leq k \leq m} L_k(w), \quad F_k(w) = N_F(R(w_1 \cdots w_k)).$$

An *exterior typing* σ assigns either type to every F which this procedure leaves unmatched. Every other F retains the type of its internally matched burger. Let $Z^\sigma(0) = 0$ and give each burger the increment e_i of its type and each resolved order the increment $-e_i$. We interpolate linearly between integer times. The exterior typing may depend on the whole word and on auxiliary data. Every genuine initial past induces an exterior typing.

Lemma 4.1 (Deterministic reduction identities). *For every finite word and every exterior typing,*

$$L_k(w) = \sum_{i=1}^2 \left(Z_i^\sigma(k) - 2 \min_{0 \leq j \leq k} Z_i^\sigma(j) \right). \quad (4.1)$$

Consequently,

$$M_m(w) \leq 2 \sum_{i=1}^2 \left(\max_{0 \leq j \leq m} Z_i^\sigma(j) - \min_{0 \leq j \leq m} Z_i^\sigma(j) \right), \quad (4.2)$$

$$M_m(w) \leq 6 \max_{i \in \{1,2\}, 0 \leq j \leq m} |Z_i^\sigma(j)|, \quad (4.3)$$

$$\max_{0 \leq j, k \leq m} \|Z^\sigma(k) - Z^\sigma(j)\|_1 \leq 2M_m(w). \quad (4.4)$$

For finite words u, v ,

$$|R(uv)| \leq |R(u)| + |R(v)|. \quad (4.5)$$

Finally, $F_k(w)$ is nondecreasing in k , and for any two exterior typings σ, τ ,

$$\max_{0 \leq k \leq m} \|Z^\sigma(k) - Z^\tau(k)\|_1 \leq 2F_m(w). \quad (4.6)$$

Proof. Every burger produced inside the word is fresher than every burger in an initial past. Induction over symbols therefore shows that the available internal burgers and their internal matching are independent of that past. A typed order uses an internal burger of its type whenever one is available. An F uses the freshest internal burger whenever the internal stack is nonempty. A step which instead uses the past does not change the internal stack. In particular, each externally unmatched F occurs when the entire internal burger stack is empty. Replacing only these F 's by arbitrary typed orders therefore preserves every internal pair and every unpaired position.

In the resulting fully typed word let $O_i(k)$ and $B_i(k)$ count the unpaired orders and burgers of type i . An order is unpaired exactly when the corresponding coordinate reaches a new negative running minimum. Thus

$$O_i(k) = - \min_{0 \leq j \leq k} Z_i^\sigma(j), \quad B_i(k) = Z_i^\sigma(k) - \min_{0 \leq j \leq k} Z_i^\sigma(j).$$

Summing $O_i + B_i$ proves Equation (4.1). Since $Z^\sigma(0) = 0$, its summands are bounded by twice the corresponding coordinate oscillations and by three times the maximum absolute coordinate height. This proves Equations (4.2) and (4.3). Also $Z_i^\sigma = B_i - O_i$, so $\|Z^\sigma(k)\|_1 \leq L_k(w)$; the triangle inequality proves Equation (4.4).

For Equation (4.5), process v on the retained burger stack of u . All its internal pairs persist. Its orders which were internally unpaired either remain unpaired or consume a retained burger of u . Such a consumption removes two letters from the two separate reductions; no new letter is created.

An order unpaired in a prefix stays unpaired in every longer prefix, since orders only consume earlier burgers. Hence F_k is nondecreasing. The increments for two exterior typings can differ only at these unpaired F positions, and each difference has ℓ^1 norm two. Summing the increment differences gives Equation (4.6). $\square$

Proposition 4.2 (Exponential bound from square-root tightness). *For the iid raw word with fixed $b_F \in (0, 1/2)$ there are constants $c, C > 0$ such that, for every integer $m \geq 1$ and $x \geq 0$,*

$$\mathbb{P}[M_m > x\sqrt{m}] \leq Ce^{-cx}. \quad (4.7)$$

In particular, for some $\lambda > 0$ and $K < \infty$,

$$\sup_{m \geq 1} \mathbb{E} \exp\{\lambda M_m / \sqrt{m}\} \leq K. \quad (4.8)$$

The same bounds hold conditionally on any previously observed information under which the raw block remains iid with this law.

Proof. Adjoin an independent bilateral iid past and resolve the block using that past. Sheffield's invariance principle [38, Theorem 2.5], equivalently [20, Theorem 1.5], gives tightness of the family

$$m^{-1/2} \max_{i,j \leq m} |Z_i(j)|.$$

Equation (4.3) gives tightness of the block-only random variables $M_m/\sqrt{m}$. Choose $A > 0$ such that for every $m \geq 1$, with $a_m = \lceil A\sqrt{m} \rceil$,

$$\mathbb{P}[M_m \geq a_m] \leq \frac{1}{2}.$$

Fix m , write $a = a_m$, and, in an infinite iid raw word, restart whenever the current block first has reduced length a . More precisely, let $S_0 = 0$ and

$$S_{r+1} = \inf\{t > S_r : |R(X_{S_r+1} \cdots X_t)| = a\}.$$

A single added symbol changes reduced length by one in absolute value, so these stopping times have no overshoot. They are finite almost surely: reduced length dominates absolute net burger count, which is a simple symmetric random walk. Strong Markov for the raw iid symbols makes $S_r - S_{r-1}$ iid. Put $N_m = \max\{r : S_r \leq m\}$. By Equation (4.5), every prefix up to time m has reduced length at most $a(N_m + 1)$. Therefore, for integers $r \geq 0$,

$$\mathbb{P}[M_m > ra] \leq \mathbb{P}[N_m \geq r] \leq \prod_{j=1}^r \mathbb{P}[S_j - S_{j-1} \leq m] \leq 2^{-r}.$$

Since $a_m/\sqrt{m} \leq A + 1$, take $r = \lfloor x\sqrt{m}/a_m \rfloor$ to obtain Equation (4.7) with $C = 2$ and $c = \log(2)/(A + 1)$. For $0 < \lambda < c$, integration of the tail gives $K \leq 1 + 2\lambda/(c - \lambda)$. The estimate depends only on the raw block, proving its conditional version. Iterating this conditional moment estimate also bounds the exponential moment of a sum over successively exposed fresh blocks whose lengths are chosen before the respective blocks are inspected. $\square$

Sheffield already observed that the invariance principle remains valid for an arbitrary fixed initial burger stack [38, Remark 3.17]. The next statement uses the quantitative unmatched-order estimate of [20, Lemma 3.7] to make the comparison uniform over randomized, word-dependent exterior typings.

Proposition 4.3 (Uniformity over exterior typings). *Fix $b_F \in (0, 1/2)$. Let ν_{b_F} be the law on $C([0, 1], \mathbb{R}^2)$ of Brownian motion started from zero with covariance matrix at time t equal to*

$$\frac{t}{2} \begin{pmatrix} 1 - b_F & b_F \\ b_F & 1 - b_F \end{pmatrix}.$$

For the bounded-Lipschitz distance associated with the uniform ℓ^1 path metric,

$$\sup_{\sigma} d_{\text{BL}} \left(\mathcal{L}(m^{-1/2} Z^{\sigma}(m \cdot)), \nu_{b_F} \right) \longrightarrow 0, \quad (4.9)$$

where σ ranges over all possibly randomized, word-dependent exterior typings. The convergence holds conditionally, uniformly over previously observed data which leave the block fresh and iid. A fixed finite family of deterministic disjoint blocks, whose lengths each tend to infinity, converges jointly to independent Brownian paths after applying the respective square-root normalizations.

Proof. The only additional public input is [20, Lemma 3.7, Equation (3.15)]. Put

$$\mu' = \frac{\pi}{2(\pi + \arctan(\sqrt{1 - 2b_F/b_F}))} \in (1/3, 1/2).$$

That lemma says that for each $\nu > \mu'$, $\mathbb{P}[\exists i \geq n : F_i \geq i^\nu]$ decays faster than every negative power of n . Choose $\nu \in (\mu', 1/2)$. It follows that $F_m/\sqrt{m} \rightarrow 0$ in probability. Its monotonicity, already proved in Lemma 4.1, simultaneously controls all prefixes.

Adjoin an independent bilateral past, and let τ be its genuine typing. Theorem 2.5 of [38] gives $m^{-1/2}Z^\tau(m \cdot) \Rightarrow \nu_{b_F}$. For each bounded 1-Lipschitz test with absolute value at most one, Equation (4.6) bounds the difference between its expectations under σ and τ by

$$\mathbb{E} [\min\{2, 2F_m/\sqrt{m}\}] \rightarrow 0.$$

This bound is simultaneous over all σ , proving Equation (4.9). Choose the auxiliary bilateral past independently of the previously observed data to obtain the conditional statement. For deterministic disjoint blocks choose independent auxiliary bilateral pasts for the respective blocks; their reference contours are independent, and the same deterministic error bound proves the finite-product assertion. $\square$

Remark 4.4 (Scope of fresh-block estimates). Starting a block after a stopping time in the raw-symbol filtration preserves the needed iid law. Conditioning on a block's own stopping duration, an exact terminal displacement, or a finite-word completion event generally does not. Such conditioning requires a separate change-of-density argument. The preceding statements only permit arbitrary assignments to the raw-unmatched flexible orders; they do not permit changing the type of an internally matched flexible order. All constants and limits concern fixed $b_F < 1/2$.

The raw-word estimates control contour fluctuations without conditioning on how the block is completed. We next obtain the one- and two-wall probability bounds needed for contacts and for the later address argument.

4.2 Brownian survival and landing

Let $Z = (Z_1, Z_2)$ be a centered Brownian motion with nondegenerate covariance, equal coordinate variances, and correlation $\rho = -\cos(\pi\gamma^2/4)$. A fixed change of time normalizes the variances. Write

$$\theta = \frac{\pi\gamma^2}{4} \in (\pi/2, \pi), \quad d_0 = 0, \quad d_1 = \frac{1}{2}, \quad d_2 = \frac{\pi}{2\theta}.$$

Whitening maps the positive quadrant to a wedge of opening θ . All constants in this subsection can depend on the fixed covariance.

Lemma 4.5 (A wedge survival bound). *Let W be a planar wedge of opening $\theta \in (\pi/2, \pi)$, let $\nu = \pi/\theta$, and let τ_W be the first exit time of standard Brownian motion. For $0 < t_0 \leq t \leq t_1 < \infty$ there is a constant C such that*

$$\mathbb{P}_z(\tau_W > t) \leq C \min\{1, |z|^\nu\}, \quad z \in W.$$

For one-dimensional Brownian motion killed at zero, the corresponding bound is $C \min\{1, z\}$.

Proof. Brownian scaling reduces the assertion to $t = 1$. Put $D = W \cap B(0, 1)$ and use polar coordinates (r, ϕ) , with $0 < \phi < \theta$. The probability $q(r, \phi)$ of leaving D through its circular arc before a side has the harmonic expansion

$$q(r, \phi) = \frac{4}{\pi} \sum_{\substack{m \geq 1 \\ m \text{ odd}}} \frac{r^{m\nu}}{m} \sin(m\nu\phi).$$

For $r < 1$ this is the harmonic extension of the sine series of the constant function one on the circular arc, with zero side values. One can first take partial sums on smaller concentric sectors and then use uniqueness of bounded harmonic extensions. The two corner points have harmonic measure zero. Absolute summation gives $q(r, \phi) \leq Cr^\nu$ for $r \leq 1/2$; the bound for larger r follows from $q \leq 1$.

For the exit time of D , set

$$f(\phi) = \frac{1}{2} \left(\frac{\cos(2\phi - \theta)}{\cos \theta} - 1 \right).$$

Since $\cos \theta < 0$ and $\cos(2\phi - \theta) \geq \cos \theta$, one has $f \leq 0$. The function vanishes at the two sides and $f'' + 4f = -2$. Choose $A < \infty$ such that $-f(\phi) \leq A \sin(\nu\phi)$; the ratio has finite limits at the two endpoints because both numerator and denominator vanish linearly. Then

$$v(r, \phi) = Ar^\nu \sin(\nu\phi) + r^2 f(\phi)$$

is nonnegative on D , satisfies $-\frac{1}{2}\Delta v = 1$, and obeys $v \leq Ar^\nu$. Nonnegativity follows from $\nu < 2$ and $r^2 f \geq r^\nu f$. Applying the stopped Itô formula on compact subdomains, then increasing the subdomains to D , gives $\mathbb{E}_z \tau_D \leq v(z) \leq Ar^\nu$. Therefore

$$\mathbb{P}_z(\tau_W > 1) \leq q(z) + \mathbb{P}_z(\tau_D > 1) \leq Cr^\nu.$$

The one-dimensional assertion follows from the reflection principle. $\square$

Lemma 4.6 (Uniform killed landing). *Constrain $k \in \{1, 2\}$ coordinates of Z to be nonnegative, and let τ_k be the first violation. For $0 < h \leq 1$,*

$$\sup_{z \in [0, h]^k} \mathbb{P}_z(\tau_k > 1) \leq Ch^{2d_k}, \quad (4.10)$$

$$\sup_{z \in [0, \infty)^k} \mathbb{P}_z(\tau_k > 1, Z(1) \in [0, h]^k) \leq Ch^{k+2d_k}. \quad (4.11)$$

For $k = 1$, only the constrained coordinate is included in the starting and landing vectors. The same bounds hold on any fixed compact range of positive durations, and conditionally when the starting position and the constant lower walls are measurable before the Brownian increments are sampled.

Proof. Whitening and Lemma 4.5 prove Equation (4.10). Let $p_t^D(x, y)$ be the killed transition density in the half-line or whitened wedge. Its symmetry, the semigroup identity, and the bounded free Gaussian transition density imply

$$p_1^D(x, y) = \int_D p_{1/2}^D(x, z) p_{1/2}^D(z, y) dz \leq C \int_D p_{1/2}^D(z, y) dz = C \mathbb{P}_y(\tau_D > 1/2).$$

Integrating over the whitened landing set, which has volume $O(h^k)$ and lies at distance $O(h)$ from the corner, proves Equation (4.11), uniformly in x . Scaling handles the duration range. Conditioning on past-measurable starting data leaves the same Brownian increment law, proving the last assertion. $\square$

Lemma 4.7 (Fresh lattice slots). *Fix $\delta > 0$. For every sufficiently large fixed R , a fresh raw inventory block of length $r(1 + O(R^{-1}))$ satisfies the following conditional bounds, uniformly over its raw past, once r is sufficiently large. Put $h = R^{-1/2+\delta}$ and let the lower walls be measurable before the block is exposed.*

- (i) *Starting within $O(h\sqrt{r})$ of k walls and remaining above them throughout the block has probability at most $CR^{-(1-2\delta)d_k}$, with a further arbitrarily small exponent loss allowed.*
- (ii) *Starting anywhere above the walls, remaining above them, and landing within $O(h\sqrt{r})$ of all k walls has probability at most $CR^{-(1-2\delta)(k/2+d_k)}$, with the same convention.*

The constants in the $O(h\sqrt{r})$ tolerances are fixed before R .

Proof. For fixed R , Proposition 4.3 gives uniform weak convergence of the block increment paths. Enlarge all path and landing tolerances by a fixed small multiple of h ; the corresponding closed Brownian events bound the desired limsup. Their probabilities are bounded by Lemma 4.6, with a changed constant. Uniformity over starting points in a fixed compact set follows by a finite net of starting heights and one more tolerance enlargement. For the landing test, starting points outside a sufficiently large compact set require a correspondingly large block fluctuation; Proposition 4.2 makes that probability smaller than any prescribed fixed power of R by choosing the compact set after fixing R . Starting below a wall has probability zero unless an already allowed tolerance covers the discrepancy. Finally choose the block length large enough that the uniform weak-convergence error is less than the desired small power of R . This proves the stated bounds, including conditioning on a fresh raw past. $\square$

The following generic-contact statements also apply to the protected charts. Their proofs use only the Brownian estimates just established and Gaussian increments between separated deterministic windows.

4.3 Generic Brownian contacts

Throughout this subsection, $Z = (L, R)$ is a two-sided Brownian motion with unit coordinate variances and correlation $\rho \in (0, 1)$. A common constant variance factor changes none of the assertions. Set

$$\theta = \arccos(-\rho) \in (\pi/2, \pi), \quad d_2 = \frac{\pi}{2\theta} > \frac{1}{2}.$$

For a continuous scalar function f , a *horizontal chord* is a pair $s < t$ such that

$$f(s) = f(t) = \min_{[s,t]} f.$$

A *branch triple* consists of three distinct times $a < b < c$ for which $f(a) = f(b) = f(c) = \min_{[a,c]} f$. Thus its middle time is a local minimum. All the following assertions concern nonzero chord spans and strictly positive witnessing time intervals.

Lemma 4.8 (Images of Brownian record times). *Fix a compact time interval K and $\delta > 0$. The surrounding Brownian path is retained far enough to include every arm of length δ used below. For signs $\sigma, \tau \in \{-1, 1\}$, let*

$$T_{\sigma,\tau} = \left\{ t \in K : \begin{array}{l} L(t + \sigma u) \geq L(t), \\ R(t + \tau u) \geq R(t) \end{array} \text{ for every } 0 \leq u \leq \delta \right\}.$$

If $\sigma = \tau$, the upper box dimensions of $Z(T_{\sigma,\tau})$ and of either coordinate projection are at most $2(1 - d_2) < 1$. If $\sigma \neq \tau$, these upper box dimensions are zero. Furthermore, almost surely there is no time $t \in K$ at which L has lower-wall arms of length δ on both sides and R has such an arm on either one of the two sides. The same statement holds with L, R interchanged.

Proof. We first record the required survival bounds. For fixed $v > 0$ and $0 < r \leq 1$, the reflection principle gives

$$\mathbb{P} \left[\inf_{0 \leq u \leq v} L(u) \geq -r \right] \leq C_v r.$$

Whitening the covariance sends the two lower walls to a wedge of opening θ . Lemma 4.5 and Brownian scaling give

$$\mathbb{P} \left[\inf_{0 \leq u \leq v} L(u) \geq -r, \inf_{0 \leq u \leq v} R(u) \geq -r \right] \leq C_v r^{2d_2}. \quad (4.12)$$

Only this elementary Brownian survival estimate is used here; no lattice or passage-kernel estimate is involved.

Partition K into intervals $I = [s, s + h]$ of length $h = 2^{-n}$, allowing two extra intervals at its ends. Fix $\varepsilon \in (0, 1/2)$ and put $r = h^{1/2-\varepsilon}$. Almost surely, for all sufficiently large n , both coordinate oscillations on every such interval are at most r . For example this follows from Brownian Hölder continuity with exponent $1/2 - \varepsilon/2$. Suppose such an interval contains a record time t . If a required arm points forward, then its constraint implies

$$\inf_{0 \leq u \leq \delta/2} \{Z_i(s + h + u) - Z_i(s + h)\} \geq -r;$$

if it points backward, the analogous constraint is

$$\inf_{0 \leq u \leq \delta/2} \{Z_i(s - u) - Z_i(s)\} \geq -r.$$

These implications hold when $h \leq \delta/2$. They use deterministic walls $-r$ relative to the respective cell endpoints, so their probability bounds do not condition on a selected record time.

For two arms on the same side, Equation (4.12) bounds this necessary exterior event by $C r^{2d_2}$. For arms on opposite sides the two events use disjoint time increments, and the reflection principle bounds their product by $C r^2$. In the last assertion of the Lemma, the necessary exterior event uses two coordinate arms on one side and one coordinate arm on the other; its probability is at most $C r^{2d_2+1}$.

Here is the resulting covering calculation, including the dimension needed below. If the survival power is α , the number N_h of cells satisfying the necessary exterior event obeys

$$\mathbb{E} N_h \leq C h^{-1} r^\alpha = C h^{-1+(1/2-\varepsilon)\alpha}. \quad (4.13)$$

For $\alpha \leq 2$ and every $\zeta > 0$, Markov's inequality and Borel–Cantelli imply, along the dyadic sequence,

$$N_h \leq h^{-1+(1/2-\varepsilon)\alpha-\zeta} \quad \text{eventually almost surely.}$$

Each occupied cell contributes a set of diameter at most $\sqrt{2}r$ to the Z -image and a set of diameter at most r to a coordinate image. Consequently their upper box dimensions are at most

$$\frac{1 - (1/2 - \varepsilon)\alpha + \zeta}{1/2 - \varepsilon}.$$

Letting ε, ζ decrease to zero through countable sequences gives $2 - \alpha$. Taking $\alpha = 2d_2$ or $\alpha = 2$ proves the two image assertions. If $\alpha = 2d_2 + 1 > 2$, choose ε so small that $(1/2 - \varepsilon)\alpha > 1$. The right side of Equation (4.13) is summable over dyadic h , so eventually there are no candidate cells. This proves the nonexistence assertion. $\square$

Lemma 4.9 (Generic horizontal contacts). *Almost surely, simultaneously on all compact time intervals:*

- (i) *The local minima of each coordinate are strict, countable, and have pairwise distinct values. Every scalar horizontal equivalence class has at most three distinct times.*
- (ii) *No nontrivial horizontal chord is a chord of both L and R .*
- (iii) *No member of a branch triple of one coordinate is an endpoint of a nontrivial horizontal chord of the other coordinate.*

The assertions also hold under any law having joint local absolute continuity with these Brownian paths, including their relative coordinate heights on separated windows. In particular they hold in strict interior windows of a nondegenerate correlated Brownian quadrant excursion.

Proof. We shall use the following precise source of the height densities. Take deterministic intervals $I = [p, q]$ and $J = [r, v]$ with $q < r$. The two paths

$$(Z(t) - Z(q))_{t \in I}, \quad (Z(t) - Z(r))_{t \in J}$$

and the increment $G = Z(r) - Z(q)$ are independent. The last has a nonsingular Gaussian density on $\mathbb{R}^2$. Its coordinate projections have Gaussian densities on $\mathbb{R}$. Thus, conditional on the two displayed paths, a prescribed Lebesgue-null set of relative heights is missed almost surely. Conditioning is only on increments inside the two separated windows; it is never on a putative contact or chord.

For completeness this also proves the scalar minimum facts. The minimum values on two disjoint compact rational intervals have probability zero to agree: express the left minimum relative to its right endpoint and the right minimum relative to its left endpoint, and condition on both window paths. Equality would prescribe one value for the independent scalar Gaussian increment in the gap. Taking a countable union shows that no such pair of rational intervals has equal minima. Two distinct minimizers on a compact rational interval could be enclosed in disjoint compact rational subintervals of that interval with the same minimum, so each rational compact interval has a unique minimizer. Every local minimum is the minimizer on one of these intervals. This makes it strict, makes the set countable, and also shows that distinct local minima have distinct values. If a scalar horizontal class contained four ordered visits, its middle two visits would be distinct local minima at the same height. This proves (i).

Suppose that $s < t$ were a chord in both coordinates. Choose disjoint rational windows I, J containing s, t in their interiors, and a positive rational δ small enough that the forward arms from s and the backward arms from t of length δ lie in these windows. Such choices are countable and cover every proposed nonzero chord. Define compact sets of relative heights using all these local joint record times in the respective windows:

$$A = \{Z(u) - Z(q) : u \in I \text{ has both forward arms of length } \delta\},$$

$$C = \{Z(w) - Z(r) : w \in J \text{ has both backward arms of length } \delta\}.$$

Here the record times are restricted to closed subintervals leaving the indicated arm margins; countably many choices of these subintervals suffice. By Lemma 4.8, both sets have upper box dimension at most $2(1 - d_2) < 1$. Hence their difference set $A - C$ has two-dimensional Lebesgue measure zero. Explicitly, for any $a \in (2(1 - d_2), 1)$ each set has an ϵ -cover with $O(\epsilon^{-a})$ balls; pairwise differences cover $A - C$ with $O(\epsilon^{-2a})$ balls, whose total area tends to zero. The endpoint equality $Z(s) = Z(t)$ would imply $G \in A - C$. The conditional Gaussian density therefore excludes the proposed common chord, proving (ii).

Now let $a < b < c$ be an L -branch triple, and suppose an R -chord has an endpoint $t \in \{a, b, c\}$ and another endpoint $u \neq t$. If $t = b$, the L -coordinate has lower-wall arms on both sides of b , and the R -chord supplies a lower-wall arm on the side pointing toward u . Truncate all three arms to a common positive rational length. The final assertion of Lemma 4.8 excludes this case, since $d_2 + 1/2 > 1$.

If $t = a$ or $t = c$, retain a small window around t disjoint from a small window around b . At t the L -arm points toward b , and the R -arm points toward u . Truncate both to a common positive rational length so that they lie in the outer-visit window. They can point in the same direction or in opposite directions; both possibilities are included. In relative L -heights, let A be the image of all local joint record times of this specified orientation in the outer-visit window. Lemma 4.8 makes A a one-dimensional Lebesgue-null set: its upper box dimension is at most $2(1 - d_2) < 1$ in the first case and zero in the second.

In the middle-visit window let C be the countable set of minimum values on its compact rational subintervals, measured relative to the window's reference endpoint. Since b is a local minimum, its relative L -height belongs to C . Use the right endpoint as reference for the earlier window and the left endpoint for the later window. The equality $L(t) = L(b)$ would force the scalar Gaussian increment between the windows to belong to either $A - C$ or $C - A$. Both are Lebesgue-null, being countable unions of translates of a null set. Conditional on the two window paths the gap increment has a density, so the equality has probability zero. Countably many rational windows, arm lengths, and orientations cover every outer contact. This proves (iii); interchanging the two coordinates gives the other color. The argument allows u to equal another member of the branch triple, as well as to lie between or outside its visits.

Every exclusion just proved is a null event specified in one window or in finitely many separated windows together with their relative heights. It therefore transfers under the stated joint local absolute continuity. On strict interior windows a Brownian quadrant excursion has this absolute continuity by its killed transition densities and the Markov bridge construction, first restricting to compact positive endpoint ranges and then exhausting them. The excursion's common initial/final endpoint is outside the assertion. $\square$

Remark 4.10 (Scope of the generic-contact assertion). For an L -branch triple $a < b < c$, assertion (i) also gives $L > L(b)$ on $(a, b) \cup (b, c)$ and lower values arbitrarily close just outside a and c . Indeed an additional interior visit would give four visits, while failure of an outside lower dip would make a or c a second local minimum at the height of b . Adjacent contacts are consequently approached by ordinary chords at heights decreasing to $L(b)$ from above, and the outer contact is approached by chords at heights increasing to $L(b)$ from below. This is a scalar consequence of continuity and these strict inequalities.

The proof does not exclude a time that is incident to ordinary chords of both colors with different other endpoints. It excludes a common pair and any other-color chord at a scalar branch triple. In applying the absolute-continuity clause, separate marginal absolute continuity of the window increments is insufficient: the relative heights must have the asserted joint law. No

statement is made at a singular center where that joint local absolute continuity is unavailable.

For later finite collections of viewed contour pieces, we need a version which charges every possible choice of contact time. The next argument works directly with Brownian pieces and their relative heights.

4.4 Polarity of finite Brownian chord diagrams

Fix a positive definite covariance matrix Σ whose two coordinate variances agree and whose correlation is $\rho \in (0, 1)$. Put

$$d_0 = 0, \quad d_1 = \frac{1}{2}, \quad d_2 = \frac{\pi}{2 \arccos(-\rho)}, \quad \beta = \frac{1}{2} + d_2 > 1.$$

The argument below uses only Brownian survival and landing estimates. It does not use a lattice arm estimate, a random address code, or a passage kernel.

Let $J_i = [a_i^-, a_i^+]$, $1 \leq i \leq m$, be pairwise disjoint compact intervals in increasing order, and let K_i be nonempty compact subintervals of their interiors. Fix $s > 0$ smaller than half the distance of every K_i from ∂J_i . On each J_i take

$$X_i(t) = A_i + B_i(t - a_i^-),$$

where the B_i are independent Brownian motions with covariance Σ . The vector $(A_1, \dots, A_m)$ may have any law independent of these Brownian motions. For absolute-continuity comparisons one may take the A_i to be independent nonsingular Gaussian vectors.

A *completion* of these pieces is a continuous function $f : [a_1^-, a_m^+] \rightarrow \mathbb{R}^2$ which agrees with X_i on every J_i . No probability law for the missing gaps is imposed. Let G_1, G_2 be graphs on $\{1, \dots, m\}$, with loops discarded, and write

$$e = \text{rank}(G_1) + \text{rank}(G_2), \quad \text{rank}(G_c) = m - \# \text{ components}(G_c),$$

where isolated vertices count as components. The graph is *realized* if there are $t_i \in K_i$ and a completion f such that, for each edge $\{i, j\}$ of G_c ,

$$f_c(t_i) = f_c(t_j) = \min_{[t_i \wedge t_j, t_i \vee t_j]} f_c. \quad (4.14)$$

Denote this event by $\mathcal{E}_G$. In particular, a chord tests every retained part of its span, including any intermediate retained windows. Requiring only equality of its endpoint heights would define a different event.

Lemma 4.11 (Finite-cell Brownian diagram bound). *For each $\varepsilon > 0$ there are $C_\varepsilon < \infty$ and $\delta_0 > 0$, depending on the fixed windows, m , Σ , and ε , such that for $0 < \delta < \delta_0$,*

$$\mathbb{P}(\mathcal{E}_G) \leq C_\varepsilon \delta^{\beta e - m - \varepsilon}. \quad (4.15)$$

More precisely, there are measurable events $\mathcal{F}_{G,\delta} \supset \mathcal{E}_G$ which satisfy the same bound. The constants are uniform in the law and values of the starting vectors A_i . Consequently $\mathcal{E}_G$ has probability zero whenever $\beta e > m$.

Proof. We first state the elementary Brownian estimates used in the proof. For a Brownian block of fixed duration s and $k \in \{1, 2\}$ constant lower walls, let τ_k be the first violation, in the constrained coordinates. For $0 < h \leq 1$,

$$\sup_{z \in [0, h]^k} \mathbb{P}_z(\tau_k > s) \leq C_s h^{2d_k}, \quad (4.16)$$

$$\sup_{z \in [0, \infty)^k} \mathbb{P}_z(\tau_k > s, Z(s) \in [0, h]^k) \leq C_s h^{k+2d_k}. \quad (4.17)$$

Fixed multiples of h change only C_s . Both bounds follow from Lemma 4.6, after translating the constant walls to zero and fixing the positive duration. The estimates remain valid conditionally when the walls and starting point are known before the block is exposed. An initial point below a tested wall has zero probability of satisfying the gate.

Choose $\vartheta \in (0, 1/2)$ with $2\beta e\vartheta < \varepsilon$, and put $\eta = \delta^{1/2-\vartheta}$. Cover each K_i by intervals of a deterministic mesh of width δ , retaining the finitely many mesh cells which meet K_i . For sufficiently small δ these cells, and an s -interval on either side of each cell, lie in J_i . Let $\mathcal{H}_\delta$ be the event that both coordinate oscillations on every such cell are at most η . The reflection principle and a union bound give

$$\mathbb{P}(\mathcal{H}_\delta^c) \leq C\delta^{-1} \exp\{-c\eta^2/\delta\} = C\delta^{-1} \exp\{-c\delta^{-2\vartheta}\}. \quad (4.18)$$

This bound is uniform in all starting vectors.

Fix one cell $[a_i, b_i]$ for each i . Suppose the graph is realized with $t_i \in [a_i, b_i]$, and suppose $\mathcal{H}_\delta$ holds. For one coordinate, the scalar minimum-chord relation is transitive. Indeed all endpoints in a connected component have the same height, and the union of the spans along a path between two vertices covers the interval between their times; every such span stays above that height. Thus, in each component of G_c , the chronologically consecutive selected times themselves form minimum chords. Use these sorted chains, with one comparison per unit of monochrome rank. Let k_i^- and k_i^+ be the numbers of colors with a predecessor and successor, respectively, at vertex i . Then

$$k_i^-, k_i^+ \in \{0, 1, 2\}, \quad \sum_i k_i^- = \sum_i k_i^+ = e. \quad (4.19)$$

Write $h_{i,c} = X_{i,c}(a_i)$.

Use the left slot $[a_i - s, a_i]$ and the right slot $[b_i, b_i + s]$. All these slots are disjoint. If color c has predecessor j at i , let H_c be their common actual chord height. Then $|h_{j,c} - H_c| \leq \eta$ and $|h_{i,c} - H_c| \leq \eta$. The entire left slot lies between the two actual chord endpoints, so, with the wall $w_{i,c}^- = h_{j,c} - \eta$,

$$\inf_{[a_i-s, a_i]} X_{i,c} \geq w_{i,c}^-, \quad 0 \leq X_{i,c}(a_i) - w_{i,c}^- \leq 3\eta. \quad (4.20)$$

If color c has a successor at i , the entire right slot likewise lies in that chord span. With $w_{i,c}^+ = h_{i,c} - \eta$, cell oscillation and the chord minimum give

$$\inf_{[b_i, b_i+s]} X_{i,c} \geq w_{i,c}^+, \quad 0 \leq X_{i,c}(b_i) - w_{i,c}^+ \leq 2\eta. \quad (4.21)$$

These are necessary gates involving only fixed-slot path restrictions, not a conditioning on the actual selected times or on $\mathcal{H}_\delta$.

Condition on all starting vectors and expose the independent Brownian pieces in the order $J_1, \dots, J_m$, each in increasing time. Every left wall in Equation (4.20) uses a representative in an earlier window. Every right wall in Equation (4.21) uses the already exposed left endpoint of

its cell. Their heights are therefore known before the respective slots. Applying Equations (4.16) and (4.17) and the tower property bounds the probability of all gates for this cell tuple by

$$C_m \eta^{\sum_i (k_i^- + 2d_{k_i^-} + 2d_{k_i^+})}.$$

The bound is uniform in the starting vectors, so it also holds before conditioning on them. Since $d_k \geq kd_2/2$ for $k \in \{0, 1, 2\}$, Equation (4.19) shows that the exponent is at least

$$e + 2d_2e = 2\beta e.$$

There are $O(\delta^{-m})$ tuples of cells. Let $\mathcal{F}_{G,\delta}$ be the union of the gate events for these tuples, together with $\mathcal{H}_\delta^c$. All its constituents are measurable, and the preceding deterministic implications give $\mathcal{E}_G \subset \mathcal{F}_{G,\delta}$. Consequently

$$\mathbb{P}(\mathcal{F}_{G,\delta}) \leq C\delta^{\beta e - m - 2\beta e\vartheta} + C\delta^{-1}e^{-c\delta^{-2\vartheta}}.$$

Equation (4.15) follows by the choice of ϑ ; the case $e = 0$ also follows directly from the trivial bound one. If $\beta e > m$, choose $\varepsilon \in (0, \beta e - m)$ and send δ to zero.

For completeness the realization event can also be taken measurable without selecting a completion. Replace every missing gap by affine interpolation in each coordinate. If any completion satisfies the chords, every gap traversed by a chord has both its endpoint values above that chord height, so affine interpolation also satisfies the chords. Thus a completion exists exactly when this fixed interpolation has a realizing tuple. Minimum values on compact intervals vary continuously with the path and with the interval endpoints. Taking the minimum of the total nonnegative chord discrepancies over the compact set $\prod_i K_i$ makes realization a measurable event. $\square$

Corollary 4.12 (Transfer and a finite class bound). *The polarity assertion of Lemma 4.11 holds under every law absolutely continuous with respect to the joint law of the pieces and their starting vectors described above. Under a density bound $L \leq K$, its measurable mesh-cover events obey*

$$\mathbb{Q}(\mathcal{F}_{G,\delta} \cap \{L \leq K\}) \leq KC_\varepsilon \delta^{\beta e - m - \varepsilon}.$$

Mere absolute continuity transfers polarity, without asserting the same polynomial bound for the mesh-cover events under the untruncated law.

Suppose a collection of eligible time points has a scalar chord relation in each color, and every fixed finite family of pairwise separated enlarged windows about distinct eligible points has this joint absolute-continuity property. Assume also that every tested finite chord diagram is realized by a continuous completion in a fixed order of those windows, or one of finitely many such orders. Then, almost surely, every class generated by the two chord relations has at most

$$\left\lfloor \frac{\beta}{\beta - 1} \right\rfloor \tag{4.22}$$

distinct eligible points, provided each finite connected subdiagram of that class uses only eligible points and can be localized in a countable family of such windows.

Proof. Absolute continuity transfers each null event. The quantitative truncated assertion is simply integration of the density over $\mathcal{F}_{G,\delta}$. There is no assertion of a uniform density bound without the displayed truncation.

For a connected union $G_1 \cup G_2$ on m vertices, take a spanning tree of the union. Its edges of either color form an independent set in that monochrome graph, so $e \geq m - 1$. If $m > \beta/(\beta - 1)$, then $\beta e \geq \beta(m - 1) > m$, and the diagram is polar. There are finitely many two-colored graphs on a fixed finite vertex set, and finitely many window orders. Countable buffered localization therefore excludes all such connected diagrams simultaneously. If a generated class had at least $m_0 = \lfloor \beta/(\beta - 1) \rfloor + 1$ eligible vertices, finite chains connecting them would give a finite connected graph on at least m_0 vertices. A spanning tree can be reduced by removing leaves to a connected subgraph on exactly m_0 vertices. Its edges are still actual chords, contradicting the preceding exclusion. This proves Equation (4.22). $\square$

Remark 4.13 (Separated windows on several lines). For finitely many off-center time points on finitely many lines, first localize each point in a compact rational window bounded away from any excluded center and from the other selected points on its line. Retain larger windows with positive buffers. The argument applies whenever their joint pair-path restrictions, including relative starting-height vectors, are absolutely continuous with respect to independent Brownian pieces with full-dimensional starting-height density. An unrecorded common additive vector is harmless: the chord event is translation invariant, and one can add an independent Gaussian common vector when comparing full starting-height laws. The chronological order must be fixed in the finite diagram or enumerated among its finitely many possibilities.

The completion requirement is substantive. It includes lower-wall conditions throughout every retained portion of a chord span; mere endpoint equalities or endpoint-only arms do not supply these tests. The assertion also requires joint absolute continuity of the pieces and offsets, rather than just Brownian marginal laws or a density for the offsets in isolation. No claim is made at a pinned center lacking this joint absolute continuity, or about chains through such a center.

5 Compact path records and protected charts

We now record discrete paths without assuming that their rescaled lengths or parametrizations are tight. Compactness gives subsequential path data. A protected chart then reconstructs every relevant primal incidence from finitely many raw word blocks and exact relative lattice heights. Together with the continuum sewing, the Brownian genericity and raw-word estimates established above supply the probabilistic inputs to this reconstruction.

5.1 Compact records of paths

A path is recorded by its endpoint times, compact range, and cost, together with finitely many ordered marks. This gives a compact state space before an endpoint modulus is available.

Fix a bounded time window $I = [-T, T]$, a deterministic positive distance denominator s_m , and the grid $m^{-1}\mathbb{Z} \cap I$. A vertex occurring on a graph path may be represented by any tour time at that vertex in the window. Both endpoint representatives of each used edge are retained. For a path with r marked consecutive subpaths, record:

$$(t_0, \dots, t_r; K_1, \dots, K_r; c_1, \dots, c_r; c).$$

Here $t_i \in I$ are the endpoint representatives, K_i is the finite set of representing times used in the i -th subpath, including its endpoints, c_i is an upper bound on its length divided by s_m , and $c \geq \sum_i c_i$. Costs lie in the compact space $[0, \infty]$. An empty subpath has the singleton range consisting of its endpoint representative. At repeated marks one uses the same chosen endpoint

representative. A zero-length change of representative retains both times and their equality as a vertex of the chart graph. Used step identities and equivalences are always those of this graph. Winding and transverse-crossing tests additionally retain their finite ordered marks and open ports; they are not inferred from an unordered compact range alone.

Let $\mathcal{K}(I)$ be the space of nonempty compact subsets of I , with Hausdorff distance. Taking the closure of all such records gives a closed subset of

$$\mathcal{X}_{T,r} = I^{r+1} \times \mathcal{K}(I)^r \times [0, \infty]^{r+1}.$$

An empty set of admissible records is also allowed. We retain these closed sets simultaneously for every integer T, r , every finite list of rational time restrictions on the marks and ranges, and every finite chart graph defined below.

Lemma 5.1 (Certificate compactness). *These observations take values in a compact metrizable product space. Along any sequence for which the rescaled contour laws converge, they admit a joint subsequential limit with the contours. In a joint almost surely convergent realization:*

- (i) *Every limiting finite certificate is approximated by actual discrete path records obeying any prescribed open loss in its endpoint, range, and cost restrictions.*
- (ii) *Every sequence of admissible discrete records has a subsequential limiting certificate.*
- (iii) *Finitely many ordered subpaths of one discrete path may be extracted simultaneously. If its limiting total cost is finite, the sum of the limiting subpath costs is no larger than that total cost.*
- (iv) *The projected range of a limiting path is connected. Its endpoints are the projections of its recorded endpoint times.*

There is no concatenation assertion for independently extracted certificates.

Proof. The hyperspace of closed subsets of a compact metric space is compact, with the empty set adjoined as an isolated point. The countable product is compact and metrizable. Joint tightness with the contours and a Skorokhod realization therefore give convergence of all the closed observation sets.

For a point of a limiting closed set, Hausdorff convergence supplies points in the approximating closed sets. A second approximation replaces those points by actual records. Any open restriction containing the limiting record is eventually satisfied. This proves (i). Compactness proves (ii). Recording the marked subpaths as one tuple proves (iii); addition is continuous where the finite total cost bounds all entries, and the inequalities pass to the limit there.

For (iv), consecutive path vertices have asymptotically equal projected locations, by the contour chord rules and continuity of η ; the same holds for different representatives of one vertex. Join their projected locations inside balls of vanishing radius. The resulting connected compact sets have Hausdorff limits equal to the projected recorded ranges. Hausdorff limits of nonempty connected compact sets are connected. This argument is purely topological and puts no bound on the length of the small joining pieces. $\square$

A spatial confinement condition always means compact confinement in an open set, or confinement after an arbitrarily small open enlargement. This convention is needed because limiting records are closed data. For example, a crossing from an inner circle to an outer circle is recorded

with a margin on both sides. First and last passages through intermediate bands are then available from the simultaneous subpath records. A limiting certificate is *attainable* in this sense of approximation with open loss.

The eventual internal passage rule $F_U(x, y)$ is the infimum of finite costs of attainable certificates with endpoints x, y and range compactly contained in U . At this stage it is only a passage functional. Its concatenation, continuity, and comparison with D_h are proved later from small circuits and open ports.

Definition 5.2 (Universal endpoint observations). For each finite chart graph, retain in addition the following countable family of variables. Given two finite unions J_1, J_2 of rational time intervals, select vertices having at least one representative in the respective unions. A rational chart confinement specifies the allowed vertex and step indices by a finite union of such intervals; the confined graph consists of the specified vertices and those specified edges whose endpoints are both allowed. Take the supremum over the selected endpoint pairs of their minimal distance in this confined graph, divided by s_m . A disconnected pair has cost infinity; an empty supremum is zero. Also retain finite refinements with several endpoint sets and several confinements.

These variables lie in a countable product of $[0, \infty]$ and hence preserve compactness. They are measurable observations of the finite chart graph. They are retained independently of existential path certificates: convergence of the latter alone does not transfer a statement uniform over arbitrary approaching discrete endpoints. Once a time or spatial endpoint modulus is proved, rational time covers and open confinement losses let it be expressed using [Theorem 5.2](#).

5.2 Words, instructions, and stable comparisons

Before specifying a chart, we show how finitely many external flexible choices can be fixed from its limiting contour. In this subsection the lattice time unit is $N \rightarrow \infty$: times are divided by N and heights by $\sqrt{N}$.

Write the inventory alphabet as $\mathcal{A} = \{h, c, H, C, F\}$, where lower-case letters produce a burger, upper-case H, C order the corresponding type, and F orders the freshest available burger. Burger identities include their creation times. A *local reduction* of a finite word starts with two empty stacks, performs every possible match with a burger produced in the word, and records an order which cannot be filled locally. An externally unresolved F is assigned a type only after this reduction. A finite instruction σ specifies that type as a function of a finite rational partition of time. Denote the resulting two-coordinate increment path by Z_N^σ . Its net increment is an exact integer function of the raw word and σ .

Lemma 5.3 (Input-independent local reduction). *The matches made by local reduction are independent of the stacks supplied before the word. Whenever an F is externally unresolved, every available burger at that step was produced before the beginning of the word. Consequently its two actual candidates, if both exist, precede that beginning.*

Proof. A burger produced inside the word is newer than every input burger. Whenever a compatible locally produced burger is available, a rigid order uses the newest such burger of its type, and an F uses the newest of all locally produced burgers. These choices do not depend on the input. Induction through the word proves both assertions. In particular, removing an input burger at an externally unresolved order does not change either local stack. $\square$

The Brownian genericity proved earlier gives the stable comparison needed for external flexible decisions. A coordinate- i horizontal pair is a pair $s < t$ such that $z_i(s) = z_i(t) = \inf_{[s,t]} z_i$. A triple

means three distinct times with the same coordinate value and no lower value between them. For the correlated Brownian motion in this paper, and for its restrictions under interior excursion absolute continuity, almost surely:

- (i) there is no nontrivial horizontal pair common to both coordinates;
- (ii) a time belonging to a coordinate triple is not the endpoint of an additional nontrivial horizontal pair in the other coordinate.

Lemma 4.9 proves both assertions from Brownian wedge survival and independent-gap density. Its hypothesis is $d_2 > 1/2$; no bound on d_γ is needed.

Lemma 5.4 (Uniform finite instructions). *Let $z = (z_1, z_2)$ be a continuous path on $[A, v]$ satisfying (i)–(ii) above. Fix $A < u - \delta < u \leq v$, with $\delta > 0$. For $\epsilon \geq 0$, let K_ϵ consist of triples $(s_1, s_2, t) \in [A, u - \delta]^2 \times [u, v]$ such that, for $i = 1, 2$,*

$$|z_i(s_i) - z_i(t)| \leq \epsilon, \quad \inf_{[s_i, t]} z_i \geq z_i(t) - \epsilon.$$

There are $\epsilon_0 > 0$, a finite rational partition of $[u, v]$, and a type assigned to each partition interval such that the assignment equals the sign of $s_1 - s_2$ for every triple in K_{ϵ_0} . Partition boundaries may be placed in open time gaps containing no such triple. The same instruction is valid for every path sufficiently close to z in the uniform norm, with a sufficiently small additional lattice height error.

Proof. The sets K_ϵ are compact and decrease to K_0 as $\epsilon \downarrow 0$. Property (i) implies $s_1 \neq s_2$ on K_0 , so $|s_1 - s_2|$ has a positive minimum there if K_0 is nonempty. Let T_+ and T_- be the projections to t of the two closed sign parts of K_0 . They are compact. If a time belonged to both, the supplier in at least one coordinate would have two distinct choices. Those choices together with t would make a scalar triple, while the other coordinate has a horizontal pair ending at t , contrary to (ii). Thus T_+ and T_- are disjoint compact subsets of the time interval. Choose disjoint finite unions of open intervals with rational endpoints around them. Decreasing ϵ_0 if necessary makes every triple in K_{ϵ_0} belong to the corresponding sign neighborhood; otherwise a sequence of contrary triples has a limit in K_0 . The resulting finite partition gives the assertion. Uniform perturbation of the path only increases the two errors in the definition of K_ϵ by a fixed multiple of that perturbation. $\square$

Lemma 5.5 (Correctness of surrogate resolutions). *Consider finitely many ordinary word intervals, each with a left extension of positive limiting duration. Suppose all suppliers of externally unresolved orders in the used portions lie in a fixed bounded time window, and that their distance from the used time is at least a fixed positive number in rescaled units. Fix a generic limiting contour and instructions from Lemma 5.4. For all sufficiently large N , every surrogate resolved word in a sufficiently small contour tube about that limit has the actual freshest-burger resolutions on the used portions, provided its initial stacks are those represented by its past contour, its unused left part is processed with actual or already verified context-imposed resolutions, and no empty-stack order is imposed.*

Proof. If a first incorrect decision exists, all preceding resolutions are correct. Its actual candidates can therefore be read from the already correct past contour. By Lemma 5.3, both candidates precede the extension. They form approximate horizontal pairs with height error at most one lattice unit, hence belong to K_{ϵ_0} after rescaling. Lemma 5.4 gives exactly the freshness comparison prescribed by the instruction, a contradiction. Apply this argument successively to the finitely many portions. Supplier truncation is robust: a left cut whose two contour heights are strictly

below all tested heights forces every required backward first hit to occur after the cut. Strict positivity supplies the no-empty-stack condition in a finite interior excursion chart. $\square$

In this induction the preceding contour is the actual contour, or the contour of a globally imposed resolution whose preceding decisions have already been verified. A locally chosen instruction for a used portion does not assert correctness for externally unresolved orders confined to its unused extension. The distinction will matter in anchoring the reference heights in Section 7.

5.3 Protected finite charts

A chart must preserve exact incidences, even where contour convergence alone does not decide a lattice equality. We specify its raw blocks, relative heights, and finite instructions first; Section 7 will then determine the conditional law of its passage observations.

Choose nested independent Poisson time marks, and round discrete cuts to even integers. For a spatial compact set K inside an open buffer U , retain finitely many marked *used* intervals whose curve ranges lie in U , together with unused left extensions when needed. Adjacent intervals may be merged. All preimages of K , and of the specified compact neighborhood of K , must be covered by interiors of used intervals. Such finite covers exist: transience bounds their time window, the preimage of a compact set is closed in that window, and continuity allows a sufficiently fine Poisson mesh to cover it by intervals lying in U . All exact forest heights below are anchored at used starts. The raw extension is retained to compute internal matches, but its externally unmatched flexible orders may have fixed surrogate types; its genuine resolved increment is never silently substituted for that surrogate in an exact height identity.

For each scalar contour, a cut presents at most two boundary lineages. Their heights are measured relative to the start of their retained block. In addition to the raw words and cut lengths, the chart records:

- (a) the relative start-height differences needed to compare retained lineages, encoded by a forest in each coordinate;
- (b) finite unions of allowed height bands for each pair of boundary sides, and any minima of retained intervening blocks that can obstruct a candidate match; every band endpoint is either a strict rational threshold or a minimum computed from a specified retained word;
- (c) directed incoming/outgoing side labels and the finite relative order of the local traversals;
- (d) a finite time partition with the decisions for flexible orders not resolved within the extended raw block.

An unused absolute height completing the forest coordinates has no geometric role. Feasible relative differences lie on the projection of the parity lattice: at even times the coordinate sum is even. Heights inside an individual block are then computed from its raw word and its prescribed external decisions.

In an intrinsic chart, each time is labeled by its block and its time relative to that block's start. The compact record spaces above are used with the finite disjoint union of these closed block intervals in place of I ; the same hyperspace argument applies. Rational endpoint and confinement tests use these block coordinates. Absolute starts on the global tour and omitted-gap durations are exterior data. Conditional on the full contour they push intrinsic records into the global time window, but they are not inputs to a spatially local kernel.

For a fixed finite choice of these instructions and lattice parameters, the resulting graph and all its certificate and universal endpoint observations are functions of the retained raw words. The

instructions are part of the input to the reference experiment; their validity in an actual word is proved below and in the guard construction.

For a scalar coordinate X and a retained block $I_i = [a_i, b_i]$, put $m_i = \min_{I_i} X$. Its left and right *ports* are the contour-tree lineages from the visits at a_i and b_i down to height m_i . They are compact intervals of length coordinates; their spatial images are contained in $\eta(I_i)$. A contact with an earlier block lies on the left port, and one with a later block lies on the right port. Several times representing the same port height are distinguished inside the retained raw block.

Local field data recover scalar contacts through the two labelled families of frontier arcs. Retaining the labels is essential: an equality of spatial points can also be caused by the other contour coordinate.

Lemma 5.6 (Local coloured port data). *For finitely many retained blocks whose closures lie in a buffered local traversal, the local fields and the traversal's side labels determine the following data: the scalar port relations in the protected region, the connected components of the port-overlap graph, and the relative block-start heights within each such component. The only order information required in this statement is the finite relative order of the directed local traversals.*

Proof. Use the labelled frontier strands in [Theorem 3.2\(iv\)](#). In the imaginary-geometry construction, strands of the same labelled family merge on meeting. Their shared arcs give scalar contour gluing with the corresponding side convention; see [\[24, Section 2.1.3 and Lemma 2.4\]](#). The block boundaries are the first and last frontier portions of the clocked local traversal, so these labels do not require an absolute exterior contour height.

For each pair of ports, take the closure of their correspondence on nontrivial common arcs of the *same* labelled family, with the incoming and outgoing sides specified. Initially this is a relation on their length-coordinate intervals. For an ordered block pair $i < j$, pull it back to endpoint times by the right-port map $s \mapsto X(b_i) - X(s)$ and the left-port map $t \mapsto X(a_j) - X(t)$; restrict to times passing the retained suffix and prefix minimum tests, which ensure membership in these ports. Denote this pulled-back closed relation by E_{ij} . Every such common arc gives a scalar contact. The scalar equal-minimum relation is closed, so taking the closure adds no false scalar contacts. Conversely, a scalar contact between interior times of two retained blocks is approached by ordinary scalar chords with endpoints in those same blocks. For an ordinary excursion pair use nearby higher levels. At a branch triple use higher levels for the adjacent pairs and lower levels for the outer pair. Choose levels avoiding the countable set of scalar local-minimum heights. Strict Brownian local minima and their distinct heights justify these approximations. Thus every protected scalar contact belongs to E_{ij} . This uses [Theorem 4.9](#) for the Brownian sewing event and the scalar approximation argument detailed in [Theorem 6.1](#).

The relation is not defined by arbitrary equality of spatial points: that equality can be caused through the other colour. Keeping the labelled common arcs is what distinguishes the two relations. Two tree lineages intersect in a lineage, so the common length coordinates of one pair of ports form an interval. There are finitely many ports. Consequently their relations have finitely many such pieces, even if the associated surfaces have infinitely many beads or the strands have infinitely many individual points of contact.

The real field gives the quantum-length coordinates on the retained arcs. Evaluating the two coordinates at a common arc point determines a single additive difference of block-start heights. A spanning forest of the nontrivial-overlap graph stores all differences in each connected component. A protected isolated endpoint contact also joins the same component, by the ordinary-chord approximation just proved. No difference between distinct components is recovered or needed.

These are measurable local data. The strands, their side labels, local clock, and length

coordinates are measurable by the cited construction. Their common-arc correspondence is a measurable closed relation: it can be tested on compact parameter subintervals, with positive common length exhausted by rational lower bounds. Standard measurable selection on these compact relations chooses a common point; every choice gives the same additive difference. Under the countable family of reference laws a common completed-measurable version has a Borel version, by using their probability-weighted sum. Choosing a spanning forest by the fixed finite order is then measurable. $\square$

The relative heights have now been recovered within each overlap component. A possible contact can still be blocked by a minimum in an omitted time gap. The following clearance argument replaces that exterior test by a rational threshold chosen from the local port relation.

Lemma 5.7 (Clearance from untested minima). *Let K_* be a compact spatial set whose full preimage lies in the interiors of finitely many retained blocks in a bounded time window. Fix a coordinate and two blocks $i < j$ in the same overlap component $\mathcal{A}$. Translate its component-root height to zero. Test exactly the suffix of block i , the prefix of block j , and every intervening retained block in that component. The combined untested set is*

$$G_{ij} = \bigcup_{\ell=i}^{j-1} [b_\ell, a_{\ell+1}] \cup \bigcup_{\substack{i < \ell < j \\ \ell \notin \mathcal{A}}} I_\ell \subset [b_i, a_j].$$

Thus it contains precisely the intervening omitted gaps, their cut endpoints, and the intervening blocks in other components. Set $m_{ij} = \min_{G_{ij}} X$.

Let C_{ij} consist of pairs (s, t) in the two blocks with $\eta(s), \eta(t) \in K_$, equal endpoint height, and every tested height at least that endpoint height. Its height set H_{ij} is compact and*

$$\text{dist}(H_{ij}, \{m_{ij}\}) > 0$$

whenever $C_{ij} \neq \emptyset$. There is a rational band test, selected using only the local port data and tested retained pieces, which decides every protected candidate's untested minimum condition. It is correct uniformly for sufficiently small contour perturbations and bounded multiples of a vanishing height-mesh error.

Proof. Interval minima are continuous in their endpoints, so C_{ij} and H_{ij} are compact. Suppose a candidate has height m_{ij} , and let $\tau \in G_{ij}$ minimize the untested contour. All heights between the candidate endpoints are then at least m_{ij} . Hence s, t, τ belong to one scalar class and $\eta(\tau) = \eta(s) \in K_*$. An omitted gap or cut cannot contain such a visit, by full-preimage retention. If τ belongs to another retained block, its scalar contact with s , approximated by common arcs as in [Theorem 5.6](#), puts that block in the same overlap component. This too contradicts its inclusion in G_{ij} . Thus $m_{ij} \notin H_{ij}$, proving positive separation.

The untested minimum is used to prove separation, but is not an input to the local selection. Instead intersect C_{ij} with the local relation E_{ij} from [Theorem 5.6](#). Call the resulting candidates good, and the other candidates bad. By the preceding separation, their respective height sets satisfy

$$H_{\text{good}} \subset (-\infty, m_{ij} - \delta], \quad H_{\text{bad}} \subset [m_{ij} + \delta, \infty)$$

for some $\delta > 0$. Both sets are compact, including the bad set. If both are nonempty, choose the first rational α in a fixed enumeration with $\sup H_{\text{good}} < \alpha < \inf H_{\text{bad}}$, and allow heights below α . If only bad candidates exist, allow none; if only good candidates exist, choose $\alpha > \sup H_{\text{good}}$. An

empty candidate set requires no test. All these choices use only local data and the finite order. Extrema and the first-admissible-rational choice are measurable; equivalently exhaust complements of closed relations by positive rational distance and take compact-set extrema.

For stability, suppose uniformly convergent contours, convergent cuts and a vanishing height mesh gave incorrectly classified protected lattice candidates. After taking endpoint subsequences, all exact retained tests pass to the limiting weak tests, so the limit lies in C_{ij} . The combined untested minima converge to m_{ij} . At every such limiting candidate both its relation to m_{ij} and its classification by the chosen band have strict and agreeing signs. This contradicts incorrect classification. The same argument covers the empty-good or empty-bad case. It proves correctness throughout one sufficiently small contour tube, rather than only along the originally sampled sequence. $\square$

We can now combine the port relations, their strict clearance, and the finite flexible-order instructions into an exact graph reconstruction.

Lemma 5.8 (Protected chart reconstruction). *On the Brownian sewing event specified in Theorem 4.9, compactly confined graph observations can be exhausted by the finite charts just described. More precisely, for $K \in U$, there is a countable collection of buffered chart descriptions with the following properties.*

- (i) *Every graph path with projected range in K , and every finite collection of its marked subpaths, is eventually represented in one of these charts. The same holds for the universal endpoint observations with an open confinement loss.*
- (ii) *Within the compact confinement, the reconstructed chart has exactly the primal incidences of the original graph for all sufficiently large scales. No unrecorded minimum in an omitted gap is used to decide one of these incidences.*
- (iii) *The continuous block input, relevant forest differences, and allowed local gluing bands are measurable from the directed local traversals, frontier lengths, and the finite global traversal order. Exterior flexible decisions are treated by the buffered comparison instructions of Theorems 5.4 and 5.5.*
- (iv) *Disjoint spatial buffers use disjoint raw blocks and distinct local observations. The countable atlas can be refined to accommodate any prescribed finite collection of observations.*

Proof. Choose nested neighborhoods

$$K \subset \text{int } K_* \in U_0 \in U_1 \in U.$$

Cover the full preimage of K_* by used intervals whose entire ranges have closure in U_0 , and give them left extensions whose entire ranges have closure in U_1 . Transience supplies a bounded enclosing window. For a bilateral discrete application enlarge the window first to coordinate descent cuts strictly below every tested height, so all needed partners are in the window. This removes the possibility of an unrecorded supplier escaping to infinity. Record every directed U_1 -traversal reaching $\overline{U}_0$. There are finitely many: each crosses a fixed positive distance, and uniform continuity on the bounded time window bounds its duration below. The dense local-traversal construction in the proof of [24, Lemma 2.4] includes them all. The real field needed to clock them is restricted to $U_1 \subset U$; a slightly larger imaginary buffer suffices to reconstruct them.

Apply Theorem 5.6 separately in the two coordinates, storing each component's offsets in a forest. Apply Theorem 5.7 to each pair of blocks in a component. Test intervening blocks only

within that component; all other blocks belong to the combined untested set in that Lemma. No protected incidence can join different components for infinitely many scales, since its limiting scalar contact would join them by [Theorem 5.6](#). There are finitely many block pairs, so one contour tube works for all their band tests.

Keep the exact discrete sewing dictionary at the retained pieces. Primal contour-vertex identifications use the appropriate equal-height weak-minimum tests. An edge supplied by the other coordinate uses a genuinely matched up-step and down-step, with the strict one-step-above-base condition in its excursion interior and the prescribed endpoint corner offsets. These tests are functions of the actual raw steps and exact lattice forest shifts. In particular a weak horizontal pair in the other coordinate is not by itself declared to be a primal edge. For example, the scalar word with heights 0, 1, 0, 1, 0 has a weak pair between its first and last visits but two separate matched excursions. Every retained suffix, prefix and intervening minimum, including a critical branch minimum, is evaluated exactly in the discrete word. Only an *untested* obstruction is replaced by a rational band. [Theorem 5.7](#) permits the bounded height-mesh errors from the step convention and proves that this replacement preserves each relevant incidence.

Exterior flexible resolutions require another step. A flexible order unresolved within its extended raw block, but lying in its used portion, has both candidate suppliers before the extension and hence at a positive limiting time separation from the used portion. Each positive-span supplier chord has the same spatial image as its decision. Since the whole used image lies in U_0 , every supplier's local traversal belongs to the finite U_1 -traversal family just retained. The no-common-chord and mixed-branch exclusions of [Theorem 4.9](#) separate the two possible supplier orders into compact disjoint sets of decision times. The finite rational instructions and first-error proof in [Theorems 5.4](#) and [5.5](#) therefore make the resolved *used* steps exact on a sufficiently small tube. No recursively enlarged collection of raw supplier blocks is needed: their continuum traversal labels and finite order supply the comparison. Externally unmatched flexible orders in the unused extension need not be genuinely typed to compute the internal matches or used increments. Their surrogate paths enter only the macroscopic reference input, using [Theorem 4.3](#); the exact density identities of [Section 7](#) keep genuine and surrogate extension increments distinct. Anchoring exact heights at used starts prevents these unused types from changing any retained incidence.

We have reconstructed every primal incidence used by a path or an ordered subpath record confined to K , with a margin inside K_* . Rational intrinsic time covers with image in K_* give the corresponding confined chart graphs for the universal endpoint tests. Their spatial interpretation uses the open loss specified above. Poisson refinements, rational bands, finite instruction lists and finite traversal orders give countably many descriptions; their durations, paths and forest shifts remain continuous or lattice input variables. Refinements accommodate any finite collection of observations. Disjoint spatial buffers have disjoint time preimages, and the unused extensions can be trimmed inside the respective buffers. Thus their retained raw blocks are disjoint, proving all four assertions. $\square$

The passage from this deterministic chart description to conditional local kernels is not automatic. In particular, relative lattice shifts cannot be replaced by continuous offsets solely because the contour converges. [Section 7](#) supplies a lattice local limit, guards external flexible decisions, and identifies the conditional kernels by a reference-block change of density.

6 From contour convergence to planar topology

This section turns uniform contour convergence into topological control of the discrete map. We first join any prescribed lifts of nearby points by paths with small projected images. These joins yield disk fillings, winding separation, and actual intersections of primal crossings. We then construct cellular neighborhoods with complete incidence layers and preserve them when a bounded contour window is capped to a sphere. No bound on graph length or normalized passage cost is asserted here.

We use the round distance d_{sph} on $\mathbb{S}^2$ and write $B(x, r)$ for a round ball; its normalization is immaterial. All round balls below have radius smaller than a fixed convexity radius. Graphs on surfaces are embedded, so distinct open edges are disjoint; loops and multiple edges are allowed.

6.1 Contour links and small-image joins

Recall the resolved tree/cotree sewing convention. A chronological mesh time specifies a primal corner. Matched up- and down-steps of the first coordinate identify primal corners in the contour tree. A matched pair in the second coordinate bounds a sewn quadrangle whose primal diagonal is an all-edge primal edge. Its endpoint corners occur at the two steps, up to one chronological mesh step. Thus a limiting second-coordinate identification must be approximated by a *matched discrete pair* before it supplies a primal edge.

Lemma 6.1 (Approximating finitely many horizontal generators). *Suppose resolved discrete contour pairs converge uniformly to a continuous contour pair on a compact chronological interval. The time and height meshes tend to zero. Assume that, in each limiting scalar contour, local minima are strict and have distinct heights. End identifications are treated by the usual excursion convention; in particular, the two ends of a positive excursion cause no additional interior zero.*

Fix a finite list of horizontal generators (s, t) , each satisfying

$$X(s) = X(t) = \min_{[s, t]} X$$

for one of the two coordinates X . Each chosen pair is contained in the interior of the interval of uniform convergence, so that both exterior sides are available. The only endpoint exception is the pair of genuine endpoints of a strictly positive excursion. For every generator there are chronological endpoints $s_j \rightarrow s$, $t_j \rightarrow t$ such that their discrete primal corners are identified, in the first-coordinate case, or are joined by one genuine all-edge primal edge, in the second-coordinate case. The choices can be made simultaneously for the finite list. More generally, if a generator is specified as a fixed finite list of such scalar generators, every member of that list has this property.

The conclusion concerns individual links. Approximations to a shared intermediate time need not be the same discrete vertex.

Proof. Write $a = X(s) = X(t)$. First suppose $X > a$ on (s, t) . For a level $a + \delta$ approaching a from above, chosen to avoid the countable set of local-minimum heights, take the component of $\{X > a + \delta\}$ containing a fixed interior compact subinterval. Its endpoints tend to s, t as $\delta \downarrow 0$. Indeed, $X - a$ has a strictly positive minimum on each compact subinterval of (s, t) , whereas its endpoint values are zero. At each boundary of the chosen component there are strictly lower values arbitrarily close on the exterior side: otherwise that boundary would be a local minimum at the excluded level. Thus, for fixed sufficiently small δ , uniform contour convergence supplies a discrete excursion above a nearby lattice level whose entrance and exit are as close to these endpoints as desired. The entrance up-step and exit down-step are a matched pair. Let δ tend to zero

sufficiently slowly with the discrete index. The sewing dictionary gives the claimed identification or primal diagonal, with endpoint errors tending to zero.

There is at most one time $r \in (s, t)$ with $X(r) = a$: two such times would be local minima of the same height. If one exists, there are values below a arbitrarily close to s on its exterior side and to t on its exterior side. Otherwise s or t would itself be a local minimum of height a , distinct from r . For $\delta > 0$, again avoiding local-minimum heights, take instead the component of $\{X > a - \delta\}$ containing $[s, t]$. The just noted exterior values show that its endpoints tend to s, t as $\delta \downarrow 0$. Apply the same discrete matched-step argument at a nearby lattice level below a . This proves the assertion for the outer pair at a scalar branch triple. Each consecutive pair at that triple has strictly positive excursion interior and is covered by the first argument. The excursion end convention is covered by the first argument as well. A constant generator needs no edge.

For finitely many prescribed generators, take one diagonal choice of the level errors and the discrete index thresholds. The number of matched-step links per prescribed scalar generator is one and does not depend on the discrete index. No rate of convergence is needed. $\square$

For Brownian contours the scalar hypothesis holds almost surely. Indeed, the minimum on a fixed interval is almost surely unique, and minima on disjoint rational intervals have distinct values. Every local minimum is the minimum on a rational interval, so a countable intersection gives strictness and distinct heights. Local absolute continuity proves the same assertion on compact interior intervals of a quadrant excursion; exhaust its open time interval and use the excursion convention at the ends.

To pass from finitely many contour links to uniform joins, we use the following deterministic assumptions. Let Σ_j be triangulated topological spheres with embedded primal graphs $G_j \subset \Sigma_j$, and let

$$\pi_j : \Sigma_j \longrightarrow \mathbb{S}^2, \quad c_j : [0, 1] \longrightarrow G_j, \quad \eta : [0, 1] \longrightarrow \mathbb{S}^2$$

be continuous maps. The chronological walk c_j visits every primal vertex, and η is onto. We require

- (T1) $\sup_t d_{\text{sph}}(\pi_j(c_j(t)), \eta(t)) \rightarrow 0$.
- (T2) Every point $x \in \Sigma_j$ can be connected to some primal vertex v by a path whose projected diameter tends to zero uniformly in x . For example, this follows if every closed triangle contains a primal vertex and $\max_\sigma \text{diam}(\pi_j(\sigma)) \rightarrow 0$.
- (T3) If $\eta(s) = \eta(t)$, then s, t are connected by a finite chain of limiting horizontal generators. For each generator (a, b) in such a chain there are $a_j \rightarrow a$, $b_j \rightarrow b$ and a primal path from $c_j(a_j)$ to $c_j(b_j)$ whose projected diameter tends to zero.

In (T3), the finite chain may depend on s, t ; no uniform bound on chain length is required. Lemma 6.1, together with vanishing projected edge diameters, supplies the link approximation in (T3).

Here is the continuous projection used for the sewn maps. Assign each primal or dual vertex the location $\eta(t)$ of one of its visits. All visits to that vertex have uniformly vanishing disagreement: otherwise a contrary sequence of pairs of visit times would converge subsequentially, and uniform contour convergence would pass their scalar equality or matching relation to a scalar chord with equal limiting locations. The same argument applies to neighboring corners and matched diagonals. Thus all corners of a triangular face receive uniformly close locations. Join them by short round arcs and fill each triangle in a small convex ball. This gives π_j with vanishing projected triangle diameter, (T1) from the chronological primal walk, and (T2) from its visited vertices. For a planar

exhaustion, apply this construction only after supplying a compact sewing that preserves the tested buffered region.

Lemma 6.2 (Uniform topological joining). *Under (T1)–(T3), for every $\epsilon > 0$ there exist $\delta > 0$ and j_0 such that, for $j \geq j_0$ and any $x, y \in \Sigma_j$ with*

$$d_{\text{sph}}(\pi_j(x), \pi_j(y)) < \delta,$$

there is a path from x to y whose projection is contained in $B(\pi_j(x), \epsilon)$. If x, y are primal vertices, the path can be taken in G_j . The image of π_j is asymptotically dense in $\mathbb{S}^2$.

Proof. First take primal vertices. If the assertion fails, there are an $\epsilon > 0$, a subsequence, and vertices $x_j = c_j(s_j)$, $y_j = c_j(t_j)$ whose projected distance tends to zero but which cannot be joined as asserted. Pass to limits $s_j \rightarrow s$, $t_j \rightarrow t$. By (T1), $\eta(s) = \eta(t) = z$. Choose one finite chain from (T3). For each of its generators use its discrete approximating path. Join an approximating endpoint to the endpoint used by the next generator by the corresponding segment of c_j . Those chronological intervals shrink to the shared limiting time, so their projected images converge uniformly to z , by (T1) and continuity of η . Use the same chronological connectors at s_j, t_j . There are only finitely many pieces. Their concatenation eventually projects into $B(z, \epsilon/2)$, a contradiction.

For arbitrary surface points, first use (T2), then the vertex version with a smaller error, and concatenate. Finally, (T1) and surjectivity of η give asymptotic density of $\pi_j(\Sigma_j)$. Notice that chronological connectors in this proof can contain unboundedly many primal edges. $\square$

6.2 Connected exteriors, disk fillings, and degree

The arbitrary-lift joins now let us fill loops without leaving a prescribed projected neighborhood. The elementary fact behind this step is that every component of the complement of a nonempty compact connected set K in a topological sphere is simply connected. Indeed, a polygonal simple loop disjoint from K has one Jordan disk disjoint from K , because K lies on one side. The Jordan–Schoenflies Theorem contracts the loop in that disk and hence in its complementary component. An arbitrary loop in the open complement is homotopic there to a polygonal loop with finitely many crossings. Split it at the crossings and contract the resulting simple loops as above. No local regularity of K is needed.

Lemma 6.3 (Exterior connection and small-loop filling). *Suppose continuous maps $\pi_j : \Sigma_j \rightarrow \mathbb{S}^2$ from topological spheres have asymptotically dense images and satisfy the uniform joining conclusion of Lemma 6.2. Fix $0 < r < R$ and $z \in \mathbb{S}^2$. For all sufficiently large j there is a nonempty compact connected set $K_j \subset \Sigma_j$ such that*

$$\pi_j^{-1}(\mathbb{S}^2 \setminus B(z, R)) \subset K_j, \quad K_j \cap \pi_j^{-1}(\overline{B(z, r)}) = \emptyset.$$

Consequently every loop projecting into $B(z, r)$ bounds a continuous disk projecting into $B(z, R)$. The index threshold can be chosen uniformly in z .

Proof. Choose $r < r_1 < r_2 < R$ and a joining error much smaller than both $r_1 - r$ and $R - r_2$. For large j , choose one anchor lift whose image lies outside $B(z, R)$. Such a lift exists by asymptotic density. The exterior of $\overline{B(z, r_2)}$ is path connected. Join the image of any prescribed lift of a point outside $B(z, R)$ to the anchor image by a target path in that exterior. Subdivide the path into sufficiently small steps. Asymptotic density supplies lifts whose images are sufficiently close to the

subdivision points; retain the prescribed lifts at the ends. Apply the uniform joining property to successive lifts. The resulting upstairs path has projection outside $\overline{B(z, r_1)}$.

Take the union of the anchor and these paths for every lift in $\pi_j^{-1}(\mathbb{S}^2 \setminus B(z, R))$, and then take its closure in Σ_j . This is a compact connected set K_j containing that entire preimage. Its projection remains outside $B(z, r_1)$, so it is disjoint from the preimage of $\overline{B(z, r)}$. Every loop in the latter preimage lies in a component of $\Sigma_j \setminus K_j$. That component is simply connected and is contained in $\pi_j^{-1}(B(z, R))$, proving the filling claim. The joining and density bounds used here are independent of z . $\square$

Proposition 6.4 (Degree and robust planar topology). *Under the hypotheses of Lemma 6.3, the maps π_j have degree $+1$ or -1 for all sufficiently large j . More precisely, for every $\epsilon > 0$ and sufficiently large j there is a continuous map $g_j : \mathbb{S}^2 \rightarrow \Sigma_j$ with*

$$\sup_{z \in \mathbb{S}^2} d_{\text{sph}}(\pi_j(g_j(z)), z) < \epsilon.$$

Loops with a nonzero projected winding difference robustly separate the corresponding lifted fibers, as specified below; in particular, opposite primal crossings of a rectangle with fixed positive margins intersect in the primal graph.

Proof of the approximate right inverse and degree. Fix a sufficiently small error ϵ . First choose a loop-image radius much smaller than ϵ , small enough that Lemma 6.3 fills loops of that radius inside an $\epsilon/3$ ball. Next choose a joining error much smaller than that loop-image radius, and then a target triangulation so fine that the endpoints of each of its edges can be joined using Lemma 6.2. For sufficiently large j , choose for each target vertex a lift with projection arbitrarily close to that vertex. Connect the chosen lifts along each target edge by the joining property. This defines g_j on the one-skeleton.

For each triangular face, the image under $\pi_j \circ g_j$ of its boundary lies in a ball of the chosen loop-image radius about one of its vertices. Lemma 6.3 extends g_j continuously over this face, with projected image inside the $\epsilon/3$ ball about that vertex. The extensions agree on shared edges. Fineness of the target mesh gives the asserted uniform error on every face. Short round geodesics give a homotopy from $\pi_j \circ g_j$ to the identity. Thus

$$\deg(\pi_j) \deg(g_j) = 1,$$

and both integer degrees are $+1$ or both are -1 . In particular π_j is onto. No embedding or injectivity of g_j is claimed. $\square$

6.3 Graph intersections forced by winding

We next use the full strength of joining arbitrary lifts: winding separates entire fibers, a property that degree alone would not give. For distinct target points a, b , the group $H_1(\mathbb{S}^2 \setminus \{a, b\}; \mathbb{Z})$ is infinite cyclic. A loop has a *nonzero winding difference about a, b* if its class in this group is nonzero. This formulation avoids choosing a point at infinity.

Lemma 6.5 (Separation of entire fibers). *Under the hypotheses of Lemma 6.3, fix $\rho > 0$. For sufficiently large j , suppose a loop $\lambda : \mathbb{S}^1 \rightarrow \Sigma_j$ satisfies*

$$\text{dist}_{\text{sph}}(\pi_j(\lambda), \{a, b\}) > \rho$$

and $\pi_j \circ \lambda$ has nonzero winding difference about a, b . Then every path joining any lift of a to any lift of b intersects λ . The loop need not be simple.

Proof. All lifts of a can be joined to any one prescribed lift by paths projecting into $B(a, \rho/3)$, using the uniform joining property. After closure, their union is compact, connected, contains the entire fiber of a , and is disjoint from λ . The same holds for b . If an upstairs path connected the prescribed lifts while avoiding λ , its union with these two compact connected sets would be a compact connected set K containing both entire fibers and disjoint from λ . The elementary complement fact above would contract λ in $\Sigma_j \setminus K$. Projection of this contraction would contract $\pi_j \circ \lambda$ in $\mathbb{S}^2 \setminus \{a, b\}$, contradicting its nonzero homology class. This proves separation even if exact fibers are disconnected: their pieces were first joined away from the loop. $\square$

We apply this separation test to transverse graph crossings. Work in a planar coordinate disk containing $[-3, 3]^2$. Fix $0 < a < 1/4$. A horizontal passage has projected image in $[-2, 2] \times [-a, a]$ and endpoints on the lines $x = -2, x = 2$. A vertical passage has projected image in $[-a, a] \times [-2, 2]$ and endpoints on the lines $y = -2, y = 2$. The same conclusions hold with small errors in these endpoints and tubes, by making the margins smaller first. Rectangles obtained by a fixed change of coordinates are treated identically.

Corollary 6.6 (Primal crossing intersection). *Assume (T1)–(T3). For sufficiently large j , any two primal graph paths meeting the preceding horizontal and vertical passage tests share a primal vertex. More generally, the assertion holds for any finite configuration of tubes and open ports whose margins permit this test after restricting to subpassages and changing coordinates.*

Proof. Call the horizontal path α and the vertical one β . Close α upstairs by a path whose projection follows an arc from its right endpoint to its left endpoint going down outside $[-a, a] \times [-2, 2]$, around the bottom of this rectangle, and up its other side. Such an upstairs path is obtained by a finite chain of the small-image joins, starting and ending at the prescribed lifts. Choose its projected error smaller than the fixed clearance from the vertical tube. It is consequently disjoint from β .

Let b_- and b_+ be the actual projected endpoints of β . The projected closed loop has winding difference one, up to sign, about b_-, b_+ . To check this without a simplicity assumption, homotope the horizontal piece inside its horizontal strip to a straight left-to-right segment, and homotope the added piece in its thin exterior tube to the chosen closing arc. These homotopies avoid both endpoint neighborhoods. The resulting rectangle loop contains b_- and excludes b_+ , giving the stated winding difference. Lemma 6.5 forces β to meet the closed loop. It cannot meet the added piece and therefore meets α . Two paths in an embedded primal graph can intersect only on a common vertex or a common edge; a common edge also gives a common vertex. Thus the intersection is an actual primal graph intersection, at which their walks may be stitched. $\square$

Remark 6.7 (Why the arbitrary-lift hypothesis is necessary). Vanishing projected mesh and existence of some nearby lifts do not suffice. The map $z \mapsto z^2$ on the Riemann sphere has degree two and admits triangulations with arbitrarily small projected mesh. Over a disk avoiding its two branch values it has two disjoint sheets. A horizontal path on one sheet and a vertical path on the other have robustly crossing projections and are disjoint upstairs. Nearby target points have suitably chosen nearby lifts, but two prescribed lifts of the same point cannot be joined with projection confined to that disk. Lemma 6.2 rules out exactly this obstruction. Asymptotic density of the image is also necessary: a constant map satisfies the small-image joining property but has degree zero.

6.4 Cellular neighborhoods and retained stars

The crossing test gives graph intersections. We also need a disk around a prescribed fiber whose nearby primal and dual incidences remain in retained time blocks. A disk preimage, or the union of its incident triangles, may have holes and pinches. We first remove these inside a larger collar, then control the necessary cell stars.

For a subset $A \subset \Sigma_j$, let $\mathcal{S}_j^0(A)$ be the union of all closed original triangles meeting A , with their faces. Let $\mathcal{S}_j^{r+1}(A)$ be the union of the closed original triangles sharing a vertex with a triangle of $\mathcal{S}_j^r(A)$. Thus only a specified finite number of full cell-star layers is retained; a proper patch is not required to contain stars of all its vertices indefinitely. Write μ_j for the maximum projected diameter of an original triangle. For each fixed r ,

$$\pi_j(\mathcal{S}_j^r(A)) \subset \{x : d_{\text{sph}}(x, \pi_j(A)) \leq (r+1)\mu_j\}. \quad (6.1)$$

Indeed, successive triangles share the image of a vertex, so their diameter bounds can be added. Vertex degrees need not be bounded.

Lemma 6.8 (A cellular disk between two lifted disks). *Assume (T1)–(T3) and $\mu_j \rightarrow 0$. If $E \Subset \text{int } F$ are closed topological disks in $\mathbb{S}^2$, then, for sufficiently large j , there is a closed topological disk $\Delta_j \subset \Sigma_j$ such that*

$$\pi_j^{-1}(E) \subset \text{int } \Delta_j, \quad \Delta_j \subset \pi_j^{-1}(\text{int } F).$$

The disk can be taken to be a subcomplex of a finite subdivision of the original triangulation. This subdivision supplies a topological boundary only; it adds no edges to the primal metric graph.

Proof. Choose nested disk collars strictly between E and F . We connect the lifted core, take a regular neighborhood, and fill its holes except for the component containing the prescribed exterior.

Connect the core. For large j , the projection is onto by Proposition 6.4. Connect every lift over E to one anchor by finite chains of small-image joins along target paths in E . Choose the joining error smaller than the first collar. The closure of the union is a compact connected set H_j containing $\pi_j^{-1}(E)$ and projecting inside that collar. The triangles meeting H_j form a connected finite subcomplex, whose full cell star lies in the next collar by Equation 6.1.

Take a regular neighborhood. Barycentrically subdivide the triangulation. Give a new vertex value zero if its original simplex belongs to the connected subcomplex, and value one otherwise. Extend linearly and let N_j be the sublevel set at $1/2$. The boundary avoids new vertices and cuts each crossed triangle in a segment, with exactly two segments meeting across each shared edge. It is therefore a disjoint union of polygonal circles, and N_j is a compact subsurface. Normalizing the barycentric weights of the zero-valued vertices retracts each piece to its zero face. These retractions agree on common faces, so N_j is connected and contains the original subcomplex in its interior. It lies in the full original cell star. Subdivide along the level segments and triangulate the resulting polygons to make N_j cellular. This resolves pinches at original vertices without modifying the primal graph.

Fill the interior holes. The exterior-connection construction of Lemma 6.3, using the intermediate disk collars in place of round disks, gives a compact connected set K_j containing $\pi_j^{-1}(\mathbb{S}^2 \setminus \text{int } F)$ and disjoint from N_j . The proof is unchanged: the exterior of a topological disk is path connected, and each of the finitely many collars has positive separation from the next. All of K_j lies in one component O_j of $\Sigma_j \setminus N_j$.

A connected proper compact subsurface of a sphere has complementary components which are open disks, one for each boundary component. This follows directly by applying the Jordan–Schoenflies Theorem to its disjoint boundary circles and using connectedness of the subsurface. Fill every complementary component of N_j except O_j , and set $\Delta_j = \Sigma_j \setminus O_j$. This is a closed disk containing H_j in its interior. It avoids K_j , hence projects inside $\text{int } F$. Filling the complementary components preserves cellularity after the same subdivision. $\square$

The cellular disk can now be placed inside retained time blocks. For each original triangle, let $I_{j,\sigma}$ be its chronological mesh interval. We use the following projection convention:

$$\max_{\sigma} \sup_{x \in \sigma, t \in I_{j,\sigma}} d_{\text{sph}}(\pi_j(x), \eta(t)) \longrightarrow 0, \quad \max_{\sigma} |I_{j,\sigma}| \longrightarrow 0. \quad (6.2)$$

The continuous projection construction above has this property. Subdivided cells inherit the interval of their original triangle.

Proposition 6.9 (Cellular sewing stability with margins). *Suppose admissible discrete contour excursions have actual sphere cell complexes, converge uniformly to contours whose finite-chain quotient is a sphere, and satisfy the link approximation and chronological conditions (T1)–(T3) and Equation 6.2. In particular, the scalar approximation hypothesis includes any separately prescribed generators at exceptional or distinguished times.*

Let $z \in \mathbb{S}^2$ have finite fiber $S = \eta^{-1}(z)$, and let J be a finite union of open retained time intervals containing S . Choose closed topological disks about z with

$$E_{-1} \subseteq \text{int } E_0 \subseteq \text{int } E_1 \subseteq \text{int } E_2 \subseteq \text{int } E_3 \subseteq \text{int } E_4, \quad \eta^{-1}(E_4) \subset J.$$

For every fixed integer $r \geq 0$ and sufficiently large j , there are nested cellular disks $\Delta_j^- \subseteq \text{int } \Delta_j^+$ such that

$$\begin{aligned} \pi_j^{-1}(E_0) &\subset \text{int } \Delta_j^- \subset \Delta_j^- \subset \pi_j^{-1}(\text{int } E_1), \\ \pi_j^{-1}(E_2) &\subset \text{int } \Delta_j^+ \subset \Delta_j^+ \subset \pi_j^{-1}(\text{int } E_3). \end{aligned}$$

Writing $A_j = \Delta_j^+ \setminus \text{int } \Delta_j^-$:

- (i) *A_j is a closed cellular annulus and contains $\pi_j^{-1}(E_2 \setminus \text{int } E_1)$.*
- (ii) *Every original triangle in $\mathcal{S}_j^r(\Delta_j^+)$ has its entire chronological interval in J . In particular, taking $r \geq 1$ retains complete primal and dual vertex stars of every original triangle meeting the disk.*
- (iii) *Every original triangle in $\mathcal{S}_j^r(A_j)$ has its chronological interval in a fixed compact subset of $J \setminus S$, up to an arbitrarily small fixed open time enlargement still contained there. It can therefore be covered by finitely many off-center time blocks, chosen with positive margins.*
- (iv) *Robust transverse primal crossings in the annular region have the actual intersections asserted by Corollary 6.6. A closed primal walk there with nonzero projected winding difference about z and a point of $\mathbb{S}^2 \setminus E_4$ contains a simple primal circuit surrounding $\pi_j^{-1}(E_0)$. Its chosen interior is contained in Δ_j^+ , so its interior and the required fixed number of cell-star layers lie in the retained patch.*

All conclusions are topological. They bound neither the length of an auxiliary joining path nor a normalized endpoint cost.

Proof. We first construct the annulus, then retain its incidence layers and extract a separating primal circuit. Apply Lemma 6.8 to (E_0, E_1) and (E_2, E_3) . Since $E_1 \subset \text{int } E_2$, the resulting disks are nested as claimed. Their difference is an annulus; the displayed inclusions give (i).

For (ii), Equation 6.1 and the margin between E_3, E_4 place all the star triangles, and their chronological images in Equation 6.2, inside E_4 for large j . The compact set $\eta^{-1}(E_4)$ is contained in the open set J . Vanishing chronological mesh therefore places each whole triangle interval in J , not just one representative time.

Every point of A_j projects outside E_0 and inside E_3 . The margins to E_{-1} and E_4 show that every triangle in its fixed star buffer has chronological images in $E_4 \setminus \text{int } E_{-1}$ for large j . The inverse image of this compact set is a compact subset of J disjoint from S . A slightly larger compact time neighborhood still in $J \setminus S$ contains the full triangle intervals. A finite interval cover, with slightly larger buffers still off S , proves (iii). Any additional finite number of incidence guards is covered by increasing r before taking j large.

For (iv), use Corollary 6.6 for transverse crossings. To obtain the surrounding circuit, decompose a closed primal walk into finitely many simple circuits. Projected winding is additive, so at least one has nonzero winding difference. By Lemma 6.5, this circuit separates a lift of z from a lift of an exterior point. It lies in A_j , whereas $\text{int } \Delta_j^-$ is connected and disjoint from it; therefore all of $\pi_j^{-1}(E_0)$ lies on its inner side. The complement of Δ_j^+ is connected, lies on the exterior side, and contains the exterior lift. The chosen circuit interior is thus contained in Δ_j^+ . Its fixed star buffers are retained by (ii). The circuit uses only edges of the original walk. $\square$

6.5 Capping while preserving local incidence

The preceding arguments apply to compact sphere sewings. To use them in a planar exhaustion, we must close a bounded contour window without changing its protected incidences. We first build admissible discrete caps. For the limiting quotient, we then use Brownian caps whose law is absolutely continuous with respect to an excursion law.

Lemma 6.10 (Axial completion as an actual sphere map). *A finite walk in $\mathbb{Z}_{\geq 0}^2$ with steps $\{(1, 0), (-1, 0), (0, 1), (0, -1)\}$, starting and ending at $(0, 0)$, defines an actual connected primal tree/cotree map on an oriented sphere. A down-step is matched to the most recent unmatched up-step of its own coordinate. The associated triangle and quadrangle cell complex, rather than the literal quotient of the chronological circle, is the discrete surface.*

For lattice endpoints a, b , an axial filler of m steps is possible only if

$$m \geq \|b - a\|_1, \quad m \equiv (b_1 - a_1) + (b_2 - a_2) \pmod{2}.$$

These conditions are sufficient with a nonnegative filler when both endpoints are nonnegative. Moreover, a continuous filler which is positive on its open time interval can be uniformly approximated, under height scale $\sqrt{n}$ and time scale n , by such fillers of prescribed positive macroscopic duration and compatible rounded endpoints. Strict positivity holds on every fixed interior time compact for all sufficiently large n .

Proof. The sphere map. The first-coordinate up-steps and their stack matches give a rooted plane tree. Thicken it on the sphere. Its complement is a disk following the tree contour, with corners marked by the intervening second-coordinate steps. Draw an arc in this disk for each matched second-coordinate pair. Stack matching makes the arcs noncrossing; their endpoints are the specified primal corners, with loops and repeated endpoints allowed. Cutting along the arcs leaves disks, giving a cellular sphere embedding. The arcs are the primal diagonals of the sewn

quadrangles, and their duals form the complementary dual tree. If there are n_1, n_2 up-steps of the two colors, respectively, the map has $n_1 + 1$ vertices, $n_1 + n_2$ edges, and $n_2 + 1$ faces, consistently with this construction. The case of a one-vertex initial tree is included. Notice that the word with steps $(1, 0), (-1, 0)$ still gives a sphere map; its identically zero second contour would collapse the literal time-circle quotient to a point.

Discrete fillers. First move each coordinate monotonically from a to b . This takes $\|b - a\|_1$ steps and stays nonnegative. Any even excess is supplied by cancelling positive-then-negative pairs in one coordinate. This proves sufficiency and necessity.

Approximation of continuous fillers. Fix a polygonal time partition on which the continuous filler oscillates by less than ϵ . Round the positions at its partition points to the required parity lattice. On each block the available number of steps is of order n , whereas the endpoint displacement is of order $\sqrt{n}$. The preceding inequality and parity conditions therefore hold for large n . Use monotone coordinate moves between the rounded endpoints and cancelling pairs for the excess. The path stays in their coordinate rectangle, enlarged by one height-mesh unit. This is within a fixed multiple of ϵ of the continuous path throughout the block. Interior positivity follows from its positive minimum on compact interior time intervals. At an excursion endpoint the first two steps, or the last two steps, may be arranged to enter, or leave, the strictly positive quadrant; these modifications have vanishing time and height size. Let $\epsilon \downarrow 0$ by a diagonal choice of partitions. Even cuts and parity-preserving translations give the required compatibility at retained block boundaries. $\square$

Lemma 6.11 (Protected capping of a bounded window). *Consider a two-sided resolved contour walk and a finite set of its triangles. Fix a finite number of cell-star layers to be protected. For each coordinate, suppose its values are unbounded below on both ends of the time axis. Then one can retain a larger bounded window and complete it by nonnegative axial prefixes and suffixes to an admissible finite excursion so that the protected triangles and their prescribed full-star layers have exactly the same primal, dual, side, and corner incidences. The prefix and suffix introduce no new partner for a protected step or corner.*

At the continuum level, a compact increment block whose law is absolutely continuous with respect to the corresponding Brownian increment law can be completed, up to a common translation of each coordinate, to a positive quadrant excursion whose whole law is absolutely continuous with respect to a longer Brownian quadrant excursion. Thus a compact capping used for the projection argument can have a sphere quotient. Arbitrary deterministic positive fillers alone do not imply that quotient conclusion.

Proof. Protect the discrete incidences. Start with a bounded window containing the desired triangles. In each coordinate find a visit to the left and a visit to the right strictly below every corner height and adjacent step level used in that window. Retain a window beyond all four visits. The two colors need not attain their low values at the same times. A scalar chord or matched step from the original window to outside the enlarged one would cross its coordinate's lower visit, which is impossible. Consequently every scalar partner needed for one full incidence layer is now in the enlarged window. Repeat this finite enlargement for the prescribed number of layers, and once more to protect their incidences. In a contour coupling, take strictly positive macroscopic height margins at these lower visits; uniform convergence preserves the inequalities, including adjacent lattice levels.

Keep this entire final window unchanged. Translate each coordinate by a sufficiently large lattice constant so that the window lies in the positive quadrant, respecting the parity lattice. Attach nonnegative axial paths from zero to its initial point and from its final point back to zero.

Lemma 6.10 gives an actual sphere map. All relations inside the retained window use the same equalities and intervening minima, and the retained lower visits exclude any partner in a cap. This proves the exact incidence claim. Independent choices of lower visits in the two coordinates suffice throughout.

Preserve the continuum sphere quotient. Let W be an increment block of duration T , and consider a Brownian quadrant excursion of duration $T + 2$. The law of its middle increment block, from time 1 to time $T + 1$, has a strictly positive density with respect to Brownian increments of duration T . Indeed, the Markov bridge decomposition writes that density, up to normalization, as

$$g(w) = \int_{\mathbb{R}_{>0}^2} e_1(a) e_1(a + w(T)) \mathbf{1}_{\{a + w(t) \in \mathbb{R}_{>0}^2 \text{ for all } 0 \leq t \leq T\}} da,$$

where e_1 is the positive entrance density, equal to the exit density by reversal, with the fixed Brownian covariance understood. These densities have Gaussian decay times a fixed harmonic power. The integral is finite and is positive for every bounded continuous w , since all sufficiently large coordinate translations place its range strictly inside the quadrant. This is also the usual killed Brownian bridge decomposition: the indicator supplies the killing during the middle block.

Sample the start height and the two caps from this excursion law conditionally on its middle increments being W . Since the given law of W is absolutely continuous with respect to Brownian increments, and $g > 0$, the resulting entire excursion law is absolutely continuous with respect to the reference excursion law. Its Radon–Nikodym derivative is the density of the prescribed middle increments divided by the normalized middle-increment density of the reference excursion. The middle block is reproduced exactly up to translation. Its caps are continuous and positive in their interiors. Theorem 3.2(ii)–(iii) gives the sphere quotient, and Lemma 6.10 approximates these caps by admissible axial fillers while retaining the actual middle steps. A countable choice among deterministic bounded windows does not change the absolute-continuity assertion, after conditioning on each chosen window of positive probability. $\square$

We finally identify the protected neighborhood before and after capping. Suppose a neighborhood in the original proper planar quotient has its *entire* chronological preimage in the initial protected time window, inside all retained lower barriers. Its topology agrees with that of the corresponding capped neighborhood. Every first scalar link from such a fiber is preserved and has no new exterior partner; induction along finite chains therefore gives the same complete fibers. A saturated open chronological set has open image under either quotient map, and the two restrictions are quotients by the same equivalence relation. They are consequently homeomorphic. This is the local identification used in an exhaustion; it does not identify arbitrary unprotected cap geometry with the original plane. For distinguished auxiliary profiles whose laws are singular at prescribed times, use their separately proved Moore quotient criterion and generator approximations in Proposition 6.9; the Brownian absolute-continuity completion is not asserted for those singular profiles.

7 Conditional local passage kernels

A protected chart reconstructs its graph from raw words and exact lattice heights. We now identify the conditional law of its passage observations. The proof has three steps: smooth the lattice heights with independent backward ladder blocks; compare the actual word law with an independent raw-block experiment by an exact density; and select spatial charts from a countable atlas using

the continuum fields. This argument applies even when the observation is a discontinuous function of the contour.

Throughout this section, $N \rightarrow \infty$ is the time unit. A lattice time is divided by N , a contour height by $\sqrt{N}$, and a graph cost by an arbitrary deterministic number $\mathfrak{c}_N > 0$. The results apply simultaneously along a prescribed subsequence and a diagonal of finitely many time and cost units. We use the certificate and universal-test spaces of Lemma 5.1 and Definition 5.2. More generally, the discrete observation may take values in any fixed compact metrizable space E .

The augmentation by canonical continuum coordinates uses the following convention. Given the complete contour and cut data $\mathcal{C}$, every additional coordinate not measurably reconstructed from $\mathcal{C}$ is sampled from its canonical conditional law independently of the entire actual and auxiliary-reference observation array. This applies also to any unreconstructed imaginary-field completion, stationary phase, or coordinate gauge. All augmented experiments and admissible laws below use this conditional-independent construction.

7.1 Backward ladder smoothing

We next supply independent noise in the exact height parameters. The noise must smooth lattice atoms, not merely have a weak Brownian limit. Backward ladder blocks provide both a local limit and the conditional path laws needed when the complete contour is retained.

Read an iid word backwards from a fixed time until its reduced suffix first contains a retained type-1 burger. Call the inspected segment a *ladder block*. Its leftmost burger is still available at the right endpoint. Consequently every F in the segment resolves internally. Its forward net displacement is $(1, Y)$ and its duration is T . Continue backwards to obtain successive ladder blocks. Since the stopping rule only inspects the letters already read, the blocks are iid. They are independent of the uninspected past. Their concatenation agrees with the resolved contour of the original word.

In the raw time and height units fixed above, let $\sigma_q^2 = (1 - b_F)/2$ be the variance of each limiting Brownian coordinate per unit time. The covariance per unit time is $b_F/2 = \rho\sigma_q^2$, as in the inventory invariance principle. For $h > 0$, put

$$g_h(t, y) = \frac{h}{2\pi\sigma_q^2\sqrt{1 - \rho^2}t^2} \exp\left\{-\frac{h^2}{2\sigma_q^2 t} - \frac{(y - \rho h)^2}{2\sigma_q^2(1 - \rho^2)t}\right\}, \quad t > 0, y \in \mathbb{R}. \quad (7.1)$$

This is the density of the duration and the forward second-coordinate displacement when the backward Brownian first coordinate first reaches $-h$.

Lemma 7.1 (Ladder local limit and conditional paths). *Let (T_j, Y_j) be successive ladder-block increments. If $m_N/\sqrt{N} \rightarrow h \in (0, \infty)$, then*

$$\mathbb{P}\left[\sum_{j=1}^{m_N}(T_j, Y_j) = (k, \ell)\right] = 2N^{-3/2}(g_h(k/N, \ell/\sqrt{N}) + o(1)) \quad (7.2)$$

on the feasible lattice $k + \ell \equiv m_N \pmod{2}$. The absolute $o(1)$ error is uniform in (k, ℓ) and, on compact subsets, in h . The corresponding rescaled contour, conditional on the displayed endpoint, converges to the Brownian first-passage bridge. This conditional convergence is uniform on compact subsets of $\{(h, t, y) : h > 0, t > 0\}$, for bounded continuous path tests. The bridge kernel is continuous there and has full support on compatible first-passage path tubes.

Proof. The iid-block contour invariance principle, applied backwards, and continuity of Brownian first passage at a positive level imply

$$\left(m^{-2} \sum_{j \leq m} T_j, m^{-1} \sum_{j \leq m} Y_j \right) \implies (\tau_1, V_1),$$

with density g_1 . The same argument gives the stopped contour path. Brownian first passage is continuous under uniform convergence on compact sets at almost every Brownian path: before each strictly earlier time the level has not been reached, and after the hitting time it is crossed arbitrarily soon. Truncation to a large time window removes no limiting mass.

Let $\phi(s, t) = \mathbb{E} e^{i(sT_1 + tY_1)}$. The characteristic-function powers converge uniformly on compact sets under the anisotropic scaling $(s, t) \mapsto (m^{-2}s, m^{-1}t)$. The limiting modulus is strictly below one away from the origin. To obtain a uniform bound near zero, set $r = \sqrt{|s|} + |t|$ and choose m comparable to r^{-1} . The rescaled frequency stays in a compact annulus. Uniform convergence of the powers there yields

$$1 - |\phi(s, t)| \geq c(\sqrt{|s|} + |t|). \quad (7.3)$$

There are no further periodic obstructions. Indeed the ladder-block support contains

$$(1, 0), \quad (2, 1), \quad (2, -1), \quad (3, 0),$$

realized respectively by h , hc , hC , and hcC . The differences generate the lattice $\{(k, \ell) : k + \ell \text{ is even}\}$, and every ladder block has $T + Y$ odd. Hence the only points of modulus one on the frequency torus are $(0, 0)$ and (π, π) . Away from neighborhoods of these points, the powers decay exponentially. Fourier inversion, using Equation (7.3), now gives an integrable majorant $\exp\{-c(\sqrt{|s|} + |t|)\}$ in the rescaled frequencies. Dominated convergence proves Equation (7.2); the second dual point gives the factor two. The same domination proves the stated uniformities.

For conditional paths, fix $\varepsilon \in (0, 1/2)$ and reserve the last $\lfloor \varepsilon m_N \rfloor$ ladder blocks. Conditional on the unreserved stopped path, the probability of the required remaining endpoint is a ladder local-limit mass. Dividing by the total endpoint mass, which is bounded below on the specified compact sets after multiplication by $N^{3/2}$, gives a bounded continuous limiting tilt for the unreserved path. For fixed ε , the density $g_{\varepsilon h}$ extends continuously by zero to nonpositive times and is bounded. Thus the unreserved path converges to the corresponding initial portion of the Brownian bridge.

The omitted portion is negligible as $\varepsilon \downarrow 0$. Its conditional probability of a specified duration or oscillation event is bounded by a constant times its unconditional probability: use the uniform upper local-limit bound for the unreserved endpoint and the positive lower bound for the total endpoint. The constant is uniform for $\varepsilon < 1/2$ on the compact terminal parameter set. Unconditional first-passage convergence and Brownian scaling make those probabilities tend to zero. Holding the unreserved path constant after its stopping time then changes the full stopped path by at most the omitted oscillation. This proves the conditional convergence and its sequential uniformity. The Brownian Markov transition densities in the half-plane are strictly positive; inserting finitely many intermediate open balls and then refining them proves full support of each compatible path tube. The same density formulas and the preceding truncation prove continuity of the bridge kernel. $\square$

Corollary 7.2 (Noise with a free first displacement). *Fix $a > 0$ and choose H_N independently and uniformly from $[a\sqrt{N}, 2a\sqrt{N}] \cap \mathbb{Z}$. Concatenate H_N ladder blocks. Writing T_N, Y_N for their duration and second displacement, one has*

$$\mathbb{P}[(T_N, H_N, Y_N) = (k, m, \ell)] = \frac{2}{aN^2} (g_{m/\sqrt{N}}(k/N, \ell/\sqrt{N}) + o(1)) \quad (7.4)$$

uniformly on compact interior parameter sets of $k + m + \ell \equiv 0 \pmod{2}$. The conditional contour kernel has the continuity and support properties in Lemma 7.1. Conditional on a fixed duration $k \sim tN$, the free pair (H_N, Y_N) has a positive continuous lattice density of order N^{-1} on $m + \ell \equiv k \pmod{2}$.

Proof. Multiply Equation (7.2) by $|[a\sqrt{N}, 2a\sqrt{N}] \cap \mathbb{Z}|^{-1}$. To obtain the duration marginal, apply the one-dimensional Fourier argument directly to $\phi(s, 0)$. Its support has span one since $T = 1$ and $T = 2$ both have positive probability. It gives

$$\mathbb{P} \left[\sum_{j \leq m} T_j = k \right] = N^{-1} \left(f_{m/\sqrt{N}}(k/N) + o(1) \right), \quad f_h(t) = \frac{h}{\sigma_q \sqrt{2\pi} t^{3/2}} e^{-h^2/(2\sigma_q^2 t)},$$

uniformly on the required compact sets. Averaging the $O(\sqrt{N})$ values of m with their normalized uniform weights yields the positive duration density $a^{-1} \int_a^{2a} f_h(t) dh$. Thus no absolute two-dimensional LLT error is summed over an unbounded displacement range. Dividing Equation (7.4) by this duration mass proves the last assertion. The path assertions are unchanged by conditioning on H_N . $\square$

7.2 Reference experiments and macroscopic density changes

In the reference law, retained raw blocks and relative height parameters are independent. We first extract their joint observation kernel. An actual experiment will have the same kernel if its change of density becomes a function of the macroscopic input alone.

A finite reference chart consists of disjoint raw word blocks $[a_i, b_i]$, with left extensions $[a_i, u_i]$ and used portions $[u_i, b_i]$; their full and extension durations; a finite instruction σ on the used portions; and integer differences of the heights at the *used starts* u_i . In each coordinate those differences are indexed by a forest. At even used starts the two heights have even sum. We denote the projected lattice of the retained forest differences by Λ_N . It is important that this is the image of the feasible height lattice, not an unrestricted coordinate lattice.

Local reduction is performed on each whole raw block. Its internally resolved matches and the list of its externally unresolved orders do not depend on types assigned to earlier external orders, by Lemma 5.3. The instruction σ supplies types only to externally unresolved F 's in the used portion. For recording a full contour of the raw block, give extension-only external F 's a fixed deterministic surrogate type. Subtract the surrogate path's value at u_i so that it is pinned at the used start. Exact used-portion heights are then reconstructed from x_i , the raw block, and σ , without using the actual extension increment. All local incidence tests inspect these used heights and steps. Altering only the extension-only surrogate types therefore leaves the observation unchanged. Each such alteration changes the full contour by at most twice the raw block's unidentified- F count, which is $o_{\mathbb{P}}(\sqrt{N})$ under the independent reference law.

Sample independent raw blocks conditional on their durations. Sample their retained shifts independently of the words with a positive smooth discretized density on Λ_N . Durations can be deterministic with positive limiting values, or have a positive continuous joint density on an open set of admissible cuts. A nonproduct duration density changes only the macroscopic input measure. All intrinsic chart times are recorded as a block label and a time relative to its used start u_i . Its paths are pinned to height zero there, and the flexible instructions use finite partitions in these relative coordinates. Absolute time origins and durations of omitted gaps belong to exterior nuisance data; they are used only when pushing a chart record into the complete contour. The observation Y_N does not inspect them. Let C_N record the intrinsic block durations and used-start

offsets $u_i - a_i$, shifts and all the rescaled block increment paths; let $Y_N \in E$ be any measurable observation of the raw blocks, exact parameters, and instruction. Tightness permits a common subsequence along which

$$(C_N, Y_N) \implies \nu(dc, dy) = \mu(dc)K_c(dy). \quad (7.5)$$

Regular conditional probabilities exist since the spaces are standard Borel. We extract jointly for a countable list of charts, instructions, and observation spaces. Their consistency is inherited from their joint discrete experiments.

Lemma 7.3 (Projected lattice densities). *Let π be a fixed integer linear map from a full-rank affine height lattice onto its image. The pushforward of a smooth compactly supported discretized density under π has a uniform Riemann-sum limit on compact sets of image parameters. The limit is the integral of the original density on the corresponding real fibers, with fixed covolume factors. These factors do not depend on the image parameter.*

Proof. Choose integer bases for the lattice, its kernel, and its image, or use Smith normal form. Each nonempty lattice fiber is a translate of the same kernel lattice. In these coordinates the assertion is the uniform Riemann-sum approximation of a smooth function on parallel affine fibers. On compact supports uniform continuity controls every cell error. The determinants of the three fixed bases give the covolume factors. $\square$

Lemma 7.4 (Macroscopic changes of density). *Suppose Equation (7.5) holds. Include nuisance data V_N among the reference variables, with $(C_N, Y_N, V_N) \implies \eta(dc, dv)K_c(dy)$. In particular, such data may be sampled independently conditional on C_N from kernels converging weakly, uniformly on compact input sets, to a continuous kernel $\lambda_c(dv)$. Then $\eta(dc, dv) = \mu(dc)\lambda_c(dv)$; this follows first for finite products of continuous tests and then by approximation. Let $e_N \rightarrow e$ be frozen data. Suppose on a bounded continuity cell the unnormalized density of an actual experiment is $W_N^{e_N}$ and there are positive factors $b_N(e_N)$ such that*

$$\mathbb{E}_{\text{ref}} |b_N(e_N)W_N^{e_N} - W^e(C_N, V_N)| \longrightarrow 0.$$

Assume W^e is bounded and continuous almost everywhere for the limiting input law, and $0 < \int W^e d\eta < \infty$. Then the actual limit has conditional observation kernel K_c , given (c, v) . The assertion remains true if a further macroscopic variable is a continuous function of (c, v, e) and contains c among its data. It also holds with random frozen data if the hypotheses hold along every convergent sequence in a full-measure set of their limiting values.

Proof. For a bounded continuous test f , the density error bounds the error in its weighted numerator by $\|f\|_\infty$ times the displayed L^1 error. Weak convergence and almost-everywhere continuity then give

$$\int f(c, y, v) W^e(c, v) \eta(dc, dv) K_c(dy).$$

The same calculation with $f = 1$ gives the strictly positive limiting denominator. Division and disintegration prove the kernel statement. The pushforward assertion follows because the density depends only on the macroscopic variables. For random frozen data, couple their convergent sequence almost surely and use the asserted sequential hypothesis, then integrate. Bounded test functions permit dominated convergence. Compact truncations are removed only after controlling their omitted mass under the actual experiment. $\square$

Smooth positive changes of the reference duration and shift densities satisfy this lemma directly. Thus they do not change K_c on their common domain. For disjoint charts sampled as a product reference experiment, the conditional kernels are products; macroscopic density changes preserve that factorization. This is an assertion about kernels after conditioning, not independence of the resolved inventory increments.

7.3 The exact resampling identity

Reserve one guard and one noise inside each omitted gap. The guard makes the later flexible comparisons stable; the noise supplies the free height displacement. Factoring the raw letter weights will then give an exact change of density, including for the finite empty-word law.

We first describe a deterministic decomposition pattern. There are r retained extended blocks, in chronological order, with positive limiting gaps before them. Choose a further gap after the last block when an exact terminal height is to be fixed. In a deterministic slot strictly inside each such gap, start a backwards search from an auxiliary cut. Read a guard of $\lceil a\sqrt{N} \rceil$ ladder blocks and then a noise with randomized depth in $[a\sqrt{N}, 2a\sqrt{N}]$. In forward order the noise precedes the guard. Restrict to the event that both fit strictly inside the slot. Freeze their endpoint times and the guard words, but not the randomized noise depths. Ordinary word pieces between these marked pieces, and every other mutable interior ordinary gap, are integrated as independent raw nuisance blocks. The frozen word data are the exterior prefix, the guards, and, in the finite experiment, the exterior suffix together with its prescribed resolved types. An interior ordinary raw piece is not frozen with its original resolved path: its flexible types could change when a preceding retained block is replaced.

Every externally unresolved F after a guard, when the local reduction begins at the end of the noise, has both candidates before that end. The guard's positive duration therefore supplies the separation in Lemma 5.4. Flexible orders which can be filled from the guard are locally resolved. On a sufficiently small stable contour tube, Lemma 5.5 makes all remaining ordinary increments exact functions of their raw words and the fixed instructions, independent of the noise displacements. For this global verification, the entire ordinary portion following a guard and preceding the next noise is a used portion in Lemma 5.5; in particular it includes any raw extension belonging to a smaller spatial chart. Orders internally resolved from this guard and ordinary portion have their input-independent types. The remaining orders have the positive supplier separation just described. These global instructions are only tools for the exact resampling identity; they are not extra data of the spatial observation. Figure 1 shows the placement of a used anchor relative to its independent raw block and the reserved noise.

Let $x_i \in \mathbb{Z}^2$ be the actual height at the used start u_i of retained block i . After freezing the outside initial height, the guards, cut positions, and the instructions, the successive noise displacements are

$$d_1 = x_1 - c_1, \quad d_i = x_i - x_{i-1} - c_i \quad (2 \leq i \leq r), \quad d_{r+1} = h_* - x_r - c_{r+1}. \quad (7.6)$$

The last term is present only when the terminal height h_* is fixed. For clarity, let A_i be the actual imposed increment from the end of noise i , through its guard and ordinary pieces and the current raw extension, to u_i . Let B_{i-1} be the actual imposed increment from u_{i-1} to the start of noise i . Then $c_i = B_{i-1} + A_i$, with the corresponding initial and terminal versions. These increments use the stable *global* instructions and the fixed guard context. They are exact functions of the raw pieces, including the current retained prefix, independent of the free noise displacements. They are not equated with the deterministic surrogate increments used to record extension paths. Their

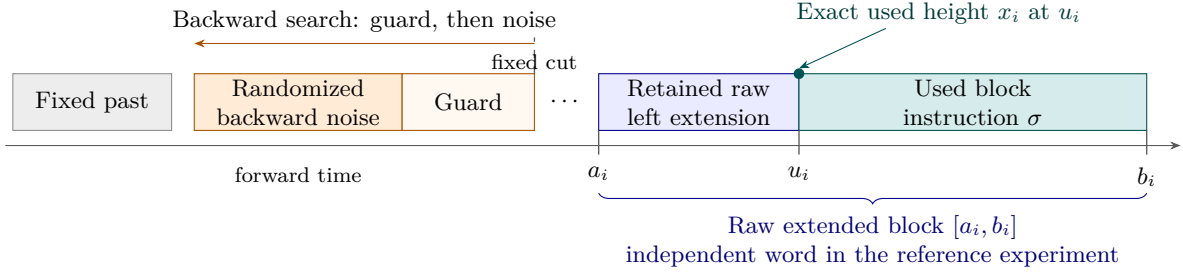


Figure 1: One block in the local resampling decomposition; ordinary pieces are suppressed. The exact height x_i is retained at the used start u_i . The local observation uses the raw word on $[a_i, b_i]$, this height, and the used-portion instruction σ . Noise paths and randomized depths are nuisance resampling data; guard words and cut times are fixed. Extension-only external F 's receive deterministic surrogate types for recording the contour, without changing the local observation.

normalized difference from the corresponding surrogate increments is bounded by the unidentified- F counts of finitely many independent raw blocks and vanishes in reference probability. The first r equations are a triangular integer affine bijection with determinant one. At even retained cuts, $x_i^{(1)} + x_i^{(2)}$ is even up to the fixed origin convention. The parity of the last equation is then automatic from the elapsed times and the fixed endpoint. Projecting the x_i to a forest introduces precisely the fixed fiber factors of Lemma 7.3.

For a fixed decomposition, let $q_{j,N}(t, d)$ be the iid probability of noise j having duration t and displacement d , including its independent randomized-depth probability. The conditional law of its raw path given those parameters is denoted by $\mathcal{B}_{j,N}^{t,d}$. If the full heights are sampled in the reference experiment with mass $r_N(x)$, the unfrozen words and full heights have, before normalization, density

$$\frac{\prod_j q_{j,N}(t_j, d_j(x))}{r_N(x)} \mathbf{1}_{\mathcal{T}_N} \quad (7.7)$$

relative to independent raw ordinary and retained blocks and the reference heights. Conditional noise paths are sampled independently from $\mathcal{B}_{j,N}^{t_j, d_j(x)}$. Here $\mathcal{T}_N$ is the chosen stable contour/parameter cell, including strict containment of the decomposition. Factors depending only on frozen observations have been omitted, since they cancel in the conditional ratio.

The finite experiment also freezes a suffix's resolved types. Equal entrance heights alone need not preserve its flexible decisions: those decisions compare burger identities. The next two statements supply the required protection near the terminal endpoint.

Lemma 7.5 (Protected prefix ranks). *Run two replacement words from the same fixed prefix. Suppose the type- i stack size never falls below an integer m_i during either replacement, and their two ending stack sizes agree. At their end the occupied ranks at most m_i agree by burger identity, for each type. Now process a common frozen suffix with prescribed flexible types. Until a first incorrect prescribed decision, all occupied ranks at most m_i continue to agree. In particular, a prescribed F decision correct in one run is correct in both whenever both current type-stack sizes are at most their respective m_i .*

Proof. Removing an original burger of rank at most m_i during a replacement would lower that stack size below m_i , which is excluded. During the suffix, equal prescribed types give equal heights. Removing above rank m_i does not change a protected rank; removing at such a rank removes the same identity in both runs. A frozen suffix addition at such a rank also has the same identity in

both runs. Induction proves agreement of all occupied protected ranks, even after the suffix dips below its original floor. At the specified F step the two available top burgers, or their absence, therefore agree exactly. Their creation-time comparison agrees as well. $\square$

Corollary 7.6 (Automatic terminal flexible decisions). *Fix an enclosing mutable interval strictly inside a positive quadrant excursion and freeze the exterior prefix and suffix. There are positive constants c_1, c_2 and a terminal collar such that, in a sufficiently small contour tube, all mutable paths stay above $m_i = \lfloor c_i \sqrt{N} \rfloor$, and all provisionally resolved suffix heights in that collar are at most m_i . If the mutable interval ends at the fixed suffix entrance heights and the imposed suffix decisions are correct before the collar, every decision in the collar is correct. Consequently the resampled word is empty precisely when its imposed heights stay nonnegative and end at zero.*

Proof. Choose each c_i below half the minimum of the limiting coordinate on the mutable interval. Its positive minimum exists by strict interior excursion positivity. The excursion coordinates tend to zero at the terminal endpoint, so choose a collar with both coordinates below $c_i/2$. Uniform convergence, with sufficiently small tube radius, gives the asserted integer inequalities. The provisional suffix has the same heights in both runs because its entrance heights and its resolved increments are fixed. Apply Lemma 7.5 at a first allegedly incorrect terminal decision. Nonnegative actual heights exclude unfilled orders, and zero final heights leave no burgers. $\square$

Remark 7.7. Equal endpoint heights and a guard alone would not prove this corollary. With prefix hc , replacing a subsequent hc by ch leaves the same endpoint heights but can change the next flexible choice after a fixed hH guard. Protected low ranks, or the separated supplier comparisons before the collar, are necessary.

Lemma 7.8 (Validity of the finite density identity). *Equation (7.7) is an exact identity for the iid experiment on a fixed decomposition and stable cell. In the empty-word experiment it is also exact after fixing the exterior prefix, the exterior suffix and its resolved types, and the resulting exact terminal height, provided the terminal consistency condition in Corollary 7.6 is imposed. The identity involves no local-limit assertion for the probability of an empty word.*

Proof. Expand the conditional probability as a finite sum over raw words and auxiliary depth choices. The iid letter weight factors over the disjoint pieces. Being a guard or a noise is a property of its own word: its successive retained type-1 burgers are detected by backwards inspection, and every F there resolves internally. Thus summing over noise words with a specified duration and displacement gives exactly $q_{j,N}$, and conditioning on those parameters gives $\mathcal{B}_{j,N}$. The guard and exterior-prefix probabilities are constant in the ratio. All interior ordinary pieces retain their independent raw letter weights. Lemma 5.5 makes the imposed ordinary resolutions genuine for every word in the stable cell. Changing variables by Equation (7.6) and dividing by the reference mass r_N proves the iid identity.

For the conditioned experiment, initially retain both indicators: the concatenated word reduces to the empty word, and its suffix has the prescribed resolved types. The empty-word normalizing constant and the raw exterior weights cancel on the frozen atom. The exact entrance height of the suffix is determined by its prescribed net increment and the terminal value zero. It is the h_* in Equation (7.6). Positivity and the stable instructions make the two retained indicators equal to one before the terminal collar; Corollary 7.6 makes them equal to one there as well. The remaining ratio is exactly Equation (7.7). $\square$

The exact identity is now established. To identify a limiting kernel, we must show that its density loses all dependence on microscopic information beyond the retained observation's macroscopic input.

Proposition 7.9 (Macroscopic form of the resampling density). *Fix a finite stable decomposition and restrict its macroscopic parameters to a compact set on which all noise durations are positive, all randomized first displacements are strictly inside their support, and the reference height density is positive. Include among the macroscopic nuisance variables the frozen contour pieces, integrated ordinary pieces, and the conditional noise contours. Fix a subsequential limiting law of the actual frozen data. The continuity cells can be chosen so that, for almost every frozen value under this law and every sequence of frozen data converging to that value, the density in Equation (7.7), after multiplication by its common power of N and fixed lattice factor, converges in the form required by Lemma 7.4. Its limit depends only on the macroscopic variables. On every cell containing a compatible strict path tube, its limiting normalizing integral is positive. Here finite frozen data range over atoms having positive probability in the chosen stable cell. In particular, a suffix-type atom comes with an actual compatible word in that cell for the protected-rank comparison.*

Proof. First keep all noise durations fixed and use full height parameters. Each noise mass is N^{-2} times a positive continuous function, up to a uniform $o(N^{-2})$, by Corollary 7.2. The reference mass for r two-dimensional heights is N^{-r} times its positive continuous density and its fixed lattice cell volume. With a terminal noise the quotient in Equation (7.7) therefore has common factor N^{-r-2} . Without that noise the corresponding factor is N^{-r} . These factors cancel in the normalized conditional law. The remaining quotient converges uniformly on the compact parameter set to the product of the translated functions in Equation (7.1) divided by the reference height density.

The translations in Equation (7.6) are exact functions of the raw words, guard context and global instructions. Their normalized limits depend only on the recorded increment paths. In particular the true extension increments in A_i need not equal the locally recorded surrogate increments: their difference is bounded by twice the corresponding raw unidentified- F counts. Unresolved flexible counts in independent raw blocks are $o_{\mathbb{P}}(\sqrt{N})$, so the macroscopic block limits are the ordinary Brownian increment laws, for every one of the finitely many instructions. The extra global extension instructions do not enter the local observation: its exact heights are anchored at u_i and it uses only the local used-portion instruction. Changing those extra instructions changes neither Y_N nor the reference limiting input, and hence does not change the extracted joint law or its kernel. Every interior ordinary gap is likewise a free raw nuisance block. Its actual resolution under the global guard context may depend on neighboring retained words, but its difference from any fixed local surrogate is bounded by twice its own unidentified- F count. The same unconditioned estimate therefore gives its macroscopic increment path. Only the exterior prefix and the finite suffix with prescribed types enter through fixed resolved paths; guards resolve internally. No conditional unidentified-letter estimate for frozen interior words is used.

Conditional noise contours converge by Lemma 7.1, uniformly on the compact endpoint set. Their exact endpoints may depend on additional raw extension data and global instructions beyond C_N . Those normalized endpoints differ by $o_{\mathbb{P}}(1)$ from the endpoints computed from C_N and the macroscopic ordinary nuisance paths. Uniform bridge-kernel convergence and continuity therefore replace the former by the latter in L^1 for bounded continuous conditional tests. This proves the limiting factorization with K_c ; finite- N conditional independence given C_N alone is not required. They can therefore be adjoined as continuous conditional nuisance kernels in Lemma 7.4. Concatenation of finitely many continuous paths with positive durations and the prescribed endpoint translations is continuous. For each of the countably many tube centers

and decomposition patterns, mix the limiting reference resampling law over the actual limiting frozen-data law. A real-valued tube-distance variable has only countably many atoms under this mixture. Choose a countable dense family of radii avoiding the union of these atom sets. Fubini's theorem then gives a full-measure set of frozen values on which all the chosen boundaries have zero conditional reference mass. Multiplication by the bounded limiting density preserves that assertion. For a sequence of frozen data converging to a value in this set, conditional bridge convergence and continuity of concatenation imply that the cell indicators converge in L^1 in an almost sure coupling. Together with the uniform density approximation this is the required L^1 macroscopic tilt approximation. This argument asserts a common full-measure set for the actual mixture; it does not assert conditional boundary continuity at every possible frozen contour.

If only forest parameters are retained, complete them by auxiliary full heights with a positive smooth density and integrate along fibers. Lemma 7.3 proves the same assertion with the correct projected lattice factor. If durations are integrated, sum the local-limit masses against the original smooth cut density. Its Riemann-sum factors appear identically in numerator and denominator. One must not freeze a randomized first-depth count in this calculation: that would remove one of the free displacement coordinates.

For positivity, take a compatible parameter/path tuple in the interior of the cell. Shrink to a product of path tubes and a small parameter neighborhood still in the cell. The endpoint densities and reference height density have positive lower bounds there. Free Brownian blocks have positive probability to follow each required tube, and the conditional noise bridges do too by Lemma 7.1. Thus the limiting weighted integral over this smaller set is positive. This argument is stable in a neighborhood of the frozen macroscopic data. Its lower bound may depend on that data and the chosen cell; no uniform lower bound as cells shrink is asserted. $\square$

7.4 Common kernels and changes of scale

The local limit turns the exact density into a macroscopic tilt on each stable cell. Exhausting those cells identifies the kernel conditional on the complete contour. We then repeat the discrete approximation argument to allow further limits and changes of quantum units.

Theorem 7.10 (Common conditional kernels on fixed charts). *Fix a finite collection of protected time charts, fixed instructions, and measurable E -valued observations of their raw words and exact parameters. Extract their independent reference experiments jointly as in Equation (7.5). Consider either the bilateral inventory word or a finite word conditioned to be empty. In the finite case keep every chart in a strict interior time interval. Use the same time and distance units in the reference and actual experiments.*

In every joint subsequential limit, conditional on the complete limiting contours, the retained cuts and marks locating the charts, and any independently adjoined surface-coordinate data, the observations have the reference kernels evaluated at their chart inputs. For charts in disjoint spatial buffers these kernels are conditionally independent. The assertion includes the universal endpoint observations, not only existential path certificates.

Proof. It suffices initially to test finitely many bounded continuous functions of the observations and of the contour on a bounded time window. Choose deterministic enclosing slots inside the positive gaps between the extended blocks. In each slot use a noise and guard as above. A countable choice of enclosing slots suffices because the gap conditions are strict. If the block cuts are Poisson cuts, retain their smooth joint density inside these enclosing slots rather than conditioning on each of their microscopic durations. Noise and guard endpoints may be conditioned on separately; they are determined in the reserved slots.

The noise and guard pieces fit with probability tending to one as $a \downarrow 0$, after $N \rightarrow \infty$. Indeed their total backwards depth is at most $3a\sqrt{N} + O(1)$. The contour invariance principle identifies their duration with a backwards Brownian first passage to a level tending to zero at an ordinary interior time. That duration tends to zero in probability. This applies under the finite excursion law by its interior absolute continuity. With a fixed, guards and noises have positive limiting durations. These durations, supplier windows, depths, and displacements can be truncated to compact interior sets with arbitrarily high actual probability.

On this truncated event, Brownian genericity and Lemma 5.4 give finitely many stable signs on a sufficiently small contour tube. Before the first reserved slot use the actual unchanged past. Between slots apply Lemma 5.5 successively. In the finite experiment, freeze the raw exterior prefix and suffix, the suffix's resolved types, guard words and endpoint cuts. The height at the end of the mutable region is then the negative of the prescribed suffix increment. Keep the randomized noise depths free. The protected-rank Corollary 7.6 handles the final collar; the separated comparison instructions handle the remaining suffix.

Disintegrate after restricting to the actual stable cell. Every frozen atom under consideration then has positive probability in that cell and supplies an original correctly resolved word satisfying its protected floors. The exact conditional distribution on each such atom is Lemma 7.8. Along convergent frozen data, Proposition 7.9 gives a macroscopic density with positive limiting denominator. The actual limiting paths are in the required support: the free blocks are Brownian paths under interior excursion absolute continuity, and the selected noises are first-passage portions with positive duration and interior randomized depth. An open tube about such a tuple contains a smaller compatible product tube of the type used in the positivity proof.

Apply Lemma 7.4. The full limiting contour on the window is reconstructed by concatenating the retained block paths, noise paths, integrated interior ordinary paths, and the fixed resolved exterior paths, with the displayed height translations and exterior absolute time origins. The retained cut and mark data locate each chart in that contour, so the pair determines its intrinsic input C . Conditioning on this pair therefore leaves the reference observation kernel unchanged. For deterministic cuts the extra cut data are redundant; for random cuts they are part of the conditioning sigma-field. In the bilateral experiment there is no terminal empty-word factor. Any further exterior contour window may be included among the nuisance pieces; unidentified flexible effects there are negligible in rescaled increments. In the finite experiment the prescribed exterior increments are exact after the terminal consistency argument.

After extracting the actual limiting frozen-data law, choose continuity cells as in Proposition 7.9. Intersect its full-measure admissible set with the generic, strict-margin set. The preceding assertions hold along every sequence of frozen data converging to a value in this intersection. Couple the actual frozen data, apply this sequential assertion almost surely, and integrate as in Lemma 7.4. Exhaust the compact parameter cutoffs, the supplier windows and the countable stable cells. Their omitted actual probabilities tend to zero. A monotone-class argument for continuous cylinder tests and increasing time windows gives conditioning on the complete contour.

For disjoint buffers, Lemma 5.8 uses disjoint raw block families. Sample their shift parameters independently in the product reference experiment. The union of their coordinate forests is again an admissible forest; the calculation above permits the actual macroscopic height constraints to couple those parameters but not their conditional observation kernels. Hence the reference product kernels remain products after conditioning on the complete contour. All arguments applied to arbitrary E -valued measurable functions of the finite chart, so they also apply to the universal endpoint variables. $\square$

Proposition 7.11 (Closure under varying laws and changes of units). *Fix the model parameter q . Form the joint array consisting of all marked certificate and universal observations under consideration, their contours and cut data, and the independent reference experiments for a countable list of fixed charts, refinements, instructions and quantum offsets. Use one common Polish array schema, with matching actual and reference readouts in every row. Retain the full canonical joint continuum input: contours, cuts and marks, decorated surface fields, and reference internal metrics on countably many fixed collars. Fix its law in the chosen units; in the bilateral applications this is the scale-covariant continuum law. Its charts have the positive durations, extension lengths, gaps and Brownian-generic inputs required by Theorem 7.10. For finite experiments require also the strict interior endpoint margins of that theorem. Call a law admissible if it is a joint subsequential law of these arrays from the discrete inventory experiments, with this continuum marginal, arbitrary deterministic positive cost units, and every fixed retained time unit tending to infinity in lattice units. The continuum coordinates belong to this same joint approximating coupling and satisfy the conditional-independent augmentation convention at the start of this section; a fixed continuum marginal alone is not the definition of admissibility.*

The admissible family is closed under further weak extraction, including extraction from a varying sequence of admissible laws. The same holds after deterministic changes of the base quantum units and relabeling the fixed offsets, provided the corresponding discrete arrays are retained and the target continuum marginal has these same genericity and strictness properties. Actual and reference coordinates use identical target time and cost units, and each chart keeps its positive durations, extensions and gaps in its own normalized coordinates. Each reference coordinate has smooth duration and shift densities in its own normalized units; it is not obtained by allowing one old density to degenerate under the changing zoom. Every resulting law has the fixed-chart common-kernel identities of Theorem 7.10; its independent disjoint-chart reference experiments give product conditional kernels. Consequently the spatial, restriction and field-density conclusions below hold anew in that law. Kernels may change with the law: the assertion identifies them with the reference kernels extracted in the new joint array. The fixed full joint continuum marginal remains the same in the further limit.

Proof. Use a metric d_w for weak convergence on the countable product of the observation spaces and their Polish contour and auxiliary spaces. Let $\mathcal{L}_j$ be admissible laws converging to $\mathcal{L}$. By admissibility, for row j choose a discrete approximant whose law is within $1/j$ of $\mathcal{L}_j$ in d_w . Choose it far enough along its defining sequence that each of the first j retained time units exceeds j lattice steps and all the prescribed finite coordinate approximations hold. The triangle inequality gives a single diagonal array converging to $\mathcal{L}$.

If a continuum coordinate has no discrete definition, retain its actual joint coupling as follows. Couple a row's discrete raw arrays to their limit almost surely, adjoin the row's canonical continuum tuple with its prescribed law conditional on the complete limiting contour and cut data, independently of the entire coupled actual and reference observation array, and attach the same tuple along the row. This gives joint convergence and preserves every discrete raw-word marginal. The resampling identity is applied to those raw coordinates; the canonical contour-to-surface and field-to-metric relations are used only in the limiting continuum readout.

Each diagonal row is still an actual raw-word experiment with its independent reference experiments. Equation (7.7) is valid there before any limit. Apply its local-limit and macroscopic transfer proof to this diagonal sequence, choosing continuity cells for its actual limiting frozen-data mixture. The retained canonical joint marginal supplies genericity, positive durations and gaps, and the required finite-volume endpoint margins in the limiting chart. Thus the free raw letters, positive-duration compact truncations and cost compactification are the same as in Theorem 7.10.

The time unit is allowed to vary with the row, and the proof imposes no restriction on the deterministic cost denominator. Thus the theorem produces the new conditional kernels. Products of independent reference laws remain products under joint weak convergence; their disintegrations give the asserted factorization. Countable selection and the same local field-density argument then give the spatial conclusions.

For a change of base units, first apply that fixed change in row j , including it in every retained actual and reference coordinate, and only then choose its sufficiently fine discrete approximant. This ensures the required divergence of every fixed rescaled lattice time unit even when the base change varies with j . Include the countable union of the row-wise relabeled coordinates, and sample each target reference density directly in its normalized units. Projection onto the retained continuum coordinates proves the last assertion. In particular no continuity of the map from a field to its internal metric is needed: the entire contour/mark/field/metric coupling is retained. For a random auxiliary continuum coordinate A beyond the complete contour and cut data $\mathcal{C}$, the extension of conditioning follows from its fixed joint law with $\mathcal{C}$. Given a bounded continuous test $\phi(A)$, approximate $\mathbb{E}[\phi(A) \mid \mathcal{C}]$ in L^1 of the fixed $\mathcal{C}$ marginal by bounded continuous functions of $\mathcal{C}$. The row-wise conditional identity, tested against bounded continuous functions of $(\mathcal{C}, Y)$, then passes to the limit with an error uniform in the row. Letting the L^1 error vanish preserves conditional independence of A and the observations given $\mathcal{C}$. Deterministically reconstructed continuum coordinates are the immediate special case. This proof uses discrete approximation and the exact density again; it does not assert weak closure of abstract conditional-independence relations or pointwise convergence of regular conditional kernels. $\square$

7.5 Spatial selection and changes of field law

We have proved the kernel identities for fixed chart descriptions. A spatial observation chooses its chart using local continuum geometry. The following conditional-expectation identity permits that choice once the evaluated selected kernel is measurable in the stated local data.

Lemma 7.12 (Countable selection of conditional kernels). *Let $\mathcal{C}$ be a standard Borel macroscopic variable and let Y_j , $j \in \mathbb{N}$, be a countable consistent family of observations. Suppose, for each j , its conditional law given $\mathcal{C}$ is $K_j(T_j(\mathcal{C}), \cdot)$. Let J be measurable with respect to $\sigma(\mathcal{C})$, and suppose the selected probability kernel*

$$Q = K_J(T_J(\mathcal{C}), \cdot)$$

is measurable with respect to a smaller macroscopic sigma-field $\mathcal{H} \subset \sigma(\mathcal{C})$. Then, for every bounded measurable f ,

$$\mathbb{E}[f(Y_J) \mid \mathcal{C}] = Qf, \quad \mathbb{E}[f(Y_J) \mid \mathcal{H}] = Qf.$$

In particular, this applies if both J and T_J are $\mathcal{H}$ -measurable. The same assertion holds for finite selections with product joint conditional kernels. Countable exhaustions of consistent observations inherit the resulting kernel.

Proof. Multiply the conditional identity for Y_j by $\mathbf{1}_{\{J=j\}}$ and sum over j . This proves the first identity. The assumed measurability of Q and the tower property give the second. Products of test functions prove the finite joint assertion, and a monotone-class argument using cylinder functions proves the exhaustion assertion. $\square$

Let h^{IG} denote the imaginary field modulo its period. Area marks can be realized as a Poisson process on $\mathbb{C} \times (0, \infty)$ with conditional intensity $\mu_h(dz) du$; marks of level at most a prescribed intensity give the nested time-cut experiments. Restrictions to disjoint spatial sets are conditionally independent given h .

Theorem 7.13 (Local spatial kernels with separate field buffers). *Let $K \Subset U$ be a compact confinement in an open real-field domain, and let V be an open imaginary-field domain containing a neighborhood of $\bar{U}$. One may alternatively first replace U by any relatively compact neighborhood of K inside it. The limiting local certificate and universal-endpoint observations in K , interpreted with open loss inside U , have conditional kernels measurable in*

$$h|_U, \quad h^{\text{IG}}|_V \pmod{\text{period}}, \quad \text{area marks in } U, \quad O, \quad (7.8)$$

where O is the finite relative global order of the buffered local traversals needed by the observation. Conditional on the complete continuum fields, curve, and marks, observations in disjoint real-field buffers have independent kernels. The imaginary buffers may be larger and may overlap for this conditional-independence assertion.

The kernels are intrinsic under changes of the quantum-surface coordinate representation. They are the same in the bilateral and strict-interior finite-excursion experiments with common time and distance units. No fixed positive separation between the test and the real-field boundary is required uniformly across all applications: any prescribed $K \Subset U$ is allowed before taking the limit.

Proof. Choose nested neighborhoods

$$K \subset \text{int } K_* \Subset U_0 \Subset U_1 \Subset U.$$

The imaginary field on V reconstructs all directed U_1 -traversals reaching $\bar{U}_0$. There are finitely many such traversals, as in Lemma 5.8. The real field on U gives their quantum-area clocks and the quantum lengths of their labelled frontier arcs. Retain their genuine relative order O . This order compares visits in distinct traversals; the directed clocks compare visits in the same traversal. No absolute origin on the global tour is required for these comparisons.

We describe the local tests used to select a protected chart. At a fixed Poisson refinement, choose finitely many used blocks inside these clocked traversals, with their closures in U_0 , and left extensions with closures in U_1 . Require their used interiors to cover every visit to K_* . This is a test on the reconstructed finite traversal family: every visit to K_* lies in one of its members. The marks in U locate the cuts along each directed clock. They therefore give the block durations and extension offsets, even though they do not give the absolute global Poisson indices of those cuts. The local frontier lengths give the two increment paths, pinned at each used start.

For these blocks, Lemma 5.6 gives the two labelled port relations and the forest differences within each overlap component. Evaluate the retained suffix, prefix and intervening minimum tests using these paths and differences. Among the protected candidates passing those tests, the port relation distinguishes the genuine contacts from the other candidates. Select the rational band which strictly separates their height sets, with the empty-set conventions in Lemma 5.7. Its existence follows from that lemma. Its selection uses the two local height sets, not the value of a minimum in an omitted gap.

The flexible-order instruction is selected from the same data. Consider the possible pairs of scalar suppliers for a decision in a used block, with each supplier at or before that block's left-extension start. Their scalar contacts are read from the labelled frontier relations. Each supplier has the same spatial image as the decision, so its visit occurs in the retained family of U_1 -traversals. The closed scalar contact relations on the finitely many closed traversal records make the candidate set compact. Positive extension duration separates its suppliers from the used decision times. The order O and the directed clocks decide which member of the pair is later. The two possible orders therefore give compact disjoint sets of decision times, by the argument of Lemma 5.4. Choose a finite rational partition of the used clock whose boundaries avoid these

sets, and prescribe the corresponding type on each part. This is the local test for an admissible instruction. A bounded global supplier window and a small tube for the complete contour are used in Lemma 5.5 to prove that the selected instruction is eventually correct in the discrete words. They are not inputs to its local selection. The same distinction applies to the omitted minima used to prove band clearance.

Thus the data in Equation (7.8) determine a finite intrinsic description: the retained block pattern, side labels, forests, rational bands and flexible instructions. They also determine its continuous input: intrinsic durations and extension offsets, used-start-pinned increment paths, and the forest differences. All choices are taken from the countable atlas and may be made by its fixed enumeration. Extension-only external F 's keep their deterministic surrogate types. Their normalized typing error vanishes under the independent reference law by Proposition 4.3; their exact types do not enter the local observation because its heights are anchored at used starts. The extra global instructions used to prove the resampling identity likewise contribute no additional input to this observation.

To apply countable selection, distinguish this intrinsic description from the global labels locating its blocks in the complete contour. Those labels are measurable from the complete contour and cut data. They need not be measurable from the local fields. The reference experiment, however, samples only the intrinsic raw blocks, durations and shifts; its observation uses no absolute tour start or omitted-gap duration. Two realizations of the same intrinsic description therefore use the same reference experiment and observation after relabelling their blocks. Smooth changes of the reference duration or shift densities leave its conditional kernel unchanged by Lemma 7.4. For the countable atlas, choose these common kernel versions simultaneously. Consequently the reference kernel evaluated at the selected chart depends only on the local intrinsic description and input just constructed.

Theorem 7.10 supplies the conditional identity for every globally located member of the atlas. Lemma 7.12 now applies because its selected evaluated kernel is local. Exhaust the finite observations and rational open losses to obtain the asserted spatial kernel. This selection is performed on the limiting continuum data, after the fixed-chart identities have been proved. Because the chart observations were extracted jointly, restriction from a larger protected chart to a smaller one is a measurable projection of that same family, so the kernels inherit restriction compatibility. Versions may be fixed for the countable atlas and the countable deterministic field changes used later, since absolute continuity preserves their common null sets.

For disjoint U the protected retained block families are disjoint. The reference experiment and Theorem 7.10 therefore give product kernels conditional on the complete continuum data. Selection of their indices by that data preserves the product identity. Overlap of the imaginary buffers is harmless here, because the entire imaginary field is conditioned on. To use independent Dirichlet field parts in a later resampling experiment one separately chooses disjoint imaginary testing domains.

The complete cone or sphere contour supplies its ambient decorated surface by Theorem 3.2(ii). Apply the local atlas directly to that surface's fields and traversals. Write ℓ for the data in Equation (7.8), $s(\ell)$ for the selected intrinsic description, and $c(\ell)$ for its intrinsic input. The same data give a map $\Phi_{s(\ell)}^\ell$ from its intrinsic time records to spatial records along the local traversals. The spatial conditional kernel is consequently

$$Q_\ell = (\Phi_{s(\ell)}^\ell)_* K_{c(\ell)}^{s(\ell)}.$$

All three constructions use the same local data in the cone and sphere settings, and Theorem 7.10 supplies the common reference kernel K_c^s in both. The clocks, quantum frontier lengths, used-start

height differences and side labels are intrinsic under conformal coordinate changes. Transporting the local fields and traversals therefore transports the spatial record while preserving its intrinsic conditional law. This proves covariance and the common cone/sphere interpretation with the actual field normalization retained. $\square$

Corollary 7.14 (Field-density transfer and local pasting). *The spatial kernels in Theorem 7.13 are unchanged under a locally absolutely continuous change of the continuum field law. For a Cameron–Martin modification, work with an actual Gaussian local reference field and a fixed compactly supported smooth real function f in its Cameron–Martin space. In the circle-average-pinned whole-plane reference used below, it suffices to choose the pinning circle outside $\text{supp } f$.*

One may then use the kernel for $h + f$ on a compact interior patch and retain the kernel for h outside a larger disjoint neighborhood on which $f = 0$. Conditional on the fields these are the prescribed product kernels. The resulting subprobability comparison is made before selecting a hit or a shortcut event. This is a comparison in the Gaussian reference law; it does not assert that an arbitrary shift preserves the global unit-area normalization of a sphere.

Proof. If a joint law has disintegration $\mu(dc)K_c(dy)$ and the field law is tilted by a density $R(c)$, its joint law is $R(c)\mu(dc)K_c(dy)$. Thus the conditional kernel is unchanged as a function of the realized macroscopic input. Bounded density approximations and truncation give the assertion for an arbitrary absolutely continuous tilt. The fields and any embedding gauge may be included in the macroscopic data: the gauge is adjoined conditionally independently of the microscopic observations given the contours.

For f in the specified Cameron–Martin space, the Cameron–Martin theorem gives the field density with the actual normalization retained. A shift with nonzero average on a fixed pinning circle would move the field to a disjoint affine support, so that case is not covered. Apply Theorem 7.13 to the shifted interior and unchanged exterior patches. Resample the area marks inside the changed region with their correct conditional intensity μ_{h+f} and retain the marks outside, where the intensity is unchanged. All chart refinements can be coupled this way. Conditional product kernels then give the stated joint law on the two separated regions; extend to the unobserved region by a regular conditional distribution. Only afterward restrict to a selected input event and bound the Cameron–Martin likelihood. In particular, a larger imaginary buffer in Theorem 7.13 does not enlarge the support required for this real-field modification. $\square$

Remark 7.15. Corollary 7.14 is a statement about conditional laws. It does not identify a passage realization at $h + f$ with a deterministic length reweighting of one at h . Weyl scaling will only be used for the known LQG metric D_h .

7.6 One-sided domination for selected local views

The finite-line argument later uses a weaker conclusion than equality of kernels. A selected word neighborhood can have bounded entropy relative to fresh iid letters. We therefore record a domination estimate which does not condition the incoming stacks to have a Brownian law.

Proposition 7.16 (Buffered slot domination). *Consider a finite chronologically ordered list of disjoint slots. In each slot a retained block has a left safety band of positive limiting duration. Conditional on the past before the first slot and on an auxiliary code or marks independent of all fresh letters, the entire list of retained cuts, durations, extension offsets and backwards guard/noise search anchors is fixed before any retained fresh letters are inspected. Independent auxiliary choices have total conditional selection weight at most one. At most one new starting-height difference per*

coordinate is retained, relative to an earlier observed block. The list of differences is a forest in each coordinate. An arbitrary forest can be put in this form by replacing each component's tree differences by differences from its chronologically earliest vertex, an integer unimodular change of coordinates. Fix a finite list of flexible-order instructions, and restrict to compact ranges of durations, shifts, ordinary increments and noise parameters on which the guard/noise construction fits in the safety bands and its endpoint density is positive. Consider the subevent on which the specified instructions are genuinely compatible.

Under fresh chronological iid sampling, conditional on any incoming past, the joint subprobability law of the raw retained blocks and their exact retained shifts on this subevent is bounded by a fixed constant times an independent raw-block reference law with positive smooth shift densities on its feasible projected lattice. The constant is uniform in N and in the incoming past after the specified compact truncations. It is multiplied only by a finite factor when a finite instruction list is allowed. Arbitrary measurable observations of the blocks obey the same domination.

Proof. Work throughout with unnormalized iid cylinder weights. At the fixed right anchor in a safety band, read backwards a guard and then a noise, so in forward order the noise precedes the guard. Expand the event as a sum over the guard stopping word g , noise duration t , ordinary nuisance words u , and retained word b . Write $w(g), w(u), w(b)$ for their raw iid letter weights; keep any anchor-mark probability as a separate factor. The randomized depth probability is included in q_N . On the compatible-instruction subevent, a prescribed retained starting-height difference is an exact affine constraint on the corresponding noise displacement coordinate. The translation is determined by the incoming history, g, u, b and the instruction. Thus the summand, after summing over noise words, is bounded by

$$w(g)w(u)w(b) q_N(t, d), \quad q_N(t, d) \leq CN^{-2},$$

with a sum over any unretained displacement coordinates. The local-limit bound is uniform over the truncated displacement range, so its translation may depend on g, u, b . All additional compatibility indicators can be dropped for this upper bound.

There are $O(N)$ admissible noise durations in the slot. Summing over them therefore leaves at most CN^{-1} for the two-dimensional free displacement. Retaining only one coordinate sums over $O(\sqrt{N})$ values of the unused coordinate and gives at most $CN^{-1/2}$. Retaining none gives a bounded factor. These are precisely the orders of the corresponding smooth reference lattice masses, whose densities are bounded below on the chosen compact target ranges. The projected parity constraints change only fixed covolume factors. For each fixed g, t , the weights of all ordinary words of the resulting lengths sum to one. The backwards guard is a stopping word read from a fixed anchor, so the sum of its stopped-cylinder weights over all allowed lengths and words is at most one. Summing the independent mark weights also contributes at most one. These statements concern raw weights: no division by the probability of a guard, fit event, or successful instruction is made. There is consequently no extra sum over possible search anchors. The bound is uniform in the incoming stacks because the fresh raw letters and the backwards ladder test inside the safety band do not depend on those stacks.

Apply this calculation in chronological order. Previously retained parameters are fixed at each step; the forest condition ensures that there is at most one new target for each smoothed coordinate. Thus no second exact constraint is imposed on the same free coordinate. Multiply the finitely many bounds and integrate the previous parameters. Sum over the finite instruction list if necessary. Pushing the dominated law through a measurable observation preserves the inequality. $\square$

Lemma 7.17 (Entropy transfer of reference null tests). *Let P_N be a possibly selected law and P_N^0 its fresh-word comparison, with $\text{Ent}(P_N \mid P_N^0) \leq C$. Suppose on events G_N the observed subprobability laws under P_N^0 are bounded by C' times reference laws Q_N . If $Q_N(A_N) \rightarrow 0$, then $P_N(A_N \cap G_N) \rightarrow 0$. The assertion remains valid with a bounded average conditional entropy after adjoining a history variable with the same marginal in the two laws.*

Proof. For $B_N = A_N \cap G_N$, domination gives $P_N^0(B_N) \leq C'Q_N(A_N) \rightarrow 0$. The binary relative-entropy inequality gives, when this probability lies in $(0, 1)$,

$$P_N(B_N) \log \frac{1}{P_N^0(B_N)} \leq \text{Ent}(P_N \mid P_N^0) + \log 2.$$

The conclusion follows. Conditional entropies integrate to joint entropy when the history marginal is unchanged, giving the last assertion. A tight, rather than deterministically bounded, conditional entropy can first be truncated. $\square$

Proposition 7.18 (Null transfer on openly fillable interior charts). *Consider selected slots as in Proposition 7.16, with no slot meeting a distinguished center time. Suppose their macroscopic input and exact lattice observations have tight absolute continuity relative to the independent reference experiment, obtained for example by Lemma 7.17. Restrict to a protected chart which is openly Brownian fillable: conditional on its retained input there is positive probability of filling the omitted pieces by ordinary Brownian or first-passage bridge pieces so that the chart, with its prescribed signs and gluing instructions, occurs in an ordinary Brownian sewing with strict unused-gap clearance.*

Any reference-null event for the local observations in that ordinary sewing is also null in the selected limit on this chart. In particular, once an almost-sure open-port crossing estimate has been proved for the ordinary bilateral passage rule in the same units, its failure on such an interior chart has probability zero in the selected limit. No absolute continuity at the omitted distinguished center is asserted.

Proof. Let $\mu(dc)$ be the retained reference input law, K_c its conditional observation kernel, and $\lambda_c(dv)$ the conditional law of the ordinary and bridge filling pieces. On a compact stable cell, write $a(c, v)$ for the positive translated endpoint-density multiplier in Proposition 7.9. Thus it includes the product of noise endpoint densities divided by the reference height density, together with any duration or ordinary-input density factors. The ordinary sewing induces, up to a common positive normalization, the retained subprobability law

$$f(c) \mu(dc) K_c(dy), \quad f(c) = \int a(c, v) \mathbf{1}_{\mathcal{T}}(c, v) \lambda_c(dv).$$

This follows from Theorem 7.10 and the exact macroscopic density: the multiplier depends on the retained input and filling, so leaves K_c unchanged. Open fillability and positivity of a on a smaller compatible tube give $f(c) > 0$. In particular the weight is the density-weighted filling integral, rather than only the unweighted probability of finding a compatible path. If B is a local failure event, measurable in the retained input and observation, which is null in the filled ordinary sewing, then

$$0 = \int f(c) K_c(B_c) \mu(dc).$$

Hence $K_c(B_c) = 0$ for μ -almost every openly fillable input. One may restrict first to $f(c) \geq \varepsilon$ and then let $\varepsilon \downarrow 0$; no uniform lower bound on filling probabilities is needed. Countable protected charts and rational open losses give a common reference-null set. Tight absolute continuity, or

Proposition 7.16 followed by Lemma 7.17, transfers that null set to the selected law. For a sequence of scale-dependent reference laws the same argument applies in each jointly extracted limit, so it also applies to mixtures over the viewing level. All retained slots avoid the distinguished times, and every filling restriction at their omitted neighborhoods has strict clearance; the proof therefore makes no claim about the law at those times. $\square$

The final proposition is used only after the ordinary crossing estimate has been proved. Its open-fillability hypothesis is a geometric obligation in the protected-surgery construction. It is not an automatic consequence of continuity of an arbitrary singular contour.

8 Comparison and rigidity of local passage laws

We first prove a comparison theorem for local passage laws. It will be used for critical FK maps and, later, for spanning-tree-decorated maps. The two applications construct their conditional laws and initialize their distance units separately. The common argument identifies a law once it has a finite positive optimal upper comparison with the known LQG metric. Theorem 8.30 identifies its infimal internal passage cost F_U as $CD_h(\cdot, \cdot; U)$, with one deterministic constant for the law. The proof proceeds through annulus comparison, equality tangents, and the finite forcing argument.

In this section only, fix

$$\sqrt{2} \leq \gamma < 2, \quad p = 1/d_\gamma, \quad \xi = \gamma p, \quad Q = 2/\gamma + \gamma/2,$$

and a number $\lambda > 1$ for the geometric sequence of quantum volumes. The applications use $\lambda = 4$ for FK and $\lambda = 2$ for spanning trees. This number changes volume units only; the Euclidean annulus radii used below are fixed geometric choices. All constants may depend on these fixed parameters. No coupling or uniformity across parameters is asserted.

Write $D_U = D_h(\cdot, \cdot; U)$ for the internal reference metric. A path has *open confinement* in U when its range is a compact subset of U . The reference metric has the properties in Theorem 3.5. The metric, locality, Weyl scaling, uniqueness and confluence inputs cited there hold for every fixed subcritical γ , including $\sqrt{2}$. For the inward box-connection proof, the estimate of [11, Lemma 3.20, Equations (3.66)–(3.67)] allows $0 < \chi < \xi(Q - 2)$; this interval remains nonempty at $\gamma = \sqrt{2}$. The same inward chain therefore supplies its face connections at that value as well. We retain the actual joint law of the fields, curve, and internal reference metrics. In particular, a field is not replaced by its class modulo real additive constants. On every fixed compact region used for Gaussian field modifications, choose one whole-plane Gaussian reference whose pinning circle lies outside that region and all its buffers. A change of field law will always keep the same local conditional passage kernel.

Lemma 8.1 (Reference continuity at the cone mark). *For the circle-average γ -quantum cone at any fixed parameter in this section, the reference metric extends to its marked point 0 and induces the Euclidean topology there. In particular,*

$$\text{diam}_{D_h}(B_\varepsilon(0)) \longrightarrow 0 \quad \text{as } \varepsilon \downarrow 0.$$

Proof. Put $r_n = 2^{-n}$, $A_n = \{r_n/2 \leq |z| \leq r_n\}$ and $A_n^+ = \{r_n/4 < |z| < 2r_n\}$. For $n \geq 2$, the cone field on these annuli is $G_1 - \gamma \log |z|$. Write $H(r) = h_r(0) + Q \log r$. The radial cone definition gives

$$H(r_n) = -(Q - \gamma)n \log 2 + o(n) \quad \text{almost surely.}$$

Conformal covariance and the reference metric's constant Weyl rule give

$$\text{diam}_{D_h(\cdot, \cdot; A_n^+)}(A_n) = e^{\xi H(r_n)} Z_n,$$

where each Z_n has the same marginal law: the internal diameter of $\{1/2 \leq |z| \leq 1\}$ inside $\{1/4 < |z| < 2\}$ for the field $G_1 - \gamma \log |z|$. This variable has a positive moment by the reference internal-diameter moment bound of [11, Theorem 1.8]. The logarithmic drift is bounded on this fixed annulus, so the reference Weyl rule preserves that moment bound. No independence among the Z_n is needed.

Since $Q - \gamma > 0$, Markov's inequality and Borel–Cantelli bound Z_n by $e^{\delta n}$ eventually, with $0 < \delta < \xi(Q - \gamma) \log 2$. The displayed internal diameters are therefore summable. Adjacent closed annuli meet on a circle; joining through these circles shows that the ambient diameter of $B_{r_n}(0) \setminus \{0\}$ is at most twice the tail sum. All sequences approaching 0 consequently define the same reference completion point, and that tail sum tends to zero. A positive crossing lower bound in any fixed compact annulus separates this completion point from every other point and prevents a sequence staying outside a Euclidean neighborhood of 0 from approaching it in the metric. This proves the claimed topology and diameter limit. $\square$

Here is the precise interface used by the common proofs.

Definition 8.2 (Families of local passage laws). A family of local passage laws consists of laws of the reference continuum input together with countable closed collections of path certificates and their local restrictions. It has the following properties.

- (i) Certificates retain endpoints, compact ranges, every finite number of ordered subpaths, and nonnegative additive cost bounds. Finite certificates have actual path realizations with arbitrary open loss; bounded-cost realizing paths have subsequential certificates. Individual edge costs tend to zero, and ordered marking retains the subdivision of any bounded-cost path at successive cost increments δ with overshoot tending to zero. The finite separator constructions used below are valid for these paths, with actual primal intersections whenever the separators are primal circuits.
- (ii) The law of a restriction compactly inside an open real-field patch U is a kernel from the actual real field on U , local imaginary traversals in a possibly larger buffer, and local Poisson area marks. It may also depend on the finite genuine order of those traversals. The feasible orders have positive conditional probabilities depending only on the unlocked imaginary input. Their reference law is the same at every quantum index. An order-free kernel is the special case of one possible order. Restrictions in disjoint real-field patches have product kernels conditional on the complete continuum input. These kernels are compatible under restriction and countable exhaustion.
- (iii) The same local kernels are used after fixed locally equivalent field changes and intrinsic coordinate changes. Local area marks have their correct conditional quantum-area intensities; they may be integrated out using their independent restrictions. If S_a multiplies all recorded costs by a and $\mathcal{K}$ is a local kernel, write

$$c_v = \gamma^{-1} \log v, \quad \mathcal{K}^{(v)}(g, \hat{g}, o) = (S_{v^{-p}})_\# \mathcal{K}(g + c_v, \hat{g}, o).$$

At each fixed volume v these indexed laws are limits of the corresponding actual time-, height-, and cost-transformed discrete experiments. The affine and volume identities are the ones stated in Equations (8.2)–(8.4).

- (iv) For each sequence of laws and changes of base units used in the argument, the model supplies uniform local tightness and modulus bounds for the coordinates it retains. These bounds give relative compactness and a jointly convergent diagonal discrete subsequence. In it, all marked observations, reference experiments, fixed quantum offsets, and actual joint field/internal-metric readouts can be retained under further extraction. This requirement includes the varying laws selected in amplification and equality-tangent extraction, with the bounds uniform along each selected indexed sequence; it makes no compactness claim for uncontrolled sequences of laws. Each fixed retained time unit still tends to infinity in lattice steps. Additional continuum coordinates are drawn from their canonical conditional law independently of the entire raw actual and reference observation array given the full contour and cuts. The local kernel identities hold anew in each resulting law, with its own extracted reference kernels. Existing uniform comparison bounds pass through these marked extractions.
- (v) Whenever a finite positive upper comparison has been obtained, continuum certificates concatenate with arbitrarily small cost loss and open confinement, and their infimal internal costs are locally continuous. The comparison bound holds uniformly at all fixed quantum offsets and in further extractions to which it is applied.
- (vi) If there is a nontrivial order input, buffered directed traversals have the local reconstruction, rational-point visit and transience properties of Lemma 8.24. The complete dependency disks of its local determination events can be used for independent Gaussian trials. In the one-order case this condition is vacuous.

All restrictions and full observation spaces are standard Borel. Thus prescribed separated restrictions with their product conditional law can be completed by a regular conditional distribution to a full sample.

Comparison assertions are tested on a countable exhaustion of ordinary buffered charts. When transferring their optimal constants between a surface law and Gaussian reference laws, the model application must also verify that its actual coordinate and quantum-unit transformations cover these charts. Absolute continuity on one patch alone is not a whole-surface assertion. Singular marked endpoints, when present, are handled by the separate endpoint or completion argument.

These are requirements on the actual coupled laws, not consequences of contour convergence alone. The FK kernels of Theorem 7.13 and Proposition 7.11 supply the local and extraction properties. Their continuum joining property will follow from the small-circuit lemma below and the FK normalization argument. The tree application verifies the same interface using its independent-walk kernels and its own stronger joining estimate. In neither application does this interface assume a Weyl rule for the unidentified passage law.

8.1 Separating circuits and port splicing

The certificate operations used in this subsection are simultaneous marked subpath extraction, the additive cost inequality, realization with open ports, and robust planar separation. Each invocation requires these operations for the actual primal graph paths. Euclidean intersections of interpolations alone do not suffice.

Definition 8.3 (Upper comparison with ports). An upper comparison C means the following. If P is a nonconstant D -rectifiable path, U is an open neighborhood of its range, and O_0, O_1 are open neighborhoods of its endpoints, there is a passage certificate confined to U , with endpoints in O_0, O_1 , of cost at most $C \text{len}_D(P) + \epsilon$, for every $\epsilon > 0$. The assertion is required simultaneously for finite lists of such tests. A lower comparison c means that every certificate with endpoints x, y and cost a satisfies $a \geq cD(x, y)$.

Here and subsequently an infimum over certificates with a specified pair of endpoints refers to the closed certificate collection, not to the limit of the distance between an arbitrarily prescribed pair of discrete vertices.

Lemma 8.4 (Splicing by small circuits). *Suppose that for every $K \Subset U$ and every $\epsilon > 0$, every $x \in K$ is surrounded by a primal separating circuit of cost at most ϵ , whose range is contained in $U \cap B(x, \epsilon)$. Circuits are required to be realizable with strict separation from a smaller neighborhood of x . Assume the certificate operations stated above. Then nonconstant certificates ending at the same point concatenate with arbitrary cost loss and open confinement. If an upper comparison $C < \infty$ also holds, the infimal cost F_U satisfies*

$$F_U(x, y) \leq CD_U(x, y), \quad x, y \in U, \quad (8.1)$$

and is locally continuous in its two endpoints. These conclusions do not bound the cost of escaping from a prescribed discrete vertex.

Proof. Consider certificates from x to z and from z to y , with $x, y \neq z$. Choose a circuit surrounding z that avoids x and y and has cost less than ϵ . In sufficiently accurate realizations both paths must cross this circuit: their endpoints near z lie on its inner side and their other endpoints on its outer side. Robust separation makes the crossings actual common primal vertices. Cut at such vertices and join along the circuit. The resulting cost is at most the sum of the original costs plus ϵ . Simultaneous marked extraction retains that inequality. Letting the circuit shrink proves concatenation; the cases with a constant certificate are dealt with by the cost-zero diagonal.

For a path P compactly confined to U , apply the upper comparison with endpoint neighborhoods shrinking to its endpoints. Extract the resulting certificates in a fixed compact confinement, first enlarging the range of P slightly within U . Their endpoints become exact and their costs are at most $C \text{len}_D(P)$. Taking the infimum over such paths proves Equation (8.1). The triangle inequality just proved and symmetry give

$$|F_U(x, y) - F_U(x', y')| \leq C\{D_U(x, x') + D_U(y, y')\}$$

whenever these quantities are finite. On a compact subset of U , D_U and D agree for sufficiently close pairs: the compact set has positive D -distance from the complement of U , and a sufficiently short near-minimizing path cannot exit U . Continuity follows. $\square$

Lemma 8.5 (Finite separator chains). *Let $K \Subset U$. For each integer n let $\mathcal{B}_n$ be a finite collection of closed Jordan domains of Euclidean diameters at most $2\epsilon_n$, where $\epsilon_n \downarrow 0$, with weights $w_B > 0$. Each domain has a designated open hole compactly in its interior, and the holes cover K . The domains and the buffered boundary connections below lie in U .*

Suppose a continuous path P in K is charged at least aw_B whenever, starting inside the hole of B , it exits B . Suppose also that any two points of ∂B can be connected along the boundary, or in its buffered band, with cost at most Aw_B . Then there is a connection between points within $4\epsilon_n$ of the endpoints of P , with cost at most (A/a) times its total charge. The connection uses only the buffered bands of disks selected along P . Its first and last errors are Euclidean errors.

Proof. At the current point of P , choose a disk whose hole contains it and stop at its first exit. Repeat, stopping before an incomplete last traversal. Completed traversal intervals are disjoint. They are finite in number when the total charge is finite, since each costs at least $a \min_{B \in \mathcal{B}_n} w_B > 0$. Their weights therefore sum to at most the total charge divided by a .

Consecutive selected domains intersect. Delete domains contained in others, and replace each domain in the original chain by a maximal selected domain containing it. The resulting intersection graph connects a domain containing the initial point to a domain containing the final completed exit. Choose a simple graph path between them. Two consecutive maximal Jordan domains have intersecting boundaries: by the Jordan separation theorem, otherwise one of the two intersecting domains would contain the other. Follow the boundary connections between intersection points. No retained boundary is used twice, so the cost is at most A times the sum of the selected weights. Containment deletion moves an endpoint by at most $2\epsilon_n$, and an incomplete last disk adds at most $2\epsilon_n$. If no traversal is completed, both endpoints are in one disk and only this endpoint error remains. $\square$

The same proof applies on a finite planar map when the domains are the filled interiors of simple primal circuits and “diameter” means the diameter of their projected images. The boundary intersections are then actual primal vertices. This version is the one used for circuit quantiles; a cheap surrounding circuit need not pass through every point of a prescribed Euclidean circle. The projection and exterior-connectivity argument must establish the asserted size bound for the filled interiors. When the separators are initially winding certificates, fix their finite family before realizing them in the graph. Lift a fine deterministic mesh approximation to the reference path by the approximate right inverse in the projection argument. Its projected path follows the reference path uniformly. A first exit from a filled primal circuit forces the reference path to traverse the corresponding buffered band, up to a vanishing spatial error. Strict band margins preserve its positive D charge. Perform the finite separator-chain construction on the map, take the realization limit, and only then refine the annulus family. This avoids assuming that the projection of a limiting primal circuit is a Jordan curve.

There is also a direct continuum version, used for the tree laws. Take Euclidean disks as the Jordan domains and assume the stated boundary connections are continuum certificate connections with arbitrary open loss. The same finite chain concatenates at its circle intersections by the already available continuum joining property. It has the same cost bound and vanishing endpoint errors. No primal circuit reconstruction is needed in this version. When an upper comparison has been obtained, local continuity removes its continuum endpoint errors; alternatively retain open endpoint neighborhoods until the final extraction, as in the port formulation below.

Corollary 8.6 (Comparison from annuli). *Assume the hypotheses of Lemma 8.5 hold simultaneously at every point of every compact subset, along arbitrarily fine families. If the charged quantity is D -length and the boundary connections are passage connections, then the port upper comparison is A/a . If the charged quantity is certificate cost and the boundary connections are D -connections, then every certificate compactly in U costs at least $(a/A)D_U(x, y)$.*

Proof. The first assertion follows by applying Lemma 8.5 to a near-minimizing D_U -path and then sending n to infinity. For the second, perform the construction on discrete realizing paths at each fixed finite disk family. Move stopping circles within their open buffers to preserve strict crossings. If the number of completed traversals were unbounded, extract any prescribed finite number of marked traversals. The additive cost inequality would give a cost greater than the total budget once their number exceeds that budget divided by $a \min_B w_B$. Thus finitely many marked pieces

suffice. The D -connection supplied by the lemma has cost at most A/a times the budget, and its endpoint errors tend to zero in D_U . This proves the lower assertion. Only D -continuity removes endpoint errors in this direction. $\square$

Lemma 8.7 (Positive lower comparisons are intrinsic). *Suppose normalized edge lengths tend to zero, marked cost subdivisions are available, and $c > 0$ is a lower comparison. A certificate of cost a compactly confined to U satisfies $a \geq cD_U(x, y)$.*

Proof. Mark approximating paths at rational fractions of their total costs. An individual overshoot is at most one edge length and tends to zero. After simultaneous extraction, marks $x_t, t \in \mathbb{Q} \cap [0, 1]$, satisfy $D(x_s, x_t) \leq a|s - t|/c$. Their Lipschitz extension is a path of D -length at most a/c in the retained compact subset of U , with the required endpoints. The definition of D_U gives the result. $\square$

8.2 Annuli at absolute quantum scales

We align deterministic quantum offsets with the coarse radial averages of the field. Gaussian estimates control large jumps and harmonic profiles; fresh local trials and a mesh union bound then supply annuli around every point.

Fix the parameter throughout this subsection. Put $d = \log \lambda$, $v_i = \lambda^{-i}$, and $c_i = \gamma^{-1} \log v_i$. For a local passage kernel $\mathcal{K}$, let S_b multiply all recorded costs by b , and set

$$\mathcal{K}_i(g, \hat{g}, o) = (S_{v_i^{-p}})_\# \mathcal{K}(g + c_i, \hat{g}, o). \quad (8.2)$$

By Definition 8.2(iii)–(iv), this law is realized by the retained discrete offset experiment at time Nv_i with cost denominator $B_N v_i^p$, followed by the fixed local field-density change. The arrays are included before extraction, and every fixed retained time unit tends to infinity in lattice units. Each further extraction uses its own common reference kernels. The indexed kernels satisfy the following common intrinsic coordinate rule. In an affine chart $\phi(z) = z_0 + rz$, the conditional law of the pulled-back physical observation, with costs multiplied by v_i^{-p} , is

$$\mathcal{K}_i(h \circ \phi + Q \log r - c_i, \hat{h} \circ \phi, o).$$

Conditional on the whole fields, the actual observations in disjoint buffered domains have product kernels. The only additional input o is the finite exterior order of the locally directed traversals relevant to the buffered test. These identities use common versions for the countable indices and deterministic affine charts below, and transfer under local absolute continuity. They are supplied by the common-kernel construction. For the reference metric, its established covariance gives separately

$$v_i^{-p} \phi^* D_h = v_i^{-p} D_{g+c_i} = D_g, \quad g = h \circ \phi + Q \log r - c_i. \quad (8.3)$$

Thus no Weyl rule for the passage kernel is used. For a general fixed volume $v > 0$, define $\mathcal{K}^{(v)}$ by the same formula with $c_v = \gamma^{-1} \log v$. If $M > 0$ and $c_M = \gamma^{-1} \log M$, then directly from the definition,

$$(S_{M^p})_\# \mathcal{K}^{(Mv)}(g + b, \hat{g}, o) = \mathcal{K}^{(v)}(g + b + c_M, \hat{g}, o). \quad (8.4)$$

This is the exact conversion used when a shortcut at volume v is inserted into a shell aligned to Mv .

Choose fixed buffered annuli

$$A = \{1/64 < |z| < 8\}, \quad A' = \{1/32 < |z| < 6\}, \quad S = \{1/16 < |z| < 4\}.$$

The full local field input of each test is restricted to S ; its certificate and metric observations have further positive clearance inside that input region. Slight fixed changes to these annuli allow any prescribed finite collection of such buffers. The central hole for coverage will be $B(0, 1/128)$.

Lemma 8.8 (Finite exterior orders). *Suppose the feasible orders of the buffered local traversals form a finite random set $\mathcal{O}(\hat{g})$, with conditional probabilities $w_o(\hat{g}) > 0$. Assume these probabilities depend only on the unlocked local imaginary input, whose reference law is independent of the index i . For any family of local events with genuine-order failure probability at most ε , the event that candidate observations for every feasible order pass has failure probability at most*

$$\mathbb{P}\left[\min_{o \in \mathcal{O}(\hat{g})} w_o(\hat{g}) < \tau\right] + \frac{\varepsilon}{\tau}, \quad \tau > 0. \quad (8.5)$$

The bound tends to zero uniformly in i as $\varepsilon \downarrow 0$, with τ chosen first. For disjoint domains, the candidates may be adjoined so that their vectors are conditionally independent given the exterior fields and the actual observation is the component corresponding to its actual exterior order.

Proof. Given the local fields, write p_i^o for the failure probability under $\mathcal{K}_i^o$. The genuine-order failure probability is $\mathbb{E} \sum_o w_o p_i^o$. Adjoin conditionally independent candidate draws for all feasible orders. On the event that every positive weight is at least τ , the union bound gives

$$\sum_o p_i^o \leq \tau^{-1} \sum_o w_o p_i^o.$$

This proves Equation (8.5). The minimum positive weight is strictly positive almost surely because there are finitely many orders. Its distribution is independent of i .

For the coupling claim, first condition on the whole fields and retain the actual draw at its actual order. Sample the other components independently from their specified local kernels. The conditional law of the full vector is the product of all the feasible-order kernels, regardless of which component was retained. This product depends only on the local fields. The assumed product law for actual observations in disjoint domains therefore gives the claimed factorization after integrating the independent interior fields. An order of conditional weight zero is excluded; the actual order has positive conditional weight almost surely. $\square$

The finiteness in this lemma is a buffered statement. A transient continuous curve visits a bounded buffer during a bounded time interval. Every distinct traversal reaching the smaller closed testing region must cover the positive distance to the buffer boundary. Uniform continuity on that bounded time interval gives a positive minimum duration for such a crossing, so only finitely many traversals occur. Their local directed pieces are supplied by the buffered imaginary-geometry reconstruction. Their possible global orders are therefore a finite set of permutations. This argument does not make the actual global order locally measurable.

Lemma 8.9 (Same-index likelihood). *Let μ be the law of two independent zero-boundary GFFs on A , with the imaginary field optionally viewed modulo its fixed period. Let $\mathcal{L}_i$ be any measurable local observation kernel from the fields on S . If a pair of Cameron–Martin shifts has total Dirichlet norm at most R , then every field-and-observation event E satisfies*

$$(\mu_u \otimes \mathcal{L}_i)(E) \leq e^{R^2/2} (\mu \otimes \mathcal{L}_i)(E)^{1/2}. \quad (8.6)$$

The bound holds conditionally when the index and shifts are exterior measurable. A harmonic residual with bounded $H^2(A')$ norm can be used in place of the shift on S .

Proof. The Cameron–Martin density is $Z_u = \exp((h, u)_\nabla - \|u\|_\nabla^2/2)$, with the sum of the two Dirichlet inner products understood. Hence $\mathbb{E}Z_u^2 = \exp(\|u\|_\nabla^2)$. Adjoining the same kernel on both sides leaves the density equal to Z_u . Cauchy–Schwarz proves the bound. Conditioning fixes the index and shifts. Multiply a harmonic residual by a fixed smooth cutoff supported in A' and equal to one on S ; its Dirichlet norm is bounded by a fixed multiple of its $H^2(A')$ norm. For the imaginary field, first choose the residual constant in one fixed interval of period length. $\square$

The reference with zero boundary values on the annulus can replace any fixed locally equivalent reference on a disk or the whole plane, uniformly over arbitrary kernels and indices. Indeed, if $Z = d\nu/d\mu$ on the observed field restriction, then

$$(\nu \otimes \mathcal{L}_i)(E) \leq T(\mu \otimes \mathcal{L}_i)(E) + \mathbb{E}_\mu[Z \mathbf{1}_{\{Z > T\}}]. \quad (8.7)$$

Thus arbitrarily small reference failure probabilities transfer with a single modulus of absolute continuity. This applies also to the kernel that first samples the feasible order with weights w_o and then its observation. Combining this observation with Lemma 8.8, arbitrarily small genuine-order failure probabilities give arbitrarily small failure probabilities for the all-order event under the annular reference.

Lemma 8.10 (Gaussian profiles). *Let h be a zero-boundary GFF on a bounded disk V and let $K \Subset V$. There exist $T > 0$, $c_* > 0$, $\theta > 0$, and finite constants C_0, C_1 with the following property. Suppose $t_{j+1} - t_j = T$, $r_j = e^{-t_j}$, and the deterministic centers satisfy $|z_j - x| \leq c_* r_j$ for some $x \in K$; all the annuli are contained in a fixed compact subset of V . Let H_j be the outer boundary circle average at radius $8r_j$, and let h_j^{har} be the harmonic part of h in $z_j + r_j A$. Set*

$$\Phi_j(z) = h_j^{\text{har}}(z_j + r_j z) - H_j.$$

For every string of m such scales,

$$\text{Var } H_j = t_j + O(1), \quad (8.8)$$

$$\mathbb{E} \exp \left(\theta \sum_j \|\Phi_j\|_{H^2(A')}^2 \right) \leq e^{C_0 m}, \quad (8.9)$$

$$\mathbb{E} \exp \left(\theta \sum_j |H_{j+1} - H_j|^2 \right) \leq e^{C_1 m}. \quad (8.10)$$

The same profile estimate holds for the independent imaginary field after subtracting its coarse average.

Proof. Take T large and c_* small so that the closed annuli are disjoint and each finer outer disk lies in the preceding inner hole. On the fixed compact subset, the covariance kernel is $G_V(u, v) = -\log |u - v| + g_V(u, v)$, where g_V is smooth and separately harmonic. Averaging nested circles gives

$$\text{Cov}(H_j, H_k) = \min(t_j, t_k) - \log 8 + g_V(z_j, z_k).$$

The logarithmic term has independent disjoint increments and increment variance T . The mixed differences of the smooth term are bounded by $C|z_{j+1} - z_j||z_{k+1} - z_k| \leq Cr_j r_k$. Therefore the covariance operator of the increment vector is uniformly bounded and its trace is $O(m)$.

For the profiles, represent $\Phi_j(z)$ as pairing h with the signed measure $\omega_j^z - \sigma_j$, where ω_j^z is harmonic measure on the two annulus boundaries and σ_j is uniform measure on the outer circle.

These measures have total mass zero and bounded total variation. The same properties hold for normalized spatial derivatives through order two: on A' the differentiated annular Poisson series is bounded by a convergent geometric series. At distinct scales $j < k$, the logarithmic potential of the scale- j signed measure has gradient bounded by C/r_j on the finer hole. Cancellation against the scale- k measure bounds the covariance by Cr_k/r_j . Cancellation in both variables bounds the smooth contribution by Cr_jr_k . These estimates hold also for the indicated derivatives. On one scale, logarithmic singularities are integrable against the smooth boundary densities, so the diagonal variances and their integrals over A' are bounded.

Consequently the centered Gaussian vector (Φ_j) in $\bigoplus_{j=1}^m H^2(A')$ has covariance operator bounded by a constant and covariance trace at most Cm . The operator bound follows from the summable estimate $Ce^{-T|j-k|}$ and Cauchy–Schwarz. If the covariance eigenvalues satisfy $\lambda_n \leq C$ and $\sum_n \lambda_n \leq Cm$, then for $\theta < 1/(4C)$,

$$\mathbb{E}e^{\theta\|X\|^2} = \prod_n (1 - 2\theta\lambda_n)^{-1/2} \leq \exp\left(2\theta \sum_n \lambda_n\right).$$

Apply this to the profiles and increments. For the whole-plane imaginary field the additive normalization terms disappear under the zero-mass pairings, giving the same estimates. Its phase modulo the fixed period does not enter the oscillatory profiles. $\square$

Lemma 8.11 (Harmonic profiles in trial disks). *Let h be a zero-boundary GFF on a bounded disk V and fix a compact subset of V . For disks $B(z, r)$ contained in that subset, write $h = h_{z,r}^0 + h_{z,r}^{\text{har}}$ for the Markov decomposition. If $0 < \kappa < 1$ is fixed, the profiles*

$$\Phi_{z,r}(x) = h_{z,r}^{\text{har}}(z + rx) - h_{z,r}^{\text{har}}(z), \quad |x| < \kappa,$$

have uniformly bounded exponential moments of a sufficiently small multiple of their squared $H^2(B_\kappa)$ norms. The bound is independent of the disk radius and of the number of disjoint disks chosen. The same holds for the independent imaginary field after its coarse constant is removed.

Proof. Write $G_V(u, v) = -\log|u - v| + g_V(u, v)$ as above. Subtracting the zero-boundary disk Green function and then the harmonic value at the center gives

$$\begin{aligned} \text{Cov}(\Phi_{z,r}(x), \Phi_{z,r}(y)) &= g_V(z + rx, z + ry) - g_V(z + rx, z) \\ &\quad - g_V(z, z + ry) + g_V(z, z) - \log|1 - x\bar{y}|. \end{aligned}$$

In particular the divergent coarse variance $-\log r$ has canceled. The displayed kernel and the derivatives through order two in each variable are bounded on the fixed smaller disk, uniformly in z, r . The covariance operator of each H^2 profile is therefore bounded and has bounded trace. The Gaussian eigenvalue calculation in Lemma 8.10 proves the claim. This is a marginal bound for each disk; no independence of their harmonic profiles is asserted or needed. The imaginary-field normalization disappears in the same centered differences. $\square$

Lemma 8.12 (All-point absolute quantum-scale annuli). *Assume the common intrinsic local-kernel properties stated above. For the quantitative assertion, let the real field be a zero-boundary GFF on a bounded disk V , with the independent stationary-phase imaginary field. Use only annuli whose fixed buffers lie in a compact subset of V . Fix $0 < \alpha \leq 1$, a compact $K \Subset V$, and the buffered testing geometry. There exist constants*

$$0 < a < B, \quad L, \beta, \varepsilon_*, c_{\text{mesh}}, c_0, C_0 > 0$$

and N_0 such that the following holds uniformly. For each $N \geq N_0$, let $I_N \subset \{N, \dots, 2N\}$ be deterministic with $|I_N| \geq \alpha N$. At every $i \in I_N$, let E_i be a local event whose genuine-order probability in the fixed unit reference experiment is at least $1 - \varepsilon_*$. With probability at least $1 - C_0 e^{-c_0 N}$, every $x \in K$ belongs to the central hole of at least βN pairwise disjoint buffered annuli passing their assigned events. Their radii lie on one fixed geometric sequence in $[e^{-BN}, e^{-aN}]$, their centers lie on deterministic meshes of spacing $c_{\text{mesh}} r$, and each assigned triple (z, r, i) satisfies

$$|\gamma(Q \log r + H(z, r)) + i \log \lambda| \leq L. \quad (8.11)$$

The physical observations are tested after affine pullback and division of costs by v_i^p . For each x , the selected annuli have distinct assigned indices. The statement also holds for $v_i = M \lambda^{-i}$ with any fixed $M > 0$, after increasing N_0 as needed and replacing Equation (8.11) by $|\gamma(Q \log r + H(z, r)) + i \log \lambda - \log M| \leq L$.

Proof. Choose $0 < \epsilon < Q - 2$ and then a, B with strict margins

$$\gamma(Q + 2 + \epsilon)a < \frac{1}{2}d, \quad \gamma(Q - 2 - \epsilon)B > 3d. \quad (8.12)$$

Use the separation T of Lemma 8.10, with $t_j = jT$ from the first j satisfying $t_j \geq aN$ to the last satisfying $t_j \leq BN$. There are at most $C_2 N$ scales. At every scale take a deterministic square mesh of spacing cr_j in a fixed neighborhood of K . Here $c > 0$ is small enough for Lemma 8.10 and the hole geometry. For each point y of the finest mesh, follow its nearest center $z_j(y)$ on each coarser mesh. These centers are deterministic and satisfy $|z_j(y) - y| \leq cr_j$ after absorbing a fixed mesh factor into c .

At either endpoint scale t , there are $O(e^{2t})$ distinct endpoint centers. Equation (8.8) and a Gaussian union bound show that the probability of any violation of $|H_j| \leq (2 + \epsilon)t_j$ there is at most

$$O(e^{2t}) \exp\left(-\frac{(2 + \epsilon)^2 t^2}{2(t + O(1))}\right) = O\left(e^{-(2\epsilon + \epsilon^2/2)t + O(1)}\right).$$

This is exponentially small in N . The endpoint meshes are counted at their own scales, not at the finest scale. Put $\Lambda_j = \gamma(Q t_j - H_j)$. For all sufficiently large N , Equation (8.12) gives $\Lambda_{\text{first}} < Nd$ and $\Lambda_{\text{last}} > 2Nd$ simultaneously for all ancestries.

Fix one ancestry. Assign each target $id, i \in I_N$, to its first upward crossing, the first index j with $\Lambda_j \geq id$. No monotonicity of (Λ_j) is required. Fix an exponent $b > 2B + 4$. Equation (8.10), with the fixed drift γQT absorbed, gives constants $\theta_1, C_3 > 0$ such that

$$\mathbb{E} \exp\left(\theta_1 \sum_j |\Lambda_j - \Lambda_{j-1}|^2\right) \leq e^{C_3 N}.$$

For $L \geq 2d$, an upward jump $s > L$ receives at most $1 + s/d \leq 2s/d$ targets. Therefore the number of labels assigned to jumps larger than L is bounded by

$$\frac{2}{Ld} \sum_j |\Lambda_j - \Lambda_{j-1}|^2. \quad (8.13)$$

Choose L large enough that this number exceeds $\alpha N/8$ with probability at most e^{-bN} . This is possible by exponential Markov applied to the preceding quadratic moment. At every remaining jump, alignment holds with error L , and at most $J = \lceil L/d \rceil + 2$ targets are assigned to one shell.

Condition on the exterior of the union of the disjoint open annuli of this ancestry, for both fields. For the imaginary field retain the lifted exterior differences, with its remaining common constant taken modulo its period. Each annulus then contains independent Dirichlet parts. The outer circle averages and harmonic extensions are exterior measurable; boundary averages can equivalently be defined as limits from outside the open annuli. Thus the first-crossing indices and their large-jump exclusions are exterior measurable as well.

Let $\widehat{\Phi}_j$ be the imaginary oscillatory harmonic profile. After J has been fixed, Equation (8.9) allows a cutoff R so large that

$$\mathbb{P}\left[\#\{j : \|\Phi_j\|_{H^2(A')} + \|\widehat{\Phi}_j\|_{H^2(A')} > R\} > \frac{\alpha N}{8J}\right] \leq e^{-bN}.$$

Indeed, the event forces the quadratic sum of the two profile norms to exceed a constant times $R^2\alpha N/J$, and exponential Markov applies. Discarding such profiles loses at most another $\alpha N/8$ labels. At least $3\alpha N/4$ favorable labels remain. Keep one label by a deterministic rule at each remaining shell. For $\nu = \alpha/(2J)$ there are at least νN distinct retained trials.

At a retained trial with index i , the normalized real field equals its fresh Dirichlet part plus

$$\Phi_j + H_j - Qt_j + \gamma^{-1}id.$$

Alignment bounds the displayed constant by L/γ . The imaginary residual constant is chosen in one period. The profile cutoff therefore gives a deterministic bound $R_1 = R_1(R, L)$ on the pair of Cameron–Martin extensions used in Lemma 8.9. The common intrinsic kernel identity identifies the conditional physical test with the same index- i reference kernel under this shifted field law.

Use Lemma 8.8 and Equation (8.7) to require that the all-order reference event has failure probability at most δ_* . Then Lemma 8.9 bounds every retained conditional failure probability by $e^{R_1^2/2}\sqrt{\delta_*}$. Choose $\eta > 0$ so small that

$$2^{C_2N}\eta^{\nu N/2} \leq e^{-bN},$$

then choose δ_* so that this conditional failure bound is at most η , and finally choose the genuine-order threshold ε_* small enough to imply that value of δ_* . Conditional independence of the fields and candidate kernel vectors implies that fewer than $\nu N/2$ successes has probability at most $2^{C_2N}\eta^{\nu N/2}$. This calculation conditions on the selected indices and profiles; neither is required to be independent of the exterior. Every successful all-order event ensures success of the actual observation, whatever its global order.

This auxiliary coupling may be constructed separately for each fixed ancestry: it bounds that ancestry's actual failure event. No compatible coupling of counterfactual candidates in overlapping ancestries is needed. There are $O(e^{2BN})$ finest-mesh points. A union bound over them for large-jump, profile, and actual-trial failures is exponentially small because $b > 2B + 4$. Add the already bounded endpoint failure. Every $x \in K$ is within cr_{last} of a finest-mesh point. Reducing c once puts it in the central hole of every selected annulus in that point's ancestry. Set $\beta = \nu/2$.

Replacing v_i by $M\lambda^{-i}$ replaces each target by $id - \log M$. The fixed endpoint margins absorb this shift for all sufficiently large N ; all increment, profile, and likelihood estimates are unchanged. $\square$

The uniformity in Lemma 8.12 is over each deterministic choice of the index set and test family, not simultaneous over all test families on one sample. Along a fixed sequence of eligible N , its failure probabilities are summable. Hence coverage eventually holds almost surely. A fixed local

absolutely continuous change of the joint field-and-kernel law preserves this eventual assertion, and in particular the probability-tending-to-one conclusion. Without a quantitative density bound, the exponential rate itself need not be preserved.

The order of constants is: geometry and α ; then ϵ, a, B, T, b ; then L, J ; then R ; then the desired conditional failure probability; and finally the genuine-order reference threshold. All constants may depend on the fixed parameter. Internal reference metric regularity can be included among the local events, using Equation (8.3).

8.3 Optimal comparison constants and equality tangents

The strategy of optimal comparison constants, local field changes, and strict geodesic savings follows Gwynne and Miller [23, Section 1.5 and Sections 3–5]. The argument below works with the conditional passage laws just constructed and proves the required rigidity for those laws.

For a fixed joint law, define its deterministic optimal constants by

$$\begin{aligned} C &= \inf\{u > 0 : \text{upper comparison } u \text{ holds almost surely}\}, \\ c &= \sup\{u \geq 0 : \text{lower comparison } u \text{ holds almost surely}\}. \end{aligned} \tag{8.14}$$

Both constants are deterministic functions of the joint law. Countable rational port, cost, and confinement tests make the events measurable. Intersecting the probability-one events for constants decreasing to C and increasing to c , and using arbitrary cost loss, shows that the optimal constants themselves are valid whenever $C < \infty$. The separate model initializations will give $0 \leq c \leq C < \infty$ and $C > 0$ before rigidity is applied.

Lemma 8.13 (Transport of comparison constants). *For common intrinsic kernels, almost-sure local comparisons on corresponding buffered charts are unchanged by a fixed locally equivalent field law. Their optimal deterministic constants, defined over the corresponding countable chart families, are unchanged as well. They have the same numerical constants in the indexed kernels of Equation (8.2), when both costs and reference lengths are expressed in the indicated quantum units.*

Proof. The joint density change is a function of the fields alone: adjoining their same conditional passage kernel does not alter the Radon–Nikodym derivative. Thus a measurable local comparison event of probability one keeps probability one. Use a countable collection of rational ports, confinements, and cost tolerances, and then continuum joining, to obtain the simultaneous assertion. Inverse absolute continuity proves that a strict improvement under either law would improve the other law, so the optimal constants coincide. For a surface application, this last assertion uses the actual chart-covering and quantum-unit identities stipulated above. It does not infer a comparison outside the tested charts from a single local absolute-continuity statement.

For the lower constant, the local event is the intrinsic comparison. An ambient lower bound $c > 0$ implies it by Lemma 8.7; for $c = 0$ it is automatic. Conversely, divide a compact certificate into finitely many marked pieces within a rational chart cover. The local intrinsic bounds, additive cost inequality, and ambient triangle inequality give the same ambient lower bound for that certificate. Hence the ambient optimal definition and these measurable local tests are equivalent.

A fixed additive real-field change on the observation patch is locally absolutely continuous under the stated Gaussian reference: extend it smoothly across its buffer with support disjoint from the fixed pinning circle. The extension is a Cameron–Martin direction of that actual Gaussian law. In Equation (8.2), multiply the comparison under $g + c_i$ by v_i^{-p} . The reference Weyl identity gives $v_i^{-p} D_{g+c_i} = D_g$. The numerical comparison constant therefore does not change. Countably many fixed indices can be handled simultaneously. This argument transports an almost-sure assertion; it does not identify individual passage samples before and after the field change. $\square$

We record explicitly the assumptions of the next implication. An indexed family has the *annulus transfer property* if, for each fixed $\vartheta > 0$, sufficiently high probability of a buffered shell event at at least ϑN labels in $[N, 2N]$ gives, with probability tending to one, arbitrarily fine simultaneous good annuli at every point of a fixed compact set. The weight of a label i is $w_i = \lambda^{-ip}$. All reference regularity requirements below must be allowed in this transfer. In the application this is the absolute quantum-scale statement, and it requires the common spatial passage kernels; it is not a consequence of convergence of the contours alone.

Proposition 8.14 (Amplification). *Suppose the family has the annulus transfer property, the certificate operations and continuum joining above, and finite optimal comparisons $0 \leq c < C < \infty$. Suppose the comparisons with the same numerical constants hold at every indexed quantum scale and pass to further extractions. Then:*

- (i) *There are fixed shell collars and $b > 0$ such that, for each $\delta > 0$ and all sufficiently large N , more than three quarters of the labels $i \in [N, 2N]$ have probability at least b of a geodesic traversal P with*

$$F_W(P(0), P(1)) \geq (C - \delta) \text{len}_D(P).$$

Here P crosses a fixed closed band, minimizes in its fixed open collar W , and its entire range is contained in that closed band. Its endpoints have fixed positive Euclidean separation. The collars and b do not depend on δ .

- (ii) *For each $c_1 \in (c, C)$ there are $b_1 > 0$, $\rho > 0$ and fixed rectangular collars such that more than three quarters of the labels in every sufficiently late window have probability at least b_1 of a rectangular certificate of cost*

$$a < c_1 D_{R^+}(x, y).$$

Its endpoints lie on opposite faces of a rectangle R of crossed width at least ρ and aspect ratio between $1/2$ and 2 , and its entire range is contained in the closed rectangle R . This closed envelope is part of the retained witness. Its slightly larger open collar is used for certificate realizations, and R^+ is a further open rectangular denominator collar. One may impose deterministic positive lower and finite upper bounds on the denominator, together with compact internal metric moduli in a further collar.

Proof. Fix the annulus threshold for favorable fraction $1/4$. On a reference regularity event of sufficiently high probability, all separated crossings in the fixed shell cost at least $a_0 > 0$ in D , and surrounding reference circuits have cost at most $A_0 < \infty$. These choices do not involve δ . Choose b smaller than one quarter of the allowable failure probability.

If (i) fails, at least one quarter of the labels in arbitrarily late windows have near-saturation probability less than b . On the complementary good event, every indicated geodesic traversal can be followed at a saving of at least $\delta a_0 w_i$ relative to the old upper comparison. Annulus transfer makes these events available simultaneously at every point. Stop a compact reference geodesic successively when it exits good surrounding separators. In each completed interval retain the subarc from its last visit to the inner testing circle until its first subsequent visit to the outer testing circle. This is minimizing in W , and has length at least $a_0 w_i$. The retained intervals are disjoint.

The reference separator chain gives

$$\text{len}_D(P) \leq A_0 \sum_i w_i + o(1).$$

Replacing the retained subarcs and following the gaps at upper factor C therefore gives total cost at most

$$C \text{len}_D(P) - \delta a_0 \sum_i w_i + o(1).$$

At each fixed finite family of separators choose the total realization and splicing error arbitrarily small. Then shrink the separators. This improves the upper factor to any number strictly above $C - \delta a_0/A_0$, contradicting optimality.

For (ii), first choose a fixed small square size in the middle of the shell. On every hole-to-outer crossing, mark a point in this middle region, follow the path until it first exits the square centered there, and choose the exit face. Start the marked piece at its last visit to the parallel midline. It lies in the resulting half-square and crosses opposite faces; its width and its collar are fixed positive fractions of the shell radius. A high-probability reference event bounds the D -distance of every pair of such opposite-face endpoints below by $a_0 w_i$: these pairs have uniformly positive Euclidean separation on a compact set.

If shortcut probability were small on at least one quarter of the labels, annulus transfer would make every marked rectangular piece cost at least $c_1 D_{R^+}(x, y)$. For $c > 0$, Lemma 8.7 gives the old intrinsic lower bound on the same piece and on every gap; for $c = 0$ this assertion is automatic. Thus its extra cost is at least $(c_1 - c)a_0 w_i$. The positive extra charge bounds the number of marked pieces at a fixed finite annulus family even when $c = 0$, by simultaneous marked extraction and the additive cost inequality. Reference separator chaining now improves the lower comparison to any number strictly below $c + (c_1 - c)a_0/A_0$, a contradiction.

The marked half-square itself is retained as the closed envelope; thus the shortcut event throughout this contradiction is the stronger event in (ii). Use an intermediate threshold between c and c_1 before selecting rational outward collar widths and gridded parameter bins. Keep the actual envelope and its opposite-face endpoints inside each bin, rather than replacing their range by the enlarged collar. Strict slack preserves the inequality. Finally, compactness of the family of reference rectangles, continuity of D , and internal box connection bounds give a reference regularity event of probability greater than $1 - b_1/2$ with the additional bounds in (ii). Intersecting it with each witness event retains probability at least $b_1/2$. $\square$

Proposition 8.15 (Equality tangent). *In addition to Proposition 8.14, suppose the common-kernel family is closed under joint extraction of its fixed quantum offsets and its full marked certificate records, as in Definition 8.2(iv). Suppose its reference internal metrics on fixed collars retain their actual joint law in these extractions. Then a further law has, separately with positive probability, a saturated nonconstant geodesic segment at upper factor C and a rectangular shortcut of ratio at most $C - \delta_0$, where $\delta_0 > 0$ is deterministic. The shortcut retains its closed rectangular envelope and its endpoints on opposite envelope faces, as in Proposition 8.14(ii). Under local field equivalence and strong confluence, the saturation event can be written for deterministic anchors $a_0 \neq a_1$, rational $0 < t_1 < t_2 < 1$, and a fixed bounded open U as*

$$P([t_1, t_2]) \Subset U, \quad F_U(P(t_1), P(t_2)) = C(t_2 - t_1)D(a_0, a_1), \quad (8.15)$$

where P is the unique fixed-anchor geodesic, parametrized by fractional length. Only the public strong-confluence assertion that a nonempty geodesic subarc contains a segment of a geodesic between points of a deterministic countable dense set is used.

Proof. Choose $c < c_1 < c_2 < C$ and $\delta_k \downarrow 0$. The two sets of labels in Proposition 8.14 intersect in more than half of every sufficiently late window. Choose a common label tending to infinity.

Retain the two witnesses separately, using a dummy witness off each event. Their simultaneous occurrence is unnecessary. The extraction property in Definition 8.2(iv) realizes the further law by a diagonal discrete array with these witnesses, all fixed offsets, and the common reference experiments retained. Its kernels are the newly extracted reference kernels; they need not be limits of kernel values in the preceding laws.

The near-saturation paths have their entire ranges in a fixed closed band K and minimize in its open collar W . Retain $D_W|_{K \times K}$ as well as the paths. On compact reference regularity sets, their lengths lie in a fixed interval $[a, A] \subset (0, \infty)$ and their parametrizations have a common Euclidean modulus. Under a joint coupling, a subsequence converges uniformly to a nonconstant D_W -geodesic P , since the identities $D_W(P_k(s), P_k(t)) = |s - t| \text{len}_D(P_k)$ pass to the limit. Exhausting these regularity sets loses arbitrarily little probability.

If a limiting competitor in W were strictly cheaper than $C \text{len}_D(P)$, its compact range would lie in a member of a fixed countable exhaustion of W . Hausdorff attainability of the full certificate records realizes it in the approximating laws, with nearby endpoints and almost the same cost. The common upper comparison and the continuum joining property correct those endpoints at vanishing cost, uniformly on the retained compact reference regularity set. This contradicts the near-saturation inequalities. Thus the limiting path is saturated. Any cheaper competitor for an interior subsegment could be joined to the unchanged remaining pieces at factor C , contradicting saturation of the full path; every interior subsegment is saturated as well.

For the low witness retain its closed rectangular envelope R_k , its endpoints, its certificate range, and its two outer collars. The envelope centers and side lengths lie in a compact parameter set with positive lower width; the possible orientations form a finite set. Uniform reference moduli and the slack $c_1 < c_2$ allow a finite gridded library for the denominator alone: choose a fixed open rectangle Ω inside the original denominator collar, still containing R_k with uniform positive clearance, and retain the internal metric on Ω and a further fixed collar. Shrinking the denominator can only increase its intrinsic distance. Extract the actual envelope parameters as well as the marked certificate and endpoints. Along $R_k \rightarrow R$, closed range inclusion and the opposite-face endpoint conditions pass to the limit. The limiting range therefore remains inside R itself, regardless of the open collar used to realize it. Its ratio relative to D_Ω is at most $c_2 = C - \delta_0$. Retaining this envelope will allow the forcing construction to preserve the certificate on a thin real-field patch still contained in the denominator domain. If its lower comparison were $C > 0$, Lemma 8.7 would contradict this witness.

Transfer to a whole-plane field in a buffered patch using the same local kernels. Choose its actual Gaussian reference to be the whole-plane GFF G_R pinned on $\partial B_R(0)$, with this fixed circle strictly outside the buffered patch. All later candidate supports are compactly contained in that patch, after discarding finitely many coarse scales. Thus every catalogue shift has zero average on the pinning circle and belongs to the Cameron–Martin space of this one reference law. The annular Markov trials use Dirichlet interiors conditioned on their exterior harmonic parts; no change of density across the pinning circle is required. Choose an interior segment shorter than its D -distance to the patch boundary; it is an ambient geodesic because any exiting competitor is longer. Strong confluence places a nonconstant part of it on a geodesic between two points of a fixed countable dense set. Fixed-pair uniqueness holds simultaneously for these pairs. Choose rational fractional times inside the common segment and an open finite union of rational disks with closure inside the saturation neighborhood. This is a countable union of events covering a positive-probability event; one choice has positive probability and gives Equation (8.15). $\square$

8.4 Local forcing and rigidity

Assume $c < C$ and fix the equality tangent from Proposition 8.15, with the deterministic anchors, fractional times and open set U in Equation (8.15). For each sample let P be its fixed-anchor reference geodesic and set

$$R = \frac{F_U(P(t_1), P(t_2))}{(t_2 - t_1)D(a_0, a_1)}, \quad (8.16)$$

with $R = +\infty$ if either endpoint lies outside U . The saturation event has positive probability and gives $R = C$. We will construct output samples with $C - \epsilon_N \leq R < C$, where $\epsilon_N \downarrow 0$, and show that their total probability stays bounded below. This is impossible for a random variable under one fixed law.

The equality tangent also has a separate positive-probability rectangular shortcut event. It need not occur together with saturation. We will sample a shortcut in a small patch, retain the old passage observations outside a larger patch, and change the real field by a function f that is constant on the shortcut patch and zero on the preserved exterior. The change will force the new reference geodesic through corridors near both ends of the shortcut, where short passage joins make the shortcut available. Write $D' = D_{h+f}$ for the output reference metric, P' for its fixed-anchor geodesic, and F' for the completed output passage rule.

There are two probabilistic requirements. After restricting the source event by a fixed likelihood bound, each output subprobability measure must be dominated by the same fixed multiple of the original law. Also, many modifications of one source sample must not produce too much mass at the same output. The next two lemmas isolate these requirements. Their application will count small balls hit by the old saturated geodesic on the source side and traversals of unchanged shells by the new geodesic on the output side.

Lemma 8.16 (Restricted Cameron–Martin transfer). *Let*

$$\mu(dh, d\eta, dZ) = G(dh)\rho(d\eta)K(h, \eta; dZ)$$

be a joint law on standard Borel spaces, with Gaussian real field law G . Fix a deterministic Cameron–Martin shift f . Suppose two separated passage restrictions have a conditional product kernel, the first depending on a patch where $f = 0$ and the second on a patch where $f = b$. Retain the first from the old sample and sample the second independently, conditional on the fields, at the exact shifted field $h + b$. Assume locality identifies their joint kernel with the corresponding restrictions of $K(h + f, \eta; \cdot)$.

There is a stochastic completion of these restrictions into an output $(h + f, \eta, Z')$ with unselected law $G_f(dh')\rho(d\eta)K(h', \eta; dZ')$. For every selection event A in the enlarged source space contained in

$$\exp\left\{W_f(h) + \frac{1}{2}\|f\|_{\mathcal{H}}^2\right\} \leq \Lambda,$$

its output subprobability measure ν_f satisfies $\nu_f \leq \Lambda\mu$.

Proof. Use the target regular conditional law of the full passage object given the two retained restrictions and the output fields. Before imposing A , the locality and product hypotheses give exactly the claimed output law. Its density relative to μ is the Gaussian shift density

$$L_f(h') = \exp\{W_f(h') - \frac{1}{2}\|f\|_{\mathcal{H}}^2\}.$$

At $h' = h + f$ this is $\exp\{W_f(h) + \frac{1}{2}\|f\|_{\mathcal{H}}^2\}$. Restriction to A decreases the unselected output measure and confines it to the set where this density is at most Λ . This proves the assertion, regardless of whether A is measurable from the output. $\square$

Lemma 8.17 (Weighted output multiplicity). *Let μ be a probability measure. For each N let I_N be finite, with deterministic weights $w_\beta > 0$, subprobability measures ν_β , and measurable sets H_N, B_β . Suppose, with constants independent of N ,*

$$\sum_{\beta \in I_N} w_\beta \nu_\beta(1) \geq \alpha N, \quad \nu_\beta \leq \Lambda \mu, \quad \nu_\beta(H_N \cap B_\beta) = \nu_\beta(1),$$

and, μ -almost surely on H_N ,

$$\sum_{\beta \in I_N} w_\beta \mathbf{1}_{B_\beta} \leq KN.$$

Then $\mu(H_N) \geq \alpha/(\Lambda K)$. In particular these assumptions cannot hold for $H_N = \{C - \epsilon_N \leq R < C\}$ when R is one fixed-law random variable and $\epsilon_N \rightarrow 0$.

Proof. Domination, support, and Tonelli's theorem give

$$\alpha N \leq \sum_{\beta} w_\beta \nu_\beta(H_N \cap B_\beta) \leq \Lambda \int \mathbf{1}_{H_N} \sum_{\beta} w_\beta \mathbf{1}_{B_\beta} d\mu \leq \Lambda KN \mu(H_N).$$

For the last assertion, $\mathbf{1}_{H_N} \rightarrow 0$ pointwise, including on $\{R = C\}$, so bounded convergence yields a contradiction. $\square$

Remark 8.18 (Source mass and observable output counts). Suppose every point of a positive-length middle arc is covered by at least cN good holes, and a hit ball of deterministic weight w_β accounts for at most $K_0 w_\beta$ of arc length. Integrating along the arc gives an order- N lower bound for the weighted number of hits. This coverage statement must already include all likelihood clipping and other selection restrictions.

For the output bound, require B_β to be a measurable condition on the output fixed-anchor geodesic, including a traversal of new length at least κw_β in a deterministic enlarged shell. Bounded overlap at each spatial radius, $O(N)$ radii, boundedly many aligned volume labels, and a fixed finite recipe list imply

$$\sum_{\beta} w_\beta \mathbf{1}_{B_\beta} \leq K_1 N \text{len}_{D'}(P').$$

A deterministic output length cutoff turns this into the hypothesis of Lemma 8.17. Such a cutoff follows from an old length cutoff and a uniform bound on the shift amplitude. The events B_β cannot refer to an old geodesic or an unobserved preimage.

The remaining work is to construct outputs to which these two lemmas apply. A deterministic field modification will first force the new geodesic through corridors near both ends of a preserved shortcut. Uniform stability as the modified patch shrinks will then place its fixed-anchor ratio arbitrarily close to C . Finally we will make the forcing geometry available in the annuli supplied by Lemma 8.12.

Lemma 8.19 (Deterministic gate forcing). *Let D be a proper length metric on $\mathbb{C}$ inducing its usual topology, let K be a closed disk, and let $a_0, a_1 \notin K$. Let ℓ_j be the infimum of the D -lengths of paths from a_j to ∂K before entering K , and choose exterior minimizers x_j . Suppose*

$$D(a_0, a_1) \geq \ell_0 + \ell_1 + g, \quad g > 0.$$

Let $D_f = e^{\xi f} \cdot D$, where f is bounded, continuous, and supported in the interior of K . Fix $B, e, d > 0$, $C > 0$, and $0 < \delta < C$, with

$$B + 2e < g, \quad 4Ce < \delta d.$$

Assume the following deterministic conditions.

- (i) A competitor from a_0 to a_1 has D_f -length at most $\ell_0 + \ell_1 + B$.
- (ii) There are effective open regions O, T_0, T_1, N , with incidence graph $O - T_0 - N - T_1 - O$. The exterior of K is contained in O , $f \geq 0$ on O , and the outer gates $A_j = O \cap T_j$ admit D -connections to x_j of length at most e . Every curve between its first and last visits to ∂K with D_f -length at most B admits an itinerary in this graph. Intervening visits to an isolated inner region are impossible.
- (iii) There are $q_0, q_1 \in \Omega$, with $N \Subset \Omega$, such that $d = D_b(q_0, q_1; \Omega)$, where $D_b = e^{\xi b} D$. Every point of the terminal gate $G_j = N \cap T_j$ can be joined to q_j inside Ω at D_b -cost at most e , and $f \geq b$ on $N \setminus (T_0 \cup T_1)$.
- (iv) Every complete corridor passage of D_f -length at most B from its outer gate to its terminal gate, or in the reverse direction, contains a point z with $D_f(z, q_j; W) \leq e$. Here W is one fixed open neighborhood containing the two marker-to- q_j connections and the preserved certificate in (v), all with compact confinement. It need not contain the exterior region O or the anchor approaches.
- (v) A passage rule F' has upper comparison C against D_f inside W , admits concatenation with arbitrarily small open loss, and contains a preserved certificate from q_0 to q_1 in W with cost at most $(C - \delta)d$.

Then every D_f -geodesic from a_0 to a_1 has a subsegment with endpoints $z_0, z_1 \in W$ such that

$$F'_W(z_0, z_1) < CD_f(z_0, z_1).$$

The same assertion holds for approximate exterior minimizers after subtracting their errors from g and allocating them within e .

Proof. Write P' for a new geodesic. It must enter K , since otherwise its length is an old path length, contradicting (i) and $B < g$. Its portion between the first and last visits to ∂K has length at most B : its exterior prefix and suffix have costs at least ℓ_0 and ℓ_1 .

Gate contacts below may lie in the closures of the open overlaps. The outer D -distance bounds extend to these contacts by continuity. For terminal contacts, $\overline{N} \subset \Omega$, and the internal metric $D_b(\cdot, \cdot; \Omega)$ is locally continuous: sufficiently close points are joined by near-minimizing paths contained in a fixed small ball of Ω . Thus its distance bound to q_j also extends from G_j to $\overline{G}_j$. Realizing these infimal bounds may add an arbitrarily small error, which the two strict displayed budget inequalities absorb.

The exterior gap also gives $D(x_0, x_1) \geq g$, by concatenating the two minimizing exterior approaches with a connection between x_0 and x_1 . The first corridor visited by P' is T_0 . Indeed, if it were T_1 , the preceding part lies in O , where its old cost is at most its new cost. Its cost is at most $\ell_0 + B$, since the final exterior approach costs at least ℓ_1 . Appending the gate correction and the old exterior approach from x_1 would give an old path of length at most $\ell_0 + \ell_1 + B + e$, a contradiction. The last corridor is similarly T_1 . A path remaining in O is excluded in the same way. No excursion in O can connect the two outer gates: such an excursion, of cost at most B , would give $D(x_0, x_1) \leq B + 2e < g$.

The finite incidence graph consequently contains an inner $T_0 - N - T_1$ passage along P' . For precision, a finite subdivision of the compact curve subordinate to the open region cover gives a finite region itinerary; repetitions can be retained or erased. There is no need to sum corrections over repeated gate visits. The first full approach to N uses T_0 , and the last full exit from N uses T_1 . Condition (iv) supplies points z_0, z_1 on these passages, in this order, with $D_f(z_j, q_j; W) \leq e$.

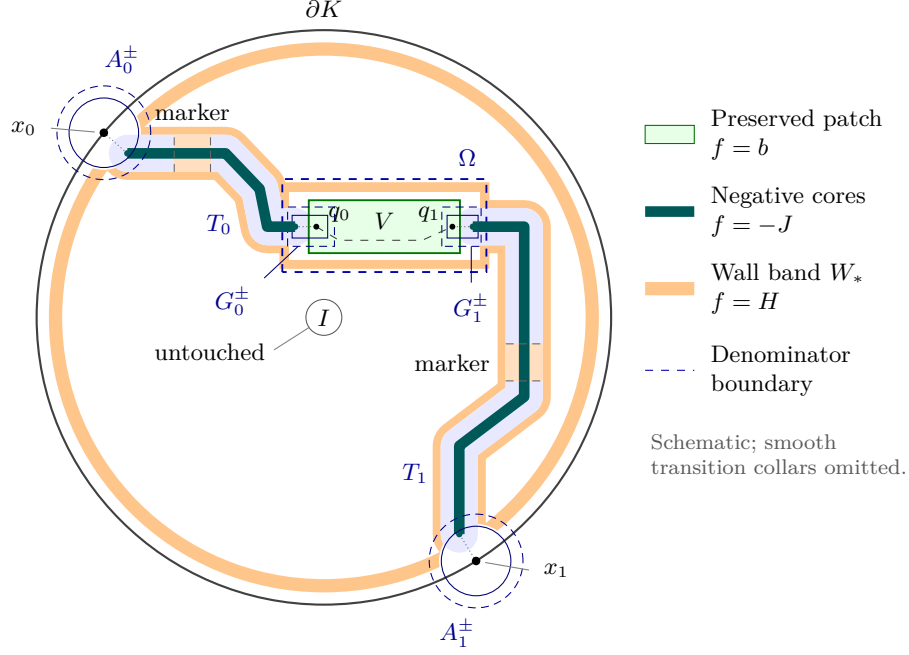


Figure 2: The fixed geometry in Lemma 8.20. The preservation patch V and its terminal caps lie inside the denominator Ω , away from the untouched disk I . Two disjoint polygonal tubes connect the outer caps at $x_0, x_1 \in \partial K$ to opposite faces of V . Their negative cores stop inside the safe caps; the dashed preserved certificate joins q_0 to q_1 and remains wholly in V . For $D_f = e^{\xi f} \cdot D$, the wall band prevents a path of D_f -length at most B from crossing the forbidden gap. The two marked cross-sections force every such complete tube passage to approach its core: the marker has height H outside the core collar. Exact offset boundaries and smoothing collars are suppressed.

Between these two points there is a central passage from the last T_0 contact to the first following T_1 contact. This passage lies in N , avoids the effective corridors in its interior, and has multiplier at least $e^{\xi b}$. Correcting its terminal points inside Ω by (iii) shows that its new length is at least $d - 2e$. In particular

$$D_f(z_0, z_1) = \text{len}_{D_f}(P'[z_0, z_1]) \geq d - 2e.$$

Upper comparison on the two short approach joins and the preserved certificate now give

$$F'_W(z_0, z_1) \leq (C - \delta)d + 2Ce < C(d - 2e) \leq CD_f(z_0, z_1).$$

The strict middle inequality leaves room for the arbitrarily small concatenation losses. □

Figure 2 displays the preservation patch, two corridors, and transverse markers used in the construction below.

Lemma 8.20 (Construction of the finite forcing catalogue). *Let $K = \overline{B_2(0)}$ and $I = \overline{B_{1/8}(0)}$. Fix a finite set of Euclidean configurations of the following type.*

- (a) *A rectangular preservation patch V and a larger rectangular denominator domain Ω satisfy $\overline{V} \subset \Omega \Subset B_{3/2}(0) \setminus \overline{B_{1/4}(0)}$. Two endpoint ports lie near opposite faces of the rectangle inside V .*

- (b) Disjoint terminal caps $G_j^- \Subset G_j^+ \Subset \Omega$, $j = 0, 1$, contain those ports. Each G_j^- also contains a fixed stop point $s_j \notin \bar{V}$, which can approach the corresponding face from its outer side. The caps have been chosen by a sufficiently fine finite gridding of the rectangle and its two endpoint ports.
- (c) Disjoint outer caps $A_j^- \Subset A_j^+$ surround two separated gridded points of ∂K . A fixed point $r_j \in A_j^- \cap \text{int } K$ lies a positive distance inside K . The possible exterior attachment points $x_j \in \partial K$ lie in a compact subset of A_j^- .

All indicated containments and separations have strict Euclidean margins. The endpoint ports and the terminal stops can, for example, be obtained by taking a thin collar of a gridded rectangle and cap boxes straddling its opposite faces. Fix a finite set $\mathcal{B} \subset \mathbb{R}$ of exact constant shifts.

For each configuration one can choose finitely many fixed polygonal core domains and marker rectangles as described below. Suppose the reference metric D and the chosen endpoints q_j satisfy the following deterministic bounds, uniformly for that finite family, with positive numbers u, w, e, A, K_*, m_* and a modulus $\omega(t) \rightarrow 0$.

- (i) Every point of A_j^+ has an old local connection to x_j of length at most e ; there is a connection from x_j to r_j in A_j^- of length at most $e/3$.
- (ii) Every point of G_j^+ has a D_b -connection to q_j in Ω of length at most e , where $D_b = e^{\xi_b} D$. There is a connection from s_j to q_j in G_j^- of D_b -length at most $e/3$.
- (iii) There is a D_b -connection from q_0 to q_1 inside V of length at most Au .
- (iv) Each point of the closed core domain can be connected to its terminal stop, within that core, at old cost at most $K_* w$. The outer stop has such a connection too.
- (v) The fixed forbidden-gap and marker crossings have old length at least $m_* w$. Points at Euclidean distance at most t from a core can be connected to that core inside its prescribed open collar at old cost at most $\omega(t)w$.

Fix a local budget B with $Au + 2e \leq B$. Then there is a finite deterministic list of smooth functions f , depending only on the geometric list, these bounds, and $\mathcal{B}$, with the following properties. Each function is supported compactly in $B_2(0) \setminus I$, is exactly b on V , and satisfies the network, gate, and cheap-approach hypotheses of Lemma 8.19 for the indicated configuration and budget B , with all marker-to-terminal joins and the preserved certificate compactly confined to the bounded neighborhood $W = B_3(0)$. It also provides a competitor of cost at most $\ell_0 + \ell_1 + B$ whenever the exterior approaches to x_0, x_1 have old costs ℓ_0, ℓ_1 . The list has uniformly bounded amplitudes and Dirichlet energies, including after any Euclidean translation and dilation.

Proof. We first make all Euclidean choices; no metric-dependent curve is used to choose the shape of a shift.

Two embedded corridors. Extend the short prescribed initial and terminal pieces to an arc joining ∂K to the corresponding face of ∂V , inside $\text{int } K \setminus (I \cup \bar{V})$. Choose the first arc simple. An arc joining distinct boundary components of this planar region does not disconnect its interior after cutting along the arc: the outer boundary and the rectangular boundary become one boundary component, while the omitted disk remains another. Thus the second pair of prescribed ends can be joined in the cut region. Approximate both arcs by simple polygonal arcs, keeping the

prescribed short end pieces and positive clearance from each other, the omitted disk, and the rectangle except in their terminal caps. Trim their ends to obtain disjoint arcs from r_j to s_j .

One can require the arcs to avoid the other attachment and terminal cap by choosing those caps small first. At a terminal cap approach along the outward normal to its specified face. Since the input configurations form a finite list, these choices can be made once for each member. Their minimum positive clearance is then positive.

Nested neighborhoods and the region graph. Choose three nested regular neighborhoods

$$O^0 \subset O^1 \subset O^2, \quad N^0 \Subset N^1 \Subset N^2, \quad T_j^0 \Subset T_j^1 \Subset T_j^2,$$

where O^k contains the exterior of K and a thin inner collar, N^k surrounds $\bar{V}$, and T_j^k is a regular tube around the j th polygonal arc with end bulbs. For the unbounded sets O^k , require positive separation between their finite boundary components. Choose the neighborhoods sufficiently thin that

$$\bar{N}^2 \subset \Omega, \quad O^2 \cap N^2 = \emptyset, \quad T_0^2 \cap T_1^2 = \emptyset,$$

and their only overlaps satisfy

$$O^2 \cap T_j^2 \subset A_j^+, \quad N^2 \cap T_j^2 \subset G_j^+.$$

All these incidences persist under a further small thickening. This follows directly from the positive clearances of the arcs away from their four prescribed end caps. Make the end bulbs large enough that the smaller safe caps A_j^- and G_j^- lie in the union of the smallest neighborhoods, with positive margin. There is no requirement that the larger effective caps be cheap in the modified metric.

Write

$$S^k = O^k \cup N^k \cup T_0^k \cup T_1^k.$$

These sets avoid a fixed neighborhood of I , and there is $\rho_* > 0$ such that the Euclidean ρ_* -neighborhood of $\bar{S}^0$ lies in S^2 . The middle layer

$$W_* = \{z : \rho_*/3 \leq \text{dist}(z, S^0) \leq 2\rho_*/3\}$$

is compactly contained in $\text{int } K \setminus I$ and disjoint from all the safe caps and the preservation patch. A curve starting in S^0 and leaving S^2 has a subcurve in W_* whose endpoints are separated by at least $\rho_*/3$. This is the forbidden-gap crossing in (v).

The nerve of the cover of S^2 by its four regions is precisely the cycle

$$O^2 - T_0^2 - N^2 - T_1^2 - O^2.$$

Any compact curve contained in S^2 admits a finite subdivision subordinate to this open cover and hence a finite itinerary in that graph. This statement allows arbitrarily many incidental boundary contacts.

Core domains and marked sections. Choose closed polygonal core domains $C_j \subset T_j^0$ around the trimmed arcs, with the following margins:

$$C_j \cap O^2 = \emptyset, \quad C_j \cap \bar{V} = \emptyset,$$

and both stops r_j, s_j lie in C_j . The outer collar is chosen thinner than the depth of r_j ; the inner stop lies outside $\bar{V}$ by assumption. The core connection bounds in (iv) concern these fixed closed

domains. They may be obtained from the internal box estimates on a finite rectangular cover, including access to the flat faces.

On a straight interior piece of each polygonal tube, away from its end bulbs and from $O^2 \cup N^2$, select a closed longitudinal marker rectangle spanning the full closed transverse section of T_j^2 between its two end cross-sections. The tube there is an ordinary Euclidean strip. A path in T_j^2 from its outer overlap to its terminal overlap must cross this rectangle between two cross-sections a fixed positive distance apart. One can see this by using the longitudinal coordinate in the regular tube; retain the subcurve between its last visit to the first cross-section and its first subsequent visit to the second. This is the marker crossing in (v).

A smooth wall and constant plateaux. Choose $H > \max\{0, \max \mathcal{B}\} + 1$ so large that

$$e^{\xi H} m_* w > B.$$

Choose a smooth baseline a , compactly supported in $\text{int } K \setminus I$, which is zero on the safe outer caps and near ∂K , equals b on a neighborhood of $\bar{V} \cup \bar{G}_0^- \cup \bar{G}_1^-$, and equals H on a neighborhood of W_* and on both marker rectangles. Require also

$$a \geq 0 \text{ on } O^2, \quad a \geq b \text{ on } N^2, \quad \min\{0, b\} \leq a \leq H.$$

These requirements have disjoint closed cores and positive margins. Explicitly, first choose a cutoff ζ equal to one on the preservation patch and the safe terminal caps, supported in S^0 and away from O^2 , the marker rectangles, the wall layer, and the safe outer caps. Next choose ψ equal to one on N^2 , the support of ζ , the marker rectangles, and W_* , and zero on the safe outer caps and near the forbidden support boundaries. The formula

$$a = H\psi + (b - H)\zeta$$

has all the stated properties, with the cutoffs chosen between zero and one.

Choose $J > \max\{0, -\min \mathcal{B}\} + 1$ so that

$$e^{-\xi J} K_* w < e/3.$$

Then choose $\rho > 0$ smaller than all the fixed geometric margins, and so small that

$$e^{\xi H} \omega(\rho) w < e/3.$$

Let χ_j be smooth, equal to one on C_j , and supported in its ρ -neighborhood. These supports are disjoint, lie inside S^0 , and avoid $O^2 \cup \bar{V} \cup W_*$. Set $\chi = \chi_0 + \chi_1$ and define

$$f = (1 - \chi)a - J\chi.$$

This is a smooth compactly supported shift. It equals $-J$ on the cores, b on V , and H on W_* . On the safe terminal caps $f \leq b$, and on the safe outer caps $f \leq 0$. Moreover,

$$f \geq 0 \text{ on } O^2, \quad f \geq b \text{ on } N^2 \setminus (T_0^2 \cup T_1^2).$$

The local budget gives the itinerary. A curve starting at ∂K starts in S^0 . If a curve of total new length at most B leaves S^2 , it crosses W_* and pays more than B by the choice of H . Thus every local-budget portion between the first and last visits to ∂K stays in S^2 . The finite-cover argument above gives its claimed itinerary. The omitted disk supplies no bypass, since it lies outside S^2 . The outer and terminal gate-error conditions follow from their containments in A_j^+ and G_j^+ and (i)–(ii).

A marked passage gives cheap endpoint access. On a marker rectangle, $a = H$. Outside the ρ -neighborhood of its core one has $\chi = 0$ and therefore $f = H$. A complete corridor passage avoiding that core neighborhood would have an all-wall marker crossing and cost more than B . Every complete passage with the local budget consequently contains a point z at distance at most ρ from the core. By (v), this point connects to the core at new cost at most $e^{\xi H} \omega(\rho) w < e/3$. By (iv), it then reaches s_j through the core at cost less than $e/3$. The safe terminal cap has shift at most b , so (ii) gives a final connection to q_j at cost at most $e/3$. Thus

$$D_f(z, q_j; \text{prescribed neighborhood}) \leq e.$$

These marker points supply cheap approach joins. The different effective gate contacts are used only for the old D_b lower-bound corrections in the forcing lemma. The prescribed core collars and terminal-cap paths lie inside $B_2(0)$, and the retained certificate lies inside V . Consequently the shortcut assembled from these joins and that certificate has compact confinement in $W = B_3(0)$. The outer region and the long anchor approaches play no part in this confinement assertion.

The competitor and finiteness. Follow the unchanged exterior approaches, the safe outer cap connections, the negative cores, the safe terminal cap connections, and the connection through V from (iii). The six approach contributions have total cost at most $2e$, so the resulting competitor has length at most $\ell_0 + \ell_1 + Au + 2e \leq \ell_0 + \ell_1 + B$.

Every region, core, marker, cutoff, and height was selected from deterministic data after the finite geometric list and uniform metric bounds had been fixed. Hence these functions form a finite list. Their amplitudes are bounded by $\max\{H, J, \max_{b \in \mathcal{B}} |b|\}$. Smoothness and compact support give a finite common Dirichlet-energy bound. Finally,

$$\int_{\mathbb{C}} \left| \nabla f \left(\frac{z - z_0}{r} \right) \right|^2 \frac{dz}{r^2} = \int_{\mathbb{C}} |\nabla f(w)|^2 dw,$$

so the energy of $z \mapsto f((z - z_0)/r)$ is unchanged by translation and positive dilation. $\square$

Remark 8.21 (Which bounds are chosen when). The preservation patch, terminal caps, and their error bounds are chosen first, followed by the thin-domain connection bound A . Next choose the fixed ratio w/u large enough for the exterior gap and budget conditions. The outer attachment grid and all Euclidean tubes are then fixed. The new core and gap bounds K_*, m_*, ω are truncated on those finitely many domains. These later truncations only affect H, J, ρ ; they do not alter A or the terminal caps. Thus the construction has no circular dependence of the upper compartment budget on the later routing constants.

Lemma 8.22 (Uniform stability for a shrinking modification). *Let D be a proper geodesic metric inducing the Euclidean topology and having a unique geodesic P between fixed distinct anchors a_0, a_1 , parametrized by fractions of its length. Fix $0 < t_1 < t_2 < 1$ and an open set U containing $P([t_1, t_2])$ with positive clearance. Suppose the old passage rule has upper comparison C in open subsets of U and*

$$F_U(P(t_1), P(t_2)) = C(t_2 - t_1)D(a_0, a_1).$$

Consider bounded shifts f with one common amplitude bound, supported in balls whose radii tend uniformly to zero and whose centers lie at vanishing distance from a fixed compact middle subarc of $P((t_1, t_2))$. Let F' be any output passage rule with upper comparison C against $D_f = e^{\xi f} \cdot D$. Suppose it preserves all old exterior certificate restrictions outside a slightly enlarged support ball.

Assume the certificate system permits marked subpath extraction with additive nonnegative costs and concatenation with arbitrarily small open loss.

Uniformly over these modifications and all output fillings, every fractionally parametrized D_f -geodesic P' satisfies

$$\sup_{t \in [0,1]} D(P'(t), P(t)) \longrightarrow 0, \quad D_f(a_0, a_1) \longrightarrow D(a_0, a_1),$$

and

$$\frac{F'_U(P'(t_1), P'(t_2))}{(t_2 - t_1)D_f(a_0, a_1)} \geq C - o(1).$$

If a strict shortcut along P' is contained in the shrinking modification neighborhood, then for all sufficiently small supports this ratio is strictly less than C .

Proof. Choose constants $0 < m \leq 1 \leq M < \infty$ with $mD \leq D_f \leq MD$. Properness puts all the relevant geodesics in one compact D -ball. Let ω_n be the supremum of the old D -diameters of suitably enlarged possible support balls. The Euclidean radii tend uniformly to zero, so continuity gives $\omega_n \rightarrow 0$.

Replace the portion of an old geodesic between its first and last support visits by an old minimizing connection. Its new length is at most $M\omega_n$. For the reverse comparison, cut a new geodesic at its first and last support visits and insert an old minimizing connection of cost at most ω_n . Outside the support the lengths coincide. Thus

$$|D_f(x, y) - D(x, y)| \leq (M + 1)\omega_n$$

for the endpoints under consideration. Moreover, the new geodesic segment spanning all its support visits has new length at most $M\omega_n$, and old length at most $m^{-1}M\omega_n$. Hence the old length of the entire new geodesic tends to $D(a_0, a_1)$.

The new geodesics are uniformly Lipschitz in D under their fractional parametrizations. Every subsequential limit is an old minimizing path; equivalently, the uniform distance comparison shows that each of its subsegments has the appropriate fractional distance. Uniqueness identifies the limit with P with its specified parametrization. A contradiction-subsequence argument proves uniformity over all support locations and bounded shifts. Write the uniform endpoint discrepancy as $\eta_n \rightarrow 0$.

Put $q_0 = P(t_1)$, $q_1 = P(t_2)$ and $q'_0 = P'(t_1)$, $q'_1 = P'(t_2)$. Consider an arbitrary output certificate from q'_0 to q'_1 compactly in U . If it enters the modification neighborhood, cut its approximants at their first and last visits to a ball strictly larger than the ball removed from the preserved exterior patch. Marked subpath extraction gives retained prefix and suffix certificates in that open exterior patch, with total cost no larger than the original cost. Transfer them to the old sample. Join their cut points at old passage cost at most $C\omega_n + o(1)$, and correct the two endpoints at total cost at most $2C\eta_n + o(1)$. All corrections lie in U , by its clearance and the fact that short old geodesics remain in small neighborhoods. For a certificate avoiding the larger ball, transfer it directly and make only the endpoint corrections. The retained pieces use their own positive clearance from ∂U ; a common clearance for all competing certificates is unnecessary.

Taking infima yields, with $\ell = (t_2 - t_1)D(a_0, a_1)$,

$$C\ell \leq F'_U(q'_0, q'_1) + C(\omega_n + 2\eta_n).$$

The new denominator ℓ' tends to $\ell > 0$, proving the ratio bound. Finally, the old middle subarc is separated from $P([0, t_1] \cup [t_2, 1])$. Uniform convergence therefore places every visit of P' to

the shrinking modification neighborhood strictly between t_1 and t_2 . Concatenating the strict shortcut with upper-comparison paths along the remaining geodesic portions gives the strict upper inequality. $\square$

Remark 8.23 (Deterministic errors before counting). The error in Lemma 8.22 is a random function of the incoming sample, uniform over all candidate modifications. Its almost sure convergence to zero allows a deterministic sequence $\epsilon_N \rightarrow 0$ for which the error exceeds ϵ_N with probability tending to zero. Intersect the event that the error is at most ϵ_N with the positive-probability saturation event and the every-point good-cover event before forming the source submeasures. A positive lower cutoff for the interval length ensures that the ratio error is controlled. This gives the deterministic common-law sets H_N required by Lemma 8.17.

It remains to make a successful catalogue available with high probability in a reference shell. When the passage kernel uses the genuine order of the local imaginary traversals, an independent trial must determine that order from its own dependency disk. The next lemma provides such disks; for an order-free kernel this step is unnecessary.

Lemma 8.24 (A locally certified imaginary order). *Let $\hat{h}$ be the whole-plane imaginary field modulo its angular period, and let η be its Lebesgue-parameterized space-filling curve. Fix a compact set $K \Subset U$, with U bounded and open. The following public properties suffice:*

- (i) *the rational-point visit times of η are dense;*
- (ii) *its full directed U -traversals indexed by rational points, parametrized by Lebesgue area from their entry times, are determined from $\hat{h}$ on an arbitrarily small enlargement of U , and the same holds for bounded disks;*
- (iii) *each rational spatial point is visited once almost surely.*

There are events D_R , measurable from the imaginary field on a slightly enlarged ball of radius R , such that $\mathbb{P}(D_R) \rightarrow 1$ and, on D_R , the genuine order of all U -traversals meeting K is determined by that restriction.

Proof. Transience confines the visits to $\bar{U}$ to a compact time interval. Uniform continuity gives a positive lower duration for each U -traversal reaching K , because it must cross $\text{dist}(K, \partial U) > 0$. There are therefore only finitely many such traversals. Property (i) ensures that each contains a rational-point visit. In the locally reconstructed collection select the first rational representative in a fixed enumeration for each traversal; this selection does not use their chronological order.

Apply (ii) again to the disk B_R . For every selected rational representative, reconstruct its full directed B_R -traversal, with time zero at entry. Let D_R be the event that all these records are identical. It is measurable from the buffered imaginary-field restriction. Distinct traversal intervals cannot have identical records: they have positive time length and hence positive area, whereas ranges of disjoint time intervals overlap in zero area. On D_R , the common directed parametrization and (iii) specify the order of the representatives exactly, and therefore the order of their distinct small-domain traversals.

Almost surely the finitely many representative visit times have a finite minimum and maximum. By continuity, the image of the interval between them is compact and lies in B_R for all sufficiently large R . They then lie in one common B_R -traversal, so D_R holds eventually. This proves the exhaustion. $\square$

Remark 8.25 (Public inputs for the preceding lemma). Properties (i) and (ii) are given in Section 2.1.3 and Lemma 2.4 of Gwynne–Miller–Sheffield [24]. For (iii), area parametrization makes the set of multiply visited spatial points Lebesgue-null: cover pairs of distinct visit times by disjoint rational time intervals, whose ranges have zero-area intersection. Translation invariance of the whole-plane curve modulo time translation then gives zero probability that any fixed spatial point is multiply visited, and a countable intersection handles rational points. The stationary angular-phase convention for the imaginary field is used here. The argument uses density of rational *visit times*; positive area of an excursion range alone would not imply that it contains a rational point.

Proposition 8.26 (Conditional availability of a finite catalogue). *Consider favorable normalization indices v and a family of passage kernels. Suppose:*

- (i) *For each open preservation patch V , the restricted certificate kernel is a function of $h|_V$, the local imaginary traversal data, and their finite genuine order. The order depends only on the imaginary field, and the kernels are compatible under restriction to smaller open patches. Separated restrictions have conditional product kernels.*
- (ii) *Under a common normalized background-field law, at every favorable index there is probability at least $\beta > 0$ of a strict low-ratio certificate whose entire range lies in a retained closed rectangular envelope and whose endpoints lie on its opposite faces, with common geometric slack and a denominator in a fixed buffered domain. The fixed reference-metric moduli and finite-domain connection bounds are almost surely finite.*
- (iii) *The exact affine and volume-normalization identities of the joint kernel family transport each trial to the corresponding favorable reference index after its real coarse constant, spatial logarithm, and fixed volume displacement have been compensated. These are identities of transported fields and deterministically normalized observations, not a Weyl rule for a fixed passage object.*
- (iv) *The normalized GFF harmonic profiles, after removing the real coarse constant and reducing the imaginary constant modulo its period, have uniformly tight interior smooth norms on fixed-ratio buffered trial disks.*

For every $\theta > 0$, one can choose finitely many disjoint buffered trials, a finite deterministic grid of exact real shifts, and the metric bounds for a finite forcing catalogue so that, uniformly over favorable indices, the completed catalogue is available with probability at least $1 - \theta$. The good event can include fixed bounds on all the resulting Cameron–Martin pairings. No continuity of passage costs under rounding a field shift is required.

Proof. First truncate the normalized small-box and face modulus at a loss smaller than $\beta/10$. Choose a small mesh width t using this common internal box modulus and the prescribed terminal error budget. Grid both the closed-envelope parameters and the positions of the two endpoints. For each resulting finite bin, choose an outward rectangular preservation collar V containing every envelope in the bin with positive margin. Each indicated face of V is within $O(t)$ of the corresponding envelope face. Choose terminal caps of diameter $O(t)$ straddling these two faces, containing the endpoint bins and fixed stop points outside $\bar{V}$. The internal box modulus on their collars gives the desired arbitrarily small terminal connection errors. The entire certificate range is in V because it was retained inside its closed envelope. Choose an intermediate denominator domain containing $\bar{V}$ inside the original one. Its intrinsic denominator is at least the old denominator, so the strict low-ratio bound remains valid.

The possible endpoints lie in deterministic compact subsets of finitely many newly thin domains. Their internal reference connection diameters are finite almost surely. Fix an upper bound A' with a further loss smaller than $\beta/10$. These cap sizes, thin domains, and A' remain fixed during all later tube truncations.

For a singleton order input, take its determination event to be the whole sample space. Otherwise choose a compact $K \Subset U$ covering the prepared local traversals. By Lemma 8.24, a fixed imaginary buffer makes the sum of order-determination failure probabilities over the finite geometry bins smaller than $\beta/10$. Write $E_{v,a}$ for the prepared success event in bin a at favorable index v , intersected with its determination event, and put $E_v = \bigcup_a E_{v,a}$. The genuine order on each of these events is a function of the buffered imaginary field. Thus the success union E_v is local and retains a common positive probability, say β_1 . No individual bin is required to have positive probability.

Choose a larger fixed conditioning disk around the complete dependency region. Local absolute continuity transfers the reference fields to a pair of Dirichlet GFFs on this disk, restricted to its interior. The field density is independent of the index. Truncate this single density to conclude that the success union E_v has probability at least $\beta_2 > 0$ under this fixed Dirichlet reference law, uniformly in the index.

For an affine trial copy, condition on the fields outside its conditioning disk. The fields inside are Dirichlet fields plus exterior-measurable harmonic functions. On an acceptable profile event, subtract the real coarse constant and reduce the imaginary constant modulo its period. Smooth cutoffs extend the remaining profiles on the dependency region to Cameron–Martin functions of uniformly bounded energy. The same remains true after adding any real residual constant in $[-1, 1]$.

Let P be the fixed Dirichlet reference law, including an independent auxiliary kernel seed, and Q one such conditional law. The passage kernel is the same function of the field inputs under both laws. A uniform inverse likelihood second moment gives, for the success union E_v ,

$$\beta_2^2 \leq Q(E_v) \mathbb{E}_Q \left[\left(\frac{dP}{dQ} \right)^2 \right], \quad Q(E_v) \geq \beta_* := \beta_2^2 / K_* > 0.$$

The profile cutoff and β_* are independent of the number of trial copies: the normalized domains and buffer ratios are fixed, and (iv) controls their harmonic oscillations uniformly.

Choose that cutoff so the expected fraction of unacceptable profiles is below $\theta/100$. Markov's inequality implies that at least half the trials are acceptable except with probability at most $\theta/50$. Choose J with

$$(1 - \beta_*)^{J/2} < \theta/10,$$

and pack J deterministic disjoint copies of the entire conditioning disk into the prescribed shell subregion. All spatial ratios are now fixed. Before choosing M , allocate a fixed portion of the remaining failure probability to the outer reference shell: fix a positive normalized crossing lower bound a , a finite boundary-connection upper bound, and a reference connection modulus there. The normalized reference field law and shell geometry are fixed, so these choices do not depend on M . Write ε for the already fixed normalized terminal error budget. Choose the fixed quantum separation M required by the forcing budget, using this a , the already fixed A' and cap errors; in particular impose $A' + 4\varepsilon < aM^p$ in the units of Lemma 8.29. Assumption (iii) absorbs its logarithmic displacement into the desired compensation, so it does not change β_* .

The J desired compensations are exterior-measurable. Their normalized vector is uniformly tight: the favorable index cancels in (iii), leaving only the fixed affine charts, their coarse field

values, and the now fixed displacement from M . Truncate this vector to a compact interval with loss at most $\theta/10$, and cover that interval by a finite grid $\mathcal{B}$ of mesh at most two. Select the nearest grid point using only exterior data. Its residual lies in $[-1, 1]$, precisely the range in the preceding likelihood bound.

For every exact $b \in \mathcal{B}$, draw a fresh auxiliary observation array for the prepared patches from the common kernel at the exactly shifted real field, and restrict it to each V . Use independent seeds across disks and grid values, conditional on the fields. For the selected grid point declare success on the union of its events $E_{v,a}$; select the first successful bin in the fixed finite ordering. On that bin's determination event the genuine order equals the locally computed order. Hence the success indicator is a function of that disk's field restriction, its independent seed, and the exterior-measurable grid choice. The indicators are conditionally independent given the exterior of the disjoint disks. Each acceptable disk has success probability at least β_* , and the chance of no success is at most $(1 - \beta_*)^{J/2}$.

The successful certificate is kept at the exact grid value at which it was drawn. Rounding has been handled by changing the field law in a uniform absolute-continuity estimate; no claim about persistence of an individual certificate under rounding has been made.

Finally impose the remaining regularity and finite internal bounds for all the gridded routing, core, gap, and access domains, with total additional failure smaller than the unused portion of θ . These later bounds affect only the wall and core heights and the smoothing widths. They do not change the cap modulus or A' . After the resulting finite smooth recipe list is fixed, truncate its finitely many Gaussian linear pairings. For a smooth compactly supported recipe, this linear functional is a distributional pairing against its Laplacian, so it depends only on the real field on the recipe support. Clipping is therefore part of the same local shell event. The uniformly bounded energies give a fixed clipping threshold with arbitrarily small failure. Allocating the remaining losses proves the claimed probability. $\square$

Remark 8.27 (The real and imaginary dependency patches differ). The larger order buffer in the preceding proof is an *imaginary*-field dependency region. A real-field modification leaves this input unchanged. By the restriction compatibility in assumption (i), a retained certificate compactly in V uses the real field only on V , even if it was originally sampled as part of a larger auxiliary passage object. Thus a recipe equal to b on V retains precisely the required kernel after modifying the real field elsewhere. The reference-metric denominator in a larger domain is used for the deterministic lower bound and need not be retained as an unchanged output passage observation.

This use of restriction compatibility is the reason the real and imaginary dependency patches are specified separately.

Proposition 8.28 (Rigidity from local forcing). *Suppose the comparison family satisfies the hypotheses of Proposition 8.15. Assume in addition that the positive-probability rectangular shortcuts can be amplified to a high-probability shell event with a finite deterministic catalogue satisfying Lemma 8.19. The event must include the following data, with constants independent of the scale:*

- (a) *regular upper and lower reference bounds of order $w = (M\lambda^{-i})^p$, and a shortcut with denominator in $[a'\lambda^{-ip}, A'\lambda^{-ip}]$;*
- (b) *shifts of uniformly bounded amplitude and Dirichlet energy, constant on the complete real-field dependency patch of the retained shortcut, vanishing on the preserved exterior patch;*
- (c) *the likelihood clipping of Lemma 8.16, imposed before applying Lemma 8.12;*

(d) *output-observable shell traversal events with a lower reference cost κw and the bounded overlap and alignment multiplicities in Remark 8.18.*

Assume the exact local product kernels permit the completion and preservation in Lemma 8.16. Then the optimal comparisons coincide, $c = C$, and $F_U = CD_U$ for every open U .

Proof. If $c < C$, take the equality tangent and the positive-probability event in Equation (8.15). Restrict its interval length below by a deterministic positive number and its total anchor length above by a deterministic finite number; some such restriction still has positive probability. Apply Lemma 8.12 to the favorable finer labels given by Proposition 8.14, with volume multiplier M and with the full good event (a)–(c).

On the retained middle subarc, every point lies in at least βN good holes. A good ball accounts for at most $K_0 w$ of geodesic length: the time span between its first and last visits is bounded by a reference boundary connection, since a geodesic minimizes between any two of its points. Integration over the subarc gives

$$\sum_{\text{good hit balls}} w \geq K_0^{-1} \beta N \text{len}_D(\text{middle subarc}).$$

Choose a recipe by a fixed ordering among the successful members of the finite catalogue. This partitions source events and adds no probability conditioning factor. Intersect with the deterministic uniform-error events of Remark 8.23; their lost probability tends to zero. The expected source weight is still at least $\alpha_0 N$ for some $\alpha_0 > 0$.

For each deterministic candidate index, perform its fixed shift and complete the passage law before restricting to the selected source event. Lemma 8.16 gives an output submeasure $\nu_\beta \leq \Lambda \mu$ with the same fixed common law μ . Gate forcing gives a strict shortcut on the output geodesic. Shrinking-modification stability puts the gain in the same fixed fractional interval and shows that the output ratio from Equation (8.16) belongs to

$$H_N = \{C - \epsilon_N \leq R < C\}, \quad \epsilon_N \downarrow 0.$$

Here R is the same function of the output fields and passage data, and μ is the common dominating law.

The output traversal tests in (d), bounded overlap at each spatial radius, $O(N)$ radii, and boundedly many aligned labels and recipes give total output weight at most $K_1 N$ times its anchor length. The source length cutoff and the common amplitude bound give a deterministic output length cutoff. Thus the hypotheses of Lemma 8.17 hold and force $\mu(H_N) \geq \alpha_0 / (\Lambda K_2) > 0$. Bounded convergence gives $\mu(H_N) \rightarrow 0$, a contradiction.

We have $c = C > 0$. The upper bound is Equation (8.1), and the lower bound follows from Lemma 8.7. Therefore $F_U = CD_U$. $\square$

The catalogue-availability hypothesis in Proposition 8.28 combines Proposition 8.26 with Lemma 8.20. The real-field dependency patch must fit inside the denominator collar: preserving a shortcut on a much larger constant-shift patch could allow a bypass around that collar. An enlarged imaginary-field buffer is harmless because the imaginary field is unchanged. This distinction is part of the local-kernel requirement in this application.

Lemma 8.29 (Verification of the forcing hypotheses). *Let a family satisfy Definition 8.2, the reference metric inputs above, and the finite positive comparison hypotheses of Proposition 8.15. Then its equality tangent satisfies the catalogue, clipping, completion, and output-count hypotheses of Proposition 8.28.*

Proof. The low event is Proposition 8.14(ii), with its closed rectangular envelope and actual internal denominator metric. The kernel restriction and product properties give condition (i) of Proposition 8.26; in the one-order case its local order event is the whole sample space. Otherwise Definition 8.2(vi) and Lemma 8.24 provide that event using a larger imaginary buffer. The exact volume and affine identities give condition (iii), and Lemma 8.11 gives condition (iv). The reference internal box and face bounds give condition (ii). Thus the complete local catalogue, including fresh observations at each exact shift and likelihood clipping, has probability as close to one as required by Lemma 8.12.

We verify explicitly how a hit of the old reference geodesic supplies the deterministic forcing data. Write $u = \lambda^{-ip}$ and $w = M^p u$. Include in the good event a lower reference crossing bound aw between the central hole and the outer circle, a surrounding boundary connection bound Aw , and fixed reference connection moduli. On a deep hit, the reference geodesic pays at least aw between first entry and last exit from the outer circle. Its exterior pieces pay at least the two exterior infima ℓ_0, ℓ_1 . Hence

$$D(a_0, a_1) \geq \ell_0 + \ell_1 + aw.$$

Its exterior minimizing landing points have reference distance at least aw , by concatenating their exterior approaches. The truncated reference modulus separates them by a fixed positive Euclidean amount, so they fit a finite outer-cap grid with the required small errors.

Let the prepared low denominator satisfy $d \geq a'u$. Choose $e = \varepsilon u$ with $4C\varepsilon < \delta_0 a'$. The thin preservation patches, terminal caps, and their connection bound $A_1 u$ have already been fixed in the catalogue-availability proof. Choose the fixed M so large that

$$A_1 u + 4e < aw, \quad B = A_1 u + 2e.$$

All later core, gap and access bounds concern a finite family of new domains and affect only wall heights, core heights and smoothing widths. Lemma 8.20 therefore applies without changing A_1 or the terminal errors. Its budget and gate conditions give $B + 2e < aw$ and $4Ce < \delta_0 d$, precisely the strict inequalities of Lemma 8.19. The resulting strict shortcut has compact confinement in the image of $B_3(0)$ around the hit ball. These neighborhoods shrink uniformly to the old compact middle subarc and eventually lie in its fixed saturation neighborhood U .

Fix the Gaussian pinning circle outside that neighborhood and all the late candidate supports, as in the equality-tangent proof. Every deterministic recipe is then a Cameron–Martin direction of this same actual Gaussian law. Its real shift is constant on the complete preservation patch and vanishes on the preserved exterior patch; the larger imaginary buffer is unchanged. Resample the local area marks with their shifted intensity. The conditional product kernel and standard Borel completion in Definition 8.2 give exactly the unselected construction of Lemma 8.16. Its finite Gaussian pairing cutoffs are part of the catalogue event before annulus coverage or a geodesic hit is selected.

For clarity, the required output count can use a shell where the shift is identically zero. Include among the local reference tests a lower bound κw for every crossing of $B_{3r}(z) \setminus \overline{B_{2r}(z)}$. Every forced reference geodesic enters the compartment inside $B_{2r}(z)$ from anchors outside $B_{4r}(z)$, and crosses this unchanged shell. Its output length inside $B_{4r}(z)$ is therefore at least κw . Define the output event by this length bound, the inherited deterministic anchor-length cutoff, and the

alignment condition on the outer circle average. That average is unchanged by the recipe supported in $B_{2r}(z)$, so every generated output satisfies the event. These are functions only of the output field, its fixed-anchor geodesic and the deterministic candidate index.

For each fixed ball, alignment allows at most $2 + \lceil 2L/\log \lambda \rceil$ volume labels, and there are only finitely many recipes. The center mesh has bounded overlap at each radius, and Lemma 8.12 uses $O(N)$ radii. Charging the output event's weight to its geodesic length in $B_{4r}(z)$ thus bounds the total output weight by a constant times N times the output anchor length. The source length cutoff and uniform shift amplitude bound this last length deterministically. This proves (d) of Proposition 8.28; the preceding paragraphs prove (a)–(c) and its completion hypothesis. Uniform stability follows from Lemma 8.22. $\square$

Theorem 8.30 (Identification of local passage laws). *Fix $\gamma \in [\sqrt{2}, 2)$ and $\lambda > 1$. Let a family of local passage laws satisfy Definition 8.2 and the reference metric inputs stated above. Suppose a law in this family has a finite positive deterministic optimal upper comparison $0 < C < \infty$, with its comparison bounds retained at its fixed quantum offsets and further extractions. Then its optimal lower comparison equals C , and*

$$F_U(x, y) = CD_h(x, y; U)$$

on every ordinary open chart U , with compact confinement and exhaustion as above. The multiplier C is deterministic for the entire joint law. This theorem gives no estimate for a prescribed discrete endpoint.

Proof. If the lower optimal constant c were smaller than C , the common annulus lemma, comparison transport, and amplification would give the equality tangent of Proposition 8.15. Lemma 8.29 supplies every additional hypothesis of Proposition 8.28, whose weighted output argument is a contradiction. Thus $c = C > 0$. The intrinsic lower bound is Lemma 8.7; the upper bound follows from continuum joining and port comparison, as in Equation (8.1). Countable compact exhaustion gives the simultaneous assertion on open charts. The constants in Equation (8.14) are deterministic properties of the whole law; proving them equal is what excludes an additional sample-dependent multiplier. $\square$

Proposition 8.31 (Two-sided shell tests control relative units). *Work in the physical passage and annulus-transfer framework of Definition 8.2 and Lemma 8.12, with the comparisons of Corollary 8.6 available. Let G_i , $i \geq i_0$, be own-unit rules with positive deterministic normalization coefficients b_i . Assume that, for every base i , its actual relative-offset laws under the intrinsic transport and quantum-unit identities (8.2)–(8.4) are $(b_{i+\ell}/b_i)G_{i+\ell}$ in the base i reference units. In particular these are the conditional kernels on the physical annuli to which the annulus-transfer estimates apply. Suppose every G_i satisfies, with probability above the annulus transfer threshold, fixed shell tests with constants $0 < m < M < \infty$: lower crossing bounds m for the passage rule and reference metric, surrounding upper bounds M for both, and an upper crossing bound M for each with endpoints to which the other's lower bound applies. Then there is $T < \infty$, depending only on m, M and the fixed geometry, such that $T^{-2} \leq b_i/b_j \leq T^2$ for all $i, j \geq i_0$.*

Proof. Take $T > 4(1 + M/m)^2$, increasing it for any fixed geometric enlargement. If at least one quarter of the labels in arbitrarily late windows satisfy $b_{i+\ell} > Tb_i$, their passage lower bounds are mTw_ℓ . The reference upper bounds are Mw_ℓ . Annulus transfer and Corollary 8.6 give the deterministic lower comparison $G_i \geq (mT/M)D$. The base shell test has positive probability of a passage crossing of cost at most M and reference distance at least m , contradicting $m^2T/M > M$.

If one quarter instead satisfy $b_{i+\ell} < b_i/T$, the same argument in the other direction gives upper comparison $M/(mT)$. A base reference crossing of length at most M , whose passage crossing cost is at least m , gives the same contradiction. Thus each of these two exceptional sets has fewer than one quarter of the labels in all sufficiently late windows. Lemma 8.32 concludes the proof. Almost-sure comparisons are obtained by choosing a subsequence of annulus windows with summable failure probabilities, and then exhausting countably many buffered tests. $\square$

Lemma 8.32 (Overlap of one-sided density bounds). *Let $b_i > 0$ and $T > 1$. Suppose that, for each fixed i , for all sufficiently large N , each of the sets*

$$\{k \in [i + N, i + 2N] : b_k > Tb_i\}, \quad \{k \in [i + N, i + 2N] : b_k < b_i/T\}$$

has fewer than $(N + 1)/4$ elements. Then $T^{-2} \leq b_i/b_j \leq T^2$ for every i, j .

Proof. For large N , the windows based at i, j have $N - O(|i - j|)$ common indices. In their overlap, the conditions $b_k \leq Tb_i$ and $b_k \geq b_j/T$ have a common index, since their excluded sets together have fewer than $(N + 1)/2$ elements. Hence $b_j \leq T^2 b_i$. Interchange i, j . $\square$

9 Circuit normalization for critical FK maps

We now return to the fixed FK parameter $q \in (0, 4)$ and use the quantum volume base $\lambda = 4$. Our deterministic distance denominator will be a quantile of the length of a circuit surrounding the cone mark. This circuit test has no prescribed lattice endpoint. Lower crossing tests first produce record scales whose earlier quantiles have the reference power-law upper bound in the limit. Annulus comparison and rigidity identify the resulting passage laws. Their two-sided shell tests then control the relative cost units, allowing a first-failure argument to propagate a polynomial ratio bound to all scales.

9.1 Circuit quantiles and coherent quantum offsets

Use the circle-average embedding of the Brownian cone, with its mark at 0, initially modulo rotations. Fix three nested buffered winding bands with strict margins which are concentric round annuli about 0, with their coarse-test compact annulus and fixed real-field buffers contained in $\{1/8 < |z| < 7/8\}$. Thus these local tests avoid both the cone mark and the circle-average pinning circle $|z| = 1$. The following choice fixes their finite-resolution law once and for all. Let P be the bilateral iid raw-word law, and let B be the normalized two-sided Brownian contour law. For every integer $k \geq 0$, choose a joint law $P_k(dW, dZ^k)$ with marginals P and B such that the resolved contour of W , divided by $\sqrt{4^k}$ and viewed at time unit 4^k , differs from Z^k by a quantity tending to zero in probability in the compact-uniform topology. The invariance principle and a coupling metric for weak convergence give these choices. They are fixed before any quantile or subsequence is selected.

Fix measurable readouts of a Brownian contour for the intrinsic decorated surface, an arbitrary measurable rotational section of its circle-average coordinate representative, its curve, and its field and internal metrics on a countable chart family. Only contour-determined readouts are meant here. Fix also the deterministic mesh projection convention of Section 6, chosen to commute with rotations, an ordering of the countably many finite primal circuits, and the open middle-band winding test. For example, select vertex representatives by their raw time ordering, project them by η , and interpolate a projected edge by a straight segment; every step of this convention commutes

with rotation. Under P_k , project the mesh using the curve readout of Z^k . Define X_k as the least genuine unscaled all-edge length of a primal circuit whose projected range satisfies this fixed test, using ∞ if the collection is empty. The first minimizer in the fixed ordering may be marked. Countable infima make X_k measurable. All later appearances of its law refer to this exact chosen P_k marginal. The band range and winding requirements are invariant under rotation, so X_k and its first raw-circuit minimizer do not depend on the chosen rotational section. We do not assert that the field in an arbitrary such section has the standard embedded cone law.

Fix $\alpha \in (0, 1)$ and, when $\mathbb{P}[X_j < \infty] > \alpha$, put

$$A_j = \inf\{m \in \mathbb{N} : m \geq 1, \mathbb{P}[X_j \leq m] \geq \alpha\}. \quad (9.1)$$

Since the finite costs are integers, this gives

$$\mathbb{P}[X_j \leq A_j] \geq \alpha, \quad \mathbb{P}[X_j < A_j] < \alpha. \quad (9.2)$$

In particular, no continuity of the distribution at its quantile is required. The hypothesis $\mathbb{P}[X_j < \infty] > \alpha$ ensures that this quantile is finite. Lemma 9.4 makes the inequality hold for every sufficiently large j ; assign arbitrary finite positive values to the finitely many remaining A_j when defining the normalization sequence. All quantile-slack arguments below concern the sufficiently large indices.

To compare these quantiles across scales, their defining circuit variables must occur in one actual word experiment. The next lemma preserves each previously fixed marginal while constructing the joint offset array. It also retains the field and reference metric in the same quantum units as the passage observations.

Lemma 9.1 (One word and exact offset quantile marginals). *For a sequence of base indices $j \rightarrow \infty$, the fixed-offset arrays can be sampled with one common iid word so that the coordinate at offset i has exactly the chosen X_{j-i} law. In every finite offset family, the auxiliary Brownian contours converge jointly to the canonical time rescalings of one common contour. Their actual joint contour-determined spatial, field and internal-metric readouts can be retained with this same consistency.*

Proof. Sample one word $W \sim P$. Conditional on that word, draw the entire countable family $(Z^k)_{k \geq 0}$ once from the product of the fixed kernels $P_k(dZ \mid W)$. Every offset array reuses the coordinate Z^{j-i} from this family. Use the previously fixed definition of X_{j-i} on that same word and auxiliary contour. Its marginal is exactly the one defining the quantile, even though different offset coordinates are dependent through W .

For $v_i = 4^{-i}$, write $(T_i Z)(t) = v_i^{-1/2} Z(v_i t)$. Both Z^{j-i} and $T_i Z^j$ approximate the same resolved word in offset units. Thus their compact-uniform difference tends to zero in probability for every fixed i . A finite union handles any fixed offset family. Both arguments of each canonical readout have the same fixed Brownian marginal. For a measurable readout into a Polish space, Lusin's theorem makes it uniformly continuous on a compact set of marginal probability at least $1 - \epsilon$. Consequently readouts of these two close arguments differ by a quantity tending to zero in probability, after a loss at most 2ϵ . Let $\epsilon \downarrow 0$, apply the argument to finite subarrays, and diagonalize over the countable chart family. This retains the joint field/metric readout, rather than only its separate marginal laws. At this stage its arbitrary rotational sections are identified by quantum-surface equivalence.

The standard residual rotation is now sampled once, uniformly, independently of the common contour and the entire observation array. This restores the standard embedded cone law. The

concentric winding tests, and the Euclidean-advance tests in the concentric compact annulus used below, are unchanged pointwise; consequently the exact finite circuit laws and the spatial-advance test laws used in Lemma 9.4 are preserved. An imaginary-field completion not determined by the contour is likewise adjoined once, conditional on the entire common limiting contour and this rotation, independently of the entire actual and reference observation array under the convention of Proposition 7.11. Its copies at other offsets are transported from that same draw. More explicitly, let ϕ_i map the circle-average representative at offset i to the common base representative, choosing its orientation by transport, and put $c_i = \gamma^{-1} \log v_i$. It is a positive dilation after this common choice of orientation. The actual readouts satisfy

$$h_i = h_0 \circ \phi_i + Q \log |\phi_i'| - c_i, \quad \eta_i(t) = \phi_i^{-1} \eta_0(v_i t), \quad D_i = v_i^{-p} \phi_i^* D_0.$$

The same identities hold for every internal metric on its transported domain. Intrinsic affine covariance of the local kernels then gives the indexed kernel $\mathcal{K}_i$ in these units. These are transported annuli, which need not coincide as Euclidean sets in the base chart. In unchanged numerical cost units, if $F_0 = C D_0$, the pulled-back rule satisfies $\phi_i^* F_0 = C v_i^p D_i$. Thus only the deterministic quantum-unit factor enters the offset comparison; the random coordinate dilation supplies no additional normalization. Unrelated rotations or imaginary-field draws at different offsets are not used to infer coherence.

The finite variables Z^k serve only to define the quantile and its projection readouts. The exact raw-chart resampling identities are proved before the limiting spatial readout, and no conditional independence of finite microscopic observations given Z^k is asserted. In the limit the contour-determined readouts are the measurable functions just identified; the additional variables have the stipulated conditional independence. These are precisely the auxiliary-conditioning conditions used by Proposition 7.11. $\square$

Annulus comparison uses local Gaussian field changes. We next transfer the indexed passage laws to an outside-pinned Gaussian reference, keeping their numerical cost units. The joint offsets then extend the comparison from ordinary patches of the normalization disk to the whole cone atlas.

Lemma 9.2 (Actual Gaussian references and the ordinary cone atlas). *The FK local kernels on buffered ordinary patches of the normalized cone disk transfer to an actual Gaussian field with a fixed pinning circle outside every observation buffer. The local field density is the same for all indexed kernels attached to that field marginal. If a comparison holds with the same numerical constant for all the indexed Gaussian laws, it holds on every ordinary cone chart. Conversely, a strict local improvement on the cone would improve the Gaussian comparison. These assertions retain the real additive constant and the complete joint observation law.*

Proof. Inside its normalization disk the cone field is $G_1 - \gamma \log |z|$, where G_1 has zero unit-circle average [13, Definition 4.10 and the following discussion]. Under the deterministic coordinate map $z \mapsto z/R$, this is

$$G_R(z) - \gamma \log |z| + (\gamma - Q) \log R, \quad G_R(z) = G_1(z/R).$$

On a compact ordinary patch in $B_R \setminus \{0\}$ the displayed drift has a smooth finite-energy extension supported away from both 0 and the pinning circle. Cameron–Martin gives mutual absolute continuity with the actual G_R restriction. To put a prescribed buffered shell in such a patch, first map it by a fixed positive affine map into $B_1 \setminus \{0\}$. The pulled-back pinning circle remains outside

the complete buffer; the drift, including $Q \log |\phi'|$, again has a Cameron–Martin extension. The pin need not be centered on the shell.

Any two fixed exterior pins give equivalent restrictions on a smaller common patch. Indeed, on an intervening domain W their domain Markov decompositions have an independent zero-boundary GFF and a harmonic part. On $K \Subset W$ each harmonic part admits a finite-energy cutoff in W . Conditional Cameron–Martin equivalence with the zero-boundary restriction, followed by integration, proves the assertion. No circle average is reset in this comparison. The independent stationary-phase imaginary field has the same local law under these affine maps. Adjoining the same local kernels, with their genuine order and correct local area marks, preserves each field density. If $Z = d\nu/d\mu$, then for any indexed kernel and event E ,

$$(\nu \otimes \mathcal{K}_i)(E) \leq T(\mu \otimes \mathcal{K}_i)(E) + \mathbb{E}_\mu[Z \mathbf{1}_{\{Z > T\}}].$$

The tail is independent of i . Thus uniform probability bounds transfer by truncating this fixed density; a quantitative exponential rate under an arbitrary equivalent field law is not asserted.

For the full atlas use the actual common offsets of Lemma 9.1, including each fixed negative offset. Put $c_v = \gamma^{-1} \log v$ and

$$r_v = \sup\{r > 0 : h_r(0) + Q \log r = c_v\}, \quad h^{(v)} = h(r_v \cdot) + Q \log r_v - c_v.$$

The circle-average cone law and its quantum scaling [13, Proposition 4.13(i)] give the original joint background law at every fixed v . The imaginary field is pulled back from the same sample; its stationary phase and independence of the real field preserve this joint law. The matching raw time, height, cost, and reference arrays were transformed together before extraction. Their intrinsic readouts therefore obey

$$F_U^{(v)} = v^{-p} \phi_v^* F_{\phi_v(U)}, \quad D_{h^{(v)}}(\cdot, \cdot; U) = v^{-p} \phi_v^* D_h(\cdot, \cdot; \phi_v(U)), \quad \phi_v(z) = r_v z,$$

and the local kernel in the new units is $\mathcal{K}^{(v)}$. These are identities for the transported observations, with fresh reference kernels identified by Proposition 7.11. They are not field-shift identities for an individual passage sample.

Set $H(r) = h_r(0) + Q \log r$. The radial cone construction gives $H(r) \rightarrow -\infty$ as $r \downarrow 0$, and H is continuous. Hence $\sup_{0 < r \leq R} H(r) < \infty$ for each finite R . Since $c_{4^m} \rightarrow \infty$, it follows that $r_{4^m} \rightarrow \infty$. For each fixed m , the comparisons under the indexed Gaussian kernel transfer to every ordinary patch inside the normalized cone disk. Pulling them back cancels the two factors 4^{-mp} and covers $B_{r_{4^m}} \setminus \{0\}$. A countable intersection over m and a countable chart exhaustion gives the whole ordinary cone atlas. This uses an exact joint quantum-unit identity, without conditioning on a random dilation and then applying local absolute continuity. For the reverse assertion, transfer a local comparison on ordinary patches of the unit disk to the Gaussian reference by mutual absolute continuity and intrinsic affine transport. Lemma 8.13 then preserves its numerical constant at every fixed quantum index. $\square$

Lemma 9.3 (Noncollapse of a quantile-normalized upper rule). *Suppose a sequence normalized by A_j has a certificate limit, winding realizations pass from the narrow band to the middle band with arbitrary cost loss, and the limiting rule can follow every rectifiable path at zero cost with open ports. If the reference narrow band has a finite rectifiable surrounding circuit almost surely, these assumptions contradict Equation (9.2).*

Proof. Cover a reference surrounding circuit by finitely many transverse port tests in the narrow band. Zero upper comparison gives certificates of arbitrarily small total cost; robust transverse

intersections stitch them into a primal winding circuit. Exhaust its finite number of ports, reference length, and strict margins. For every $\delta > 0$, this gives, with limiting probability at least $1 - \delta$, a middle-band circuit of cost less than $A_j/2$ for all sufficiently large j . Choose $\delta < 1 - \alpha$. Its probability is then strictly greater than α , contradicting $\mathbb{P}[X_j < A_j] < \alpha$. $\square$

The preceding lemma excludes a zero upper comparison after an upper comparison has been constructed. A single quantile does not by itself give tight lower band costs, a finite upper comparison, or any estimate for nearby discrete endpoints.

Lemma 9.4 (Concrete finite-resolution tests). *Assume the framework's contour coupling, marked certificate extraction, and robust primal projection topology on compact sets. The winding variables X_j can then be chosen so that $\mathbb{P}[X_j < \infty] \rightarrow 1$ and $A_j \rightarrow \infty$. There is a fixed finite collection of compact spatial-advance tests that satisfies the crossing-test hypotheses of Proposition 9.5.*

Proof. Keep the three concentric winding bands fixed above, and choose a concentric compact annulus K , bounded away from the cone mark, containing the widest band in its interior. Finite disk covers of a slightly larger annulus supply all required local chart patches. Choose $\delta > 0$ smaller than the Euclidean diameter forced by winding around the inner hole. A deterministic polygonal loop in the narrowest band has an approximate primal lift by the projection topology. Its lift is a finite graph walk, stays in the middle band, and has nonzero winding. Extract a simple surrounding subcircuit. Thus, on the events where the projection error is smaller than the fixed margins, X_j is finite. These events have probability tending to one. Time windows are first enlarged to contain the preimages of the annulus; transience gives this with arbitrarily high probability. No cost bound for the lift is asserted.

Define Y_j to be the infimal number of edges of a primal path whose projected range lies in K and whose projected endpoints are at Euclidean distance at least δ . Use ∞ for an empty collection. Slightly larger compact confinements and slightly smaller separations provide strict versions. The contour coupling and the closed chord relation make the largest projected diameter of a single edge tend to zero, uniformly on the relevant compact time windows. A path of at most k edges has diameter at most k times this quantity. Therefore, for each fixed k , $\mathbb{P}[Y_j \leq k] \rightarrow 0$. Every winding circuit tested by X_j contains an admissible path for Y_j , so $Y_j \leq X_j$ and $A_j \rightarrow \infty$.

At any proposed cost unit b_j , failure of the test $\mathbb{P}[Y_j \geq b_j] \geq 1 - \eta$ retains, with probability at least η , a path of normalized cost at most one and endpoints separated by at least δ . The endpoints and range lie in fixed compact sets. Marked extraction retains a certificate with that separation and finite cost. First choose a sufficiently large time window that its lost probability is less than $\eta/2$, then pass to the extraction and exhaust windows. The fixed spatial margins retain the separated endpoints and compact confinement.

Choose a buffered annulus inside the interior of K with radial crossing separation greater than δ . Its primal traversal tests are included among these spatial-advance tests. Translations and the fixed affine changes needed to place the annulus use a finite collection of slightly enlarged versions of K . The common local kernels transfer their high-probability tests to the fixed GFF reference: the field change of density is fixed, so Equation (8.7) allows η to be chosen small enough for the annulus threshold. Uniform bounded-edge advance gives the required initial runs, and the retained finite-cost witness gives the required first-failure contradiction. These are precisely the test conditions of Proposition 9.5. $\square$

9.2 Deterministic scale arguments

We initialize a positive lower barrier, obtain rigidity at record scales, control relative quantiles, and propagate the comparison by excluding a first failure.

Proposition 9.5 (Lower-barrier initialization). *Assume the following assertions for arbitrary deterministic cost normalizations and the same fixed buffered tests.*

- (a) *Every sequence of finite-cost crossing witnesses has a subsequential certificate retaining a fixed positive spatial advance. The common-kernel and annulus transfer properties hold in these limits.*
- (b) *At unscaled cost, the largest projected advance of every path using at most k edges tends to zero in probability, for each fixed k , uniformly in the compact testing region.*
- (c) *A winding circuit used in X_j contains a subpath tested by a fixed finite collection of these crossing tests. The probability and cost thresholds have strict margins, so failure of their high-probability lower test retains a finite-cost witness with probability bounded away from zero.*

Then, for the finite circuit quantiles in Equation (9.1),

$$\limsup_{j \rightarrow \infty} j^{-1} \log_4 A_j \geq p.$$

Proof. Fix $0 < \varepsilon < p$. Start at j_0 with denominator one and use $b_j = 4^{(p-\varepsilon)(j-j_0)}$ thereafter. Choose the finite lower test's failure probability smaller than the annulus threshold, α , and $1 - \alpha$, distributing it among the finitely many crossing geometries. By (b), every fixed initial run passes as $j_0 \rightarrow \infty$.

Suppose there are arbitrarily late starts for which a first failure $f(j_0)$ occurs. Then $f(j_0) - j_0 \rightarrow \infty$. Extract at the failed scale in units $b_{f(j_0)}$. At each fixed preceding offset i the lower crossing threshold, expressed in reference metric units, is

$$\frac{b_{f(j_0)-i}}{b_{f(j_0)} 4^{-ip}} = 4^{\varepsilon i}.$$

For any $T < \infty$, all sufficiently large labels therefore have barriers of at least Tw_i , with reference surrounding costs at most $A_0 w_i$ on a sufficiently high probability regularity event. Annulus transfer and Corollary 8.6 force every nonconstant retained certificate to have cost at least $(T/A_0)D$ between its endpoints. Take the countable intersection over integer T . This excludes every finite-cost certificate with positive spatial advance, contrary to the failed-test witness from (a) and (c).

Consequently a sufficiently late start passes forever. With the chosen probability margins, (c) gives $\mathbb{P}[X_j < b_j] < \alpha$ at every subsequent level, and hence $A_j \geq b_j$ (integer rounding changes this by at most one). This yields the stated lower growth after letting $\varepsilon \downarrow 0$. $\square$

For the FK maps, Lemma 9.4 supplies these strict crossing tests and their retained finite-cost failure witnesses.

Lemma 9.6 (Record scales). *If $A_j \in (0, \infty)$ and $\limsup_{j \rightarrow \infty} j^{-1} \log_4 A_j \geq p$, then there are $r_k \uparrow \infty$ and $t_k \uparrow p$, $t_k < p$, such that*

$$\frac{A_{r_k-i}}{A_{r_k}} \leq 4^{-t_k i}, \quad 0 \leq i \leq r_k.$$

Proof. For every fixed $t < p$, the sequence $4^{-tj} A_j$ is unbounded and has arbitrarily late record maxima. At a record r , $4^{-t(r-i)} A_{r-i} \leq 4^{-tr} A_r$. Choose successively larger records for a sequence $t_k \uparrow p$. $\square$

Lemma 9.7 (Continuum comparison at the cone mark). *Suppose the FK family has a common finite upper comparison at all fixed quantum indices on the ordinary cone atlas, and arbitrarily small realizable primal separating circuits at every ordinary point, with costs tending to zero and strict winding margins. Then almost surely the cone mark is surrounded by realizable primal circuits whose diameters and normalized costs tend to zero. The port comparison, continuum concatenation, and any intrinsic lower comparison extend across that mark with the same constants.*

Proof. In the fixed narrow reference annulus choose an almost surely finite rectifiable surrounding circuit, with a finite length bound Y . The upper comparison on this ordinary annulus, finite transverse port tests, and small primal separators at its ordinary points realize a winding primal circuit in the middle annulus, with cost at most $CY + 1$. Only this fixed annular construction is used here.

Apply it in each of the exact units $v = 4^{-m}$ retained in Lemma 9.1. In base units the circuit has cost at most $v^p(CY_v + 1)$ and lies in a fixed multiple of $B_{r_v}(0)$, where Y_v has the same finite marginal law as Y . The radial construction in [13, Definitions 4.5 and 4.10, Remark 4.4] gives $H(r) \rightarrow +\infty$ as $r \rightarrow \infty$ and $H(r) \rightarrow -\infty$ as $r \downarrow 0$, since $Q - \gamma > 0$. Consequently $\inf_{r \geq \varepsilon} H(r) > -\infty$ for every $\varepsilon > 0$, and $r_v \rightarrow 0$ as $v \downarrow 0$. Select a deterministic subsequence m_k so that the probabilities of a radius larger than 2^{-k} and of a cost larger than 2^{-k} are each summable in k . Borel–Cantelli gives the asserted circuits, with strict winding margins and simultaneous realizations. No independence of the different offsets is required.

For a compact reference path meeting 0, use the first and last crossings of one of these circuits. The exterior initial and final subpaths are compactly away from 0, so their port upper bounds are already known. Join their realizations along the primal circuit, whose extra cost tends to zero. If an endpoint is 0, choose the circuit inside its prescribed open endpoint port. This proves the same upper port comparison for all paths. The circuits at 0 and the existing circuits at ordinary points now give Lemma 8.4 on the whole chart.

For the lower comparison, retain the ordered prefix and suffix outside $B_\varepsilon(0)$ of a realizing certificate. The omitted middle cost is nonnegative. The known lower bounds for the two exterior pieces give the lower bound for its endpoints up to $c \text{diam}_D(B_\varepsilon(0))$, which tends to zero by Lemma 8.1. The same argument with just one exterior piece treats an endpoint at 0; if both endpoints are 0 the bound is zero. First extract these markings with open margins, then let $\varepsilon \downarrow 0$. This proves the ambient lower bound everywhere, and Lemma 8.7 makes it intrinsic. This is a statement about continuum certificates. $\square$

Proposition 9.8 (Comparison at exact-power records). *Assume the certificate, spatial-kernel, coordinate-transport, and extraction properties used above. Choose the quantile level α sufficiently close to one for Lemma 8.12, including reference crossing lower bounds in its local event. Suppose a joint extraction in units A_j has*

$$\limsup_j A_{j-i}/A_j \leq K4^{-ip}, \quad i \geq 0,$$

where $K < \infty$ is deterministic. Then its port rule has a finite positive deterministic optimal upper comparison C . The same numerical upper bound is valid at every fixed quantum index and on the full cone atlas before changing its own cost normalization. If the family satisfies Definition 8.2, then this rule satisfies $F_U = CD_U$.

Proof. At label i the wide-band upper circuit test has, with probability at least α and arbitrary cost slack, a surrounding circuit of cost at most $K4^{-ip}$. Include a reference lower crossing bound a_04^{-ip} , with failure probability chosen before α . The common coordinate kernels make this the corresponding fixed-index test under the local reference field law; fixed changes of density are handled by Equation (8.7). Lemma 8.12 supplies arbitrarily fine circuits at every point of each compact set. Apply the actual-primal-circuit version of Lemma 8.5, charging a reference path's complete traversals. It gives upper factor K/a_0 , up to the fixed geometric constants. The circuit costs also tend uniformly to zero, so Lemma 8.4 applies.

The same bound holds for every fixed forward index $v_0 = 4^m$. Indeed, its sufficiently fine indexed laws satisfy

$$(\mathcal{K}^{(4^m)})^{(4^{-\ell})} = \mathcal{K}^{(4^{m-\ell})}, \quad \ell \geq m.$$

This tail belongs to the backward family with the same shell-test bounds. Deleting finitely many labels does not change the favorable density in arbitrarily late windows or the resulting comparison ratio. Thus one finite bound works for every fixed m , without a rate uniform in m . The same fine annuli provide the small primal circuits at ordinary points at every fixed index. Lemma 9.2 transfers these assertions to the ordinary cone atlas, and Lemma 9.7 extends it across the mark. The upper optimum cannot be zero by Lemma 9.3. Its optimal constants have the same numerical values in the indexed laws by the reverse transfer in Lemma 9.2. Theorem 8.30 gives equality on ordinary charts, and the same marked extension gives the stated intrinsic conclusion. $\square$

The preceding comparison identifies a limiting passage law. We now return to the winding tests that chose its distance unit: their quantile slack bounds both the multiplier and the quantiles at every fixed offset.

Lemma 9.9 (Quantile bounds in a rigid law). *Suppose a joint extraction, including any fixed finite set of quantum offsets, has $F_U = CD_U$ in base cost units, where $C \in (0, \infty)$. Assume the narrow, middle, and wide winding tests used in Equation (9.1) have the following strict passage rules: upper realizations in the narrow band pass to the middle band, and middle-band circuits have retained certificates in the wide band. Use the fixed reference law after each quantum-unit change. Then there are $0 < m_* < M_* < \infty$, independent of this rigid law and the fixed offset i , such that*

$$m_*C4^{-ip} \leq \liminf \frac{A_{j-i}}{A_j} \leq \limsup \frac{A_{j-i}}{A_j} \leq M_*C4^{-ip}. \quad (9.3)$$

In particular $M_^{-1} \leq C \leq m_*^{-1}$ when the base is normalized by its own quantile.*

Proof. In the wide reference band, every winding circuit has two points at a fixed positive Euclidean separation. By D -continuity on the compact band, the infimum Y_- of the distances of all such pairs is strictly positive almost surely. In the narrow reference band there is an almost surely finite reference rectifiable surrounding circuit; let $Y_+ < \infty$ be any measurably chosen upper length bound. Choose $m_* > 0$ sufficiently small and $M_* < \infty$ sufficiently large that

$$\mathbb{P}[Y_- > 2m_*] > 1 - \alpha/2, \quad \mathbb{P}[Y_+ < M_*/2] > (1 + \alpha)/2.$$

The reference laws of these variables are the same at every fixed offset in their own quantum units.

If A_{j-i}/A_j had a subsequential limit below m_*C4^{-ip} , the probability- α quantile upper event would retain a wide-band certificate below this cost with probability at least α . The lower comparison and the Y_- bound permit this with probability less than $\alpha/2$, a contradiction. This argument also rules out a zero limiting ratio.

Conversely, on the event $Y_+ < M_*/2$, upper comparison and transverse port stitching realize a middle-band circuit below $M_*C4^{-ip}A_j$, with strict cost and probability margins. The quantile definition forces A_{j-i} below this bound for all sufficiently large j along the extraction. It also rules out an infinite limiting ratio. All costs and probabilities were separated from their thresholds, so no continuity at a quantile atom was used. Set $i = 0$ to obtain the bounds on C . $\square$

Lemma 9.10 (Polynomial bound after propagation). *Suppose there are $J, m \in \mathbb{N}$, $R \geq 1$, and $s > 0$ such that for $j \geq J + m$,*

$$A_{j-1}/A_j \leq R, \quad A_{j-m}/A_j \leq 4^{-sm}.$$

Then, after enlarging a finite deterministic constant K ,

$$\frac{B(n')}{B(n)} \leq K(n'/n)^s \quad (1 \leq n' \leq n), \quad B(n) = A_{\lfloor \log_4 n \rfloor}. \quad (9.4)$$

In particular $B(n) \rightarrow \infty$.

Proof. For indices beyond the finite exceptional range write $j - q = am + b$ with $0 \leq b < m$. Iteration gives $A_q/A_j \leq 4^{-sam}R^b \leq (R4^s)^{m-1}4^{-s(j-q)}$. The finitely many earlier indices can be joined through $J + m$ and absorbed in the constant. If $q = \lfloor \log_4 n' \rfloor$ and $j = \lfloor \log_4 n \rfloor$, then $4^{q-j} \leq 4n'/n$. This proves Equation (9.4). Taking $n' = 1$ gives $B(n) \geq K^{-1}B(1)n^s$. $\square$

Proposition 9.11 (First-failure propagation). *Assume the preceding framework and common-kernel hypotheses, including closure under varying-law extraction with all fixed-offset reference experiments retained as in Proposition 7.11. Assume that the robust lower tests of Proposition 9.5 and the finite catalogue of Proposition 8.28 have been verified for the FK certificate family. Assume also that the strict winding transfer rules of Lemma 9.9 hold at all fixed offsets. Then there are deterministic J, m, R, K and $s = p/2 > 0$ such that all sufficiently large scales satisfy adjacent ratio bounds in $[R^{-1}, R]$, $A_{j-m}/A_j \leq 4^{-sm}$, and the high-probability lower shell tests. Consequently Equation (9.4) holds.*

Proof. A neighborhood of the rigid laws. We describe the required law neighborhood, since a first-failure argument needs more than a pointwise quantile bound. Use the product topology of the full marked certificate records, the contours and the reference internal metrics on countably many buffered domains. Include the auxiliary reference experiments that ensure closure of the common-kernel identities by Proposition 7.11. The cost coordinates are compactified; the reference marginal is fixed and tight. Thus these joint laws are precompact under the assumed framework.

Let $\mathcal{R}$ be the closure of the quantile-normalized rigid laws arising in these extractions. Lemma 9.9 puts their deterministic factors in a fixed compact subset of $(0, \infty)$. The closure still consists of rigid laws. Indeed, along a convergent subsequence their deterministic factors converge. Marked extraction retains the intrinsic lower comparison. Finite port realizations and the uniform reference moduli retain the upper comparison and its small-circuit splicing. These arguments are the same two directions used in Proposition 8.15. Hence $\mathcal{R}$ is compact and its laws all have the same uniform comparison bounds.

Choose coarse shell constants $0 < m_0 < M_0 < \infty$ so loosely that every law of $\mathcal{R}$ passes the tests in Proposition 8.31 with strict cost, probability, and confinement margins. Such constants are obtained from the fixed reference law and the compact interval of factors. Upper tests ask for finitely many strict port or winding realizations in an open margin; lower tests exclude a closed set of bounded-cost certificates with separated endpoints in compact confinement. Retaining a slightly

wider band and using a slightly weaker cost threshold makes these tests stable under nearby laws. More precisely, after the geometric slack has been fixed their good events are open in the retained data topology. For an upper test this is the existence of a certificate satisfying strict cost, port, and open-confinement conditions. An upper winding test uses a finite marked recipe of robust transverse port crossings; it does not ask for equality of two continuously perturbed loop endpoints. For a lower test the good event is disjointness from a closed bounded-cost certificate set with compact confinement and separated endpoints. For each $\mu \in \mathcal{R}$, choose a bounded continuous function f below the indicator of each such good event, with $\int f d\mu > 1 - \epsilon/2$. The neighborhood $\{\nu : \int f d\nu > 1 - 3\epsilon/4\}$ has closure contained in the laws assigning that good event probability at least $1 - 3\epsilon/4 > 1 - \epsilon$. Finite time and chart cutoffs lose arbitrarily little probability, since their continuum marginal is fixed. Intersect the finitely many neighborhoods for the tests. Compactness then gives a finite union $\mathcal{O}$ covering $\mathcal{R}$, whose closure passes all the weaker coarse tests.

By Lemma 9.9, every rigid joint limit has all fixed ratios in

$$\frac{m_*}{M_*} 4^{-ip} \leq A_{j-i}/A_j \leq \frac{M_*}{m_*} 4^{-ip}.$$

Choose R strictly larger than both adjacent ratio bounds and their reciprocals. Then choose m large enough that $(M_*/m_*)4^{-mp} < \frac{1}{2}4^{-ms}$, with $s = p/2$. Call a finite scale successful if its own-unit law belongs to $\mathcal{O}$ and its adjacent and m -step ratio inequalities hold with these thresholds.

Record scales start long successful runs. Initialization and Lemma 9.6 supply record indices $r_k \rightarrow \infty$ with complete preceding tails $A_{r_k-i}/A_{r_k} \leq 4^{-t_k i}$, where $t_k \uparrow p$ and $t_k \geq s$. Proposition 9.8 and rigidity apply to every joint record extraction. Apply the same argument on a fixed finite forward enlargement, initially retaining the record's cost units. At forward offset a , its fine past offsets satisfy $A_{r_k+a-\ell}/A_{r_k} \leq 4^{-t_k(\ell-a)}$ for fixed $\ell \geq a$, whose limit is $4^{pa}4^{-p\ell}$. These bounds first give the enlarged comparison in the old record units. A zero upper factor would contradict the retained old X_{r_k} quantile by Lemma 9.3, before any bound on the new quantile is needed. Lemma 9.9 makes its own quantile ratios positive and finite, so changing to those units gives laws in $\mathcal{R}$. All numerical inequalities pass with strict margins. Therefore every fixed finite run after a sufficiently late record is successful. Diagonalization gives successful initial runs of lengths tending to infinity.

The successful tail and the first-failure extraction. Before a first failure, the full record tail remains useful. If all levels from r to j are successful and r has its record tail, iteration of the m -step inequalities and the remaining at most $m - 1$ adjacent inequalities gives

$$\frac{A_q}{A_j} \leq K_0 4^{-s(j-q)}, \quad 0 \leq q \leq j, \quad K_0 = (R4^s)^{m-1}.$$

For $q < r$, combine the same bound at r with the record tail.

Suppose unsuccessful levels occur arbitrarily late. Let f_k be the first failure after r_k , and set $j_k = f_k - 1$. The initial success runs imply $j_k - r_k \rightarrow \infty$. Extract in units A_{j_k} . Every fixed preceding own-unit law lies in $\overline{\mathcal{O}}$, and the adjacent bounds ensure that each fixed relative quantile has a positive finite limit. Write $b_i = 4^{ip} \lim_k A_{j_k-i}/A_{j_k}$, and let G_i denote the limiting passage rule at offset i in its own cost units. For fixed $i, \ell \geq 0$, viewed from base i , the cost unit at offset $i + \ell$ contributes the ratio $A_{j_k-i-\ell}/A_{j_k-i}$. Dividing by the reference length unit $4^{-\ell p}$ gives

$$4^{\ell p} \lim_k \frac{A_{j_k-i-\ell}}{A_{j_k-i}} = \frac{b_{i+\ell}}{b_i}.$$

The limiting denominators b_i are positive by the preceding adjacent bounds. The common-word coupling of Lemma 9.1 therefore makes $(b_{i+\ell}/b_i)G_{i+\ell}$ the actual physical offset family of each

base i . The assumed common-kernel framework and its annulus transfer, Lemma 8.12, supply the physical compatibility hypothesis of Proposition 8.31. Proposition 8.31 makes all b_i uniformly comparable. Consequently the past ratios satisfy an exact-power upper bound. Proposition 9.8 and rigidity make the base law rigid.

The failed scale in preceding units. Examine the failed level in the preceding units before normalizing it. It is only a fixed time enlargement. Its cost records are still available, and the fine past offsets just used give a finite upper comparison there. This upper optimum is positive: a zero upper comparison would restrict to the retained old winding annulus and contradict Lemma 9.3 for the old X_{j_k} quantile. Rigidity is therefore available before any division by the failed scale's own normalization. Lemma 9.9 then makes A_{f_k}/A_{j_k} positive and finite in the limit. The failed own-unit law therefore belongs to $\mathcal{R}$, and all its ratio tests lie strictly within the successful bounds. Since $\mathcal{O}$ is open, it is successful for large k , a contradiction. All large scales are successful. Finally Lemma 9.10 gives the stated polynomial bound. $\square$

Proposition 9.12 (Verification for the FK certificate family). *The bilateral FK certificate family constructed in the preceding framework and localization sections satisfies the hypotheses of Proposition 9.11.*

Proof. Initial tests and local kernels. Lemma 9.1 fixes the exact quantile marginals and their joint offset readouts before any normalization extraction. It also verifies the contour-determined and independent-augmentation conventions required by Proposition 7.11. Lemma 5.1 supplies full marked records, realizability, and additive cost subdivisions. Proposition 6.4 supplies primal winding and intersection, the small projected advance of a bounded number of edges, and the approximate mesh lifts. Lemma 9.4 therefore gives finite circuit quantiles, their divergence, and the concrete separated-endpoint lower tests. Its nested compact confinements and the three winding bands give the strict transfer rules at every fixed time offset.

Theorem 7.13 supplies restriction-compatible spatial kernels with arbitrarily small real-field buffers, a possibly larger imaginary buffer, and finite genuine traversal order. The fixed-chart reference experiments and Lemma 7.12 are retained jointly for all countable chart refinements and fixed quantum offsets. Proposition 7.11 proves closure for the resulting admissible laws, including the further varying-law and changing-unit extractions used in the equality tangent and first-failure argument. Its diagonal discrete approximation retains every prescribed finite array and sends every fixed rescaled time unit to infinity in lattice units. The exact density calculation and local limit theorem apply anew to that diagonal array and identify its kernels with its own reference kernels. Its retained actual joint field/internal-metric marginal is unchanged. This verifies the closure hypothesis without inferring conditional independence from weak convergence alone. Intrinsic coordinate transport is part of Theorem 7.13; Equations (8.2) and (8.4) specify all changes of cost and volume units. Corollary 7.14 supplies their density transfer and the correct resampling of local area marks. The independent real and imaginary GFF inputs and their local equivalences come from Theorem 3.2.

In particular, the indexed family in Equation (8.2) uses actual discrete offset experiments at times Nv_i and cost denominators $B_N v_i^p$, retained jointly before applying the fixed local field-density changes. This checks admissibility of the indexed laws required by Proposition 7.11; it is not deduced solely from the algebraic kernel formula.

Annulus comparison. Consequently Lemma 8.12 applies. It proves the annulus transfer needed for initialization, exact-power comparison, and amplification. Small circuits give port splicing before any endpoint continuity of F_U is used. The exact tail reindexing in Proposition 9.8 gives one numerical upper bound at every fixed quantum index. The actual atlas in Lemma 9.2, followed by

Lemma 8.13, then identifies the optimal constants of the whole joint law with those of its indexed members. The metric and confluence inputs for equality tangents are exactly Theorem 3.5.

The common interface and forcing. We have verified Definition 8.2(i)–(iv) using marked compact certificate records, the spatial product kernels, actual indexed experiments, and the diagonal closure theorem. For the FK records the compactified cost and marked path coordinates provide the required tightness before comparison; after comparison, reference metric moduli and small-circuit joining give the uniform continuum moduli used in its further extractions. The edge-cost condition follows from the divergence of the quantile units. The finite genuine order satisfies (vi) by Lemma 8.24. Once the annulus argument has provided the finite positive upper comparison, small circuits and Lemma 8.4 supply (v), before any theorem about a prescribed discrete endpoint is used.

Lemma 9.2 supplies the actual outside-pinned Gaussian law and the countable cone atlas. In particular, all recipes in the forcing argument are Cameron–Martin directions for one fixed Gaussian law, and the same local kernels are used with correctly resampled area marks. The reference inputs are Theorem 3.5. Therefore Lemma 8.29 supplies the complete catalogue, its deterministic gate budgets, unselected completion followed by likelihood clipping, source weight and observable output count. Its disk-profile bound is Lemma 8.11; it retains the real preservation patch inside the denominator collar and allows a larger unchanged imaginary buffer. Theorem 8.30 applies to every equality-tangent family, and Lemma 9.7 gives the conclusion across the cone mark. *Rigidity and propagation.* Rigidity is therefore available for every family used at records or at a proposed first failure. The remaining inputs of Proposition 9.11 were verified above, so its first-failure argument applies to the actual FK normalizations. $\square$

Corollary 9.13 (Bilateral comparison). *For the fixed FK parameter, every sequence of bilateral time units tending to infinity has a further joint extraction, normalized by $B(n)$, with*

$$F_U = CD_h(\cdot, \cdot; U)$$

simultaneously in all strict intrinsic charts. The multiplier C is deterministic for that extraction and belongs to a fixed compact subset of $(0, \infty)$. Upper comparisons follow paths inside any open clearance; lower comparisons apply to all compactly confined certificates and use the intrinsic metric of their confinement. Also $B(n) \rightarrow \infty$ and Equation (9.4) holds. This conclusion concerns crossing data and infimal continuum costs, and supplies no uniform estimate for arbitrary discrete endpoints.

Proof. Proposition 9.12 permits application of Proposition 9.11; every sufficiently late fixed-offset law passes its tests. The same common-word coupling and intrinsic indexed-kernel identities identify these offsets with the physical annular family, as required by Proposition 8.31. Apply Proposition 8.31, then Proposition 9.8 and rigidity, to any subsequential family. The quantile bounds give the stated interval for C .

For arbitrary time units $n_r \rightarrow \infty$, put

$$k_r = \lfloor \log_4 n_r \rfloor, \quad m_r = 4^{k_r}, \quad u_r = n_r / m_r.$$

Pass to a subsequence with $u_r \rightarrow u \in [1, 4]$. Before extracting the m_r -time arrays, add the countable family of fixed $u4^m$ transforms, $m \in \mathbb{Z}$, of their charts and reference experiments. Retain also the actual and reference experiments in n_r time units, sampling their reference duration and shift densities in those target units as in Proposition 7.11. Retain the full joint continuum input and its fixed- u transform, including the actual fields, curves, coordinate dilation and internal

reference metrics. Lemma 9.1, the exact joint transport in Lemma 9.2, and Proposition 7.11 permit this enlargement with one common residual rotation and one transported imaginary-field completion. The joint background law is fixed along the extraction; no continuity of its measurable reembedding as a function of u_r is required. If Z_r is the retained normalized base contour, the contour in n_r units is approximated by $u_r^{-1/2}Z_r(u_r \cdot)$. The contour modulus identifies its limit with the fixed- u Brownian rescaling.

Keep graph costs in the units $A_{k_r} = B(n_r)$. In the fixed- u coordinates, let $\phi_{u,r}$ map back to the base coordinates and let η_r be the retained base curve. The projection at n_r -time label t is

$$\phi_{u,r}^{-1}\eta_r(u_r t) = \eta_r^{(u)}((u_r/u)t), \quad \eta_r^{(u)}(t) = \phi_{u,r}^{-1}\eta_r(ut).$$

The fixed- u curve law has a tight local uniform modulus, so this projection differs uniformly on compact time windows from $\eta_r^{(u)}(t)$ by a quantity tending to zero in probability. Retain the raw marked path arrays on fixed enlarged windows. Their graph paths and costs are unchanged; only their time labels and spatial projections have been relabeled. Strict open confinement and port margins preserve upper realizations, and the reverse marked extraction retains every bounded-cost confined path. Thus the limiting passage collection is the fixed- u transport of the power-of-four extraction. This step uses geometric margins for certificates, not a bound for joining prescribed discrete endpoints.

If the latter extraction has multiplier C , its transported passage rule, with the graph costs still in the base units, satisfies

$$\tilde{F}_U(x, y) = CD_h(\phi_u(x), \phi_u(y); \phi_u(U)) = Cu^p D_{h^{(u)}}(x, y; U).$$

The multiplier Cu^p is deterministic and remains in a fixed compact subset of $(0, \infty)$ because $u \in [1, 4]$. Intrinsic scope follows from Lemma 8.7 and Equation (8.1). $\square$

10 Arm estimates and multiscale address entropy

We now bound the number of contour neighborhoods needed to describe adaptively chosen finite chord diagrams. The fresh-block estimates of Section 4 give the probability cost of a prescribed diagram. A nested interval code pays for the choices of its locations. For the fixed parameter,

$$\beta := \frac{1}{2} + \frac{2}{\gamma^2} > 1.$$

One unit of logarithmic address resolution costs one unit of entropy, whereas one unit of monochrome rank costs at least β units of probability. This strict surplus will bound total address occupancy and force an acyclic component at a suitable scale.

We first replace arbitrary scalar contact graphs by chronological comparisons with a total oscillation bound. We then construct and count the address code, apply the fresh-slot estimates, and allocate the resulting entropy budget to a uniformly sampled scale. All of these steps concern the raw inventory word; they use no endpoint metric bound.

10.1 Sorting scalar contacts with a total oscillation bound

We first give a deterministic statement. In particular, its constant does not depend on the degrees of the contact graph. A *minimum chord* of a continuous function f is a pair (u, v) such that

$$f(u) = f(v) = \min_{[u \wedge v, u \vee v]} f.$$

All graphs in the next Lemma are finite. Parallel edges and repeated chord endpoints are allowed.

Lemma 10.1 (Sorted scalar contacts). *Let f be continuous on a compact interval containing pairwise disjoint nonempty compact intervals $I_1, \dots, I_m$, ordered from left to right. Choose $t_v \in I_v$ and write*

$$h_v = f(t_v), \quad \omega_v = \max_{I_v} f - \min_{I_v} f.$$

Let G be a graph on these intervals, each of whose edges is witnessed by a minimum chord with one endpoint in each incident interval. Let C be a connected component of G , and list its vertices in increasing order as $v_1, \dots, v_k$. For $1 \leq i < k$ set

$$\mu_i = \min_{[t_{v_i}, t_{v_{i+1}}]} f, \quad E_i = h_{v_i} + h_{v_{i+1}} - 2\mu_i. \quad (10.1)$$

Then $E_i \geq 0$,

$$|h_{v_{i+1}} - h_{v_i}| \leq E_i, \quad \inf_{[t_{v_i}, t_{v_{i+1}}]} f \geq \max\{h_{v_i}, h_{v_{i+1}}\} - E_i, \quad (10.2)$$

and

$$\sum_{i=1}^{k-1} E_i \leq 6 \sum_{v \in C} \omega_v. \quad (10.3)$$

Thus the wall bound also holds on the gap between the two intervals, or on any part of that gap obtained by deleting other intervals. The same bound can be summed over all components of G .

Proof. Discard any loops of G , which do not change its connected components. We construct a finite metric tree directly from the minima of f . This also explains the contour order used in the proof.

Let $[A, B]$ contain all the intervals, and let $m_* = \min_{[A, B]} f$. For each $I_v = [a_v, b_v]$, choose a point r_v at which f attains its minimum on I_v . Let $\mathcal{T}$ be the finite set consisting of A, B , all a_v, b_v, r_v, t_v , all chord endpoints, and a point attaining m_* . Coincident marked times occur only once in this set. Write

$$M(s, t) = \min_{[s \wedge t, s \vee t]} f \quad (s, t \in \mathcal{T}).$$

For each $t \in \mathcal{T}$ take a vertical segment $\{t\} \times [m_*, f(t)]$. Identify (s, z) with (t, z) exactly when $z \leq M(s, t)$. This is an equivalence relation: the only nontrivial check follows from

$$M(s, u) \geq \min\{M(s, t), M(t, u)\},$$

because the interval between s and u is contained in the union of the intervals between s, t and between t, u .

The quotient, with distance measured along the vertical segments, is a finite rooted metric tree T . To verify this explicitly, adjoin the marked times in increasing order. The segment for a new time t_j shares with the previously constructed tree precisely the initial segment up to height $M(t_{j-1}, t_j)$: indeed

$$\max_{i < j} M(t_i, t_j) = M(t_{j-1}, t_j).$$

Its remaining segment is therefore attached at a single point. This induction constructs a tree, allowing a new segment to have length zero. Declare all marked tips and all attachment points to be vertices of T . All its segments start at the common root of height m_* . Denote the top of the segment for t by $\pi(t)$. The common part of the two root paths for $\pi(s), \pi(t)$ ends at height $M(s, t)$, so

$$d_T(\pi(s), \pi(t)) = f(s) + f(t) - 2M(s, t). \quad (10.4)$$

In particular, the endpoints of every minimum chord have the same image in T .

For $I_v = [a_v, b_v]$, put $m_v = f(r_v)$ and define the subtree

$$L_v = [\pi(a_v), \pi(b_v)]_T \cup [\pi(r_v), \pi(t_v)]_T.$$

Here and below $[x, y]_T$ denotes the unique tree arc between x and y . The point $\pi(r_v)$ lies on the first arc: it is the common ancestor of $\pi(a_v)$ and $\pi(b_v)$ at height m_v . It is also an ancestor of $\pi(t_v)$. Thus L_v is connected. If ℓ denotes the length of a finite metric tree, then

$$\begin{aligned} \ell(L_v) &\leq f(a_v) + f(b_v) - 2m_v + f(t_v) - m_v \\ &\leq 3\omega_v. \end{aligned} \tag{10.5}$$

Overlapping arcs only reduce the first quantity.

Every chord contact in I_v with a time outside I_v belongs to $[\pi(a_v), \pi(b_v)]_T$. To see this, let $u \in I_v$ be the contact and let $w < a_v$ be the other endpoint. The chord condition implies

$$M(a_v, u) = f(u), \quad m_v \leq f(u) \leq f(a_v).$$

Hence $\pi(u)$ is the point of height $f(u)$ on the root path to $\pi(a_v)$, between $\pi(r_v)$ and $\pi(a_v)$. If instead $w > b_v$, the same argument uses b_v . This proves the claim, including contacts at interval endpoints.

For each edge of G joining v to w , its common chord image therefore belongs to $L_v \cap L_w$. Graph connectivity shows that

$$U_C = \bigcup_{v \in C} L_v$$

is connected. A connected subset of a tree contains the unique arc between any two of its points: removing an omitted point of that arc would separate the two points. Consequently U_C is a subtree and contains the finite hull H of $\pi(t_{v_1}), \dots, \pi(t_{v_k})$. Moreover,

$$\ell(H) \leq \ell(U_C) \leq 3 \sum_{v \in C} \omega_v. \tag{10.6}$$

We next justify the contour tour bound for the chronologically ordered representatives. Subdivide H at every vertex of T lying in H , including all representative images; there are finitely many resulting edges. For a point x in the interior of such an edge, let λ be its absolute height. Among the components of $T \setminus \{x\}$, call the one not containing the root the distal component. The representative indices whose images belong to this component form a consecutive block. Indeed, if the images at times $s < u$ belong to it, then their common ancestor has absolute height strictly greater than λ . Thus $M(s, u) > \lambda$. For any marked time $t \in [s, u]$,

$$M(s, t) \geq M(s, u) > \lambda,$$

which puts $\pi(t)$ in the same distal component. The strict inequality is valid because x is in an edge interior and all marked images are vertices of the finite tree after subdivision.

The sequence of consecutive representative arcs can therefore cross any edge of H at most twice: it can enter and leave the consecutive block of indices on its distal side only once each. Summing edge lengths gives

$$\sum_{i=1}^{k-1} d_T(\pi(t_{v_i}), \pi(t_{v_{i+1}})) \leq 2\ell(H). \tag{10.7}$$

This is the outer-face tour inequality, with its ordering verified directly from interval minima. It also constructs the relevant planar order: root H at its point nearest the root of T , order each child subtree by its consecutive block of marked indices, and put repeated marks of an internal vertex in the intervening corners. The boundary tour of this ordered tree encounters the marks chronologically and uses each positive-length edge twice. The preceding edge-count argument does not require choosing any of those corners.

Equations (10.4), (10.6), and (10.7) give Equation (10.3). Finally,

$$E_i = (h_{v_i} - \mu_i) + (h_{v_{i+1}} - \mu_i)$$

is a sum of two nonnegative numbers. It dominates their difference in absolute value and each summand separately, proving Equation (10.2). A singleton component has an empty sum and needs no separate argument. $\square$

Corollary 10.2 (Discarding large comparison errors). *Under the hypotheses of Lemma 10.1, for every $\eta > 0$ the number of consecutive comparisons with $E_i > \eta$, summed over all components, is at most*

$$\frac{6}{\eta} \sum_{v=1}^m \omega_v.$$

All remaining comparisons have both representative height error and wall error at most η .

Proof. Sum $\eta \mathbf{1}_{\{E_i > \eta\}} \leq E_i$ and apply Equation (10.3). $\square$

Remark 10.3. The errors E_i may depend on the whole contour. The Lemma is a deterministic implication, not a claim that these errors are measurable with respect to the past of a sampling slot. In a probabilistic use one first keeps only comparisons whose error is at most a prescribed deterministic tolerance. The consequent wall may then be stated using the earlier representative height and that tolerance. Any conditioning or enumeration needed to select the retained comparisons must still be justified in the probability argument.

10.2 A noncrossing forest witnessing the contacts

The preceding proof works for the original graph components, without assuming that their witnesses form a noncrossing family. For counting purposes, the following separate construction gives a simultaneous noncrossing witness, with possible mergers of components.

Lemma 10.4 (Uncrossing and contraction). *In the setting of Lemma 10.1, one may add minimum chords whose endpoints are among the original chord endpoints and then retain an outerplanar forest on the interval vertices with the following properties.*

- (i) *The forest has one tree in each component of the enlarged chord graph. Every original graph component is contained in one of these components.*
- (ii) *Every forest edge is witnessed by a genuine minimum chord.*
- (iii) *The vertices occur on the outer boundary in their time order. There are at most 32^m possible simple forests of this form on m prescribed ordered vertices.*

If the construction is applied to the endpoints belonging to a single original connected component, it produces a noncrossing spanning tree on exactly that component's vertices. Forests obtained separately in this way need not be noncrossing with one another.

Proof. On the finite set of original chord endpoints, put $s \sim t$ when $f(s) = f(t) = M(s, t)$. This is an equivalence relation, either directly from the interval-minimum rule or from Equation (10.4). For each class, add the minimum chords between its consecutive times. They connect all times of that class, so they connect the endpoints of every original chord.

These new chords can be drawn without crossings in the upper half-plane. In fact, suppose two minimum chords have strictly interleaving endpoints $a < b < c < d$, with pairs (a, c) and (b, d) at heights h and h' . Since $b \in [a, c]$ and $c \in [b, d]$, the minimum conditions give respectively $h' \geq h$ and $h \geq h'$. Both heights agree, and all four times belong to the same equivalence class. Consecutive pairs of one class do not interleave. Chords from different classes therefore do not interleave either. Open semicircular arcs with nested or disjoint time intervals as diameters do not intersect. Shared endpoints belong to the same class and cause no crossing.

Contract each interval I_v in the boundary line to a single point. This preserves an embedding with all vertices on the outer boundary. One explicit way to see this is to choose a continuous proper nondecreasing function φ on the line that is constant precisely on the nondegenerate I_v and strictly increasing elsewhere. For $y > 0$, the map

$$(x, y) \mapsto \left(\frac{\varphi(x) + yx}{1 + y}, y \right)$$

is a homeomorphism of the open upper half-plane, extends continuously to the boundary, and there makes exactly the desired contractions. Thus distinct chord interiors remain disjoint. Discard loops and keep one edge from each collection of parallel edges. In every resulting component choose a spanning tree, retaining its plane embedding. The earlier connectivity observation proves the first two assertions. Using only endpoints from a single original connected component gives the final assertion as well.

For completeness, the number of simple noncrossing forests on a fixed ordered vertex set is exponential in its size. Place the vertices on a convex m -gon in that cyclic order. Any noncrossing forest extends to a triangulation after missing polygon sides are added and the remaining faces are triangulated. For $m \geq 3$ there are fewer than 4^m triangulations, by the Catalan recursion, and each has fewer than $3m$ edges including the polygon boundary. Its edge subsets number at most 2^{3m} . Their product is at most 32^m . The cases $m = 1, 2$ satisfy the same bound directly. $\square$

Remark 10.5. A simultaneous noncrossing forest can require merging original components. For example, two specified chords with interleaving endpoints have a common contour height and allow additional true chords between their endpoints, even if the specified graph initially placed the two pairs in different components. Such mergers are explicit in Lemma 10.4; they must not be silently treated as preservation of the original partition. Adding these true constraints cannot decrease the rank of the monochrome relation. Lemma 10.1 itself needs no mergers.

10.3 Sparse projected addresses and disjoint sampling slots

We now give a finite formulation of the address count. Throughout this subsection, $\mathbb{P}$ denotes the raw bilateral inventory-word law. Additional randomness used to select finite witness lists may be included, provided that it does not change the word's marginal law. A subsequent selection law $\mathbb{Q}_N$ will be required to satisfy

$$\frac{d\mathbb{Q}_N}{d\mathbb{P}} \leq \exp(k_N), \quad k_N = o(N). \quad (10.8)$$

The independent grid displacement introduced below has the same conditional law under both measures.

Definition 10.6 (Projected address forest). Fix an integer $R \geq 8$ and write $\lambda = \log R$. Let I be a fixed interval of length H , where H is a fixed constant multiple of the base time unit n . Let

$$J = \left\lfloor \frac{bN}{\lambda} \right\rfloor, \quad \delta_j = HR^{-j}, \quad 0 \leq j \leq J.$$

All retained time units are assumed to diverge. Select a finite list of markers (t, ℓ) with $t \in I$ and $0 \leq \ell \leq J$. The list, including the exact positions, terminal levels, and all incidences, may be a function of the entire word. *After this selection*, sample U uniformly on $[0, H)$, independently of everything already selected. Use the nested half-open grids

$$\mathcal{G}_j(U) = \{[U + k\delta_j, U + (k+1)\delta_j) : k \in \mathbb{Z}\}.$$

A marker (t, ℓ) occupies the containing cell at each level $j \leq \ell$. Let V_j be the set of occupied cells, and put

$$m_j = |V_j|, \quad T = \sum_{j=0}^J m_j, \quad A = \lambda T.$$

The occupied cells, with the containing-cell parent relation, form the *projected address forest*. Write M for the total number of marker entries, with multiplicity; entries terminating at an internal forest vertex are retained as terminal labels there.

There are h_0 distinguished bases, each marked through level J . At a probe level j_p , a witness list consists of finitely many chains from these bases. Every chain endpoint and intermediate position is marked through j_p . Every link is either a chord of one of the two colors, or a time jump satisfying

$$|s - t| \leq K\delta_{j_p}.$$

Here K is a fixed finite list bound, enlarged to bound the number of marker and edge entries per probe as well. Thus each chain projects in its entirety to every coarser level. Exact endpoints within their terminal cells will not be part of the address code.

Only a bounded number of root cells can meet I ; with the convention in Definition 10.6, there are at most two. Their integer names are determined by I and U , so no unspecified macroscopic time coordinate must be encoded. All the sampling slots below are contained in a fixed deterministic enlargement of I . The bilateral word supplies their chronological increments, including increments outside I if a slot meets an endpoint of that interval.

Lemma 10.7 (Projected graph ranks). *At level j , retain all occupied boxes, and place the projected colored chords between them, including isolated vertices. Let c_j be the number of components of the union graph, let b_j be the number of occupied base boxes, and let D_j count active jump entries whose two endpoints lie in different level- j boxes. If e_j is the sum of the two monochrome graph ranks, then*

$$c_j \leq b_j + D_j, \quad e_j \geq m_j - c_j \geq m_j - b_j - D_j. \quad (10.9)$$

These counts are made before any sampling slots are discarded.

Proof. Include the projected jump edges temporarily. Every occupied vertex then has a chain to a base: a marker arises in a fully retained chain, or is itself a base. The enlarged graph therefore has at most b_j components. Removing the D_j jump edges increases the component count by at most D_j . For the rank inequality, take a spanning forest of the remaining union graph. Its edges of either color form a forest in that monochrome graph, so their number is bounded by that color's rank. Summing the two bounds gives $m_j - c_j \leq e_j$. $\square$

Lemma 10.8 (Regular slots). *For every $\varepsilon > 0$, one can choose a fixed sufficiently large R such that, conditional on any selected marker system, with probability at least $7/8$ over U , all but*

$$C_R M + \varepsilon T \quad (10.10)$$

of its occupied vertex occurrences admit the following disjoint paired slots. At a retained vertex $v \in V_j$, let t_v be the midpoint of its cell and put $r_j = R^2 \delta_j$. The slots are

$$S_v^- = [t_v - r_j, t_v - 2r_j/R], \quad S_v^+ = [t_v + 2r_j/R, t_v + r_j]. \quad (10.11)$$

All these intervals, for every retained vertex, level, and sign, are pairwise disjoint. Their locations are determined by the forest and U . The constant C_R is independent of the positions, the number of probe levels, and the finite exploration cutoff except through M .

Proof. Call a vertex critical if it is a root, has a number of children different from one, or carries a terminal entry. There are at most M leaves and at most M terminating vertices. For a finite forest,

$$\sum_{v: \deg^+(v) \geq 2} (\deg^+(v) - 1) = \#\{\text{leaves}\} - \#\{\text{roots}\},$$

so there are at most M branching vertices as well. Thus at most $4M$ vertices are critical. Fix an integer $g = 4$ and discard all vertices at forest distance at most g from a critical vertex. Since the degree is at most $R + 1$, this discards at most

$$C_R M, \quad C_R = 4 \sum_{k=0}^g (R + 1)^k.$$

For a remaining vertex $v \in V_j$, its ancestor Q_v at depth $j - g$ exists. The intervening path is unary. In particular, v is the only occupied depth- j descendant of Q_v . Additionally discard v if its cell is at distance less than $3r_j$ from ∂Q_v . Call this a boundary deletion. We first prove disjointness for the vertices left by these two deletion rules.

Take distinct retained vertices $v \in V_j$ and $w \in V_k$, with $k \geq j$. If $k = j$, the cell of w is outside Q_v , since v is its sole occupied depth- j descendant. Hence $|t_w - t_v| \geq 3r_j$, whereas the sum of outer slot radii is $2r_j$. If $k > j$ and w descends from v , then

$$|t_w - t_v| \leq \delta_j, \quad r_k \leq r_j/R, \quad \delta_j + r_j/R = r_j(R^{-2} + R^{-1}) < 2r_j/R.$$

Both slots at w therefore lie strictly inside the inner gap of the slots at v . If $k > j$ and w does not descend from v , its occupied ancestor at depth j lies outside Q_v . Thus t_w is outside Q_v , $|t_w - t_v| \geq 3r_j$, and the sum of outer radii is at most $r_j + r_j/R < 2r_j$. This proves disjointness in every case. The two slots at a single vertex are disjoint by their definition.

It remains to count boundary deletions without imposing any distribution on the marker positions. Condition on all these positions and terminal levels. At each depth j , let s_j be the occupancy count for a fixed auxiliary grid of width δ_j , with no random displacement, and let $S = \sum_j s_j$. Each cell of either of two grids of the same width meets at most two cells of the other grid. It follows, for every U , that

$$S/2 \leq T(U) \leq 2S. \quad (10.12)$$

Choose one active marker in each occupied auxiliary cell. If an occupied primary cell at depth $j \geq g$ is boundary-deleted, a chosen representative in an auxiliary cell meeting it lies within

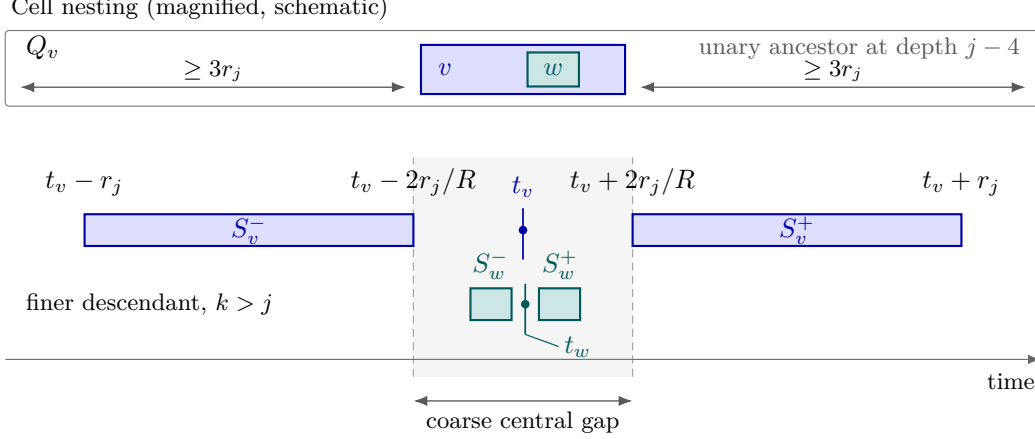


Figure 3: Regular sampling slots from Lemma 10.8. The inset shows cell containment; the two slot rows share one time axis and are vertically separated for visibility. Here $r_j = R^2\delta_j$, the unary ancestor Q_v lies at depth $j - 4$, v is its sole occupied depth- j descendant, and $\text{dist}(v, \partial Q_v) \geq 3r_j$. For a finer descendant $w \subset v$ at depth $k > j$, $|t_w - t_v| + r_k \leq \delta_j + r_j/R = r_j(R^{-2} + R^{-1}) < 2r_j/R$; thus both finer slots lie strictly inside the coarse central gap, although they need not lie inside the cell v . For an incomparable retained node at depth $k \geq j$, $|t_w - t_v| \geq 3r_j > r_j + r_k$. These inequalities give disjointness in both cases.

$3r_j + 3\delta_j$ of a depth- $(j - g)$ boundary. The displacement modulo δ_{j-g} is uniform. The probability of this event for a fixed representative is at most

$$\frac{2(3r_j + 3\delta_j)}{\delta_{j-g}} \leq 12R^{2-g}.$$

An auxiliary cell meets at most two primary cells. If $B(U)$ is the number of boundary deletions, linearity of expectation gives

$$\mathbb{E}_U B(U) \leq 24R^{2-g}S.$$

By Equation (10.12) and Markov's inequality,

$$\mathbb{P}_U\{B(U) > \varepsilon T(U)\} \leq \frac{48R^{2-g}}{\varepsilon}. \quad (10.13)$$

For $g = 4$, choose R large enough that this is at most $1/8$. Together with the structural deletions this proves Equation (10.10). $\square$

Figure 3 illustrates the nested-slot separation and the padding inside the unary ancestor.

Remark 10.9 (Other fixed test bands). The same construction applies to a fixed finite family of test bands contained in $a_*r \leq |t - t_v| \leq A_*r$, where $0 < a_* < A_* < \infty$. First refine the location boxes until their width is at most $a_*r/4$, and sample generations far enough apart that the next outer radius is at most $a_*r/4$. This uses a fixed larger grid ratio or a fixed residue class of generations; an independent uniform choice of the residue preserves uniform sampling of levels. Increase the boundary padding beyond $2A_*r$ and then increase the fixed ancestor depth g if necessary. A finer descendant's bands are now inside the coarser inner gap, while an incomparable descendant lies outside the padded ancestor. The preceding disjointness proof applies verbatim to these two explicit inequalities. The structural and boundary counts have the same form, with constants

depending on the fixed bands. For integer word times, rounding slot boundaries inward preserves disjointness and changes normalized lengths by a vanishing amount.

Lemma 10.10 (Integrated connectivity errors). *For each fixed K, R , a jump entry at terminal level ℓ contributes at most $C_{K,R} < \infty$ to the expected sum of its separated coarse projections. Consequently, if M_{jump} is the number of jump entries,*

$$\mathbb{E}_U \sum_{j=0}^J D_j \leq C_{K,R} M_{\text{jump}}. \quad (10.14)$$

If the bases lie in an interval of length at most CHe^{-N} and $j_N = \lfloor N/\lambda \rfloor$, then also

$$\mathbb{E}_U \sum_{j=0}^{j_N} (b_j - 1) \leq C'_{C,R}. \quad (10.15)$$

For a sparse family with $P = O(N/L)$ probes, $L \rightarrow \infty$, and fixed list bound K , there is a deterministic $d_N = o(N)$ such that, uniformly in the selected witness data,

$$\mathbb{P}_U \left\{ \sum_{j=0}^J D_j + \sum_{j=0}^{j_N} (b_j - 1) > d_N \right\} = o(1). \quad (10.16)$$

Moreover $C_R M = o(N)$. Together with Lemma 10.8, these assertions hold simultaneously on a set of displacements of conditional probability at least $1/2$ for all sufficiently large N .

Proof. Two fixed times at distance d lie in different uniformly displaced cells of width δ_j with probability at most $\min\{1, d/\delta_j\}$. Therefore a jump of length at most $K\delta_\ell$ has expected total separated projections at most

$$\sum_{j=0}^{\ell} \min\{1, KR^{j-\ell}\} \leq \sum_{a=0}^{\infty} \min\{1, KR^{-a}\} =: C_{K,R} < \infty.$$

Linearity proves Equation (10.14). For Equation (10.15), list the bases in increasing order and join consecutive bases by artificial edges for counting only. At every depth, $b_j - 1$ is bounded by the number of these edges that cross a cell boundary. Their total Euclidean length is at most CHe^{-N} . Hence

$$\mathbb{E}_U (b_j - 1) \leq Ce^{-N} R^j, \quad \sum_{j=0}^{j_N} Ce^{-N} R^j \leq \frac{CR}{R-1}.$$

For fixed K , both M and M_{jump} are bounded by a fixed multiple of $P + h_0$. Choose $d_N = o(N)$ so slowly that $(P + h_0 + 1)/d_N \rightarrow 0$. Markov's inequality gives Equation (10.16); the bound $C_R M = o(N)$ follows from $P = o(N)$. The last assertion follows by intersecting this event with the event of Lemma 10.8. $\square$

Lemma 10.11 (Stability under an independent grid displacement). *Fix a marker system with M entries before sampling its displacement. At level j , partition its active entries without using a grid: first group entries at the same time, then join consecutive distinct active positions whenever their gap is strictly less than δ_j . Denote the resulting partition by $\mathcal{P}_j^*$, and let $\mathcal{P}_j(U)$ be the partition into occupied cells of the displaced grid. Put*

$$\Delta(U) = \#\{0 \leq j \leq J : \mathcal{P}_j(U) \neq \mathcal{P}_j^*\}.$$

For an independent uniform displacement,

$$\mathbb{E}_U[\Delta(U) \mid \text{marker system}] \leq \frac{2R}{R-1}M. \quad (10.17)$$

At every level where the partitions agree, the entire projected colored graph, including its base labels, is the same up to cell names. In particular, if $M \leq M_N = o(N)$, two independent uniform displacements give the same projected graphs outside $o(N)$ levels with probability tending to one, uniformly over the selected marker system.

Proof. If $M = 0$, all partitions are empty and the assertions are immediate. Keep the labels and incidences of coincident entries when grouping their positions. As j increases, entries and hence distinct active positions can only disappear. Initially there are at most $M - 1$ consecutive-position pairs; deleting one position creates at most one new pair. Thus at most $2M - 1$ distinct consecutive pairs occur over all levels. Deleting all positions with a common terminal level one at a time only enlarges this bound.

Every grid partition refines $\mathcal{P}_j^*$: positions in the same half-open cell have total span strictly less than δ_j , and so do all their successive gaps. The partitions can differ only if a consecutive active pair with gap $d < \delta_j$ is split by a cell boundary. For a fixed pair this has probability d/δ_j , since U modulo δ_j is uniform. For each positive gap,

$$\sum_{j: \delta_j > d} \frac{d}{\delta_j} \leq \frac{R}{R-1}.$$

The same bound holds after restricting to the levels at which the pair is consecutive and active. A union bound over pairs and levels proves Equation (10.17).

All active edge incidences and base labels were fixed before the grid, so equal marker partitions give identical projected graph data. Choose a deterministic $q_N = o(N)$ with $M_N/q_N \rightarrow 0$. Markov's inequality bounds the conditional probability of $\Delta(U) > q_N$ by $2RM_N/((R-1)q_N) = o(1)$. Apply this to each of two independent displacements. Outside that exceptional event they both agree with $\mathcal{P}_j^*$ except at at most $2q_N$ levels. The estimate is conditional on arbitrary marker data, so it also holds under any selection law sampled before the displacements. $\square$

Remark 10.12 (Why the order of selection matters). The witnesses in Definition 10.6 must precede U . For example, take two terminal positions $U - \eta$ and $U + \eta$ with $\eta \ll \delta_J$. They occupy distinct unary branches on opposite sides of the persistent boundary U at all $J+1$ levels, but the same-level paired slots overlap. At least half the $2(J+1)$ nodes must be deleted, although there are only two terminal entries. This contradicts a bound $C_RM + \varepsilon T$ with $\varepsilon < 1/2$ for large J . Our conditional estimates avoid this obstruction. We keep U independent in the probability argument, rather than treating a sample-dependent favorable displacement as free fixed information.

10.4 Counting the address codes

Lemma 10.13 (Address code count). *Fix U and the distinct probe depths $j_1, \dots, j_P$. Include the base entries in the probe list at depth J , adding that depth if needed, and enlarge K to cover those entries. The resulting probe depths remain distinct. Suppose each list has at most K marker and edge entries. There is a coding of the projected forests and their list incidences with the following bound, uniformly in the total occupancy T :*

$$\log \#\{\text{codes of occupancy } T\} \leq (\lambda + C_0)T + KP \log(1 + T/P) + CPK \log(K+1) + O(1). \quad (10.18)$$

The constant C_0 does not depend on K . A family of additional annotations with at most $C_a^{m_j}$ choices at depth j , for a fixed C_a , only increases C_0 by $\log C_a$. In particular, for fixed K and $P = o(N)$,

$$\log \#\{\text{codes of occupancy } T\} \leq (\lambda + C_0)T + o(T + N), \quad (10.19)$$

uniformly in T . A sufficient condition for a growing bound $K = K_N$ and probe stride L is

$$\frac{K_N \log(K_N L)}{L} \rightarrow 0. \quad (10.20)$$

The codes can also be assigned subprobability weights at an additional cost $2 \log(T + 1) + O(1)$.

Proof. An ordered forest with T vertices and at most two roots has at most $C4^T$ possible shapes. One way to see this is to attach a new root above its roots and use the depth-first parenthesis word of the resulting ordered tree; its length is $2(T + 1)$, and a balanced parenthesis word determines the tree. Each nonroot cell has one of R child offsets in its parent, so the shape has at most R^T geometric assignments. The occupied root cells have a bounded presence mask among the cells meeting I , whose names are already fixed by I, U . This proves the bound $C(4R)^T$ for the forest before labels. Its vertices' depths follow from its parent relation.

At probe depth j_p , name a position by its ordered index among the m_{j_p} occupied cells; use one extra index for an absent list entry. Thus the position entries have at most $(1 + m_{j_p})^K$ possibilities. Repeated indices record coincident positions. Names in different probe lists do not require global forest indices: each local index specifies a forest vertex, whose ancestors are already determined by the forest. A correspondence to a distinguished base uses that base's fixed name, and its depth- J address determines its coarser addresses. The list's chain order, colors, endpoint incidences, and base names have at most $\exp(CK \log(K + 1))$ possibilities. The count includes absent entries and shorter lists. It never specifies an exact endpoint within its terminal cell.

Multiply these choices and use the distinctness of probe depths to obtain $\sum_p m_{j_p} \leq T$. Concavity gives

$$\sum_{p=1}^P \log(1 + m_{j_p}) \leq P \log(1 + T/P),$$

which proves Equation (10.18). The same estimate bounds the maximum number of list decorations over all forest shapes, so variation of the individual m_j between shapes causes no extra factor.

For $0 < \eta \leq K$ and $z \geq 0$,

$$K \log(1 + z) \leq \eta z + K \log(K/\eta).$$

Indeed the maximum of $\log(1 + z) - (\eta/K)z$ occurs at $z = K/\eta - 1$ and is at most $\log(K/\eta)$. Hence

$$\log \#\{\text{codes of occupancy } T\} \leq (\lambda + C_0 + \eta)T + KP \log(K/\eta) + CPK \log(K + 1) + O(1). \quad (10.21)$$

For fixed K , let η tend to zero sufficiently slowly while $P = o(N)$. This yields Equation (10.19) uniformly in T . Under Equation (10.20), for example $\eta_N = 1/\log L$ for large L makes both remainders in Equation (10.21) equal to $o(N)$. No condition $L \gg \log N$ is used. If the probe depths themselves must be encoded among the $J + 1$ levels, their extra cost is at most $\log \binom{J+1}{P} \leq P \log(e(J + 1)/P) = o(N)$ for $P = O(N/L)$; alternatively their independent mesh displacement can be conditioned on.

Finally give T a weight proportional to $(T + 1)^{-2}$ and distribute that weight uniformly among its finite enumeration of codes. The weights sum to at most one, and the negative logarithm of each weight is the code count bound plus $2 \log(T + 1) + O(1)$. $\square$

We call a displacement *favorable* if it satisfies the boundary bound of Lemma 10.8 and the connectivity bound of Lemma 10.10. For fixed R, K , its conditional probability given the selected witness system is at least $1/2$ for large N . At a favorable displacement the structural deletions are $o(N)$ and

$$E := \sum_{j=0}^J e_j \geq T - h_0(J+1) - d_N. \quad (10.22)$$

For earlier scales, Equation (10.15) also gives the stronger replacement of h_0 by one, up to an integrated $o(N)$ error. The simpler bound in Equation (10.22) suffices for the following occupancy estimate.

The code now specifies every box, component, and sampling slot. It remains to bound the probability that its required contacts occur. We first control the sum of contour oscillations across its prescribed intervals.

10.5 Oscillations and the probability of a fixed code

Lemma 10.14 (An exponential sum bound for prescribed intervals). *Consider a deterministic finite family $\mathcal{I}$ of intervals whose lengths belong to a geometric family with ratio $R > 1$. Suppose that at each length at most K intervals cover a point. There are constants $c, C > 0$, depending on R, K and the raw-word law, such that*

$$\mathbb{P} \left\{ \sum_{I \in \mathcal{I}} \frac{\text{osc}_I Z}{\sqrt{|I|}} > u|\mathcal{I}| \right\} \leq \exp\{-cu|\mathcal{I}| + C|\mathcal{I}|\}.$$

Here coordinate oscillations can be summed, and the estimate holds for all exterior typings. Integer intervals of bounded length can be handled by increasing the constant. If K is fixed and $R \geq 2$, the constants can be chosen uniformly in R .

Proof. Cut the union of the intervals at all their endpoints. There are at most $2|\mathcal{I}|$ nonempty atoms J . Proposition 4.2 gives independent dominating variables $Y_J = 2M_{|J|}/\sqrt{|J|}$, with a common exponential-moment bound, for the oscillations on these disjoint fresh raw pieces. Subadditivity of oscillation gives

$$\sum_{I \in \mathcal{I}} \frac{\text{osc}_I Z}{\sqrt{|I|}} \leq \sum_J a_J Y_J, \quad a_J = \sqrt{|J|} \sum_{I \supset J} |I|^{-1/2}.$$

If I_0 is the shortest interval containing J , then

$$a_J \leq K \sqrt{|J|/|I_0|} \sum_{v \geq 0} R^{-v/2} \leq \frac{K}{1 - R^{-1/2}}.$$

Apply the exponential-moment bound with a parameter divided by this last constant, and use Markov's inequality. This proves the claim. The bound on a_J is uniform for $R \geq 2$, proving the asserted uniformity. $\square$

Proposition 10.15 (Fixed-code arm test). *Let $\lambda = \log R$. In the finite shifted address model, fix a code, let m_j be its number of occupied boxes at level j , and let e_j be the sum of its two monochrome ranks. Put*

$$T = \sum_j m_j, \quad E = \sum_j e_j,$$

and let M be the number of terminal entries. For each $\zeta > 0$, one can choose a fixed sufficiently large R and sufficiently small fixed deletion tolerances so that, as the minimum sampling radius tends to infinity,

$$\mathbb{P}\{\text{the code's chord constraints are realizable}\} \leq \exp\{-\beta\lambda E + \zeta\lambda T + C_{R,\zeta}M\}. \quad (10.23)$$

The statement is for codes with the regular-slot deletion bound of Lemma 10.8. It is uniform over such finite codes and over arbitrary exact chord endpoints inside their recorded boxes. Small fixed errors in sampling radii or inward integer rounding are absorbed by increasing the minimum permitted radius.

Proof. Choose a deterministic time before all the code's representative points and sampling slots, and anchor both contour coordinates at zero there. A common coordinate translation changes none of the chord constraints or slot tests. With this convention the heights observed while exposing the finite word window are measurable in its chronological raw filtration, even when the original bilateral contours had been anchored at time zero inside that window.

For each coordinate and each projected chord component, apply the sorted comparison lemma to the occupied intervals of that component. It gives a comparison between consecutive occupied intervals, one per unit of monochrome rank, with a total height-discrepancy bound of at most six times the sum of the relevant box oscillations. A box is used in at most one component of each color. Therefore the sum of all normalized comparison discrepancies is controlled by a fixed multiple of the normalized box oscillations.

Include also the intervals between each regular-node representative midpoint and its adjacent sampling-slot endpoint. Their lengths are comparable to r/R at a node of sampling radius r . The same-scale overlaps are bounded independently of R : regular centers are separated at the scale r , whereas the box intervals themselves have disjoint interiors. Thus Lemma 10.14 applies, separately to a bounded number of geometric length families. For fixed $0 < \epsilon < \delta$, outside an event of probability at most

$$\exp\{-cR^\epsilon T + CT\},$$

the sum of all these normalized oscillations is at most $R^\epsilon T$. Discard each comparison whose assigned discrepancy exceeds $R^\delta \sqrt{r/R}$, and each node-side with an excessive local oscillation. At most $CR^{\epsilon-\delta}T$ comparison incidences are lost. The originally irregular nodes lose at most $C\epsilon_0 T + C_R M$ incidences: after sorting, a node has at most one predecessor and one successor per color. Choosing R large makes the first two relative losses as small as required.

All remaining required slot conditions now have fixed deterministic tolerance $CR^\delta \sqrt{r/R}$. On a left slot the target wall is the already observed representative height in the previous occupied box of the same component, minus that tolerance. The comparison bound makes survival above this wall and landing near it necessary. The previous box lies strictly before the slot, by regular-node separation. On a right slot use the current midpoint height minus the same tolerance; that height is already observed before this slot. Its initial discrepancy is small by the retained local oscillation condition. We do not choose a random wall using the unobserved actual comparison error.

Expose all slots in chronological order. Lemma 10.8 makes them disjoint, and the intervening raw words can also be exposed in chronological order. Each slot is fresh conditional on this history. If k colors are incident on a retained left slot, Lemma 4.7 gives logarithmic charge at least

$$(k/2 + d_k - O(\delta) - o(1))\lambda.$$

For a right slot it gives $(d_k - O(\delta) - o(1))\lambda$. The $o(1)$ is uniform once the minimum radius is sufficiently large for the fixed R . Summed incoming incidences give $E/2$ before deletions. There

are $2E$ side incidences in total, and $d_k \geq kd_2/2$ for $k = 0, 1, 2$. The undeleted leading survival charge is therefore at least d_2E . Deletions lose only a bounded charge per incidence. The choices of deleted incidences have at most $\exp(CT)$ possibilities, which are summed rather than conditioned upon.

Choose δ sufficiently small in terms of ζ , then choose R large enough that C/λ , the deletion fractions, and the fixed Brownian constants divided by λ fit within the remaining error budget. The oscillation-exception exponent can simultaneously be made larger than $3\beta\lambda T$ in absolute value, since $R^\epsilon/\log R \rightarrow \infty$. As $E \leq 2T$, that exception is absorbed into Equation (10.23) after an arbitrarily small enlargement of its error budget. Finally take the minimum radius large enough for the uniform lattice-slot estimates. Chronological multiplication and the finite sum over discard patterns give the proposition. $\square$

10.6 The address occupancy bound

The fixed-code probability bound outweighs the address count because $\beta > 1$. We combine them before returning to the independent grid, so a favorable displacement is never treated as uncharged selected data.

Proposition 10.16 (Exponential occupancy bound). *Put $\beta = 1/2 + d_2 > 1$. Assume the fixed-code test of Proposition 10.15 in the following quantitative form: for every sufficiently small $\zeta > 0$, choose a fixed boundary tolerance sufficiently small and a fixed R sufficiently large so that, once the minimum radius is sufficiently large, every favorable admissible decorated code $\mathbf{c}$ satisfies*

$$\mathbb{P}\{\mathbf{c} \text{ is realized}\} \leq \exp\{-\beta\lambda E(\mathbf{c}) + \zeta\lambda T(\mathbf{c}) + C_{R,\zeta}M(\mathbf{c})\}. \quad (10.24)$$

This estimate is required uniformly in the fixed displacement U . The realization event includes all the actual marked chord constraints. The fractional boundary deletions are absorbed into the $\zeta\lambda T$ term, and the structural deletions into the explicit terminal term, using the bounded number of sorted-chain charges at each deleted vertex. The fixed-code estimate includes its oscillation exceptions. No such estimate is asserted here for codes whose slots are predominantly irregular.

Then there are $C, c, C_1 > 0$, independent of any fixed finite exploration cutoff, such that for all sufficiently large N and all $u \geq 0$,

$$\mathbb{P}\{A \geq CN + u\} \leq C_1 \exp(-cN - cu). \quad (10.25)$$

The estimate holds for any word-dependent selection of witness lists made before U . It also holds for a fixed reference grid, and with A replaced by its maximum over grid displacements. The threshold in “sufficiently large” may depend on the fixed exploration cutoff. If Equation (10.8) holds, then

$$\mathbb{E}_{\mathbb{Q}_N} A = O(N). \quad (10.26)$$

The corresponding code has length at most a deterministic constant times N with $\mathbb{Q}_N$ -probability tending to one. At an independently and uniformly sampled level $j \in \{0, \dots, J\}$, $\mathbb{E}_{\mathbb{Q}_N} m_j = O(1)$.

Proof. Fix U . Use Equation (10.21) with a small fixed $\eta > 0$ and then let the sparse terminal error be $o(N)$. For every favorable code of total occupancy T , combine Equations (10.24) and (10.22). Summing over its possible codes gives

$$\begin{aligned} & \mathbb{P}\{\text{some favorable realized code has occupancy } T\} \\ & \leq \exp\{-(\beta - 1 - \zeta)\lambda - C_0 - \eta\}T + \beta\lambda h_0(J + 1) + \beta\lambda d_N + C_{R,\zeta}M + o(N)\}. \end{aligned}$$

Choose fixed R large enough and then the error tolerances small enough that

$$\delta := \beta - 1 - \zeta - (C_0 + \eta)/\lambda > 0.$$

Since $\lambda(J+1) = bN + O(1)$, $d_N = o(N)$, and $M = o(N)$, there is a fixed $B < \infty$ such that the last probability is at most

$$\exp\{-\delta\lambda T + BN\}. \quad (10.27)$$

Here B and δ are independent of the fixed list cutoff: its contribution was confined to the sparse $o(N)$ terms. Geometric summation over $\lambda T \geq x$ shows, uniformly in U ,

$$\mathbb{P}\{\text{favorable and } A \geq x\} \leq C_2 \exp(BN - \delta x).$$

For $C_* = (B+2)/\delta$, this gives

$$\mathbb{P}\{\text{favorable and } A \geq C_*N + u\} \leq C_2 \exp(-2N - \delta u). \quad (10.28)$$

The same bound holds after integrating over the independent U .

The event of a favorable displacement has only a positive conditional probability; we do not give its complement an exponential estimate. Instead compare all shifts of the same realized witness system. The same-width cell intersection argument gives

$$T(U) \leq 2T(V) \quad \text{for all displacements } U, V.$$

Let $A_{\max} = \max_U \lambda T(U)$. This maximum is measurable: there are finitely many markers and levels, so only finitely many marker-boundary crossings can change the occupancy as U varies in $[0, H)$. If $A_{\max} \geq x$, every displacement has $A \geq x/2$, and at least half of the displacements are favorable. Conditional on the complete witness system, this proves

$$\mathbf{1}_{\{A_{\max} \geq x\}} \leq 2\mathbb{P}_U\{\text{favorable and } A \geq x/2 \mid \text{witness system}\}.$$

Integrate and apply Equation (10.28). Increasing the constant multiplying N proves Equation (10.25) for $A_{\max}$, and therefore for every particular grid. All word-dependent choices have been included in the finite code enumeration. The auxiliary favorable shift has never been selected as unencoded sample-dependent information.

Under Equation (10.8), the tail is bounded above by

$$\mathbb{Q}_N\{A \geq CN + u\} \leq \min\{1, C_1 \exp(k_N - cN - cu)\}.$$

Since $k_N = o(N)$, integration in u gives

$$\mathbb{E}_{\mathbb{Q}_N} A \leq CN + C_1 c^{-1} \exp(k_N - cN) = O(N).$$

Equation (10.21), including its optional integer-size code, then gives the stated deterministic $O(N)$ code cutoff outside an event of probability tending to zero. Finally, for an independently uniform level,

$$\mathbb{E}_{\mathbb{Q}_N} m_j = \frac{\mathbb{E}_{\mathbb{Q}_N} T}{J+1} = O(1),$$

since $J+1$ is comparable to N for fixed R, b . □

Remark 10.17 (Finite cutoffs and the order of limits). Fix the desired error tolerance first, then choose R and the padding constants. For each fixed finite exploration bound, let N and the minimum time unit diverge, and let the probe stride satisfy $L \rightarrow \infty$ and $L = o(N)$. Structural, jump, and terminal code errors are then $o(N)$. A subsequent slow diagonal may increase the exploration and list bounds, retaining Equation (10.20) and the vanishing jump error. For example, the constant in Equation (10.14) is $O(1 + \log_R K)$, so that the displayed growing-list condition also controls that error when the jump tolerance is bounded by K_N . The coefficient in the occupancy bound remains fixed; only the speed of the diagonal changes. The construction counts all projected addresses, including intermediate markers from different probe levels. It assumes no sparsity of the original candidate locations at a finest scale. Producing a saturated, jointly realizable finite-line system from the occupancy bound is a separate step.

10.7 The credit contradiction and entropy of viewed blocks

For a connected component C of a union of two colored graphs, let $e(C)$ be the sum of the ranks of the two colored graphs restricted to C , where rank is the number of vertices minus the number of connected components, including isolated vertices. We say that C has a *rank cycle* when $e(C) \geq |C|$, and that it is *acyclic* when $e(C) = |C| - 1$. The union spanning-tree argument in Lemma 10.7 shows that these alternatives exhaust the possibilities. Thus an acyclic component has exactly the minimum total rank needed for connectivity.

Proposition 10.18 (Rank credit). *Let $\varepsilon = \beta - 1 > 0$. Consider the projected witness systems of the address construction on logarithmic levels $[0, bN]$, with h bases in an interval of length $O(ne^{-N})$. Suppose their pairwise time distances exceed the largest address-cell width used on $[aN, bN]$, where $1 < a < b$; thus they occupy distinct boxes there for every displacement. All selected laws satisfy $\mathbb{Q}_N \leq e^{o(N)}\mathbb{P}$. Then the following combination of events has probability tending to zero: outside a vanishing fraction of levels in $[aN, bN]$, every connected component containing a base has a rank cycle.*

Here $a - 1$ is chosen sufficiently small relative to $\varepsilon(b - a)$, and then h sufficiently large, with all choices fixed before the scaling limit. Witness-list cutoffs may increase only along the sparse diagonals allowed by the address construction.

Proof. At a level with m occupied boxes, c union components, and total monochrome rank e , one has

$$e \geq m - c, \quad m - \beta e \leq \beta c - \varepsilon m.$$

The integrated component excess caused by projected jumps is $o(N)$ on favorable grids. Before level N , all base cells can be charged as one component up to $o(N)$; hence the possible positive credit there is at most $C_0N + o(N)$, independently of h . On $[N, aN]$, there are at most h base components, so positive credit is at most $C_1(a - 1)hN + o(N)$. The constants include fixed endpoint-rounding and grid-step changes.

At every level on which all base components have rank cycles, $e \geq m$ after the negligible connectivity errors. The separated bases occupy at least h boxes on $[aN, bN]$, so the credit there is at most $-\varepsilon h$ per unit log length. At exceptional levels the positive credit can still be bounded by a constant times the number of acyclic base components; a bound on the maximum m is not needed. Proposition 10.16 absorbs all small relative errors in the integrated occupancy into an arbitrarily small multiple of N .

Choose $a - 1$ such that $C_1(a - 1) < \varepsilon(b - a)/4$, and then choose h with $C_0 < \varepsilon(b - a)h/4$. The total leading address budget minus arm charge is then at most $-cN$ for some $c > 0$, even

after the permitted errors. Summing Equation (10.23) over the length-weighted codes gives an exponentially small probability for the stated cycle event *and a favorable displacement*, under $\mathbb{P}$. Multiplying by $e^{o(N)}$ gives the same vanishing conclusion under $\mathbb{Q}_N$.

We now remove favorability without assuming that cyclicity is invariant under a change of grid. Let $\eta_N \downarrow 0$ and denote by $E_N(U)$ the cycle event with at most $\eta_N N$ exceptional generations for the independent displacement U . Generation count and logarithmic length differ by the fixed factor λ . Lemma 10.11 supplies $q_N = o(N)$ and a uniform conditional error $\alpha_N = o(1)$ such that $\Delta(U) \leq q_N$ outside that error. Let E_N^* be the corresponding cycle event for the grid-independent partitions $\mathcal{P}_j^*$, with at most $\eta_N N + q_N$ exceptions. Then

$$\mathbb{Q}_N(E_N(U)) \leq \mathbb{Q}_N(E_N^*) + \alpha_N.$$

Sample a second independent displacement V . Conditional on the marker system, V is favorable and has $\Delta(V) \leq q_N$ with probability at least $1/2 - \alpha_N$. On E_N^* this gives the projected cycle event $E'_N(V)$ with at most $\eta_N N + 2q_N = o(N)$ exceptional generations. Consequently

$$(1/2 - \alpha_N)\mathbb{Q}_N(E_N^*) \leq \mathbb{Q}_N(E'_N(V) \cap \{V \text{ favorable}\}).$$

The preceding favorable-grid estimate applies to this right side. Indeed, on each added exceptional level the positive credit is at most $\beta(h + D_j)$, by Equation (10.9); the extra levels and the favorable-grid sum $\sum_j D_j$ are both $o(N)$. No maximum-occupancy bound is needed for this error. The right side therefore tends to zero, proving the claim for U . The conditional grid estimates are uniform in the preselected data; their errors are not multiplied by the selected density. $\square$

Lemma 10.19 (Entropy allocation for a finite address code). *Let W have a product raw-word law $\mathbb{P}$, including an independent past if needed. Let $\mathbb{Q} \leq e^\kappa \mathbb{P}$, and let C be a countably valued code chosen using W and independent auxiliary randomness. Suppose C has code lengths $\ell(c)$ with $\sum_c e^{-\ell(c)} \leq 1$. For each fixed code, let finitely many disjoint deterministic word slots be ordered chronologically. Then the sum of their expected conditional relative entropies with respect to fresh iid raw blocks, averaged over C , is at most*

$$\kappa + \mathbb{E}_{\mathbb{Q}} \ell(C). \tag{10.29}$$

Auxiliary continuous grid shifts can be kept as independent side information and integrated; no code length for such an independent shift is charged.

Proof. Let $\pi_c = e^{-\ell(c)}$ and place any missing mass at an unused code. Compare the joint law of (C, W) under $\mathbb{Q}$ to the law in which C has probabilities π_c and W independently has law $\mathbb{P}$. The chain rule gives

$$D(\mathbb{Q}_{C,W} \| \pi \otimes \mathbb{P}) = D(\mathbb{Q}_W \| \mathbb{P}) + \mathbb{E}_{\mathbb{Q}_W} D(\mathbb{Q}_{C|W} \| \pi) \leq \kappa + \mathbb{E}_{\mathbb{Q}} \ell(C),$$

because the conditional Shannon entropy of C is nonnegative. Disintegrate instead first over C and then over the chronological raw pieces. Product structure under the reference law makes the reference conditional law in every slot a fresh iid block. Discarding the nonnegative entropy terms for the gaps, past, and other pieces leaves the asserted upper bound for the slots. This proof also applies to finite restrictions of an infinite past, followed by monotone increase of the observed sigma fields.

For an independent grid displacement U , apply the argument conditionally on U and integrate. The original word law is independent of U , and the selected law before grid sampling has the same bounded-density property. We do not condition on a sample-dependent favorable displacement before this calculation. $\square$

Corollary 10.20 (Central oscillations at intermediate levels). *For a projected address code, assign to every occupied node v an interval of length at most Ar_v centered within its location cell, where A is fixed and r_v is its viewing radius. Centers rounded using the terminal address are also allowed, provided they lie in the specified cell. Write*

$$S_v = \frac{\text{osc}_{I_v} Z}{\sqrt{r_v}}.$$

Suppose $\mathbb{E}_{\mathbb{Q}} \sum_j m_j = O(N)$, the expected code length is $O(N)$, and the selected density bound is $e^{O(N)}$. Then

$$\mathbb{E}_{\mathbb{Q}} \sum_v S_v = O(N).$$

In particular, the sum of the central oscillation amplitudes of the occupied lines has bounded first moment at a uniform intermediate viewing level. The constants may depend on the fixed grid ratios, padding, and A , but not on the exploration-list cutoff.

Proof. Replace each I_v by the deterministic interval

$$\hat{I}_v = [\inf v - Ar_v/2, \sup v + Ar_v/2],$$

where v denotes the code's location cell. This interval contains I_v regardless of the chosen in-cell center. Its length is exactly $(A + R^{-2})r_v$. At a fixed level, location cells have common length r_v/R^2 , so these enclosures have overlap multiplicity at most $C_A R^2$. Across levels their lengths are geometric. The proof of Lemma 10.14 gives constants $\theta, c > 0$ such that, for every fixed code,

$$\mathbb{E}_{\mathbb{P}} \exp \left\{ \theta \sum_v S_v - c \sum_j m_j \right\} \leq 1.$$

The same holds under the product reference in which the code has its Kraft weights and is independent of the raw word. The entropy variational inequality and Lemma 10.19 imply

$$\theta \mathbb{E}_{\mathbb{Q}} \sum_v S_v \leq c \mathbb{E}_{\mathbb{Q}} \sum_j m_j + \kappa + \mathbb{E}_{\mathbb{Q}} \ell(C) = O(N).$$

Division by the number of intermediate levels proves the last claim. $\square$

Remark 10.21 (Arbitrary off-center bands at a coarsened level). The entropy allocation is not restricted to seed probe levels. Fix finitely many bands $ar \leq |t - t_v| \leq Ar$, with $0 < a < A < \infty$, at the intermediate viewing radius r . Refine the location cells so their length is at most $ar/8$, retain only a residue class of levels with successive radius ratio greater than $8A/a$, and increase the ancestor padding in Lemma 10.8. The same ancestor/descendant proof then makes the enlarged test bands pairwise disjoint: descendants lie in the inner hole, and incomparable nodes lie outside the padded ancestor. The residue class is chosen uniformly and independently, so averaging over it preserves the uniform-level marginal. Its spacing is fixed once the finite band request is fixed.

For origins rounded from terminal address cells, slow coarsening makes the rounding error $o(r)$. The deterministic enlarged bands therefore contain every actual requested block, for large enough coarsening. One applies Lemma 10.19 to the enlarged raw slots and then takes the requested observations as measurable functions there. Structural and boundary deletions are bounded by the same marker and fractional-occupancy errors. A continuous logarithmic phase, if used, is integrated as independent reference side information; equivalently, stationarity under the fixed grid dilation already suffices for the zero-or-infinite conclusions about pinned heights and first sign-change times.

11 Finite contour views and protected annuli

We now construct cheap circuits around selected discrete vertices. The inputs are the address estimates of Section 10, the local passage kernels, and the bilateral crossing comparison. The output, Proposition 11.30, applies whenever the contour contacts seen from a base form an acyclic component.

The proof has three parts. First, finitely many contour neighborhoods cover every bounded exploration from the bases at a typical logarithmic scale. One code records these neighborhoods at all seed scales; an independent coarsening with fixed origins then gives stationary limiting profiles. Second, their contact classes determine a finite auxiliary sphere containing an annulus whose incidences agree with the original map. Third, the annular crossing tests are expressed entirely in retained off-center data and transferred from ordinary Brownian completions. This last step estimates open-port crossings only; access from the surrounded vertex is the task of Section 12.

11.1 Probability spaces and entropy transfer

For an index j , let P_j be the law of the entire bilateral raw inventory word on its countable product space, resolved using its bilateral past. Entropy calculations below are first made on finite observations and then extended by increasing those observations. Auxiliary grid translations and sampling variables are independent under P_j . Let $N_j \rightarrow \infty$, let n_j be the base time unit, and put $r_j(x) = n_j e^{-x}$. We only use levels for which $n_j e^{-bN_j} \rightarrow \infty$. The selected law Q_j satisfies

$$Q_j \ll P_j, \quad \frac{dQ_j}{dP_j} \leq e^{\kappa_j}, \quad \kappa_j = o(N_j). \quad (11.1)$$

Conditioning on an event of probability $e^{-o(N_j)}$ has this form. The bounded-density hypothesis gives the address first-moment estimates used below.

We will use bilateral recurrence below every fixed scalar height. Here is its inventory justification. Every letter has a finite match almost surely, by [38, Proposition 2.2]. At a fixed integer cut the past stack contains infinitely many burgers of each type: if it contained exactly k of one type with positive probability, the independent next $k+1$ letters could all be rigid orders of that type, contradicting finite matching. The k th remaining burger of a given type, counted from the freshest, has birth baseline $-k$ relative to that coordinate at the cut. Its birth realizes this level in the past, and its eventual matched order realizes it in the future. Taking all integer cuts and k proves the asserted recurrence simultaneously, including at selected origins. Absolute continuity in Equation (11.1) preserves this probability-one property.

Lemma 11.1 (Macroscopic cutoffs before selection). *Fix a compact starting time interval in a bilateral contour coupling, a sequence $n_j \rightarrow \infty$, and deterministic $\rho_j \downarrow 0$. For every $\epsilon > 0$ there are deterministic nested intervals $[-C_d n_j, C_d n_j]$, one for each integer $d \geq 1$, and deterministic $D_j \rightarrow \infty$, such that events $\mathcal{T}_j$ with $\limsup_j P_j(\mathcal{T}_j^c) \leq 2\epsilon$ have these properties simultaneously for every $d \leq D_j$:*

- (i) *Every exploration from the starting interval with at most d steps, each either a scalar link or a time jump of magnitude at most $d n_j$, stays in the corresponding interval $[-C_d n_j, C_d n_j]$. The statement includes the endpoints of actual first- and last-return witnesses.*
- (ii) *If its jumps have magnitude at most $d \rho_j n_j$, all its projected points lie within α_j of its base, where $\alpha_j \downarrow 0$ is deterministic.*

The events and their full arrays of cutoffs are chosen before any bad-level, bad-base, or bounded-density selection. They hold for all seed levels satisfying $r_j(x) \leq \rho_j n_j$. After selection one may slow the actual exploration cutoff beneath D_j , and may slow it further to accommodate any fixed finite family of composed explorations. No new typicality event is intersected after selection.

Proof. First fix d . Start with a bounded macroscopic time interval I_0 containing every permitted base. Enlarge it by d on each side to allow a time jump of length dn_j . On this enlarged interval the limiting Brownian coordinates have finite minima. In each coordinate choose one time to its left and one to its right at which that coordinate is below its minimum on the enlarged interval by more than three. Such times exist almost surely by scalar Brownian recurrence. They need not be the same times for the two coordinates. Let I_1 contain the enlarged interval and all four chosen times. A scalar partner of a source in the enlarged interval cannot lie beyond its corresponding lower barrier: the intervening minimum would then be strictly smaller than the source height. Repeat this enlargement and barrier construction $d + 1$ times. It gives an almost surely finite random interval containing every such chain, uniformly over all its choices of source, color, and endpoint.

Choose a deterministic C_d large enough that these finitely many strict barriers lie in $[-C_d, C_d]$ except with probability at most $\epsilon 2^{-d-2}$. Increase the constants to make them nested. Compact-uniform contour convergence preserves all the barrier inequalities with a margin exceeding one for all sufficiently large j on this event. A discrete scalar link changes the tested endpoint level by at most one lattice step, whose normalized size tends to zero. Thus the same argument confines the actual discrete chains. The barriers also confine their last-upcrossing and first-downcrossing searches whenever these are used to realize such a source chord. This is simultaneous containment over the continuum of source times, not a separate probability estimate for each seed.

For (ii), keep d fixed and use its deterministic enclosing interval. If uniform spatial collapse failed, some subsequence would have a base and a chain of at most d steps with a projected separation bounded below. Pass to a further subsequence on which all their rescaled times converge in that compact interval, repeating vertices to keep the chain length fixed. The endpoints of a time jump have the same limit because $d\rho_j \rightarrow 0$. The endpoints of an actual scalar link limit to a scalar horizontal pair, by compact-uniform contour convergence and its endpoint/minimum test. They have the same curve image by Theorem 3.2(iii). Continuity of the curve then makes every successive pair of projected chain points have the same limiting image, contradicting the claimed separation. This compactness argument is uniform over all bases, all seed levels, and all choices of a bounded chain. In particular it uses only $r_j(x) \leq \rho_j n_j$, not the number of tested levels.

For each fixed finite collection $d \leq k$, the preceding convergence and barrier events imply their simultaneous conclusions with spatial error at most $1/k$ and failure at most $\epsilon + o_j(1)$; the sum of the displayed barrier budgets is less than $\epsilon/2$. Choose deterministic thresholds j_k increasing so that for $j \geq j_k$ the remaining error is at most ϵ , and put $D_j = k$ between successive thresholds. Let $\mathcal{T}_j$ be that finite intersection and $\alpha_j = 1/D_j$. This defines a master event with the asserted probability before any subsequent selection. Additional deterministic slow-growth requirements on D_j may be imposed by increasing j_k .

If a selected law is supported on $\mathcal{T}_j$, all these assertions hold under it by support, without transferring a small ordinary probability through its density bound. Every later fixed exploration or composition bound is eventually below D_j . A diagonal chosen after that law may only decrease this cutoff, retaining each earlier window C_d separately. For a prescribed open spatial neighborhood with a fixed positive margin about the permitted bases, (ii) gives the required confinement once α_j is below that margin. $\square$

Whenever we apply the address estimates below, the selected laws are supported on one of these preselection master events for a fixed compact starting interval, with ρ_j bounding every used time scale divided by n_j . This is the finite-chain typicality cutoff. The loss 2ϵ is removed before forming the bad-configuration contradiction.

Lemma 11.2 (Coding and disjoint slots). *Let C_j be a countable code with lengths $\ell_j(c)$ satisfying $\sum_c e^{-\ell_j(c)} \leq 1$. Conditional on a code, suppose there are $T_j \geq cN_j$ disjoint chronological raw-word slots, their boundaries and the observations assigned to them being fixed by the code. Suppose $E_{Q_j} \ell_j(C_j) \leq KN_j$. Let I_j be uniform among the slots, independently after all other sampling. The mean relative entropy of the conditional law of the selected raw slot, given its preceding observations and the code, with respect to its iid raw-word law is at most $(K + o(1))/c$.*

Proof. Apply Lemma 10.19 to the selected law, the code C_j , and its chronological disjoint slots. It bounds the sum of their expected conditional relative entropies by $\kappa_j + E_{Q_j} \ell_j(C_j) \leq \kappa_j + KN_j$. If only a coarser preceding history is retained, data processing gives the same bound for that history. Since I_j is uniform after sampling and $T_j \geq cN_j$, its mean conditional relative entropy is at most $(\kappa_j + KN_j)/(cN_j) = (K + o(1))/c$. No independence of the resolved walk from its past is asserted or used. $\square$

Lemma 11.3 (Null-test transfer). *If $Q \ll P$ and $D(Q\|P) \leq H$, then, for an event E with $0 < P(E) < 1$,*

$$Q(E) \leq \frac{H + \log 2}{\log(1/P(E))}. \quad (11.2)$$

Consequently uniformly bounded entropy transfers events whose reference probabilities tend to zero. The same conclusion holds for averaged conditional entropies and uniformly vanishing conditional reference probabilities.

Proof. Data processing to the indicator of E gives $D(Q\|P) \geq q \log(1/p) - \log 2$, where $q = Q(E)$ and $p = P(E)$. For conditional laws apply this inequality to each section and integrate. $\square$

The conclusion is tight absolute continuity, not a bounded likelihood ratio. For lattice passage observations Proposition 7.16 supplies the following additional input: after fixing finitely many compatible guard and flexible-order instructions, the iid slot experiment must have a subprobability law dominated by a fixed multiple of a reference experiment with independent raw blocks and smoothly sampled shifts on the feasible lattice. If that domination has error η , a reference event of probability u has iid-slot probability at most $Mu + \eta$; Equation (11.2) applies to this bound. One first lets $j \rightarrow \infty$, then $u \downarrow 0$, and finally $\eta \downarrow 0$ by exhausting the guards and compact parameter ranges. Brownian weak convergence of the shifts does not supply this lattice domination.

11.2 Simultaneous packing and finite-line saturation

The occupation estimate bounds the number of separated contour neighborhoods visible from the bases. We need those neighborhoods and their connecting links to occur in one realization. The following packing construction provides this simultaneous choice.

Here all randomness of the selected environment is sampled before the viewing level. Let I_j be a deterministic logarithmic interval of length $T_j \asymp N_j$, and let Y_j be uniform on I_j , independently of that environment. For $k \geq 1$ let $\mathcal{R}_{j,k}(x)$ be the lattice times reachable from the bases using at most k links or time jumps, each jump having magnitude at most $kr_j(x)$. Recurrence below every fixed scalar level in both directions makes each lattice scalar class finite, hence each bounded-depth

reachable set is finite. Enclosing chronological windows are only cutoffs for coding and are removed; a link search has no bound on its chronological duration. These sets increase with k and satisfy

$$y \leq x \implies \mathcal{R}_{j,k}(x) \subseteq \mathcal{R}_{j,k}(y). \quad (11.3)$$

All intermediate points of a witness belong to its reachable set. A ℓ -bounded exploration started from $\mathcal{R}_{j,k}(x)$ stays in $\mathcal{R}_{j,g(k,\ell)}(x)$ for some deterministic g ; for example, $g(k,\ell) = 2(k + \ell + 1)$ suffices. Define

$$P_{j,k,H}(x) = \max\{|A| : A \subseteq \mathcal{R}_{j,k}(x), \quad |u - v| > Hr_j(x) \text{ for distinct } u, v \in A\}.$$

Lemma 11.4 (The required consequence of address occupation). *Suppose the projected-address estimate holds, with a constant C independent of fixed k and M , for simultaneous lists formed as follows. At the levels of an independently translated mesh of stride L , choose at most M terminals in $\mathcal{R}_{j,k}$ and include their access witnesses. Project each list through the preceding coarser cell of length L . Its active-box count m_j satisfies*

$$\int_{I_j} m_j(x) dx \leq (C + o_L(1))T_j$$

with probability tending to one, or with any prescribed fixed probability loss which can subsequently be removed. The projected boxes have diameter at most $cr_j(x)$, and the marker errors divided by T_j vanish as $L \rightarrow \infty$ for each fixed k, M . Then

$$\sup_{k,M} \lim_{H \rightarrow \infty} \limsup_{j \rightarrow \infty} E_{Q_j} \min\{P_{j,k,H}(Y_j), M\} \leq C. \quad (11.4)$$

Proof. Choose $L = L(H) \rightarrow \infty$ slowly enough that $He^{-L} > 2c$. At each seed y take a maximal H -packing, truncated at M . Throughout its coarser cell $[y - L, y]$, the chosen terminals remain in different projected boxes, since the time unit grows by at most e^L . Thus their count contributes at least L times itself to $\int m_j$, apart from the two end cells. Averaging the independent mesh translation converts the sum of these seed counts into the uniform-level integral. On the exceptional event bound the truncated packing by M , without attempting to bound m_j . Divide by T_j , let $j \rightarrow \infty$, remove the prescribed probability loss, and then let H and $L(H)$ tend to infinity. The end-cell loss is at most $2ML/T_j$ and vanishes in the first limit. $\square$

Proposition 11.5 (Saturated centers and actual colored witnesses). *Assume Equation (11.4). There is a subsequence and deterministic $H_j, k_j, L_j \rightarrow \infty$ such that $L_j = o(T_j)$ and $L_j = o(\log H_j)$ with the following properties. Choose a finite set of centers $\mathcal{C}_j$ at the seed level Y_j , then independently choose $U_j \sim \text{Unif}[0, L_j]$ and put $X_j = Y_j - U_j$. The centers have a finite limiting count K with $EK \leq C$, remain mutually divergent in $r_j(X_j)$ units, and have tight access depths. At both $V_j = Y_j$ and $V_j = X_j$, for every fixed k ,*

$$\lim_{A \rightarrow \infty} \limsup_{j \rightarrow \infty} Q_j \left[\sup_{t \in \mathcal{R}_{j,k}(V_j)} \frac{\text{dist}(t, \mathcal{C}_j)}{r_j(V_j)} > A \right] = 0. \quad (11.5)$$

Finite access witnesses and finitely many actual colored edges can be chosen simultaneously so that their access depths are tight and all their within-line time offsets collapse after coarsening. The limiting colored graph contains every edge which is witnessed by an actual link at bounded reach and bounded line coordinates at the coarsened level.

Proof. Take a countable joint subsequential limit of all truncated packing counts for integer k, H, M . On this common product probability space write $P_{k,H}$ for the extended-integer limits. Their monotonicities are preserved. Set $A_k = \lim_{H \rightarrow \infty} P_{k,H}$ and $A = \lim_{k \rightarrow \infty} A_k$. Equation (11.4), first with finite truncations, gives $EA \leq C$. Thus A is finite, and the increasing integer sequence A_k equals A eventually on each sample.

At a candidate seed level, construct an H_j -separated greedy set by processing $\mathcal{R}_{j,1}, \dots, \mathcal{R}_{j,k_j}$ in that order. Within each stage add points until maximal, using a fixed measurable ordering of the finite lattice set. Let $G_{j,k}$ be its size after stage k . Maximality and the nearest-center assignment imply

$$P_{j,k,3H_j} \leq G_{j,k} \leq P_{j,k,H_j}, \quad k \leq k_j.$$

Choose H_j and k_j sufficiently slowly that $G_{j,k} \Rightarrow A_k$ for each fixed k , while the final count $K_j = G_{j,k_j} \Rightarrow A$, jointly with the original fixed-cutoff array. Explicitly, at stage d choose a finite height cutoff large enough to approximate the first d limiting counts and their threefold cutoffs, then choose j_d so their joint laws for $j \geq j_d$ are within $1/d$ of those limits; let the stage index tend to infinity only through these thresholds. A further slow stage choice handles the depth cutoff. No convergence rate is required.

Let D_j be the largest access depth of a selected center. The greedy construction gives $\{D_j > k\} = \{G_{j,k} < K_j\}$. In the joint integer limits the two counts are ordered and have marginals A_k and A . Since these marginals approach the same finite law, their probability of inequality tends to zero. Thus D_j is tight.

In a further joint limit write K for the limit of K_j and retain the entire fixed packing array. On $D_j \leq d$ the selected centers are an H -packing in $\mathcal{R}_{j,d}$ for every fixed H and all large j . Taking j , then H , then d to infinity shows $K \leq A$ almost surely. Their laws agree, so $K = A$ almost surely. For fixed k, d, H , a point of $\mathcal{R}_{j,k}$ at distance exceeding $Hr_j(Y_j)$ from all centers, on $D_j \leq d$, adds one point to their packing. Therefore

$$Q_j[\text{coverage fails at } H] \leq Q_j[D_j > d] + Q_j[P_{j,\max(k,d),H}(Y_j) \geq K_j + 1].$$

In the joint limit the second probability tends to zero as $H \rightarrow \infty$, because its packing count decreases to $A_{\max(k,d)} \leq A = K$. Let $d \rightarrow \infty$. This proves the seed version of Equation (11.5).

Choose $L_j \rightarrow \infty$ with $L_j = o(T_j \wedge \log H_j)$. The old centers remain separated by $H_j e^{-L_j} \rightarrow \infty$ in the new units, and their access depth stays tight by Equation (11.3). Independence of the seed level from the selected environment gives

$$\|\mathcal{L}(\omega_j, X_j) - \mathcal{L}(\omega_j, Y_j)\|_{\text{TV}} \leq L_j/T_j. \quad (11.6)$$

To verify the inequality, condition on ω_j, U_j and translate the uniform interval for Y_j . Use an outer buffer of length L_j , or discard its vanishing relative length. The full limiting packing count B at X_j consequently has the law of A . Tight access and preserved separation give $K \leq B$. Equality of the finite laws gives $K = B$. The preceding extra-point argument now proves coarse coverage. It also covers bounded explorations started in compact ranges around centers, by the composition property of reachability.

We detail the simultaneous edge selection. Label centers chronologically. For each fixed d , assign points of $\mathcal{R}_{j,d}(Y_j)$ to their nearest center and record all color-and-unordered-label-pair bits witnessed by actual links with both endpoints in that set. Call the graph $G_{j,d}$. With centers fixed, these graphs increase with d . Coverage and diverging separations make each endpoint's containing label unique after a tight coordinate truncation. A countable joint extraction, first truncating K_j , gives increasing graphs G_d on K labels. Their union G stabilizes because it has at most $2\binom{K}{2}$ bits.

A slow cutoff attains this union in law. For each bit choose its witness at its first available depth. Monotone finite stabilization implies that the maximum of these first depths is tight, exactly as for D_j . Include all endpoints and access chains in the same list; they occur in one environment, so this is simultaneous realization. Seed coverage makes all their within-line offsets tight. Since $U_j \rightarrow \infty$ in probability, those offsets and every bounded seed jump vanish in coarse units.

Finally recompute the identical greedy-center and earliest-edge recipe at X_j . Equation (11.6) gives the same limiting joint profile of line and edge counts. The recomputed centers and the old centers match uniquely in chronological order: each system has tight access, covers the other by coarse coverage, and has divergent separations; the equal saturated counts turn the resulting injection into a bijection. All preserved seed edges belong to the recomputed full coarse graph. The two finite edge counts have the same law, so inclusion forces equality. Any additional bounded-reach actual coarse edge could be united with the existing witnesses and would add a bit, a contradiction. All auxiliary graph cutoffs are incorporated into the earlier slow diagonal. After any fixed probability truncation their lists are bounded and eligible for Lemma 11.4. $\square$

Before drawing U_j , round each selected origin to a deterministic even representative of its terminal address cell. The displacement is $O(r_j(Y_j))$, so it is an additional bounded seed jump, tends to zero in coarse units, and does not alter the preceding conclusions. The rounded origin is measurable from the code. Thus observing a block at a fixed offset from it incurs no unrecorded cost for specifying an exact lattice time. This rounding is performed before the origins are frozen.

We have obtained time coverage and actual-link saturation. To use them under the selected law, we next put every seed recipe in one code. This will pay for observations at a uniformly chosen scale while keeping its origins fixed during coarsening.

Lemma 11.6 (One code for all seed cells). *The finite-line law of Proposition 11.5 can be realized, up to a total-variation error tending to zero, by first coding the same seed recipe at every seed of one sparse translated mesh and only then selecting a seed cell and its coarsened level. On a fixed finite-chain typicality cutoff, this joint code has expected length $O(N_j)$ and expected total address occupancy $O(N_j)$. For every fixed finite off-origin band request, its selected-view conditional slot entropies have bounded sum after a deletion of arbitrarily small probability. The selected sum of normalized central oscillations has bounded first moment. Origins remain frozen before the uniform coarsening.*

Figure 4 shows the sampling order and the seed-generation trials used in the entropy average.

Proof. Seed recipes and one code. Fix the address ratio R of Definition 10.6, and put $\lambda = \log R$. Round L_j to an integer multiple $m_j^* \lambda$, retaining $L_j \rightarrow \infty$ and $L_j = o(T_j \wedge \log H_j)$. Sample an independent logarithmic mesh displacement $V_j \sim \text{Unif}[0, L_j]$. Retain the complete cells $[y_p - L_j, y_p]$ in I_j , where the endpoints y_p run through this translated mesh. Their number P_j^* satisfies $P_j^* L_j = T_j + O(L_j)$.

At every y_p , run exactly the deterministic greedy-center and earliest-edge recipe in Proposition 11.5, with the same fixed orderings and diagonal cutoffs. Temporarily truncate at an integer d_j : retain the recipe only if its center count, all access depths, all earliest-edge witness depths, and all witness-endpoint distances from their assigned centers in seed units are at most d_j . Otherwise replace that seed's list by the first distinguished base. Include every access chain, every chosen actual colored link, and a time-jump entry associating each endpoint of an earliest-edge witness with its assigned seed center. The latter jump has length at most $d_j r_j(y_p)$. These entries record the same center labels as the saturated graph, rather than merely their separate access from bases. There are at most $2^{\binom{d_j}{2}}$ colored bits, and each access chain has at most d_j entries. Thus a

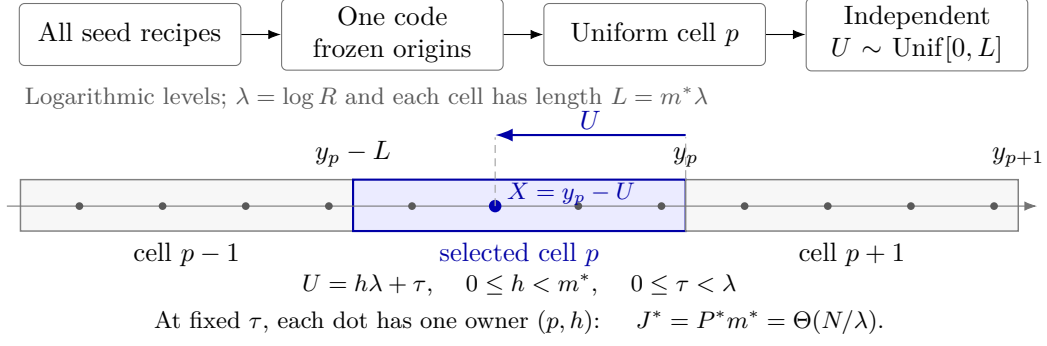


Figure 4: Sampling in Lemma 11.6, with the sequence index omitted. All seed recipes are chosen before the independent chronological grid shift, the common address code, and origin rounding. The uniform cell and coarsening are sampled afterwards; the selected time origins remain fixed as U varies. The dots show the coarsened levels at one fixed fractional phase $\tau = U \bmod \lambda$. Each is assigned to one seed cell, so the $O(N)$ code budget is averaged over J^* trials. All complete cells participate, including those assigned the fallback recipe.

deterministic bound $\Lambda_j = C(d_j + 1)^3$ bounds the total marker and edge entries per seed and its jump tolerance. It includes the distinguished base names.

The enclosing chronological interval is part of the master event in Lemma 11.1. Use its window indexed by a deterministic composition bound $b(d_j)$ for all the retained access chains, association jumps, and the finitely many current band requests. Choose the actual cutoff slowly enough that $b(d_j) \leq D_j$. Write $\hat{C}_j = C_{b(d_j)}$. A seed leaving $[-\hat{C}_j n_j, \hat{C}_j n_j]$ is also replaced by the base. Every fixed recipe is eventually covered, since the selected laws are already supported on the master event. No ordinary high-probability chronological bound is transferred through the growing density bound in Equation (11.1).

All lists have now been chosen, including the association jumps. Only next sample the independent chronological grid displacement of Definition 10.6. Form *one* projected forest from their union, marking every chain through its seed probe depth and the distinguished bases through the final depth. The seed depth is rounded to the nearest address generation; its time unit differs by a factor at most R , and different seeds still have distinct depths. Round origins to the fixed even representatives inside their coded terminal cells. This is a function of the resulting code and grid; it makes no new exploration or witness choice. All projections of the original chains and association jumps are already present at every ancestor generation.

Code and occupation budgets. Choose $d_j \rightarrow \infty$ sufficiently slowly that

$$\frac{\Lambda_j \log(\Lambda_j L_j)}{L_j} \rightarrow 0. \quad (11.7)$$

Also require $\Lambda_j \log(2\hat{C}_j)/L_j \rightarrow 0$, by slowing d_j further. At each fixed cutoff the address root interval is a fixed multiple of n_j , as required in Definition 10.6. Along the diagonal its enlarged root interval adds at most $O(1 + \log_R(2\hat{C}_j))$ ancestors per marker. The crude total extra occupancy and code cost is therefore $O(\Lambda_j(N_j/L_j) \log(2\hat{C}_j)) = o(N_j)$. For every fixed ancestor padding g , the total number of markers and the structural deletion error are then

$$M_j \leq \Lambda_j P_j^* + O(1) = O(\Lambda_j N_j / L_j) = o(N_j), \quad C_{R,g} M_j = o(N_j).$$

Lemma 10.10 bounds the integrated jump error by $O_R((1 + \log_R \Lambda_j) \Lambda_j N_j / L_j) = o(N_j)$. This count includes the endpoint-association jumps. Write T_j for the forest's total occupied-node count. The exact code-count bound of Lemma 10.13 is

$$(\lambda + C_0) \mathsf{T}_j + \Lambda_j P_j^* \log(1 + \mathsf{T}_j / P_j^*) + C P_j^* \Lambda_j \log(\Lambda_j + 1) + O(\log(\mathsf{T}_j + 1)).$$

Equation (11.7) and Equation (10.21) bound this by $C_R \mathsf{T}_j + o(N_j) + O(1)$. Proposition 10.16 applies to these simultaneous lists before the grid displacement; its constant is independent of the fixed list cutoff. Its exponential tail and Equation (11.1) give

$$E_{Q_j} \mathsf{T}_j = O(N_j), \quad E_{Q_j} \ell_j(C_j) = O(N_j). \quad (11.8)$$

The Kraft code includes its occupancy-size cost. Enlarging the slow diagonal keeps these estimates and every prescribed fixed-cutoff convergence. No event requiring *all* seed recipes to pass their truncation has been conditioned on.

The selected view law. Choose p uniformly among the P_j^* complete cells and, independently, $U_j \sim \text{Unif}[0, L_j]$. Set $Y_j = y_p$ and $X_j = Y_j - U_j$. Averaging the independent mesh displacement and the uniform cell makes the law of Y_j differ from the uniform law on I_j by $O(L_j/T_j)$ in total variation, jointly with the word and all its single-seed recipe data. To see this, extend a bounded test function by zero outside I_j and average its mesh sum; every interior point receives density $1/(P_j^* L_j)$, and the discarded end intervals have total length at most $2L_j$. The denominator differs from T_j by at most $2L_j$. The mesh is independent of the word. Tightness of the single-seed center count, first witness depths, and centered endpoint offsets therefore makes the probability of fallback at this uniformly selected seed tend to zero as the cutoffs are exhausted. This proves the claimed equality of selected laws up to vanishing error. Choosing only among successful seeds would not have this property.

One trial at each generation. Write

$$U_j = h\lambda + \tau, \quad h \in \{0, \dots, m_j^* - 1\}, \quad \tau \in [0, \lambda).$$

The phase τ is independent of the entire code and is sampled after it. Conditional on this phase and the mesh, the pair (p, h) is uniform on $P_j^* m_j^*$ possibilities. The levels $y_p - h\lambda$ are all distinct consecutive address generations, up to their common fixed fractional phase. Consequently

$$J_j^* = P_j^* m_j^* = \Theta(N_j/\lambda). \quad (11.9)$$

At such a generation use the frozen origins belonging to its unique seed cell. Distinct selected centers are separated by $H_j e^{-L_j} \rightarrow \infty$ in viewing units, so they occupy distinct location nodes for all large j . A fallback uses only one base. Thus the selected center occurrences inject into occupied forest nodes at each generation. One does not count a new copy of a node for each other seed whose chain also projects there.

Central oscillations. Corollary 10.20 and Equation (11.8) give $E_{Q_j} \sum_v S_v = O(N_j)$ for the same forest. The injection of the selected centers into its nodes and division by J_j^* give the asserted bounded first moment of the selected central amplitudes. The estimate holds on every fixed enlargement of the selected time neighborhoods. In particular, normalized start-height differences between blocks on the same line are tight before any passage transfer or continuity conclusion is used.

Disjoint slots and entropy transfer. For a fixed finite band request, use the enlarged bands of Remark 10.21. The bounded phase factor e^τ and the bounded seed-depth rounding factor only

change their fixed inner and outer radii. Remove a fixed number of generations near each seed endpoint; their selected probability is $O(1/L_j)$. The terminal rounding and association offsets then vanish in view units: for example $Q_j(d_j e^{-U_j} > \epsilon) \leq \log^+(d_j/\epsilon)/L_j \rightarrow 0$. For each fixed residue class with sufficiently spaced generations, the regular-node construction gives disjoint enlarged chronological slots. Use an arbitrarily large *fixed* ancestor depth g in that construction. Its boundary-deletion proof gives, for a constant $C_{A,R}$ depending on the requested bands,

$$\frac{E_{Q_j}\{\text{deleted selected center occurrences}\}}{J_j^*} \leq o_j(1) + C_{A,R} R^{-g} \frac{E_{Q_j} \mathbb{T}_j}{J_j^*}.$$

Indeed, conditional on the marker list, boundary loss is bounded by a constant times R^{-g} times the occupancy of a fixed reference grid, which is within a factor two of every displaced-grid occupancy. The structural loss is $C_{R,g} M_j = o(N_j)$. Sending $j \rightarrow \infty$ first and then $g \rightarrow \infty$ makes the selected deletion probability arbitrarily small, with R and the underlying view law unchanged.

Given the code and the independent side information, all surviving slot boundaries are fixed. For one fixed residue, group these disjoint raw slots by their trial (p, h) , assigning no slots to a deleted request. Let h_i be the expected relative entropy of the i th chronological raw-slot kernel, conditional on the code and its entire preceding raw history, relative to a fresh iid block. Averaging over the code, Lemma 10.19 gives

$$\sum_i h_i \leq B_j := \kappa_j + E_{Q_j} \ell_j(C_j) = O(N_j).$$

There is a direct comparison retaining incoming stacks. For a trial $t = (p, h)$ with slot set S_t , keep the selected law's code marginal and all its chronological gap and nonselected-slot kernels, but replace precisely the kernels in S_t by fresh iid blocks. Call this comparison $R_{j,t}$. If $H_i(C_j, \mathcal{H}_i)$ is the conditional relative entropy before slot i , so that $h_i = E_{Q_j} H_i$, the chronological chain rule gives

$$D(Q_j \| R_{j,t}) = E_{Q_j} \sum_{i \in S_t(C_j)} H_i(C_j, \mathcal{H}_i).$$

Keep the code-dependent trial membership inside this expectation. Use finite observations first and then increase the observed history. Under $R_{j,t}$, each requested slot is fresh conditional on its actual incoming history, so the reference domination of Proposition 7.16 applies on each fixed compatible-instruction subevent. No joint conditioning on future gap histories is made; such histories could encode the earlier target slots.

Sum over trials and then the fixed finite residue classes, and divide by J_j^* in Equation (11.9). Every slot belongs to at most one trial within its residue, so the mean comparison entropy is $O(1)$. The denominator counts all seed-generation trials, including deleted or empty requests. Averaging over trials counts each generation once, regardless of its number of lines. This comparison includes any fixed finite family of the selected lines. Lemma 11.3 and the reference-slot domination transfer null tests on every fixed compact range of retained shifts, on those compatible-instruction subevents. Whenever these shifts are tight under the selected law, the compact restriction can then be removed. The separate continuum argument in Lemma 11.7 below requires no flexible-order instruction. First send the reference failure probability to zero, and then remove the arbitrary deletion probability. This divides the *all-seed* code budget by order N_j levels.

Stationarity. All seed recipes, the code, and its rounded origins precede U_j ; conditioning on them leaves U_j uniform. The fixed-origin total-variation estimate of Lemma 11.8 is therefore unchanged. $\square$

11.3 Continuum laws in fresh slots

The raw-slot entropy comparison gives the first Brownian null-event statements without compatible passage-chart instructions. A fresh band before each retained block supplies a Gaussian increment to its starting height. We use this smoothing only for continuum paths and heights; exact lattice observations will require the separate guarded comparison.

Lemma 11.7 (Continuum transfer through fresh safety bands). *Consider a fixed finite list of chronological disjoint raw-word slots at a common time scale $r_N \rightarrow \infty$. Their boundaries and retained block cuts are fixed by a code and auxiliary side information before the fresh letters are sampled. Each retained block has a left safety band inside its slot whose rescaled duration is bounded above and away from zero; retained durations satisfy the same bounds. Let R_N refresh these slots with iid raw words conditional on their actual incoming histories, keeping all other chronological kernels evaluated at the evolving past.*

Record the retained increment paths and a fixed forest of relative block-start heights in each coordinate, in units $\sqrt{r_N}$. Every subsequential continuum law under R_N , restricted to finite recorded differences, is locally dominated by independent Brownian increment paths and independent forest differences with positive smooth densities. The reference retains the marginal law of the finite duration and forest parameters. Domination on a compact set of differences is uniform in the incoming histories.

If selected laws Q_N satisfy $D(Q_N \| R_N) \leq H < \infty$, also allowing this bound after adjoining and averaging the sampled trial, their retained increment paths are tight and every joint continuum limit has the same absolute continuity on finite differences. The assertion applies to a selected subprobability on which those differences are tight. It is a statement about continuum contour data, not about arbitrary scale-dependent lattice observations.

Proof. Fix the forest pattern. In each of its components replace the tree differences by differences from its chronologically earliest block. This is an invertible integer linear change of coordinates with determinant of magnitude one. At each later block there is now at most one new difference per coordinate, relative to an already observed block. A pattern chosen later can be handled by proving the assertion for each fixed pattern and then restricting the limiting measures.

Anchor the contour before all slots. Write a_i^N for the beginning of slot i , u_i^N for its retained-block start, and $K_i \subseteq \{1, 2\}$ for the coordinates with a new difference. If $p(i, c) < i$ is the earlier block used in coordinate c , set

$$A_{i,N}^c = \frac{Z_c(a_i^N) - Z_c(u_{p(i,c)}^N)}{\sqrt{r_N}}, \quad G_{i,N}^c = \frac{Z_c(u_i^N) - Z_c(a_i^N)}{\sqrt{r_N}}.$$

The first vector is measurable before the fresh slot. The new retained difference satisfies the exact identity

$$D_{i,N} = A_{i,N} + G_{i,N}.$$

Let $X_{i,N}$ be the retained increment path, parametrized on $[0, 1]$. Proposition 4.3, applied to the entire fresh safety band and retained block, gives conditional bounded-Lipschitz convergence

$$(G_{i,N}, X_{i,N}) \implies (G_i, B_i), \quad G_i \sim N(0, \tau_i \Sigma|_{K_i}), \quad B_i \sim \nu_{t_i}.$$

Here Σ is the Brownian covariance matrix, τ_i and t_i are the limiting safety and retained durations, and ν_{t_i} is the Brownian increment-path law of duration t_i . The Gaussian G_i and the path B_i are independent. Convergence is uniform over incoming stacks and compact duration ranges: the

arbitrary-typing error is uniform, and deterministic restrictions and time changes of the whole-slot limit give the displayed pair. No compatible flexible-order instructions are needed.

The density of G_i is bounded uniformly because τ_i is bounded away from zero. Give the K_i -coordinate difference an independent positive smooth reference density p_i . For a nonnegative test f supported on $|d| \leq M$ and any translation A_i ,

$$\mathbb{E}f(A_i + G_i, B_i) \leq C_M \int f(d, x) p_i(d) dd \nu_{t_i}(dx).$$

For bounded-Lipschitz tests the prelimit inequality has a uniform $o(1)$ error. Indeed translation by A_i changes neither the bound nor the Lipschitz constant of the test. Empty K_i requires only the Brownian increment-path comparison.

Apply this bound to one nonnegative bounded-Lipschitz function of *all* retained outputs, supported on a compact set of their differences. Start at the last fresh slot and work backwards. Earlier outputs are fixed at each conditional step. Integrating the current output against its independent reference law leaves a test with the same bounded-Lipschitz control in the earlier outputs. The intervening causal gap kernels integrate to one; they may depend on earlier target slots, but their future realizations are never conditioned on jointly. This backward iteration bounds the limiting expectation by a fixed multiple of the product Brownian and smooth-height reference expectation, including the common parameter marginal. Bounded-Lipschitz approximation then gives local domination of the joint continuum measures.

If a retained raw record includes a left extension before its used start, place the fresh safety band before the entire record. The used-start difference is then exactly $D = A + G + \Phi(X)$, where Φ is the retained extension's endpoint evaluation. Conditional on the retained Brownian path this is another translation of the same Gaussian. Endpoint evaluation is Lipschitz in the uniform path metric, so the preceding argument is unchanged.

The comparison does not assume tightness of incoming heights. Explicitly, on $|D_{i,N}|_\infty \leq M$ and $|A_{i,N}|_\infty > K$, the fresh safety fluctuation is at least $K - M$. Proposition 4.2 and its deterministic oscillation domination therefore give, uniformly over the incoming history,

$$R_N[|D_{i,N}|_\infty \leq M, |A_{i,N}|_\infty > K] \leq Ce^{-c(K-M)}, \quad K > M.$$

With a used-start extension, apply the same bound to the safety band and retained prefix together. Thus escaped incoming heights produce no unaccounted mass at finite retained differences.

Finally compactify the recorded differences to $[-\infty, \infty]$. The retained path laws are tight under R_N by the uniform fresh-slot limit, and under Q_N by the binary entropy bound. First adjoin the sampled trial under both laws; their common trial marginal turns the averaged bound into joint entropy. Then apply data processing to the finite duration and forest parameters, retained paths, and compactified differences, projecting away the trial identifier. A joint weak extraction of these two observation laws gives, by lower semicontinuity of relative entropy,

$$D(Q^{\text{obs}} \| R^{\text{obs}}) \leq H < \infty, \quad Q^{\text{obs}} \ll R^{\text{obs}}.$$

The finite-difference part of R^{obs} has the local domination proved above. Exhausting finite compact difference sets proves the selected assertion. For a restriction by an additional event, adjoin its indicator in the extraction; its limiting observation submeasure is bounded by Q^{obs} and inherits the same absolute continuity on its finite-difference part. Undo the forest coordinate change. Countably many buffered requests give the simultaneous off-origin null-event assertions used below. This reasoning uses weak limits of contour paths; it supplies no prelimit domination for discontinuous lattice passage costs. $\square$

11.4 Frozen origins and dilation stationarity

We next extract the contour profiles around the chosen centers. Their origins are fixed before the viewing radius changes. This gives a stationary dilation law and, together with the oscillation estimate, continuity even at a selected origin.

The all-seed construction supplies two different controls for this step. The expected sum of the selected central oscillations is bounded. For each fixed off-origin band request, the selected experiment also has bounded average relative entropy with respect to the trial law $R_{j,t}$, which refreshes only the requested slots and leaves the other chronological kernels unchanged, evaluated at the evolving past. We first use the fresh-slot Brownian limit to control continuum increments and finite relative heights. Exact lattice passage observations will additionally require the compatible-instruction domination of Proposition 7.16. For the later annular crossings we must also express each test using only the retained data and prove that those data admit an ordinary filling of positive weight. The following extraction establishes the contour and height control needed for these geometric steps.

For a finite ordered list of times $\mathbf{t} = (t_1, \dots, t_k)$ and a pair of contours Z , define the centered paths at time unit r by

$$Z^{i,r}(u) = r^{-1/2}(Z(t_i + ru) - Z(t_i)).$$

The view also records the relative center heights in each coordinate in $[-\infty, \infty]$, the quantities $r^{-1/2}(\min_{[t_i, t_\ell]} Z^c - Z^c(t_i)) \in [-\infty, 0]$ for every ordered center pair and coordinate c , and the closed scalar minimum relations on every finite collection of bounded time windows. Chronological ordering labels the lines. Dilation never changes these labels. On a fixed line count the dilation action is

$$(\mathcal{D}_v Z)^i(u) = e^{-v/2} Z^i(e^v u), \quad \mathcal{D}_v H_{i\ell} = e^{-v/2} H_{i\ell}.$$

The relation data are transformed by the same change of time variables.

Lemma 11.8 (Uniform coarsening with fixed origins). *Let a seed level Y_j , a word, and a finite origin list $\mathbf{t}_j$ be sampled first, with arbitrary dependence. Independently sample $U_j \sim \text{Unif}[0, L_j]$, where $L_j \rightarrow \infty$, and form the view at time unit $r_j(Y_j - U_j)$. For every fixed $v \in \mathbb{R}$, its law and the law of its $\mathcal{D}_v$ transform differ in total variation by at most $|v|/L_j$, provided both are defined. If boundary truncation fails with probability δ_j , add $2\delta_j$ to this bound. Every weak limit on which the dilation action is continuous is dilation stationary.*

Proof. Condition on everything sampled before U_j . Replacing the time unit by e^v times itself replaces U_j by $U_j + v$, with the same origins. Uniform measures on $[0, L_j]$ and $[v, L_j + v]$ differ in total variation by at most $|v|/L_j$. Pushforward cannot increase total variation. Integrate the conditional inequality and remove the boundary exceptional events. For a weak limit, test bounded continuous functions and their compositions with $\mathcal{D}_v$. First take rational v ; continuity gives all real v . $\square$

The freezing of the origins is necessary. For example, let $f(t) = \sqrt{|t|}$, set $r = e^{-Y}$ and choose $t(Y) = r$. Then $[f(t(Y) + ru) - f(t(Y))]/\sqrt{r} = \sqrt{|1+u|} - 1$ for every Y . Uniform sampling of Y therefore gives a constant law which is not dilation stationary. Lemma 11.8 keeps the origins fixed during coarsening, while Proposition 11.5 ensures that they continue to cover the required exploration.

Lemma 11.9 (Consequences of pinned stationarity). *Let a finite chronologically labeled line system have a dilation-stationary law. Assume its centered coordinate paths are continuous, and assume that on each line distinct off-origin local minima of one coordinate almost surely have distinct heights. Then:*

- (i) *every recorded center-height difference belongs to $\{-\infty, 0, +\infty\}$ almost surely, and every recorded normalized whole-center-gap minimum belongs to $\{-\infty, 0\}$;*
- (ii) *on each time half-line, a coordinate either is strictly positive everywhere relative to its value at the origin, or takes strictly negative values arbitrarily close to the origin;*
- (iii) *if a coordinate is strictly positive sufficiently far out on a half-line, it is strictly positive on the entire open half-line.*

These conclusions hold simultaneously for the finite random system.

Proof. For (i), the law of H equals the law of $e^{-v/2}H$. The disjoint geometric annuli $(e^m, e^{m+1}]$, $m \in \mathbb{Z}$, have equal probability. Each therefore has probability zero. Apply the same argument to $-H$, and to each normalized minimum, which transforms by the same scalar multiplier.

Consider the positive half-line and put $\tau = \inf\{t > 0 : Z(t) < 0\}$, with the infimum of the empty set equal to ∞ . Its law is invariant under multiplication by e^v , so $\tau \in \{0, \infty\}$ almost surely by the same argument. If $\tau = \infty$, the path is nonnegative. Any zero at positive time is then a local minimum. There can be at most one such zero, by the distinct-minima hypothesis. The time of that sole zero, if it exists, again has a multiplicatively invariant distribution on $(0, \infty)$, so it cannot exist. This proves (ii). For (iii), apply the argument to $\sup\{t > 0 : Z(t) \leq 0\}$, with the empty-set convention zero. Eventual strict positivity makes this supremum finite, so invariance forces it to be zero. Reflect time for the other half-line. Truncate the line count, apply the argument to each coordinate and each label, and then remove the truncation. $\square$

The distinct-minima hypothesis follows from joint off-origin Brownian absolute continuity on countably many separated compact intervals. The next Lemma establishes continuity at the origins, and Lemma 11.12 identifies the limiting center contacts.

Lemma 11.10 (Continuity at the frozen origins). *Use the all-seed sampling construction of Lemma 11.6. Suppose the code and oscillation estimates on that marked address list give $E_{Q_j} \sum_v S_v \leq CN_j$, where the sum runs over its projected nodes and S_v is the contour oscillation on a fixed enlargement of the node box divided by the square root of its time radius. Suppose also that the disjoint-slot entropy estimate applies to every fixed finite family of time bands at positive distance from the rounded origins and to their recorded forest shifts. Then the finite-line extraction can be made with continuous centered contours. On off-origin blocks, every joint continuum law of increments and any recorded forest of tight finite relative heights is absolutely continuous with respect to independent Brownian blocks and smoothly distributed forest differences. In particular the dilation-stationary limit satisfies Lemma 11.9.*

Proof. Amplitude and off-origin control. Division by the order- N_j trial count in Equation (11.9), using the selected-node injection of Lemma 11.6, bounds the expected sum over lines of the normalized oscillations on every fixed enlargement. The projected boxes contain the rounded origins and the requested bounded time intervals. Thus, for every fixed $A < \infty$, the centered amplitudes on $[-A, A]$ are tight. Relative start heights between blocks on the same line are differences of these amplitudes and are tight as well. This is the source of the compact height truncation; it is not an inference about the entire selected past.

Truncate the line count and fix finitely many bands with rational endpoints avoiding zero. Coarse rounding errors are covered by slightly enlarging these bands. Along each unary part of the projected address tree, choose a fixed residue class of logarithmic levels sufficiently spaced that their enlarged bands are disjoint. Increase the ancestor padding by the fixed amount required for these bands. The regular-slot estimate has arbitrarily small deletion probability by the fixed-padding exhaustion in Lemma 11.6; the residue is chosen uniformly and independently, preserving the level marginal. Conditional on the code and independent phase, the surviving raw-word slots have fixed disjoint boundaries. Give each tested block a deterministic left safety band of positive rescaled duration within its enlarged off-origin slot. Apply Lemma 11.7 to the freshened trial experiment of Lemma 11.6. It gives punctured compact-uniform tightness and joint continuum absolute continuity without choosing flexible-order instructions. The Gaussian safety increment smooths each new forest difference; the following retained increment path is Brownian in the reference limit. The bounded average trial entropy transfers its null events. Tightness of the tested finite differences allows their compact ranges to be exhausted. The same argument applies to every additional fixed forest whose differences are tight on the restricted subprobability under study. Exact lattice observations will instead use Proposition 7.16 on compatible protected charts later in the argument.

Control at the origins. For the fixed unit enlargement write the preceding expectation bound as C' , and put $M_j(\delta) = \sum_i \sup_{|u| \leq \delta} |Z_j^i(u)|$. Apply Lemma 11.8 before taking a limit. For fixed $\delta, \epsilon > 0$, scaling the frozen view by δ gives

$$Q_j[M_j(\delta) > \epsilon] \leq Q_j[M_j(1) > \epsilon/\sqrt{\delta}] + O(|\log \delta|/L_j) + o_j(1) \leq C' \sqrt{\delta}/\epsilon + O(|\log \delta|/L_j) + o_j(1).$$

Thus $\lim_{\delta \downarrow 0} \limsup_j Q_j[M_j(\delta) > \epsilon] = 0$. Together with tightness on punctured compacts, this is full compact-uniform tightness, including the origins. It excludes narrow prelimit spikes which punctured convergence alone would miss. In a joint subsequential limit, Fatou's lemma gives

$$E \sum_{i=1}^K \sup_{0 < |u| \leq 1} |Z^i(u)| \leq C'.$$

Lemma 11.8 applies on the punctured-line path space. Consequently, for $0 < \delta < 1$,

$$E \sum_{i=1}^K \sup_{0 < |u| \leq \delta} |Z^i(u)| = \sqrt{\delta} E \sum_{i=1}^K \sup_{0 < |u| \leq 1} |Z^i(u)| \leq C' \sqrt{\delta}.$$

The left-hand suprema decrease as $\delta \downarrow 0$, so their limits are zero almost surely. Set $Z^i(0) = 0$. This proves continuity at every origin, together with the already obtained continuity elsewhere. Countably many bands, increasing line truncations, and the original packing and graph statistics can all be included in the same diagonal. $\square$

Corollary 11.11 (Tightness of retained height differences). *In the preceding stationary extraction, fix a coordinate and group centers whose normalized height differences have finite limits. Within each such group those limits are zero. Every finite forest of relative start heights between retained off-origin blocks in the group is tight, and its joint continuum law with the block increments has the reference absolute continuity of Lemma 11.10.*

Proof. Write Z_j for the chosen scalar contour, $c_{j,i}$ for a center, and r_j for the viewing time unit. For block starts at coordinates u and v on lines i and ℓ , their normalized height difference is

$$r_j^{-1/2} (Z_j(c_{j,\ell}) - Z_j(c_{j,i})) + Z_j^\ell(v) - Z_j^i(u).$$

The first term tends to zero within a finite relative-height group by Lemma 11.9(i). The remaining terms are tight by the compact centered-profile bounds. For a random group pattern, these assertions concern the corresponding restricted subprobability law; stationarity is used under the original law before this restriction. After truncating the line count there are finitely many group and forest choices; applying the off-origin transfer to each gives the joint assertion. No finite relative height between different groups is needed. $\square$

11.5 First returns and protection of omitted gaps

Use a joint contour coupling after Proposition 11.5 and Lemma 11.10. Write $c_{j,i}$ for the ordered centers and $r_j = r_j(X_j)$. Record center differences and all intervening minima in the extended real line. Centers $i < \ell$ have a scalar center contact when their normalized height difference tends to zero and

$$\liminf_j r_j^{-1/2} \left(\min_{[c_{j,i}, c_{j,\ell}]} Z_j - Z_j(c_{j,i}) \right) \geq 0. \quad (11.10)$$

For a given coordinate these contacts form the ordinary scalar contour equivalence relation on the finite ordered center set. Every scalar center class lies in one finite relative-height group, so Corollary 11.11 applies to its retained blocks. Our next task is to realize its contacts by actual lattice chords and to keep the omitted chronological gaps strictly above the tested levels.

Lemma 11.12 (Actual return witnesses and whole-gap clearance). *Assume each bilateral scalar contour eventually returns below each fixed level in either time direction. In the saturated stationary system:*

- (i) *Every strictly positive half-line faces another member of the same scalar center class. Each inward half-line of a scalar class is strictly positive and its two outward half-lines have negative values arbitrarily near their origins.*
- (ii) *Consecutive members of each scalar class are joined by actual lattice chords whose endpoints approach the corresponding origins in view units. Thus the scalar rank of the center class is witnessed by the simultaneously saturated actual-link graph.*
- (iii) *Fix positive neighborhoods of all enrolled origins within their retained line intervals. Between consecutive members of a scalar center class, after removing those neighborhoods, the scalar height has positive clearance above the class height on the whole omitted chronological gap, in probability: its normalized positive margin has tight reciprocal after any fixed finite geometric truncation.*

Proof. Consider a positive right half-line of center i and fix a departure time $d > 0$. Choose deterministic $\epsilon_j \downarrow 0$ sufficiently slowly to dominate contour errors on every currently fixed compact and the vanishing differences of tied center heights. Round $Z_j(c_{j,i}) + \epsilon_j \sqrt{r_j}$ to an attainable scalar lattice level h_j . Its rounding error divided by $\sqrt{r_j}$ vanishes. Let a_j be the last visit to h_j before $c_{j,i} + dr_j$, and let b_j be the first visit thereafter. The first exists because the center height is below h_j and the departure height is above it; the second exists by bilateral recurrence. The last/first convention gives

$$Z_j(a_j) = Z_j(b_j) = h_j, \quad \min_{[a_j, b_j]} Z_j = h_j.$$

This is an actual scalar chord. Strict positivity on each compact $[\eta, d]$ implies $(a_j - c_{j,i})/r_j \rightarrow 0$.

The first endpoint is reached by one bounded jump, and the second by one chord. Universal coverage in Equation (11.5) therefore puts b_j within a tight coordinate range of an enrolled line.

Choose a joint convergent subsequence of its label and coordinate, denoting the latter by y . The receiving line cannot be i , by positivity on all its positive compacts. It is a later line, say ℓ . The equality at b_j shows that the center-height difference is finite; Lemma 11.9(i) makes it zero, and the receiving profile satisfies $Z^\ell(y) = 0$. Every finite receiving coordinate $u < y$ occurs between the departure and the return for large j , so $Z^\ell(u) \geq 0$.

We claim $y = 0$. If $y < 0$, nonnegativity on $(-\infty, y)$ and distinct off-origin minima imply strict positivity sufficiently far to the left; Lemma 11.9(iii) then gives strict positivity on the entire left half-line, contradicting $Z^\ell(y) = 0$. If $y > 0$, nonnegativity on $(0, y)$ rules out the negative-near-zero alternative on the right half; strict positivity there again contradicts the zero. Hence $y = 0$ and the receiving left half is positive. The chord inequalities also give Equation (11.10), so i and ℓ are in the same scalar class. Reflect time for a positive left half.

Conversely a center contact makes the corresponding inward half-lines nonnegative, hence positive by Lemma 11.9. A positive outward half would, by the preceding return construction, produce an additional class member beyond the extreme one. The outward halves must therefore have negative values arbitrarily near zero. This proves (i).

For consecutive class members $i < \ell$, choose the errors ϵ_j also to exceed all their vanishing center-height differences. The first return from i occurs before $c_{j,\ell}$, whose height is below h_j . The preceding argument places any limiting return at an intermediate center in the same class. Consecutiveness forces it to be ℓ , and its coordinate tends to zero. These are actual finite-reach witnesses. Their finite union can be included in the edge extraction of Proposition 11.5. This proves (ii), including the rank $|A| - 1$ from consecutive pairs of a scalar class A .

For (iii), suppose instead that a whole omitted gap has normalized minimum approaching zero with probability bounded below. Record that event in a joint extraction, retaining the original unconditioned stationary law; do not condition the stationary law on the event. Take $\epsilon_j \downarrow 0$ more slowly than this minimum and the convergence errors. Start the preceding chord search at a fixed positive departure lying inside the retained interval before that gap. Its first return must occur before the low point in the omitted gap. By (ii), and by choosing the start after the last near-origin crossing of h_j , every subsequential return approaches the next class center. If the search encounters a retained intervening line, equality at the return and the minimum inequality put its center into the same class; consecutiveness excludes it. More precisely, the offending low point precedes $c_{j,\ell} - \eta r_j$ for the fixed radius $\eta > 0$ of the removed neighborhood of the next member ℓ . Hence $b_j \leq c_{j,\ell} - \eta r_j$, whereas the return argument in (ii), applied with these same slowly decreasing heights, gives $(b_j - c_{j,\ell})/r_j \rightarrow 0$. This is a contradiction. Universal coverage has already excluded escape between the enrolled lines or to an unbounded coordinate on one of them. This proves that the whole-gap minimum is bounded away from zero after a probability truncation. There are finitely many classes and gaps on each truncation, so their minimum margin has the same property. Remove the truncations and use a countable sequence of origin neighborhoods. $\square$

Corollary 11.13 (Interline partners belong to one scalar class). *An actual scalar link whose endpoints have bounded coordinates on two distinct enrolled lines has endpoints on lines in the same scalar center class in every subsequential limit.*

Proof. The equality of endpoint heights and bounded local contour oscillations make the center-height difference finite, hence zero by Lemma 11.9. The chord lower bound, with the bounded end pieces between endpoints and centers, bounds the normalized whole-center-gap minimum from below. Its limit is finite, hence zero by the same Lemma. This is Equation (11.10). $\square$

Lemma 11.14 (Protected completion of one center component). *Fix a finite combined center component from Lemma 11.12. Retain sufficiently short closed intervals about all its center visits, in their chronological order. There is a pair of positive continuous excursion contours on a finite interval containing translated copies of all these retained profiles. Its scalar center classes are exactly the prescribed ones. Each translation is common to a scalar class, and every omitted connector satisfies strict lower inequalities protecting all links sufficiently near the center class. The component-height choices and connector conditions range over a nonempty open set. They can be sampled with densities and Brownian bridges of the prescribed covariance.*

Proof. Handle the two coordinates separately for the height assignment. A scalar partition of ordered contour visits is noncrossing: two interleaving scalar contacts are at the same height and are one class. Consequently class spans are nested or disjoint. If B lies between consecutive members of A , impose $A \prec B$. These finitely many nesting inequalities have no cycle. They also agree with the original strict height ordering: the minimum inequality for A makes a center inside its span no lower than A ; equality would make it a member of A , so a distinct intervening class is strictly higher. By stationarity its original relative height is in fact $+\infty$ in view units.

Choose rational cuts on every retained half-profile. On each outward half retain a definite negative dip before its cut. The two coordinates need not have their negative dips at the same time; take the common cut after both required dips. Every inward half is positive at its cut. Let K_0 bound all retained centered oscillations. For each coordinate choose a linear ordering of its classes extending nesting, with ranks $s(A) \in \{1, \dots, k\}$, and assign

$$H_A = (4K_0 + 10)s(A) + u_A, \quad |u_A| < 1.$$

Give every profile of class A this common added height. All pieces are positive. Every enclosed class lies above an enclosing level by more than $2K_0 + 8$. The two coordinates have distinct height variables, so these choices impose no cross-coordinate compatibility equation. Equal heights of unrelated classes need not be retained.

Choose $\delta > 0$ smaller than one quarter of the finitely many inward cut heights, chosen outward dip depths, and original protected-gap margins from Lemma 11.12. Shrink the working neighborhoods of the centers so all centered profile heights there have magnitude less than δ . At a gap between consecutive retained intervals, the classes whose spans cross that gap form a nested chain. Both cut values exceed every active class height by more than 2δ : a cut in that class lies on its inward side, and every other relevant cut belongs to a strictly enclosed class. If no class spans the gap, positivity is the only lower requirement. Coordinatewise linear interpolation joins the endpoint vectors while preserving these strict bounds. A small uniform tube about it does likewise. Exterior tails join to zero with positive interior values; ordinary quadrant-excursion entrance and exit pieces can be used in place of deterministic tails.

Inward positivity and connector bounds join all members of a prescribed scalar class at their common level. Outward dips block any contact leaving its span. No additional zero-level contact occurs on a retained half: inward halves are strictly positive and every outward interval starting at the center contains a negative value. Distinct scalar classes have distinct assigned heights. Thus the prescribed center class is saturated under both scalar relations. All inequalities use strict margins, so the height variables may have a density supported in the indicated open feasible box. Brownian bridges have positive probability of satisfying the connector tubes by Lemma 11.16. Countably many rational cuts and margins suffice for the choices. $\square$

11.6 Open fillings and crossing transfer

The completed center profiles may have singular laws at their origins. We will test passages only away from those times. The next lemmas explain why a strictly compatible ordinary filling transfers a null event on the retained data, even if its filling probability is small.

Let $I_1, \dots, I_m$ be finitely many disjoint closed intervals in a bounded time interval, with positive gaps between them. A block datum consists of the increment path on each I_i and, for each coordinate, height differences along a forest on the set of blocks. A forest supplies one independent difference per non-root vertex, so its differences and one absolute height per component parameterize all block-start heights bijectively. No cyclic list of redundant differences is assigned an independent density.

Definition 11.15 (Open fillability). A finite recipe specifies finitely many strict inequalities on endpoint heights, minima of omitted gaps, and minima or maxima of retained compact subintervals. The inequalities may depend continuously on the recorded block datum. A datum is openly fillable for the recipe if one can choose the unrecorded component heights and continuous paths in the omitted gaps so that every inequality holds. The gap paths must have the endpoint values prescribed by their adjacent blocks. All interval lengths are bounded above and away from zero on each compact parameter truncation.

Lemma 11.16 (Positive fill weight). *Fix a nondegenerate two-dimensional Brownian covariance matrix. Give the unrecorded component heights an everywhere positive continuous density, and conditionally fill each positive-length gap by the corresponding Brownian bridge. Every openly fillable datum has positive conditional probability of satisfying the recipe. Openly fillable data form a measurable set which is a countable union of versions using rational parameter boxes and rational polygonal approximations with a positive inequality margin.*

Proof. For a chosen filling, finitely many strict continuous inequalities have a common positive margin. A sufficiently small open box of component heights and a sufficiently small uniform neighborhood of the chosen continuous gap paths preserve that margin. The height box has positive probability. A Brownian bridge with nondegenerate covariance has full support on the continuous paths with its endpoints: approximate the chosen path by a polygonal path, constrain the bridge at its finitely many vertices using the strictly positive Gaussian density, and constrain its bridge fluctuations on the remaining subintervals to a small uniform ball. Each of these probabilities is positive. The same statement holds uniformly on a sufficiently small box of endpoints, although no uniform lower bound over all openly fillable data is asserted. Independent filling of finitely many gaps proves positivity. Rational approximations to the height boxes, polygonal paths, and margins give the last assertion. $\square$

One may also require positivity in the quadrant on the interior filling intervals, since it is an open condition on compact subintervals. To obtain an ordinary quadrant excursion, its initial and terminal pieces are sampled from the ordinary excursion law, while the positive-time interior gaps have the usual killed-bridge densities. Their strictly positive densities on the quadrant give the same support assertion. This does not condition free Brownian motion to reproduce any distinguished singular center profile.

Lemma 11.17 (Transfer on an openly fillable chart). *Let ν be the reference law of the recorded block data, and let $K(v, do)$ be a probability kernel for the passage observations. Suppose an ordinary Brownian completion which realizes a fixed recipe induces the subprobability law*

$$f(v)\nu(dv)K(v, do), \quad f(v) > 0 \quad \text{for } \nu\text{-almost every openly fillable } v. \quad (11.11)$$

If a measurable failure test E has probability zero in that completion, then

$$\int_{\{v: v \text{ openly fillable}\}} K(v, E_v) \nu(dv) = 0.$$

Consequently any sequence of selected chart experiments having tight absolute continuity with respect to these reference experiments transfers their vanishing failure probabilities.

Proof. The assumed null event says $\int f(v)K(v, E_v)\nu(dv) = 0$. The integrand is nonnegative and f is positive on the set in question. For instance, restrict to $f \geq 1/k$ and let $k \rightarrow \infty$. This gives the assertion. The sequential version follows from the definition of tight absolute continuity, or from Lemma 11.3 together with the required iid reference domination. $\square$

The positive filling density in Equation (11.11) is supplied by Lemma 11.16. The common passage kernel and feasible-lattice domination are supplied by the local-kernel construction of Section 7.

Lemma 11.18 (Off-center law of the completion). *Suppose the original finite-line view has the joint off-origin absolute continuity specified in Lemma 11.10, with all forest differences within each scalar center class retained. The completion in Lemma 11.14 may be sampled so that every finite collection of disjoint compact time intervals avoiding its centers and common endpoint has joint absolute continuity with respect to Brownian blocks with a full-density vector of absolute starting heights.*

Proof. First fix the countable choices of center labels, scalar center classes, rational cuts, and inequality margins. Restricting to the event of this choice preserves every null assertion. For blocks in retained profiles, record a spanning tree of block-start differences within each scalar center class, separately in each coordinate. Their union is a forest. Each such class lies in a finite relative-height group, so Corollary 11.11 supplies joint absolute continuity of these differences and the block increments. Discard differences between distinct scalar center classes, even when their center heights belonged to the same finite relative-height group before completion.

For each scalar center class, adjoin the absolute start height of one reference block. Conditional on the retained profiles, this anchor is a translate of that class's continuously randomized height and hence has a density. The within-class forest differences and one anchor per class parameterize all block-start heights by an invertible linear map with determinant of magnitude one. Fubini's theorem gives absolute continuity of all starts and increments jointly. No Brownian conditional law given a singular center profile is required.

Conditionally on the retained profiles and their heights, each filler is a Brownian bridge restricted to an event of positive conditional probability. Its law is absolutely continuous with respect to the unrestricted bridge with those endpoints. Marginalizing any unused central data preserves that absolute continuity. Brownian bridge disintegration across a deterministic cut identifies the resulting reference measure on a neighborhood crossing the cut with a Brownian restriction having the same start density. Thus the assertion holds also for intervals meeting seams. The entrance and exit pieces have the same local property away from the common endpoint. Finally take the countable union over recipe choices and rational windows covering the requested compact intervals. $\square$

11.7 A deterministic sphere theorem for the completed contours

We have protected the distinguished contacts. We must also know that the entire completed contour quotient is a sphere. The following deterministic criterion separates this topological

question from the passage-cost estimates: its two hypotheses are finite fibers and an acyclic incidence graph within each fiber.

Theorem 11.19 (Finite acyclic contour sewing). *Let $L, R : [0, T] \rightarrow [0, \infty)$ be continuous, zero at the endpoints and strictly positive in between. On the circle $\mathcal{C} = [0, T]/(0 = T)$, let $\sim_c$ be the scalar contour relation*

$$s \sim_c t \iff c(s) = c(t) = \min_{u \in [s, t]} c(u), \quad c \in \{L, R\},$$

with ordered representatives and the endpoint convention. Let $\sim$ be the relation generated by the two scalar relations. Suppose:

- (i) *a finite constant M bounds the cardinality of every $\sim$ -class;*
- (ii) *for every $\sim$ -class E , the bipartite multigraph with vertices the L -classes and the R -classes in E , and one edge for each $t \in E$ joining its two scalar classes, is a tree.*

Then $\mathcal{C}/\sim$ is homeomorphic to the sphere. The quotient map is continuous and its fibers are exactly the $\sim$ -classes.

Proof. Realize $\mathcal{C}$ as the unit circle by $z(t) = \exp(2\pi it/T)$. The scalar relation is closed, since the interval minimum is a continuous function of its endpoints. Distinct scalar classes are unlinked: if $s < t < u < v$, $s \sim_c u$, and $t \sim_c v$, the minimum inequalities imply $c(s) \leq c(t) \leq c(u) = c(s)$, and all four points belong to one scalar class. Hence the closed convex hulls of distinct scalar classes are disjoint.

We check that these hulls cover their closed disk; no unstated maximality of a lamination is needed. For a circular interval $[a, b]$ write $K[a, b] = \text{conv}(z([a, b]))$. Suppose an interior point w is in no scalar-class hull. For $h > 0$, components (a, b) of $\{c > h\}$ have scalar-equivalent endpoints. If $w \in K[a, b]$, it is in the interior of that cap, since its chord edge is already in a class hull. There is such a cap at some positive h : a sufficiently long compact arc avoiding the common endpoint contains w in its cap and has positive minimum height. Let H be the supremum of these heights. Along heights increasing to H the relevant components are nested. Their limiting endpoints a, b satisfy $c(a) = c(b) = H$, $c \geq H$ on $[a, b]$, and $w \in K[a, b]$. The finite set $A = \{t \in [a, b] : c(t) = H\}$ is contained in one scalar class. The cap $K[a, b]$ is the union of $\text{conv}(z(A))$ and the caps over successive complementary intervals of A . Their separating edges belong to the former hull. Thus w , being outside that hull, lies in the interior of a cap over an interval (u, v) where $c > H$. There are $u < u' < v' < v$ with $w \in K[u', v']$; the positive minimum of $c - H$ on $[u', v']$ gives a component cap containing w at a height above H . This contradiction proves coverage. Singleton hulls cover any remaining boundary points.

The generated relation is also closed. If $s_n \sim t_n$ and $(s_n, t_n) \rightarrow (s, t)$, join each pair by a simple monochrome chain of length at most $M - 1$. Pad it with constant steps. There are finitely many color patterns; after choosing one pattern, compactness of the finite product $\mathcal{C}^M$ supplies limits for all intermediate points. Closedness of the scalar relations gives a chain from s to t .

Glue an L -disk and an R -disk along $\mathcal{C}$. For a combined class E , let H_E be the union of its scalar hulls in the respective disks. Coverage and disjointness show that these sets partition the sphere. Each is a compact connected finite union of polygons, segments, and points. The decomposition relation on the sphere is closed: if $x_n, y_n \in H_{E_n}$ converge, choose their containing hemispheres and express each as a convex combination of at most three hull vertices. After a subsequence all six vertices and their coefficients converge. Each triple remains scalar-equivalent, and all six vertices remain combined-equivalent by the closedness just proved. The limits therefore belong to one H_E .

Each H_E is nonseparating. A polygon can be strongly deformation retracted, fixing its boundary vertices, onto the star from an interior point to those vertices: triangulate from the interior point and collapse each triangle onto its two radial sides. These collapses agree where triangles and hulls meet. Segments and singleton hulls need no change. The resulting finite graph is the subdivided bipartite incidence graph, with the harmless contractions due to singleton hulls. Hypothesis (ii) makes it a tree. Thus H_E is contractible. It is a finite planar polyhedron, so Alexander duality gives $\tilde{H}_0(S^2 \setminus H_E; \mathbb{Z}) = 0$; its complement is connected.

We have a closed decomposition into compact connected nonseparating sets. It is not the one-set decomposition, because its intersections with the infinite circle have at most M points. Moore's sphere theorem in this form, as stated in [39, Theorem 1.1], gives a sphere quotient. Every hull meets the circle, so the induced map from $\mathcal{C}/\sim$ is a continuous bijection onto that sphere. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism. $\square$

Remark 11.20 (The rank in the topological criterion). For a combined class E of size m , let v be its number of scalar classes and let $e = \sum_{A \text{ scalar class in } E} (|A| - 1)$. Every visit is counted once in each color, so $e = 2m - v$. Its connected incidence multigraph has m edges and v vertices. It is a tree exactly when $e = m - 1$, and has a cycle exactly when $e \geq m$. This includes a two-edge cycle from a nontrivial common chord. Filled polygons are not themselves called trees; the deformation argument in the proof is what converts the incidence condition into nonseparation.

Corollary 11.21 (One distinguished class). *In Theorem 11.19, suppose a finite distinguished set S is exactly one combined class, its incidence graph is a tree, and no scalar relation joins S to its complement. Suppose on the complement of S and the common endpoint there is joint Brownian absolute continuity on every finite family of separated compact time intervals, including their relative starting heights. Then the completed quotient is a sphere.*

Proof. By Lemma 4.11, a separated off-center diagram is polar when $\beta e > m$. A rank cycle has $e \geq m$ and is therefore polar. Choose an integer m_0 with $\beta(m_0 - 1) > m_0$. A connected diagram with m_0 distinct visits has $e \geq m_0 - 1$ and is also polar. An equivalence class with at least m_0 visits contains such a finite connected diagram: start at one point and add new points successively along a monochrome connection. All its points lie in some finite family of disjoint rational windows avoiding the distinguished times. A countable union of the polarity assertions therefore bounds every other class by $m_0 - 1$ and excludes its cycles. The distinguished set is saturated by assumption, and the common endpoint is a singleton by strict positivity. Apply Theorem 11.19 with $M = \max\{|S|, m_0 - 1, 1\}$. $\square$

Lemma 11.22 (Localizing an annulus by its finite fiber). *Let $\eta : \mathcal{C} \rightarrow S^2$ be the quotient from Theorem 11.19, and suppose $\eta^{-1}(z) = S$ is its finite distinguished fiber. For every open $J \subset \mathcal{C}$ containing S there are nested closed topological disks about z , with an annular margin between them, whose preimages lie in J . The preimage of any compact annulus avoiding z is bounded away from S .*

Proof. The compact set $\eta(\mathcal{C} \setminus J)$ excludes z . Choose the nested disks inside its open complement using a sphere chart at z . For the last assertion, the annular preimage is compact and disjoint from the finite fiber S . $\square$

11.8 Exact retained lattice patches

We next transfer a disk around the distinguished point back to the original map. Both accepted and rejected scalar tests must be preserved; retaining only the intended chords could create additional

partners. A retained interval carries its actual resolved steps. Each coordinate has a partition of the retained intervals into tied groups. Within each group an auxiliary completion may translate that coordinate by a common integer, preserving all its exact differences. Parity restrictions are imposed on these translations. The auxiliary completion may use arbitrary admissible filler steps in the omitted intervals.

Lemma 11.23 (Exact incidence under protected gaps). *Consider an original and an auxiliary resolved contour sewing with the same retained step intervals, up to the preceding group translations. Let T be a set of retained triangles together with all triangles incident to their vertices and sides. Suppose the following conditions hold for every scalar relation needed to specify these incidences.*

- (i) *Every endpoint of the relation belongs to a retained interval, and the endpoints belong to the same translation group for that coordinate.*
- (ii) *Every intervening retained comparison is preserved: tied pieces have the same heights relative to the tested height, and other intervening pieces lie strictly above that height in both sewings.*
- (iii) *Every omitted gap between its endpoints has minimum strictly above the tested height in both sewings.*
- (iv) *Every other candidate partner of a step or corner in T is excluded in both sewings, either by unequal endpoint heights or by a retained value strictly below its tested height.*

Then the sewings have identical triangle, side, and vertex incidence on T . In particular, any embedded disk whose triangles and full vertex stars are contained in T transfers as a planar patch. A circuit and its chosen interior inside that disk transfer with their genuine all-edge lengths unchanged.

Proof. A scalar contour identification is the equality of the endpoint heights together with the statement that every intervening height is at least their common value. A step match is the corresponding adjacent-level version. Condition (i) preserves the exact endpoint equality and adjacent levels. Condition (ii) preserves every intervening retained comparison, and Condition (iii) supplies the remaining comparisons across omitted gaps. Condition (iv) excludes all false partners. Apply this argument in both coordinates to the resolved sewing instructions. These instructions specify the triangle sides, their orientations, and the cyclic incidence around vertices. The full-star assumption therefore gives an isomorphism of the relevant planar cell complexes, and of their primal map edges. It preserves the disk orientation and hence the indicated interior of each circuit. Only actual primal map edges are counted in its length. $\square$

Lemma 11.24 (Original and auxiliary patches agree). *Use Proposition 6.9 for discrete axial excursions and a limiting finite acyclic sphere sewing. Under the preceding finite-view and protected-completion conclusions, an acyclic combined center component has an auxiliary disk and an annular collar which agree with a patch of the original map, including the full stars needed to transfer paths and separation. All their time preimages lie in retained intervals. A compact annulus avoiding the central point has time preimage bounded away from every singular center.*

Proof. Truncate the line count and the centers' access depth. All times within one view unit of the relevant centers are reached by one more bounded jump. Every one of their scalar partners is reached by a further link. Equation (11.5) therefore puts *all* these partners, simultaneously, within a deterministic range $[-M, M]$ of enrolled centers, except with any prescribed small probability. Interline partners must belong to the same scalar center class by Corollary 11.13. This assertion

is uniform: if it failed with positive probability, select a violating pair, apply the uniform range bound, and extract its finite coordinates; the Corollary would give the excluded center contact. Thus partners cannot lie on a line outside the chosen combined component.

Retain intervals extending beyond $[-M, M]$ on all its lines. Choose their rational cuts after the required outward dips. The construction in Lemma 11.14 remains valid for these larger retained pieces; increase K_0 and the assigned class heights if needed. Use Lemma 11.12(iii) to truncate the reciprocals of all whole-gap margins. Choose a smaller positive δ below these margins and the outward dip depths, and shrink the working intervals about the origins until both centered coordinates there have magnitude less than δ . This is possible uniformly in the approximants by the compact-uniform tightness proved in Lemma 11.10.

Complete the discrete contours using the actual resolved words on the retained intervals. In each coordinate translate a whole tied group by one common integer. Choose these translations in a fixed parity class to approximate the assigned heights; their diffusive rounding errors vanish. In particular, different original center heights in a tied group are not forced equal at the lattice level. Fill each gap by an axial walk approximating its chosen continuous connector and staying within its open tube. This can be done with the required parity: approximate the connector first by a polygonal path, realize its finitely many displacements in sufficiently long time slots, and fill the unused steps with unit backtracks. The displacements have order $\sqrt{r_j}$ whereas each fixed positive-duration slot has order r_j steps. Adjust gap lengths by at most a bounded number of steps to satisfy parity. Refine the polygonal approximation only along a sufficiently slow diagonal. The positive margins keep all interior fillers nonnegative; axial entrance and exit pieces complete a finite excursion. These fillers are used to construct the auxiliary planar map, not to charge paths in its artificial parts.

We verify the exact incidence criterion for links beginning in the working intervals. Their tested levels differ from the relevant class height by at most $\delta\sqrt{r_j} + 1$. Every original partner is retained, by the uniform coverage just proved, and belongs to the same translation group. All intervening retained pieces of that group have their exact relative heights preserved. Any other retained class inside an active gap has original relative height $+\infty$, and the auxiliary height assignment puts it strictly above the active level as well. Original omitted gaps and auxiliary connectors both lie above the tested levels by the chosen margin. Finally the retained outward dips block links escaping the extremities of a class in either sewing. Candidates in unrelated high classes also fail the endpoint-height comparison. In the auxiliary sewing, a low-level partner cannot lie in a filler or an unrelated high class. These checks apply to full scalar pairs and to adjacent-level step matches in both coordinates. They are exactly Conditions (i)–(iv) of Lemma 11.23.

By Lemma 11.18, the off-center polarity argument in Corollary 11.21 applies to the completed contours. The central class is saturated by construction and acyclic by assumption. Its quotient is a sphere. Lemma 11.22 gives nested disks and collars whose time preimages are contained in the working intervals.

The link-approximation hypothesis of Proposition 6.9 also holds at the distinguished times. For each scalar class, use the consecutive-member return witnesses of Lemma 11.12(ii). The last departure up-step and first return down-step at their slowly vanishing attained level form a matched pair, with endpoints approaching the two center visits up to one mesh step. The common group translations and strict gap comparisons just checked preserve these matches in the auxiliary sewing. The tree/cotree dictionary supplies a primal identification or one genuine primal edge for each pair. These finitely many pairs generate the distinguished combined class.

For a generator outside that class, saturation excludes a distinguished time at its tested minimum level anywhere in its span. Its endpoints and all visits at that level are therefore

off-center, where joint Brownian absolute continuity gives strict local minima of distinct heights. The proof of Lemma 6.1 applies to this generator: nearby levels can avoid the countable off-center minimum heights and the finitely many distinguished heights. Thus all required links have approximations. This verifies (T3) without a metric bound on chronological connectors.

Apply Proposition 6.9 to the discrete approximations of these disks and collars. Its core disk has the full stars needed for incidences inside a larger retained collar; it does not assert that a proper subcomplex is closed under stars of all of its vertices. Lemma 11.23 now transfers this patch to the original map. The compact annular preimage avoids the central finite fiber, hence is bounded away from all its singular times. All truncation losses were arbitrary and may be removed in the stated order. $\square$

11.9 Annular tests determined by retained data

The auxiliary annulus is now an actual original-map patch. Its cost estimate must be a test of retained off-center data, since the common passage kernel does not condition on the omitted singular profiles. We first add the free height anchors, then protect full fibers, and finally describe a countable family of annular tests with this property.

Lemma 11.25 (Adjoining randomized class anchors). *Suppose selected laws of (c_j, y_j) have tight absolute continuity with respect to reference laws $\nu_j(dc)K_{j,c}(dy)$. After the entire selected sample, draw finitely many class-anchor heights with conditional densities bounded by a constant. On a compact height truncation, adjoining these anchors preserves tight absolute continuity with respect to the reference law times an independent positive smooth height density. The assertion also holds for the corresponding scaled lattice densities.*

Proof. It suffices to use Lebesgue measure on the compact anchor range; a positive smooth density is bounded below there. For a bad joint event E_j , let $p_j(c, y)$ be its anchor-sectional Lebesgue measure. If its reference product mass is $u_j \rightarrow 0$, Markov's inequality gives reference probability at most $\sqrt{u_j}$ to $p_j > \sqrt{u_j}$. The selected probability of this set tends to zero by tight absolute continuity. On its complement, the conditional density bound gives joint selected mass at most $C\sqrt{u_j}$. The density bound given the entire sample also holds given (c_j, y_j) by conditional expectation. Replace Lebesgue measure by the covolume-normalized counting measure for the lattice statement. Truncations of the finite anchor count and height ranges can subsequently be removed. $\square$

Lemma 11.26 (Finite guards for full source fibers). *Let f be one continuous coordinate of a completed excursion. Let J be a finite union of closed retained intervals and let P be a compact subset of $\text{int } J$. Assume that every scalar partner of every source time in P belongs to $\text{int } J$. Then finitely many strict lower bounds on the minima of the omitted gaps preserve all scalar relations involving P , and create no partner outside J , whenever the omitted pieces are replaced subject to those bounds and the retained paths are kept fixed.*

Proof. Treat a gap $G = [a, b]$ and sources to its left first. Let A_G consist of $t \in P$ with $t < a$ such that all retained values between t and a are at least $f(t)$, ignoring the omitted gaps. This set is compact: P stays a positive distance from the retained cuts, and finitely many continuous retained-interval minimum inequalities define eligibility. For $t \in A_G$ write $h = f(t)$. If any omitted gap before or including G reached height at most h , take the first such gap. All preceding retained values are at least h and all earlier gaps are above h . Its entrance cut must be strictly above h : equality would itself give a scalar partner of t at a retained cut, outside $\text{int } J$, contrary to the

hypothesis. Continuity then gives a first hit of h inside this omitted gap or at its far cut. The minimum from t to that hit is h , so it is again a forbidden partner. This contradiction proves $\min_G f > f(t)$ for every $t \in A_G$. By compactness, $\min_G f > \max_{t \in A_G} f(t)$. Choose a rational q_G strictly between these quantities. Do the same from the right, and take the larger lower bound if both source sets are nonempty. An empty eligibility set imposes no bound. Exterior gaps are treated in the same one-sided way.

Now take an endpoint pair with its source in P . If its endpoint heights differ, it is not a relation. If an intervening retained value is below the source height, it is not a relation in either completion. Otherwise the source is eligible at every intervening omitted gap, and each replacement gap has minimum above its source height by the bounds just chosen. Thus the pair is a relation in both completions exactly when its retained comparisons permit it. Similarly a new partner in an omitted gap would require a first hit of the source height in an eligible gap, which its strict lower bound forbids. This proves both preservation and absence of new external partners. The all-partners hypothesis is essential: without it, raising individual gaps could remove a genuine omitted obstruction and create a false relation. $\square$

Lemma 11.27 (Countable retained-data annular recipes). *Let a protected synthetic annulus be given by Lemma 11.24. Its crossing tests can be represented by a member k of a countable family of recipes such that:*

- (i) *the recipe input c consists only of retained off-center block increments, their forest shifts, randomized class anchors, durations, and fixed finite gluing and resolution instructions;*
- (ii) *its feasibility event $F_k(c)$ and crossing failure tests $E_{k,A}(c, y)$ are measurable functions of this input and its retained passage observations y ; no omitted center profile or filler variable is an argument of $E_{k,A}$;*
- (iii) *on F_k , every compatible completion satisfying the recipe's strict gap conditions induces the same local quotient and the same cyclic tube-and-port pattern, by identity of retained time labels;*
- (iv) *the actual synthetic annulus has such a feasible recipe, and its successful recorded crossings force the required circuit in that synthetic annulus and in the original patch.*

For each fixed recipe, the fresh reference slots are fixed before their raw letters are sampled. A successful recipe may be selected afterwards by taking a countable union, followed by a finite probability truncation.

Proof. Full-fiber protection. Choose a compact annular core inside larger protected collars, all avoiding the central point. Their full time preimages avoid the finitely many singular centers. Let M bound the off-center fiber sizes, as provided by Lemma 4.11. Use $M + 2$ nested compact annular collars and choose rational closed time unions $J_0 \subset \text{int } J_1 \subset \cdots \subset \text{int } J_{M+1}$ so that the core preimage lies in $\text{int } J_0$ and every full scalar partner of every time of J_ℓ belongs to $\text{int } J_{\ell+1}$. Such choices follow from the finite-fiber quotient: choose the image of J_ℓ inside the next compact collar and put its entire time preimage in the interior of $J_{\ell+1}$. Uniform continuity permits rational time covers, and only finitely many layers are needed.

Apply Lemma 11.26 in both coordinates to $P = J_M$ and $J = J_{M+1}$. Record a rational lower bound for every omitted gap in each coordinate, including gaps whose original eligibility sets are empty. In the latter case choose any rational number below the original gap minimum. For other gaps use the strict bounds supplied by the lemma. Validity on any new retained input requires each recorded bound to exceed all its eligible source heights, with supremum of an empty

set interpreted as $-\infty$. These finitely many bounds preserve every scalar relation involving J_M , including all rejected candidates, and exclude new omitted partners. The proof explicitly uses full source-fiber closure and exclusion of the cuts. It does not replace an arbitrary inactive gap by a higher path. If one instead uses the more general charts of Lemma 5.8, the equivalent recipe keeps the combined untested-minimum bands of Lemma 5.7, preserving both accepted and rejected comparisons; a responsible omitted piece for an upper minimum bound is just another finite recipe choice. Keep all included minima exact. Refine flexible instructions and left safety bands at the same time.

Any new combined chain starting in J_0 has its first M scalar steps preserved: after ℓ steps it is in J_ℓ . A simple chain with M steps would have $M + 1$ original visits, impossible by the original fiber bound. Thus the full combined fibers meeting the core, and not only their individual scalar edges, are unchanged in every allowed completion. This argument also excludes a new path through an omitted singular neighborhood.

The retained input. Record one absolute first-block anchor for each scalar center class as well. In Lemma 11.14 its height has the form of a sample-dependent translate of a uniform variable on a fixed open box. The uniform variables are sampled independently after the original sample and its fixed pre-anchor retained observation vector y . Their translations may depend on that sample, but the conditional density of the resulting anchor vector is still bounded. The vector y is the countable consistent family of fixed-recipe passage observations, not an observation selected to encode the anchors. Lemma 11.25 therefore permits adjoining these anchors to the retained input. Together with the forest differences they determine every retained absolute block-start height. All guard and gap inequalities are now functions of this augmented input and the finitely many prescribed fill constraints. No original singular center profile is retained as an additional argument of the crossing test.

A common open quotient. Let $T = J_{M+1}$, viewed as the finite disjoint union of its retained closed time blocks. For each coordinate draw a pair (t, u) exactly when $t, u \in T$, at least one endpoint belongs to J_M , their anchored heights agree, and every intervening retained value is at least that height. Include the diagonal everywhere else and take the generated equivalence relation $\mathcal{R}_c$ of the two colors. Lemma 11.26 proves that these explicit retained tests are equivalent to genuine scalar pairing in every allowed completion. No relation is inferred in an unprotected outer part of T . The preceding M -step argument makes the combined fibers meeting J_0 identical in every such completion. The generator relations are closed, since there are finitely many closed block pairs, J_M is closed, and the retained minimum tests are continuous. Their classes are bounded by Lemma 4.11 on these off-center intervals, since all their generators are genuine in a compatible completion. Hence their generated relation is closed by the finite-chain argument in Theorem 11.19. The quotient $X_c = T/\mathcal{R}_c$ is therefore compact Hausdorff. Put $O = X_c \setminus q_c(T \setminus \text{int } J_0)$, where q_c is its quotient map. Every fiber in O lies wholly in $\text{int } J_0$ and is its full true fiber in every compatible completion, by the M -step argument. Hence $q_c^{-1}(O)$ is open in T , is contained in $\text{int } T$, and is open on the entire completed contour circle. It is saturated for the true equivalence relation of every completion. The quotient property therefore makes its image an *open* sphere patch in each completion. Identical saturated open sets give a homeomorphism from O to each such patch. The core is covered by these sets, since all its full fibers lie in $\text{int } J_0$. Thus identity of time labels identifies the local ambient topology, not just a finite list of incidences. Free class anchors only enforce the recorded separation conditions; they do not change a retained scalar relation.

Countable annular recipes. There is a useful countable basis for this quotient. If $q_c : T \rightarrow X_c$ is the

projection and $J \subset T$ is a finite union of relatively open rational time intervals, put

$$O_J(c) = X_c \setminus q_c(T \setminus J).$$

This is open, since q_c is a closed map on a compact space. These sets form a basis: if $x \in V$ with V open, the compact fiber $q_c^{-1}(x)$ lies in the open set $q_c^{-1}(V)$, so finitely many rational intervals give $q_c^{-1}(x) \subset J \subset q_c^{-1}(V)$ and then $x \in O_J(c) \subset V$. For compact containment use regularity of the compact Hausdorff quotient: choose V' with $x \in V' \subset \overline{V'} \subset O_J(c)$, and then choose a rational time union J' with $q_c^{-1}(x) \subset J' \subset q_c^{-1}(V')$. It follows that $\overline{O_{J'}(c)} \subset O_J(c)$. Containment of time-interval closures alone would not justify this quotient-space containment. The recipe records the latter containment explicitly.

Place finitely many quadrilateral tubes cyclically in the annular core, with overlapping transverse passages and strict port margins. Choose an inner route corridor in each tube and a slightly larger outer quadrilateral, both within the common protected patch. Refine by finitely many basis sets to arrange

$$\text{inner compact route} \subset \text{recorded open confinement} \subseteq \text{outer quadrilateral},$$

with the recorded port closures inside its separated end-port neighborhoods. All time unions used here lie inside $\text{int } J_0$, so their full fibers are preserved. The complete recipe records only their rational time unions; the finite incidence, orientation, and cyclic-order pattern; inner and outer collars; rational inequality margins; and the earlier block and instruction lists. There are countably many such records. Include a retained-data validity event $V_k(c)$: each recorded gap bound is strictly above the supremum of its eligible source heights computed from c ; the recorded scalar pairs send J_ℓ into $\text{int } J_{\ell+1}$; and their generated classes have at most M visits. These checks are measurable compact candidate tests. For example, failure of the last check has a connected diagram of $M + 1$ distinct points, and one takes a countable union over positive separation cutoffs; all its finite retained equality and minimum tests are closed. The analogous compact source tests handle the first two checks. For arbitrary reference input satisfying V_k , the strict gap bounds alone give exact scalar relations and no outside partners for J_M ; the recorded nesting and M -step argument then give the same full core fibers in every compatible completion. Thus the common-quotient claim is not restricted to inputs originally sampled from the synthetic surface.

Feasibility from retained data. Let $F_k(c)$ assert $V_k(c)$, positive compatible fill weight, and that the specified sets in the retained local quotient satisfy these inclusions for an oriented cyclic quadrilateral pattern, with strict transverse-overlap and separation margins. In particular the entire permitted confinement sets lie inside the witness tubes; it is insufficient that they merely contain some correctly arranged tubes. Thus every allowed port-to-port path, including a minimizing one, has the prescribed topological crossing behavior. The inner routes ensure that the required port pairs can be connected inside the recorded open confinements. This is a property of X_c and the recorded sets and is therefore the same in every allowed completion. Its measurability can also be seen without choosing a coordinate chart for the singular synthetic surface. Sample an ordinary positive-weight Brownian completion as in Lemma 11.16. In that ordinary surface the pattern event is measurable, by a countable polygonal approximation in its coordinate charts with strict margins. More explicitly, for $J \subset \text{int } J_0$ its recorded open set is $S^2 \setminus \eta(S^1 \setminus J)$ in that completion. The complementary compact image is measurable from its continuous quotient curve. The nested route, confinement, port, and outer-quadrilateral conditions are therefore a countable union of measurable rational polygonal witness tests. Its indicator is constant over all compatible completions, by the local quotient identification. More explicitly, let Q_c be the measurable ordinary

filling kernel, G_k its compatible strict-fill event, and H_k the marked planar pattern event. On positive fill weight, constancy on compatible fills gives the measurable definition

$$F_k(c) = \mathbf{1}_{V_k(c)} \mathbf{1} \left\{ \int \mathbf{1}_{G_k} \mathbf{1}_{H_k} dQ_c = \int \mathbf{1}_{G_k} dQ_c > 0 \right\}.$$

No uniform positive lower bound on this fill weight is asserted. The polygonal witnesses and their positive spatial margins are existential in the pattern event, not fixed Euclidean coordinates in the recipe. Only its time unions, order, contour inequalities, and topological inclusions are fixed. Homeomorphisms of compatible completed patches preserve this existence statement even when their Euclidean coordinates differ.

Crossing tests and transfer. At the lattice level the recipe defines port vertex sets and confinement subgraphs from its fixed time-cell labels and the exact retained chart incidences. One may require that all representatives of a vertex in the reconstructed chart lie in the chosen inner time union, and use a slightly larger union for the allowed confinement. Full-fiber protection makes this an entirely local test. Record the infimal genuine all-edge cost of a path for each specified port pair inside that subgraph, with the convention $+\infty$ if no path exists. These are measurable functions of the finite raw blocks, exact shifts, and fixed instructions. Define $E_{k,A}(c, y)$ as $F_k(c)$ together with failure of one of these costs to be at most A in view units, and define $E_{k,\infty}$ as $F_k(c)$ together with at least one infinite limiting recorded cost. These tests have no argument for the omitted geometry. Open inner/outer margins permit the usual certificate limits and prevent a boundary equality from deciding the test. In every compatible completion the same recorded tubes have the same transverse-intersection pattern; their successful crossings force a circuit. In the original synthetic surface the indicated hole is the one containing the center, because that is how this feasible recipe was chosen. This last geometric fact is used only after the local crossing test has succeeded.

Finally, distinguish the two reference laws in the entropy argument. The Kraft-product reference makes the code independent of the whole raw word. The trial law $R_{j,t}$ instead keeps the selected code marginal and chronological nontrial kernels; each requested slot is fresh conditional on that code and its actual incoming history. For fixed k , the code and recipe fix the rounded origins, interval endpoints, and guard/noise search anchors before the requested letters are sampled. Additional marks have the independent laws required by Proposition 7.16, whose bound is uniform in the incoming history. Use unnormalized compatible subprobabilities in Proposition 7.16; do not condition on a rare successful guard atom. If the successful recipe was found using omitted geometry, its index is still one of the countable records above. The null assertion for every fixed $E_{k,\infty}$ can therefore be intersected over all k . A finite threshold failure need not be null; its probability tends to zero as $A \rightarrow \infty$. For such vanishing-probability statements first restrict to a finite number of successful recipes carrying all but a prescribed small probability. This proves the assertion even for adaptive recipe selection. $\square$

11.10 Crossing costs and surrounding circuits

It remains to put a tight cost bound on these annular tests. We first record the deterministic conversion from crossings to a circuit. The cost estimate will then come from ordinary sphere completions in the same distance units, before any arbitrary-endpoint bound is used.

Proposition 11.28 (Circuits from protected annuli). *Suppose an acyclic center component at a uniform level $X_j \in [aN_j, bN_j]$ admits the following construction, with probability tending to one on that event. It has a retained auxiliary disk and a compact annulus around the component point such that:*

- (i) the original lattice patch agrees with the auxiliary patch by Lemma 11.23, including the full stars needed for separation;
- (ii) the preimage of the inner disk contains a time neighborhood of each relevant base of radius $\delta_j r_j(X_j)$, where δ_j^{-1} is tight; the full preimage of the closed outer disk lies in finitely many retained intervals, each of length at most $M_j r_j(X_j)$, with both the number and M_j tight;
- (iii) finitely many tube crossings of the annulus, arranged to intersect successively and to wind around the inner disk, have total all-edge cost $C_j B(r_j(X_j))$, with C_j tight;
- (iv) all these patches lie in the required base spatial neighborhood.

Assume $1 < a_0 < a < b$, $s' < s$, and $B(r)/B(n_j) \leq K(r/n_j)^s$ on the used lattice scales. Then the original map has a surrounding circuit of cost at most $B(n_j)e^{-s'N_j}$ with probability tending to one. Its interior contains the base time neighborhood of radius $n_j e^{-bN_j}$ and its closed interior contains no uninterrupted contour time interval of length $n_j e^{-a_0 N_j}$.

Proof. In an annulus choose a finite cyclic collection of quadrilateral tubes, with successive passages transverse in their overlaps. Any crossings of their prescribed ports meet by planar separation. Their union contains a closed walk with nonzero winding around the inner disk. Removing inessential loops gives a simple surrounding circuit of no greater cost. Condition (i) transfers this circuit and its interior to the original map.

For any fixed $\eta > 0$, with loss at most $O(\eta)$ first bound C_j, M_j, δ_j^{-1} and the number of intervals by deterministic constants. The exceptional event $bN_j - X_j \leq \sqrt{N_j}$ has probability $O(N_j^{-1/2})$. Outside it, $\delta_j r_j(X_j)/(n_j e^{-bN_j}) \rightarrow \infty$, which gives the lower time bound. The circuit and its closed interior lie in the closed outer disk, whose full time preimage is covered in (ii). Hence every uninterrupted visit in the closed circuit disk, including a boundary visit or a run along its edges, lies in the union of those retained intervals. Its length is at most a constant times $r_j(X_j)$; even if the intervals touch, their total length has this bound. Since $X_j \geq aN_j$, $r_j(X_j)/(n_j e^{-a_0 N_j}) \leq e^{-(a-a_0)N_j} \rightarrow 0$. Finally

$$\frac{C_j B(r_j(X_j))}{B(n_j) e^{-s'N_j}} \leq K C_j \exp\{-(sa - s')N_j\} \rightarrow 0$$

in probability. Here $sa - s' > 0$. Remove the fixed tightness truncations by letting $\eta \downarrow 0$. Condition (iv) gives confinement. $\square$

Lemma 11.29 (Finite port costs in an ordinary sphere completion). *Fix a deterministic total excursion duration $T \in (0, \infty)$. Extract the strict-interior chart observations of ordinary empty-word excursions at time units $r_j \rightarrow \infty$, jointly with the bilateral reference experiments, using distance units $B(r_j)$ throughout. After the further extraction of Corollary 9.13, every compactly confined open-port crossing test in the punctured ordinary sphere has finite limiting passage cost whenever its two ports can be connected inside its prescribed open confinement. This holds simultaneously for the countable protected atlas and finite lists of tests with strict margins. The comparison used on a real-field domain U is with $D_h(\cdot, \cdot; U)$, the internal metric of that domain.*

Proof. The area- T field is the unit-area field plus $\gamma^{-1} \log T$. Lemma 3.3 applies also to this actual field: after its stated domination, add this deterministic constant on a neighborhood of the tested compact patch and remove it from the Gaussian reference by a Cameron–Martin cutoff. Thus, in the affine chart of a first auxiliary area pair, for every bounded U with $\bar{U} \subset \mathbb{C} \setminus \{0, 1\}$, its actual

real-field law $\mu_{T,U}$ is absolutely continuous with respect to the local cone law $\mu_{\text{cone},U}$ used in that Lemma. The cone normalization circle is chosen outside $\bar{U}$.

Retain the *entire* independent stationary-phase imaginary field and its unparameterized directed curve, with common law ν . Let Π_h be the common conditional Poisson area-mark law in U . Then

$$\mu_{T,U}(dh) \nu(d\vartheta) \Pi_h(d\pi) \ll \mu_{\text{cone},U}(dh) \nu(d\vartheta) \Pi_h(d\pi).$$

In particular the complete traversal order is retained; the finite order O required by a protected chart is a measurable function of this input. Theorem 7.13 gives the same local observation kernel on both sides, including the same spatial interpretation and all strict port and confinement tests. Appending that common kernel preserves the displayed domination. The upper comparison of Corollary 9.13 is a probability-one event for the cone input and observations, with its chosen deterministic multiplier. The reference internal metric on U is measurable from $h|_U$, so this event transfers to the sphere. No unrestricted metric depending on the omitted field is transported. The sphere observations are strict-interior observations because the full preimage of a compact patch avoiding the exploration root is a compact subset of $(0, T)$.

Intersect over bounded rational-box domains, their buffered protected charts, and rational open losses. Adjoin a second independent area pair and repeat for its three-marked marginal, as in Remark 3.4. No comparison is conditioned on the other pair. Their off-mark covers exhaust the punctured sphere. The affine covariance of the kernels and of the internal metrics makes both conclusions statements in the same quantum units, with the same bilateral multiplier and the same exploration root.

For completeness, an open-port crossing does not require joining independently extracted endpoints from these covers. In either affine representation a connected open surface confinement remains connected after removal of the two auxiliary points. Choose points of its two open ports away from those points and a compact topological route between them. The local length-metric and internal connection inputs of Theorem 3.5, applied along a finite cover of the route, give a finite-length D_h path compactly inside the punctured confinement. Enclose that path in a bounded connected finite union of rational coordinate boxes U whose closure is still inside the confinement and avoids 0, 1. Both Lemma 3.3 and Theorem 7.13 permit this domain U ; they do not require U to be a disk. The transferred upper comparison supplies one finite-cost crossing certificate in U , with its endpoints strictly inside the prescribed ports. Lemma 5.1(i) realizes it in the retained primal chart with the prescribed open losses. The full-fiber inner port and outer confinement margins in the protected atlas ensure that these records obey the recipe's time-label restrictions as well. Hence the recorded infimal port cost is finite. Apply this argument to a finite list simultaneously and sum its finite costs. This uses endpoint slack only, with no arbitrary-vertex joining input. $\square$

A fixed duration for the reference completions. For each fixed recipe and compact duration truncation in Lemma 11.27, choose a deterministic integer T larger than the maximum total retained duration, including all retained guard and noise pieces, with a positive reserve for every omitted gap and the two excursion caps. Assign these omitted durations from a compact subset of their positive simplex. Absolute origins and omitted durations are nuisance variables of the reference chart; its block increments, retained durations, forest shifts, and relative traversal order are unchanged. Reparameterizing each continuous omitted filling separately preserves every required height inequality. The killed-bridge and excursion-cap densities have positive support at these positive durations. Take the actual ordinary duration- T excursion law with the permitted cut experiment, restrict to compatible fillings, and disintegrate over the retained input. Lemma 11.16

and the positive macroscopic density of Proposition 7.9 then give the positive density-weighted fill integral. All ordinary reference completions for this recipe have total duration and quantum area T . Use empty words of total length the nearest feasible even integer to Tr_j , while retaining distance denominator $B(r_j)$ in the actual, block-reference, and completed experiments. The field addition $\gamma^{-1} \log T$ describes the reference surface's area; it does not replace that denominator by $B(Tr_j)$ or invoke Weyl scaling for the passage rule.

Proposition 11.30 (Acyclic components admit cheap circuits). *Use the simultaneous all-seed construction of Lemma 11.6, the address-occupation estimate, Corollary 10.20, and Remark 10.21. Also use Propositions 7.16 and 7.18, the bilateral interior crossing comparison of Corollary 9.13, and Proposition 6.9. Fix $1 < a_0 < a < b$ and $0 < s' < s$. For a uniform view level in $[aN_j, bN_j]$, the probability that a base belongs to an acyclic combined center component but has no genuine surrounding circuit with the cost and time bounds of Proposition 11.28 tends to zero. The circuit may be confined to any base spatial neighborhood on which the finite-chain typicality cutoff holds.*

Proof. Suppose such an acyclic event has probability bounded below along a subsequence. Include its indicator in the joint extraction; the view law itself is not conditioned after choosing the uniform coarsening. We verify in turn the retained patch, its crossing costs, and its confinement, which are the inputs to Proposition 11.28.

The retained patch. Proposition 11.5 and Lemmas 11.6 and 11.10 give the simultaneous finite system and the stationary continuous contour law. Lemma 11.12 identifies its center classes and supplies actual rank witnesses and strict whole-gap margins. Choose the base's combined component and truncate its finite data with any fixed small probability loss.

Lemmas 11.14 and 11.18, Corollary 11.21, and Lemma 11.24 give a sphere completion with an original-map disk and an annular collar about this component. Choose nested inner and outer collars with strict margins. Their time preimages are contained in the finite retained pieces; the compact annular preimage has positive distance from all singular centers. The inner disk preimage contains a positive time neighborhood of every center visit. Since each original base approaches its frozen center under coarsening, the same is true of the base. The full preimage of the closed outer disk is contained in the finite retained intervals, with the larger full-star collar of Lemma 11.24 covering boundary vertices and their incident edges. Truncating the radii, durations, and their reciprocals gives exactly the time bounds in Proposition 11.28(ii).

The crossing costs. Apply Lemma 11.27. It supplies a countable recipe index k , a retained-data feasibility event $F_k(c)$, and local failure tests $E_{k,A}(c, y)$. Although the index can have been chosen using the omitted singular geometry, the test itself has no such argument. First restrict to finitely many recipe indices with an arbitrary small probability loss. For one fixed recipe, place its finite cyclic family of crossing tubes in the annulus, with transverse overlaps and positive margins. Cover their time preimages and the suppliers needed for their local instructions by finitely many blocks strictly away from the centers. Use a smaller annular core inside a larger retained collar, so all local extensions, noise pieces, and guards remain in these off-center blocks. The strict scalar gap comparisons and ordinary off-center genericity allow finite separated flexible-order instructions. Exhaust the finite cut and guard choices and their positive margins by a countable atlas before selecting the successful chart.

For each fixed chart, retain its block increments and forest shifts in both coordinates. The selected lattice experiment has tight absolute continuity with respect to the independent-block reference experiment with smoothly distributed shifts on the feasible lattice, by the slot domination, Lemma 11.2, and Lemma 11.3. The chart is openly fillable: omit small neighborhoods of every singular center and all unused gaps, keeping the finite annular recipe. All inequalities on omitted

pieces have strict slack by the larger protected collar. The existing continuous filling witnesses feasibility, and Lemma 11.16 gives positive conditional ordinary Brownian fill weight. No exact center profile or approximate lattice tie is imposed on this reference filling.

The common-kernel statement identifies the passage observations of that ordinary completion with the reference block kernels. Every compatible completion realizes the same local quotient and port-and-tube recipe by Lemma 11.27; therefore the test remains exactly $E_{k,A}(c, y)$, independent of which omitted pieces were filled. The infinite-cost failure $E_{k,\infty}$ is the corresponding null test. Choose the ordinary reference completion at the fixed total duration specified above, in the same $B(r_j(X_j))$ view units. Lemma 11.29 makes each of its prescribed open-port crossing costs finite. Its hypotheses apply to the retained quotient tubes, whose full fibers lie in a compact interior time set, and their strict ports and confinements are unchanged by compatible filling. Thus $E_{k,\infty}$ is null in every jointly extracted ordinary reference completion. The finite sum of the recorded costs has vanishing upper tails there. Apply Lemma 11.17 and the quantitative entropy transfer to the failure probabilities at thresholds tending to infinity, first for a fixed atlas recipe and then after the finite probability truncations. This proves the tight total crossing cost required in Proposition 11.28(iii).

Transfer to the original map. The topological sewing stability and exact patch agreement preserve the transverse intersections and winding in the original map. Thus only genuine original all-edge paths are used in stitching the surrounding circuit. Every enrolled line is reached by a tight finite chain from the bases; the imposed finite-chain typicality cutoff therefore confines the retained patch and circuit to the required base neighborhood. Proposition 11.28 now supplies the cost and both strict time inequalities, contradicting the alleged failure. Let the arbitrary probability truncation losses tend to zero. $\square$

12 Uniform control of discrete endpoints

The preceding section supplies cheap surrounding circuits. We now use them to control distances from every prescribed discrete vertex. A circuit's length does not bound the cost of reaching it from the vertex it surrounds. Our deterministic covering lemma keeps these access costs in a bounded number of shorter chronological intervals. Rank credit allows the covering to be iterated, and the remaining intervals are finally traversed by the contour walk itself.

The first output is a uniform modulus for endpoints close in contour time. The second joins arbitrary representatives of the same limiting spatial point, using the finite horizontal-chain relation. Both conclusions retain a prescribed open confinement.

Throughout this section, $v_n(j)$ is the primal tour vertex at integer time j . The continuous interpolation of $j \mapsto v_n(j)$ follows a primal tree edge or stays at a vertex at each step. In particular,

$$d_{G_n}(v_n(i), v_n(j)) \leq |i - j|. \quad (12.1)$$

Whenever confinement is specified, the right side is still an upper bound if the intervening chronological walk stays in that confinement.

12.1 Covering an interval by circuit interiors

We use a simple circuit as an embedded Jordan curve consisting of primal edges, with a specified disk on its inside. In a planar chart this is the bounded disk. Length means the number of its edges, with multiplicity one on the circuit. Loops and two-edge circuits in a multigraph are allowed when their embeddings are Jordan curves.

Definition 12.1 (An available circuit). Let $v : \mathbb{Z} \rightarrow V(G)$ be a nearest-neighbor walk with allowed stays in an embedded planar graph, and interpolate it along graph edges. Fix integers $d \geq 1$ and $\ell \geq 4d$, a positive number L , and an open planar confinement U . An integer time t is good if there is a simple circuit $C_t \subset U$, of length at most L , with interior D_t , such that

- (i) the walk on $[t - d, t + d]$ is contained in D_t ;
- (ii) no chronological interval of ℓ consecutive steps is contained in $\overline{D}_t$.

All other times are bad. Condition (ii) is required only in the chronological window in which the exit searches below are made.

For vertices $x, y \in U$, let $d_{G,U}(x, y)$ denote the shortest-path distance restricted to graph paths in U , with value infinity when no such path exists.

Lemma 12.2 (Covering by circuit interiors). *Fix an integer $h \geq 1$. There is a constant $K_h < \infty$ with the following property. Let $I = [i, j] \cap \mathbb{Z}$ have length at most T . Suppose that no h bad times in I are pairwise at distance at least ℓ , and that the interpolated walk on the $K_h \ell$ enlargement of I lies in U . Then there are at most $h + 2$ pairs of integer times (i_r, j_r) , all in the $K_h \ell$ enlargement of I , satisfying*

$$|i_r - j_r| \leq K_h \ell,$$

and graph paths in U of total length at most

$$K_h(1 + T/d)L,$$

which, together with connections between $v(i_r)$ and $v(j_r)$, form a connection between $v(i)$ and $v(j)$. Consequently,

$$d_{G,U}(v(i), v(j)) \leq K_h(1 + T/d)L + \sum_{r=1}^{h+2} d_{G,U}(v(i_r), v(j_r)). \quad (12.2)$$

Unused summands can be taken to be zero. The conclusion also holds when there are no good times in I .

Proof. Choose a maximal ℓ -separated set of bad times in I . It has at most $h - 1$ elements. The union of their chronological ℓ -neighborhoods covers every bad time. Merge overlapping neighborhoods and intersect with I . Their total length is at most $2(h - 1)\ell + 2h$, and the remaining set has at most h integer intervals, all of whose times are good.

Consider one such interval $[a, b] \cap \mathbb{Z}$. Choose good times $a = t_1 < \dots < t_k = b$ with successive differences at most d , using at most $2 + (b - a)/d$ times. For each one choose an available circuit. The interiors of consecutive circuits overlap: the walk at each time between t_u and t_{u+1} belongs to both interiors on a nonempty chronological subinterval.

Among this finite family retain the disks maximal for inclusion. Every discarded disk is contained in a retained disk. The intersection graph of retained interiors is connected: a chain of overlapping original disks maps to a chain of overlapping containing maximal disks. Two distinct maximal Jordan disks which overlap have intersecting boundaries. Indeed disjoint Jordan boundaries with intersecting interiors would force one disk to contain the other. Thus the union of the retained circuit boundaries is a connected subgraph. Its number of edges is at most the sum of the lengths of the original circuits.

Select a retained disk containing the first disk, so that it contains $v(a)$ in its interior. Condition (ii) implies that the interpolated walk first reaches its boundary at an integer time in $[a, a + \ell + 1]$.

The intersection is a graph vertex, since both the walk and the circuit use edges of the same planar graph. Similarly a retained disk containing the last disk has a boundary vertex represented at a time in $[b - \ell - 1, b]$. The connected union of retained boundaries joins these two vertices at a cost no greater than the sum of the chosen circuit lengths. We have made no estimate of the cost of reaching either boundary from $v(a)$ or $v(b)$.

Do this for every remaining good interval, and concatenate the resulting boundary paths in the order of those good intervals. Within a short good interval the forward hit may follow the backward hit; orient its boundary path from the forward hit to the backward hit regardless of their time order. Each gap between these paths is represented by a pair of times separated by a removed bad interval and at most $2\ell + 2$ extra steps. Initial and final gaps obey the same bound. There are at most $h + 1$ gaps; allowing $h + 2$ covers endpoint and rounding conventions. Every gap has length at most $K_h \ell$, and every time used lies in the $K_h \ell$ enlargement of I , for a constant depending only on h . The total number of circuits is at most $2h + T/d$, which gives the stated cost after increasing K_h .

If no good interval remains, the length of I is at most the total length of the bad neighborhoods. Use the single unresolved pair (i, j) . This proves all claims, including Equation (12.2). $\square$

12.2 Rank credit bounds the separated bad times

We next show that the obstruction to the covering lemma—many separated times without an available circuit—cannot persist under the selected laws used in the recursion. All scale constants are fixed here, before choosing such a law.

Let $s > 0$ be a polynomial-ratio exponent for the crossing normalization, as in Equation (9.4):

$$\frac{B(u)}{B(v)} \leq K(u/v)^s, \quad 1 \leq u \leq v. \quad (12.3)$$

It follows, by fixing u , that $B(v) \geq cv^s$. Decreasing s preserves Equation (12.3); we henceforth take $0 < s < 1$. Choose constants in the order

$$0 < b - 1 < s/8, \quad s' = s/2, \quad 1 < a'_0 < a_0 < a < b, \quad (12.4)$$

where $a - 1$ is sufficiently small relative to $(d_2 - 1/2)(b - a)$ for the rank-credit argument. The integer h is then chosen sufficiently large for that argument. All these quantities depend on the fixed q , but not on n .

Here $b > 1$ is a log-window constant and is unrelated to the inventory parameter b_F . At level N , put, with integer rounding understood,

$$T = ne^{-N}, \quad \ell = ne^{-a_0 N}, \quad d = ne^{-bN}, \quad L = B(n)e^{-s'N}. \quad (12.5)$$

We use the preselection events of Lemma 11.1. Fix a compact chronological window, a spatial compactness and clearance cutoff, an error tolerance $\epsilon > 0$, and a deterministic largest view radius $\rho_n n$ with $\rho_n \rightarrow 0$. Before selecting a bad level or its bases, that lemma constructs a master event $\mathcal{T}_n$ and a deterministic depth bound $D_n \rightarrow \infty$. Its failure probability on the fixed compactness and clearance cutoff has limsup at most 2ϵ . On $\mathcal{T}_n$, simultaneously for every depth $d \leq D_n$, all explorations from the tested window with at most d steps, each a scalar link or a jump of size at most $d\rho_n n$, lie in the corresponding deterministic chronological enclosure and have the required spatial confinement. The event uses macroscopic contour barriers and uniform finite-chain projection control; it is chosen under the unselected law.

After conditioning on a bad configuration, the finite-list depth, enclosure, and coding cutoffs may be increased only along a slower diagonal within these preselected bounds. In particular, the sparse coding conditions of Lemma 11.6 are imposed by slowing that diagonal. An ordinary high-probability enclosure is never transferred through a subsequently chosen density $e^{o(N)}$.

Proposition 12.3 (Circuit sparsity input). *Use the constants above and the available-circuit definition with the scales in Equation (12.5). Let $E_{n,N}$ be the event that some chronological interval of length at most T , in the tested compact window, contains h pairwise ℓ -separated bad times. Along every sequence for which*

$$N \longrightarrow \infty, \quad ne^{-bN} \longrightarrow \infty,$$

there is no sequence of probability laws $\mathbb{Q}_{n,N}$ supported on $E_{n,N}$ and the preselected master events just described, with $e^{-N} \leq \rho_n \rightarrow 0$, such that

$$\mathbb{Q}_{n,N} \leq \exp(\kappa_{n,N})\mathbb{P}_n, \quad \kappa_{n,N} = o(N). \quad (12.6)$$

The same assertion holds for any fixed constant multiple of T and for sufficiently small fixed spatial neighborhoods inside the selected confinement.

Proof. Mark the h selected bad bases and finite witness diagrams before choosing the independent shifted address grids. The address estimate of Section 10 bounds the integrated box count under Equation (12.6) and charges β times the integrated monochrome rank, up to arbitrarily small relative errors. Proposition 11.30 shows that a positive probability of an acyclic base component at a uniform viewing level in $[aN, bN]$ would supply an available circuit at one of the bad bases.

We relate these center components to the projected graphs charged by the address estimate. The bases are separated by $\ell/r(X) \geq e^{(a-a_0)N}$ at a view level $X \in [aN, bN]$, so their enrolled centers are distinct. On each fixed finite truncation, Lemma 11.12(ii) supplies actual links witnessing a spanning tree of every scalar center class. Proposition 11.5 identifies these links with the graph attained by the seed recipe. The all-seed code includes its actual witnesses and the jumps associating their endpoints to their frozen centers. Outside the levels charged by the integrated jump error, these endpoints and centers occupy the same projected cells; distinct center lines occupy distinct cells. Bases also have tight seed offsets from their assigned centers. Apply the independent-grid boundary estimate to the original centers, chosen before that grid: conditional on a bounded seed offset, the crossing probability is at most a fixed multiple of $\min\{1, e^{-U_j}\}$, whose uniform-coarsening average tends to zero. Rounding an origin keeps it in the same terminal cell, and hence in the same cell at every coarser generation. Thus each base and its frozen origin occupy the same projected cell with probability tending to one.

Thus a rank cycle of a limiting base component is witnessed inside the corresponding projected base component. Adding connected vertices cannot destroy it: extend monochrome spanning forests by a union spanning tree rooted at the old component, gaining at least one unit of total rank per new vertex. All other code entries may therefore be kept. The acyclic-circuit estimate makes the probability of an acyclic base component at a uniform view level tend to zero. Averaging over the seed-generation trials and applying Markov's inequality now shows that every projected base component has a rank cycle outside $o(N)$ generations, in probability. Fixed truncations are removed along the sparse diagonal of Lemma 11.6.

The separation hypothesis of Proposition 10.18 holds for every displacement. At a fixed chronological cutoff, the largest charged cell width is $O(ne^{-aN}) = o(\ell)$. Along the permitted enlarging cutoff, the all-seed condition $\Lambda_j \log(2\hat{C}_j)/L_j \rightarrow 0$, together with $L_j = o(N_j)$, gives $\log \hat{C}_j = o(N_j)$, so the same comparison remains true.

The rank-credit estimate gives negative credit at least $(d_2 - 1/2 - o(1))(b - a)hN$ on this range, whereas the earlier possible positive credit is at most $C_0N + C_1(a - 1)hN + o(N)$. The choices in Equation (12.4), followed by the choice of h , make the former strictly greater. The selected density costs only $o(N)$, so this contradicts the address probability bound and proves the proposition. $\square$

Remark 12.4. The bounded-density hypothesis in Equation (12.6) is intentional. Conditioning on an event of probability $\exp(-o(N))$ has precisely this property, and that is the only kind of selection used below. A bound on relative entropy alone would transfer vanishing reference probabilities, but would not by itself give the uniform first-moment bounds for arbitrarily large address systems used in constructing the views. We do not make that stronger inference.

12.3 Iteration with a growing lattice cutoff

The recursion pays for circuit boundaries at each stage and passes only the short access intervals to the next stage. We must ensure that all stages are valid together, while stopping before the lattice estimates reach a bounded time scale.

For a chronological window J and spatial confinement U , write

$$\omega_n(N; J, U) = \frac{1}{B(n)} \sup_{\substack{i, j \in nJ \cap \mathbb{Z} \\ |i-j| \leq ne^{-N}}} d_{G_n, U}(v_n(i), v_n(j)).$$

Window enlargements below are measured in units of n . On the complement of $E_{n, N}$, Lemma 12.2, applied with Equation (12.5), gives for sufficiently large N

$$\omega_n(N; J, U) \leq C_h e^{-cN} + H \omega_n(a'_0 N; J^{C_h e^{-a_0 N}}, U), \quad c = s' - (b - 1) > 0, \quad (12.7)$$

where $H = h + 2$. Indeed $K_h \ell \leq ne^{-a'_0 N}$ for sufficiently large N , and the circuit payment is at most $K_h(1 + e^{(b-1)N})e^{-s'N}$. Integer rounding is absorbed in the same constants because the smallest scale d tends to infinity.

Lemma 12.5 (Uniform validity along the recursion). *Assume Proposition 12.3. Start at a deterministic $N_0 = N_0(n) \rightarrow \infty$ and set $N_j = (a'_0)^j N_0$. Apply the recursion only at those levels satisfying*

$$N_j \leq (1 - s/2) \log n.$$

After the compactness and clearance cutoffs have been exhausted, the probability that any applied level has a bad configuration tends to zero.

Proof. Suppose the probability of some bad level is bounded below by $\delta > 0$ along a subsequence on a fixed compactness and clearance cutoff. Choose $\epsilon < \delta/8$ and apply Lemma 11.1 with $\rho_n = e^{-N_0}$ to a fixed slight enlargement of the tested window. This gives one master event $\mathcal{T}_n$ for every applied level, before any level or bases are selected. The union of the bad-level events, intersected with $\mathcal{T}_n$ and the fixed cutoff, still has probability at least $\delta/2$ for all large n . Since $\sum_{j \geq 0} (j+1)^{-2} < \infty$, there is a deterministic index $j = j(n)$ for which the intersection of that level's bad event with these same preselected events has probability at least $c\delta(j+1)^{-2}$. Condition on this intersection and then choose the bad bases by the finite lattice ordering. The logarithm of the density bound is at most $C_\delta + 2\log(j+1)$, whereas $N_j = (a'_0)^j N_0$; thus the density bound is $\exp(o(N_j))$.

At every applied level,

$$ne^{-bN_j} \geq n^{1-b(1-s/2)} \rightarrow \infty,$$

since $0 < s < 1$ and $b - 1 < s/8$ imply

$$b(1 - s/2) < (1 + s/8)(1 - s/2) < 1.$$

All interval-window enlargements fit in a fixed slightly larger window: their total size is bounded by $\sum_j C_h e^{-a_0(a'_0)^j N_0}$, which tends to zero. The resulting conditional laws contradict Proposition 12.3. Letting the arbitrary preselection loss tend to zero and exhausting the compactness and clearance cutoffs proves the claim. $\square$

Proposition 12.6 (Chronological endpoint modulus). *Assume Equation (12.3) and Proposition 12.3. In a bilateral contour coupling, let J be a compact time interval whose limiting curve image has a neighborhood compactly contained in an open set U . Then, for every deterministic $\delta_n \downarrow 0$,*

$$\frac{1}{B(n)} \sup_{\substack{i,j \in nJ \cap \mathbb{Z} \\ |i-j| \leq n\delta_n}} d_{G_n,U}(v_n(i), v_n(j)) \longrightarrow 0 \quad \text{in probability.}$$

The assertion also holds locally on the event that the stated compact-containment condition is satisfied, by exhausting its clearance and the chronological and spatial compactness cutoffs.

Proof. The claim is immediate when $n\delta_n < 1$, since only identical integer times are tested. Otherwise set $N_0 = -\log \delta_n$. If $N_0 > (1 - s/2) \log n$, chronological steps give a bound $Cn^{-s/2}$ after normalization. Otherwise iterate Equation (12.7) until the first new level $N_{j_*} > (1 - s/2) \log n$. Lemma 12.5 shows that all recursions are valid with probability tending to one. Their total normalized payment is bounded by

$$\sum_{j \geq 0} H^j C_h e^{-c(a'_0)^j N_0},$$

which tends to zero. For example, after increasing N_0 , the ratio of successive terms is at most $1/2$, so the sum is at most twice its first term.

Each terminal unresolved pair has chronological separation at most $ne^{-N_{j_*}} \leq n^{s/2}$. There are at most H^{j_*} such pairs, and $j_* \leq C_1 + C_2 \log \log n$. Equations (12.1) and (12.3) bound their total normalized cost by

$$C(\log n)^{C_3} n^{-s/2} \longrightarrow 0.$$

Their chronological paths stay in U with probability tending to one by the uniform contour coupling and the stated clearance. The same is true of all earlier circuit paths by their definition. This proves the assertion, including confinement. $\square$

12.4 Joining endpoints at the same limiting point

Close spatial endpoints need not have close contour times. The sewing relation connects their limiting times by finitely many horizontal generators. Approximate each generator by a genuine discrete link, then use the chronological modulus to join its small endpoint errors.

Proposition 12.7 (Local endpoint joining). *Assume Proposition 12.6, Lemma 6.1, and the compact sewing interface (T1)–(T3) of Section 6, with vanishing projected edge diameters. In a joint extraction on such a chart, arbitrary primal-vertex endpoints whose continuum locations approach the same point can be joined at cost $o(B(n))$, with the path confined to any prescribed open neighborhood of that point. The conclusion is uniform over endpoints in a fixed compact chart, in the usual sequential sense. The same conclusion applies in a bilateral exhaustion whenever the tested vertices have tight chronological representatives in its compact sewings.*

Proof. If uniformity failed, select a violating pair with positive probability and then a further coupled subsequence. First restrict to a compact sewing; in an exhaustion, truncate the stipulated tight chronological representatives. This truncation is part of the hypothesis; it is not deduced from a passage comparison. Property (T1) gives limiting representative times t and u with $\eta(t) = \eta(u)$. Their equivalence is generated by a finite chain of horizontal coordinate identifications by (T3). For each ordinary generator, strict-height approximations yield actual primal identifications or genuine primal edges whose time representatives approach its two endpoints. At a branch visit one first splits into the finitely many consecutive ordinary generators. This is a topological statement about the discrete sewing; it estimates no chronological connector cost.

Choose closed rational time neighborhoods of all the finitely many visits whose curve images lie in the prescribed open spatial neighborhood. Truncate the number of visits and then their countable rational neighborhood choices to a finite list, with arbitrarily small probability loss. The finitely many time errors in the link approximations are then bounded, with probability tending to one, by a deterministic sequence $\delta_n \downarrow 0$, chosen sufficiently slowly. Proposition 12.6 joins the selected original representatives to the approximating link endpoints, and joins successive approximating endpoints at each visit, at cost $o(B(n))$. The finitely many genuine edge costs are also $o(B(n))$ because $B(n) \rightarrow \infty$, and their projected diameters tend to zero, so they stay inside the prescribed neighborhood. Concatenation contradicts the selected violation after the arbitrary truncation losses vanish. This proves the uniform sequential assertion. Notice that a second-coordinate scalar identification is not itself declared a unit-cost primal edge: it is the strict-height matched-step approximation, together with the chronological modulus just proved, that supplies the primal connection. $\square$

Corollary 12.8 (Bilateral endpoint modulus). *Under their respective hypotheses, the chronological and spatial endpoint conclusions of Propositions 12.6 and 12.7 hold simultaneously.*

The universal endpoint tests recorded in the certificate framework allow this conclusion to enter the common-kernel transfer. It is not inferred by reversing a semicontinuity assertion for existential path certificates.

Part II

Spanning-tree maps

13 Independent-walk encoding and local quantum surfaces

We now construct the local passage laws for spanning-tree maps, using independent walk increments and the disk topology of SLE_8 cells. These model-specific inputs will feed into Theorem 8.30. We first record the exact discrete sewing and the continuum facts needed to construct its local passage laws. Throughout the spanning-tree argument,

$$\gamma = \sqrt{2}, \quad \kappa' = \frac{16}{\gamma^2} = 8, \quad Q = \frac{2}{\gamma} + \frac{\gamma}{2} = \frac{3}{\sqrt{2}} > 2, \quad p = \frac{1}{d_\gamma}, \quad \xi = \gamma p, \quad s = \frac{p}{2}. \quad (13.1)$$

The number $d_\gamma > 2$ is the volume-growth exponent in Equation (13.4) and the dimension parameter of the LQG metric. Fix an integer k with $ks > 8$.

13.1 The graph encoded by an axial walk

Let $Z^{\text{disc}} = (L^{\text{disc}}, R^{\text{disc}})$ be a two-sided random walk on $\mathbb{Z}^2$, with independent increments uniform on $(1, 0), (-1, 0), (0, 1), (0, -1)$ and $Z_0^{\text{disc}} = 0$. Write L_i, R_i for its coordinates in this subsection. For $i \leq j$, set

$$i \sim_L j \iff L_i = L_j = \min_{i \leq r \leq j} L_r. \quad (13.2)$$

This is an equivalence relation. Its classes are the primal tree vertices; a horizontal step joins its two consecutive classes by a tree edge. The up- and down-traversals of one tree edge are identified. A vertical up-step with terminal time u is matched to the vertical down-step with terminal time $v > u$ when

$$R_{u-1} = R_v < \min_{u \leq r < v} R_r. \quad (13.3)$$

Each match adds a unit edge between the primal vertices represented by $u - 1$ and $v - 1$. Since a vertical step has constant L , either endpoint of that step represents the same primal vertex. The added edges may be loops or parallel edges.

Theorem 13.1 (Mullin encoding and its infinite version). *Rooted spherical map–spanning-tree pairs with n edges are in bijection with length- $2n$ axial walks starting at the origin, remaining in $\mathbb{Z}_{\geq 0}^2$, and ending at the origin. The reconstruction is (13.2)–(13.3), with the contour order supplying the embedding and root. Thus uniform pairs correspond exactly to the walk conditioned on this excursion event. Applying the bilateral sewing gives the uniform infinite spanning-tree decorated planar map $\mathcal{M}$.*

The finite assertion, including the root and all non-tree edges, is Mullin’s bijection in the explicit form [12, Section 2.3, Theorem 1]; see also [28, Section 3.2, Theorem 3.3]. The bilateral construction and the coordinate-contour identification are [17, Sections 3.1.1–3.1.3, Proposition 3.3]. Each excursion has probability 4^{-2n} before conditioning, which proves the asserted uniformity. A pair has $1 + \#\{\text{first-coordinate up-steps}\}$ vertices: its spanning tree has exactly one edge for each up-step.

For a finite integer interval $I = [a, b]$, define its *internal graph* G_I by the same rules, retaining every horizontal step in I , including unmatched traversals, and only vertical matches whose two steps belong to I . Let π_I be the projection of contour times to its vertices, and let d_I be its all-edge graph distance. Both are functions of $(Z_t^{\text{disc}} - Z_a^{\text{disc}})_{a \leq t \leq b}$. The equivalence relation inside I is the restriction of the bilateral relation: its minimum test already concerns exactly the interval between the two times. Hence G_I maps injectively on vertices to $\mathcal{M}$, with every edge a genuine map edge. It is generally not an induced subgraph. If $I \subset J$, every path in G_I is available in G_J . Consecutive contour vertices are connected at cost at most one, so G_I is connected.

The only discrete metric theorem imported here is the following consequence of [10, Theorem 1.6, model 2]: for the root vertex ρ of $\mathcal{M}$,

$$\lim_{R \rightarrow \infty} \frac{\log \#B_{d_{\mathcal{M}}}(\rho, R)}{\log R} = d_\gamma \quad \text{almost surely.} \quad (13.4)$$

In particular, for each $\varepsilon > 0$, the probability that this ball has more than $R^{d_\gamma + \varepsilon}$ vertices tends to zero. These distances use all primal edges.

13.2 Quantum surfaces, Brownian contours, and interval cells

A quantum surface is an equivalence class of fields in conformal charts, where a conformal map $\phi : U \rightarrow V$ changes a field h on V to $h \circ \phi + Q \log |\phi'|$ on U . Its area measure is denoted by μ_h

and its quantum boundary length by ν_h . The constant-shift conventions are

$$\mu_{h+c} = e^{\gamma c} \mu_h, \quad \nu_{h+c} = e^{\gamma c/2} \nu_h. \quad (13.5)$$

The ordinary unit-area sphere is the sphere measure of weight $4-\gamma^2$ disintegrated at $\mu_h(\mathcal{S}) = 1$, as in [13, Definition 4.21(ii)]. Its distinguished points are forgotten at the end. By [13, Proposition A.13], these points are conditional quantum-area samples, so forgetting them gives exactly the ordinary unmarked sphere law.

Theorem 13.2 (Peanosphere and interval surfaces). *The following statements hold at $\gamma = \sqrt{2}$.*

1. *A γ -quantum cone decorated by an independent space-filling SLE_8 curve η , subsequently parameterized by μ_h -area with $\eta(0) = 0$, has a bilateral boundary-length process $Z = (L, R)$ with independent Brownian coordinates. The process determines the decorated quantum surface up to its conformal coordinate equivalence.*
2. *For the ordinary unit-area sphere, the corresponding area-parameterized curve has duration one, with $\eta(0) = \eta(1)$, and its boundary-length process consists of two independent Brownian excursions. It determines the decorated sphere.*
3. *On the cone, for a deterministic finite interval $[a, b]$, the restricted increments $(Z_t - Z_a)_{a \leq t \leq b}$ determine the abstract quantum surface $\eta([a, b])$, its restricted curve, and its incoming and outgoing boundary arcs with their quantum lengths. The determination is modulo conformal maps. Abstract decorated surfaces from disjoint bilateral intervals are independent. For the sphere, the same reconstruction holds on $[\delta, 1 - \delta]$ for every fixed $0 < \delta < 1/2$, with its excursion increment law.*

The cone assertions are [13, Theorems 1.9 and 1.11]; the interval assertion is also explicitly proved in the proof of [13, Corollary 9.3], with independence in the proof of [13, Lemma 8.2]. A precise formulation of interval determination is [18, Lemma 4.9]. The sphere assertion is [34, Theorems 1.1 and 3.1]. The symmetric-interval reconstruction is proved in the final paragraph of its proof of Theorem 1.1 for $\gamma \leq \sqrt{2}$, on p. 49. The correlation is $-\cos(\pi\gamma^2/4) = 0$. A fixed deterministic normalization of contour heights matches the Brownian variance convention to the walk variance $1/2$ per coordinate per step; this choice affects neither the quantum area parameter nor the target metric law. For completeness, bilateral contour convergence follows by the ordinary central limit theorem for bounded independent increments: their covariance is $\frac{1}{2}I_2$, and disjoint time increments remain independent. The bound $\mathbb{E}|Z_b^{\text{disc}} - Z_a^{\text{disc}}|^4 \leq C|b - a|^2$ gives tightness of the linearly interpolated paths on bounded windows. The excursion version and its density estimates are addressed in Section 19.

Here are the precise topological facts used when passing from contours to cells. Two Brownian times have the same image if and only if they are identified by the equivalence relation generated by

$$L_t = L_u = \inf_{[t,u]} L \quad \text{or} \quad R_t = R_u = \inf_{[t,u]} R, \quad t \leq u. \quad (13.6)$$

Away from the common sphere endpoint, every class is a singleton, a pair, or a triple, and every nonsingleton class belongs to just one of the two coordinates. In a triple $t_1 < t_2 < t_3$, t_2 is a strict local minimum of that coordinate; different strict local minima have different heights. These statements are the Brownian identification description in [13, Section 1.3.1, Figure 1.6 and footnote 2, Proposition 1.6, Lemma 8.15]. The mixed-class exclusion includes both temporal orientations, as stated in the paragraph accompanying Figure 1.6 of [13]. Its footnote explains the

record-set argument: locally these sets have the law of zero sets of reflected Brownian motions, with time reversal for the opposite orientations, and a point is polar for the corresponding planar Brownian motion. Brownian minima on rational intervals are unique, and distinct strict local minima have distinct heights. Four scalar-equivalent times would give two such minima at the same height. Local absolute continuity transfers these facts between finite Brownian windows and excursion interiors.

These fiber facts supply the contact identifications. We will also need paths through a cell's interior. For each fixed $a < b$, the cell $H_{a,b} := \eta([a, b])$ is homeomorphic to a closed disk when $\kappa' = 8$ [10, Section 4.1.2]. Thus its interior is connected and both tips are accessible from the interior. We use this simultaneously for a prescribed countable collection of intervals and for independent Poisson endpoints. Every deterministic $t \in (a, b)$ has $\eta(t) \in H_{a,b}^\circ$ almost surely, since lying on its frontier would impose a one-sided record-minimum condition at that deterministic time. Moreover,

$$\eta^{-1}(H_{a,b}^\circ) \subset (a, b). \quad (13.7)$$

Indeed an outside visit to an interior point would, by continuity, put the image of a positive-length outside interval inside $H_{a,b}^\circ$. This contradicts area parametrization, which makes images of time intervals with disjoint interiors overlap only in zero quantum area. The same argument places both tips on the boundary. The curve is proper in the bilateral plane: $\eta(t) \rightarrow \infty$ as $t \rightarrow \pm\infty$. Hence the preimage of a planar compact set is bounded. None of these topological assertions identifies an internal metric limit at a cell boundary.

13.3 The established LQG metric

Theorem 13.3 (Reference metric). *With one deterministic normalization fixed, the γ -LQG metric D_h is a measurable function of h , is a length metric, and induces the conformal-chart topology. Its internal metric $D_h(\cdot, \cdot; U)$ is determined by $h|_U$. For continuous f ,*

$$D_{h+f}(x, y) = \inf_{P: x \rightarrow y} \int_0^{\text{len}(P; D_h)} e^{\xi f(P(t))} dt, \quad (13.8)$$

where the infimum is over D_h -rectifiable paths parameterized by D_h -length. For conformal $\phi: U \rightarrow V$,

$$D_{h \circ \phi + Q \log |\phi'|}(x, y; U) = D_h(\phi(x), \phi(y); V). \quad (13.9)$$

On the whole plane geodesics exist between all points, and for each fixed pair the geodesic is almost surely unique. The metric extends intrinsically to the cone and sphere; the unit-area sphere is almost surely a nondegenerate compact metric space.

Existence and the length, locality, and Weyl properties are [23, Theorems 1.1–1.2 and Section 1.2]; geodesic and local-extension conventions are recorded in its Sections 1.2 and 1.5, including Remark 1.5. Full conformal covariance is [22, Theorem 1.3]. In particular, for $c_v = \gamma^{-1} \log v$,

$$D_{h+c_v} = v^p D_h. \quad (13.10)$$

Only the already established metric is assigned these axioms.

We also use internal control near artificial flat faces. If K is compactly contained in a field chart and $0 < \alpha < \xi(Q - 2)$, then almost surely there are finite random C and $r_0 > 0$ such that every sufficiently small dyadic open square S meeting K , with closure in the chart and side length $r < r_0$, satisfies

$$\sup_{x, y \in S} D_h(x, y; S) \leq Cr^\alpha. \quad (13.11)$$

This follows from the simultaneous internal-square bound [11, Lemma 3.20, Equation (3.67)], using dyadic Borel–Cantelli; see also its Proposition 3.10. It controls approach to a flat face from within the square. Local absolute continuity transfers this almost-sure assertion to the charts used here. Quantitative probability bounds for a family of changes of field require separate uniform estimates.

13.4 Local fields and locally retained Poisson cells

Use h for the real quantum field and $\hat{h}$ for the independent imaginary-geometry field generating η before area parametrization. Its angular period is $2\pi\chi$, where $\chi = 2/\sqrt{\kappa} - \sqrt{\kappa}/2 = 1/\sqrt{2}$ and $\kappa = 16/\kappa' = 2$; see [24, Section 2.1.3] for this convention. Precisely, take a whole-plane GFF $\hat{h}_0$ whose average on $\partial\mathbb{D}$ is zero, independently of h , and a uniform $\Theta \in [0, 2\pi\chi)$ independently of both fields. Then $\hat{h}$ is the class of $\hat{h}_0 + \Theta$ modulo additive constants in $2\pi\chi\mathbb{Z}$. This is the standard whole-plane GFF modulo a period [33, Section 2.2.1, p. 48].

This phase convention makes the imaginary-field law invariant under every fixed positive dilation. Indeed, for $r > 0$ the field $\hat{h}_0(r \cdot)$, after subtracting its circle average on $\partial\mathbb{D}$, is a circle-average-zero whole-plane GFF. Conditional on $\hat{h}_0$, its removed average added to Θ , modulo $2\pi\chi$, is still uniform. Thus $\hat{h}(r \cdot)$ has the same law as $\hat{h}$. The domain Markov property in this convention is [33, Proposition 2.8].

We specify the joint cone normalization used for changes of quantum units. For deterministic $v > 0$, let $c_v = \gamma^{-1} \log v$ and use the positive dilation

$$r_v = \sup\{r > 0 : h_r(0) + Q \log r = c_v\},$$

where $h_r(0)$ denotes the circle average of the cone field. The circle-average construction makes $r_v \in (0, \infty)$ almost surely and measurable from h alone. Define

$$\begin{aligned} h^{(v)} &= h(r_v \cdot) + Q \log r_v - c_v, & \hat{h}^{(v)} &= \hat{h}(r_v \cdot), \\ \eta^{(v)}(t) &= r_v^{-1} \eta(vt), & Z^{(v)}(t) &= v^{-1/2} Z(vt). \end{aligned} \tag{13.12}$$

By [13, Proposition 4.13(i)], $h^{(v)}$ has the original circle-average cone law. The same normalization and curve transport are written explicitly in [24, proof of Theorem 4.4]. Conditional on h , dilation invariance and independence give the original imaginary-field law to $\hat{h}^{(v)}$. Consequently $\hat{h}^{(v)}$ is independent of $h^{(v)}$. The dilation is positive, so the imaginary-geometry coordinate change has no argument term. Its curve is therefore exactly $\eta^{(v)}$ after area parametrization. Boundary-length scaling gives $Z^{(v)}$, and the metric transforms as

$$D_{h^{(v)}}(x, y) = v^{-p} D_h(r_v x, r_v y).$$

It follows that the full joint marginal $(Z^{(v)}, h^{(v)}, \hat{h}^{(v)}, \eta^{(v)}, D_{h^{(v)}})$ is the same for every fixed v . Countably many fixed values of v use these transformations of one realization. This convention concerns the continuum background; it does not impose a scaling rule on an additional passage law.

Proposition 13.4 (Background locality). *The following properties concern the fields, curve, and marks.*

1. *On disjoint buffered subdomains of a GFF chart, conditioning on the exterior gives independent zero-boundary GFF parts and exterior-measurable harmonic parts, for each of the two fields. For changes of the real reference field, fix one pinning circle outside all the subdomains and*

their dependency buffers; the imaginary constant is interpreted modulo its angular period. On compact patches inside the cone's unit disk and away from zero, the actual normalized real-field law is locally equivalent to this GFF restriction. The sphere has the corresponding local absolute continuity modulo constants for background properties invariant under constant shifts. Its metric law will instead be transferred using the excursion encoding in Section 19.

2. *For an open U and a positive spatial buffer contained in a chart, $\hat{h}$ on that buffer determines all oriented curve segments between entrance into and exit from U . The real field gives their quantum-area parametrization and the quantum lengths of their directed frontier arcs.*
3. *Add a Poisson point process of area times of intensity ε^{-1} , independent of the surface and curve. The cells between consecutive marks which are compactly contained in U , with their directions and frontier lengths, are determined by the two local fields and the spatial marks in U . Conditional on the two fields, the spatial marks form a Poisson process of intensity $\varepsilon^{-1}\mu_h$ and have independent restrictions to disjoint open sets. Nested Poisson refinements may be used with their level labels.*

The GFF decomposition is [24, Lemmas 2.1–2.2]; the buffer in the curve assertion is exactly that of [24, Lemma 2.4]. The retained-cell statement is [8, Lemma 3.7], with the spatial marked-point data specified there. The following elementary details explain the form used here. By the mapping theorem for Poisson processes and area parametrization, the conditional intensity of the spatial marks is $\varepsilon^{-1}\mu_h$, regardless of the curve. Thus their restrictions are conditionally independent. Locally oriented curve segments locate consecutive marks whenever the entire intervening cell stays in U ; compact containment permits the stated buffer and exhaustion. Restriction of the area and frontier-length measures then supplies the local parametrizations and directed lengths. Independent additional Poisson processes give nested refinements; retaining their level labels preserves locality. These facts specify local cell data, not a conditional law for graph distances.

For the asserted transfer of field laws, in the circle-average embedding the cone field on the unit disk has the law of a whole-plane GFF minus $\gamma \log |\cdot|$, with the same circle average normalization [13, Definition 4.10]; see [10, Section 4.1.1]. On a buffered patch compactly contained in $\mathbb{D} \setminus \{0\}$, this logarithm is smooth. Extend it with a smooth cutoff supported away from the pinning circle. The Cameron–Martin theorem and the GFF domain comparison [10, Lemma 2.1] then give local equivalence for the actual field restrictions. Reference tests use such patches or their fixed affine images, with one fixed pinning circle outside the full dependency region. Intrinsic coordinate and quantum-unit identities remain available on other domains, but do not assert absolute continuity under random reanchoring. For the unit-area sphere, [7, Lemma 2.4] gives local absolute continuity modulo constants, with smooth logarithmic terms away from the marks. We use this assertion for constant-invariant background properties, not to identify an absolutely normalized passage law.

13.5 The form of confluence used below

Proposition 13.5 (Dense-pair coverage of geodesic subarcs). *Let h be a whole-plane GFF and fix a countable dense set $\mathcal{Q} \subset \mathbb{C}$. Almost surely, every nonempty open subarc of every nonconstant D_h -geodesic contains a nonconstant subsegment of the geodesic between some two points of $\mathcal{Q}$. The same local conclusion holds for geodesic subarcs in buffered field charts on the cone or sphere, with internal metrics used for the localization.*

Proof. The strong confluence theorem is [6, Theorem 2]. More specifically, its Propositions 5–6 say that a strictly interior geodesic segment is uniquely minimizing, and that sufficiently small perturbations of its endpoints force every connecting geodesic to contain a prescribed shorter interior segment. Given an open subarc of $P : [0, T] \rightarrow \mathbb{C}$, choose $s < t$ strictly inside its parameter interval and $0 < \varepsilon < (t - s)/2$. Choose points of $\mathcal{Q}$ in the endpoint neighborhoods from Proposition 6. Their geodesic contains $P([s + \varepsilon, t - \varepsilon])$. There are countably many pairs in $\mathcal{Q}$, so fixed-pair uniqueness holds for all of them on one event of probability one.

For a local chart first shorten the subarc so that it lies in a smaller domain with positive metric distance to the boundary of a larger chart. The internal version of strong confluence, proved in [6, Section 6.1, proof of Proposition 50, Equation (93)], applies there. The field absolute continuity in Proposition 13.4 transfers this local event. By shortening again, connecting geodesics between nearby endpoints are too short to reach the chart boundary, so their relevant segments are also ambient geodesics. Additive constants rescale every length equally and do not affect the conclusion. $\square$

14 Internal moments and scales of compactness

We seek deterministic scales at which internal graph distances are tight and their normalizing moment survives in the limit. The ingredients are the unconditioned bilateral walk of Theorem 13.1, independence of its increments, and the volume bound (13.4).

For an integer interval $I = [a, b]$ of length $|I| = b - a$, let G_I , d_I , and π_I be its internal graph, all-edge distance, and contour projection. Define the endpoint costs, diameters, and dyadic endpoint norms by

$$X_I = d_I(\pi_I(a), \pi_I(b)), \quad \Delta_I = \text{diam}(G_I, d_I), \quad A_j = (\mathbb{E}X_{[0, 2^j]}^k)^{1/k} \quad (j \geq 0). \quad (14.1)$$

Graph diameters refer to vertex sets. Recall that $p = 1/d_\gamma$, $s = p/2$, and the integer k is fixed with $ks > 8$; in particular, $s > 1/k$.

The contour path gives $X_I \leq |I|$. Equality holds when all increments are $(1, 0)$, an event of positive probability, so every A_j is finite and positive; in fact $A_0 = 2^{-1/k}$. Internal paths remain available when their interval is enlarged. Concatenation and Minkowski's inequality therefore give

$$X_{[a, c]} \leq X_{[a, b]} + X_{[b, c]} \quad (a \leq b \leq c), \quad A_{j+1} \leq 2A_j. \quad (14.2)$$

Costs on intervals with disjoint interiors are independent, since each uses only its own increments. Their laws depend only on the interval lengths.

14.1 Norm growth from volume

The volume-counting idea is closely related to [26, Lemma 3.2]: many vertices visited during an encoded interval must fit in one ambient metric ball. We apply it below to endpoint moments and prove the needed estimate directly.

Lemma 14.1 (Norm growth). *The deterministic sequence in (14.1) satisfies*

$$\limsup_{j \rightarrow \infty} \frac{\log_2 A_j}{j} \geq p. \quad (14.3)$$

Proof. For $0 \leq r \leq j$, let $M_{j,r}$ be the maximum of X_J over the 2^r consecutive dyadic intervals J of length 2^{j-r} in $[0, 2^j]$. For nonnegative random variables, the k th power of their maximum is at most the sum of their k th powers. Thus

$$\|M_{j,r}\|_{L^k} \leq 2^{r/k} A_{j-r}. \quad (14.4)$$

Decompose $[0, u]$, for any integer $u \in [0, 2^j]$, into dyadic intervals with at most one of each length. Their internal endpoint paths concatenate to an all-edge path in the full map. Thus, with π denoting the full-map projection,

$$\max_{0 \leq u \leq 2^j} d_{\mathcal{M}}(\pi(0), \pi(u)) \leq \sum_{r=0}^j M_{j,r}.$$

Suppose (14.3) fails. Since $1/k < p$, we may choose $1/k < \alpha < \beta < p$ and a finite C such that $A_j \leq C2^{\alpha j}$ for every j . By Minkowski's inequality and (14.4), the L^k norm of the last display's right-hand side is at most

$$C2^{\alpha j} \sum_{r=0}^j 2^{-r(\alpha-1/k)} \leq C'2^{\alpha j}.$$

Since $\alpha < \beta$, Markov's inequality places every vertex visited during $[0, 2^j]$ in the ambient ball of radius $\lceil 2^{\beta j} \rceil$ about $\pi(0)$, with probability tending to one.

The interval has $2^j/4 + o_{\mathbb{P}}(2^j)$ first-coordinate up-steps, whose terminal vertices are distinct. Different terminal heights cannot be equivalent. If two terminal heights both equal h , the base of the later step has height $h-1$ between the two terminal times, violating the minimum criterion for equivalence. This also holds in the bilateral map. The ball therefore contains at least $2^j/8$ vertices with probability tending to one. Choose $\varepsilon > 0$ with $\beta(d_\gamma + \varepsilon) < 1$. By (13.4), the same ball has at most $\lceil 2^{\beta j} \rceil^{d_\gamma + \varepsilon} = o(2^j)$ vertices with probability tending to one, a contradiction. $\square$

14.2 A quantitative dyadic modulus

A bound on earlier endpoint norms controls all internal distances and their time modulus. We keep the denominator arbitrary: the estimate will be needed before the limiting norm at a larger scale is known to be positive.

Lemma 14.2 (Moment compactness). *Fix $K < \infty$. Suppose $j \geq 0$ and a deterministic $B > 0$ satisfy*

$$\frac{A_{j-r}}{B} \leq K2^{-sr}, \quad 0 \leq r \leq j. \quad (14.5)$$

For each $T \geq 1$ there is a finite C_T , depending only on T, k, s , such that every deterministic integer interval I of length $1 \leq m \leq T2^j$ satisfies

$$\left\| \frac{\Delta_I}{B} \right\|_{L^k} \leq C_T K \left(\frac{m}{2^j} \right)^s. \quad (14.6)$$

Put $q = s - 1/k > 0$. For $2^{-j} \leq \delta \leq 1$, set

$$\Omega_{j,T}(\delta) = \frac{1}{B} \max \left\{ X_{[a,b]} : a, b \in \mathbb{Z} \cap [-T2^j, T2^j], 0 \leq b - a \leq \delta 2^j \right\}.$$

Then

$$\|\Omega_{j,T}(\delta)\|_{L^k} \leq C_T K \delta^q. \quad (14.7)$$

In particular, along any sequence $j \rightarrow \infty$ satisfying (14.5) with a common K , $B \rightarrow \infty$; the rescaled pullbacks of d_J , for a fixed bounded macroscopic confinement window J , are tight after interpolation and every limit is a continuous pseudometric. The same estimates apply on every fixed finite enlargement of the base window.

Proof. Every integer interval $[a, b]$ is the union of its maximal contained dyadic intervals, with at most two of each length. Their endpoint paths concatenate inside $[a, b]$ and hence are available in every G_I with $[a, b] \subset I$. We use the dyadic grid on the full integer line.

Diameter bound. First assume $m \leq 2^j$ and choose $r_0 \in \{0, \dots, j\}$ so that $2^{j-r_0} \leq m < 2^{j-r_0+1}$. At depth $r \geq r_0$, let $M_r(I)$ be the maximum of X_J/B over dyadic intervals $J \subset I$ of length 2^{j-r} , with the maximum of the empty collection set to zero. There are at most 2^{r-r_0+1} such intervals. Consequently

$$\|M_r(I)\|_{L^k} \leq 2^{(r-r_0+1)/k} K 2^{-sr}.$$

For any two vertices of G_I , choose contour representatives $a \leq b$ and use the preceding dyadic decomposition. It follows simultaneously for all pairs that

$$\frac{\Delta_I}{B} \leq 2 \sum_{r=r_0}^j M_r(I).$$

The geometric series gives an upper bound $2^{1+1/k} K 2^{-sr_0} / (1 - 2^{-q})$ for its L^k norm. Since $2^{-r_0} \leq m/2^j$, this proves (14.6) for $m \leq 2^j$. When $2^j < m \leq T2^j$, split I into at most $\lceil T \rceil$ pieces of length at most 2^j . Concatenation and the first bound prove the assertion after increasing C_T .

Time modulus. Choose r_0 so that $2^{-r_0} \leq \delta < 2^{-r_0+1}$. At depth $r \geq r_0$, there are at most $(2T+2)2^r$ dyadic intervals of length 2^{j-r} in the window. The L^k norm of their maximum normalized endpoint cost is at most $(2T+2)^{1/k} K 2^{-qr}$. Every interval of length at most $\delta 2^j$ has at most two dyadic pieces at each such depth, so summing this bound proves (14.7).

Compactness. Taking $r = j$ in (14.5) gives $B \geq A_0 K^{-1} 2^{sj}$, hence $B \rightarrow \infty$. On a fixed window J , the triangle inequality bounds the change in $d_J(a, b)/B$ when its arguments change by the corresponding two short-interval costs. Equation (14.7), with the harmless replacement of δ by $\delta + 2^{-j}$ for rounding, controls this change. The diameter bound gives tight values. The Arzelà–Ascoli criterion in probability therefore gives tightness of continuous interpolations, whose interpolation errors vanish. Symmetry, nonnegativity, zero diagonal, and the triangle inequality pass to every limit. The estimates used the full translated grid, so they hold on bounded enlargements as well. $\square$

14.3 Preserving the normalizing moment

Moment compactness alone does not ensure that the normalizing moment passes to the limit. Independence of disjoint walk intervals supplies the additional uniform integrability.

Lemma 14.3 (Uniform integrability without a higher moment). *Let $j_n \rightarrow \infty$ and deterministic $B_n > 0$ satisfy (14.5) with a common K . Fix $T < \infty$. For every sequence of deterministic integer intervals I_n of length at most $T2^{j_n}$, the variables $(X_{I_n}/B_n)^k$ are uniformly integrable. Consequently, if $X_{I_n}/B_n \Rightarrow X$, then*

$$\mathbb{E}X^k = \lim_{n \rightarrow \infty} \mathbb{E}(X_{I_n}/B_n)^k. \quad (14.8)$$

This conclusion also holds for any nonnegative variables bounded by the corresponding X_{I_n}/B_n , including distances between the same endpoints computed in a larger internal graph. In particular, when $B_n = A_{j_n}$ and $|I_n| = 2^{j_n}$, any weak limit of X_{I_n}/A_{j_n} has k th moment one.

Proof. Set $X_n = X_{I_n}/B_n$. Fix an integer $m \geq 2$ and split I_n into m consecutive integer intervals with lengths differing by at most one; zero-length pieces are allowed. Let $Y_{n,1}, \dots, Y_{n,m}$ be their normalized internal endpoint costs. They are independent for fixed n , and $X_n \leq S_n := \sum_i Y_{n,i}$. By Lemma 14.2, for fixed m and all sufficiently large n ,

$$\mathbb{E}Y_{n,i}^k \leq C_T K^k m^{-ks}, \quad 1 \leq i \leq m. \quad (14.9)$$

If $|I_n| < m$, the same estimate follows by increasing n : each piece has at most one step, and $B_n^{-1} = O(2^{-sj_n})$.

Expand S_n^k by the multinomial formula. The pure monomials have total expectation at most $C_T K^k m^{1-ks}$. In a mixed monomial $H_n = \prod_i Y_{n,i}^{a_i}$, the product runs over the indices with positive exponents: their sum is k , and each is at most $k-1$. Put $u = k/(k-1) > 1$. Independence, $ua_i \leq k$, and Lyapunov's inequality give

$$\mathbb{E}H_n^u = \prod_i \mathbb{E}Y_{n,i}^{ua_i} \leq \prod_i (\mathbb{E}Y_{n,i}^k)^{ua_i/k} < C_{m,T,K}.$$

Thus, for each fixed m , every mixed monomial is uniformly integrable. Also $\sup_n \mathbb{E}X_n^k < \infty$ by (14.6). Hölder's and Markov's inequalities imply $\mathbb{E}[H_n \mathbf{1}_{\{X_n > R\}}] \rightarrow 0$ uniformly in n as $R \rightarrow \infty$. Since $X_n^k \leq S_n^k$, we obtain

$$\lim_{R \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{E}[X_n^k \mathbf{1}_{\{X_n > R\}}] \leq C_T K^k m^{1-ks}.$$

Only now let $m \rightarrow \infty$; $ks > 1$ makes the right-hand side vanish. Finitely many initial n cause no problem, since each has finite k th moment. The convergence in (14.8) follows by truncating x^k and then removing the truncation. Domination proves the uniform-integrability assertion for smaller costs. Finally, stationarity and (14.1) give $\mathbb{E}(X_{I_n}/A_{j_n})^k = 1$ when $|I_n| = 2^{j_n}$, so (14.8) proves the stated normalization for limits of these internal endpoint costs. $\square$

14.4 Record scales

We now choose deterministic scales satisfying the past-ratio bound. The preceding compactness and moment conclusions will then apply simultaneously.

Lemma 14.4 (Record scales). *There are deterministic integers $j_n \rightarrow \infty$ and real numbers $t_n \uparrow p$, $s \leq t_n < p$, such that*

$$\frac{A_{j_n-i}}{A_{j_n}} \leq 2^{-t_n i}, \quad 0 \leq i \leq j_n. \quad (14.10)$$

They may be chosen beyond any prescribed sequence of lower bounds on j_n . Along these scales the moment compactness and uniform integrability lemmas hold with $K = 1$. After extracting so that every fixed-offset ratio converges, its limit a_i satisfies

$$2^{-i} \leq a_i \leq 2^{-p i} \quad (i \geq 0), \quad a_0 = 1. \quad (14.11)$$

Proof. For every $t < p$, Lemma 14.1 implies that $2^{-tj} A_j$ is unbounded. It therefore has strict record maxima at arbitrarily large indices. Choose any $t_n \uparrow p$ in $[s, p)$, and choose j_n to be a strict record for that exponent, beyond j_{n-1} and any specified lower bound. For $0 \leq i \leq j_n$, the record inequality is $2^{-t_n(j_n-i)} A_{j_n-i} \leq 2^{-t_n j_n} A_{j_n}$, which is (14.10). Compactness of each bounded ratio and a diagonal extraction give (14.11). For its lower bound, iterate (14.2) to get $A_{j_n} \leq 2^i A_{j_n-i}$ before passing to the limit. $\square$

At these records, every weak limit of the normalized unit-interval endpoint cost has k th moment one, and every fixed backward ratio is positive. The lower bound 2^{-i} does not give a uniform positive lower bound for the reference-unit ratios $2^{ip} a_i$ as i grows. It also gives no lower bound for the forward ratio A_{j+1}/A_j ; that is the normalization issue in the first-failure argument of Section 17.

We finish by recording the change of units used below. At base index j , one macroscopic unit is 2^j walk steps, contour heights are divided by $2^{j/2}$, and costs by B_j . Replacing the time unit by $v2^j$, $v = 2^{-i}$, replaces the height unit by $v^{1/2}2^{j/2}$. The reference quantum length unit changes by v^p , so costs expressed in reference metric units are divided by $v^p B_j$. This convention does not assert a scaling law for a new passage metric. The conditional-law formulation of this change of units is established in Proposition 15.9.

15 Local passage laws from independent walk blocks

The moment bounds give compactness of distances, but our later changes of field require more: a conditional law for all paths confined to a given open set. We construct that law in three steps. First retain paths together with their ranges and ordered subpaths. Next reconstruct confined paths from finitely many locally readable walk intervals. Finally compare independent interval experiments with the full walk by an explicit density and bridge calculation. Locality will mean dependence of this conditional law on both h and $\hat{h}$ in the open set; the passage object need not be a function of h . The final subsection, Section 15.5, retains one common continuum background through changes of scale.

15.1 Paths with ranges and ordered marks

Write $\overline{\mathbb{R}}_+ = [0, \infty]$, metrized by $c \mapsto c/(1+c)$, with the usual convention at infinity. For a compact metric space E , write $\mathcal{H}(E)$ for its nonempty compact subsets with the Hausdorff metric. It is a compact metric space. The empty set can be adjoined as an isolated point when recording a collection that might contain no paths.

Fix a compact time window J . At discrete resolution n , a path is a finite sequence of primal edges, with zero-cost changes between contour representatives of the same primal vertex also

permitted. Its cost is its number of primal edges divided by the deterministic denominator B_n . Here and throughout this section $B_n \rightarrow \infty$, and the interval estimates of Lemma 14.2 hold in these units. We can use either the whole sewn graph restricted to representatives in J , or one of the finite instruction graphs defined below. An edge in the latter is given the same cost as the corresponding primal edge. In an instruction graph, equality of vertices uses only its enabled first-coordinate identifications. Two window copies are not identified merely because their images coincide in a larger graph.

For every integer $m \geq 1$ retain *all* paths with $m + 1$ ordered marks, allowing repeated marks. The data of a marked path are

$$(t_0, \dots, t_m; K_1, \dots, K_m; c_1, \dots, c_m) \in J^{m+1} \times \mathcal{H}(J)^m \times \overline{\mathbb{R}}_+^m. \quad (15.1)$$

The t_i are contour representatives of the marked vertices; K_i contains all representatives used on the portion between marks $i - 1$ and i ; and c_i is any upper bound for its cost. The closure of these data is the set of *path certificates*. If a vertex has several representatives, all choices are retained, rather than choosing a preferred representative. For a finite instruction graph, replace J by the finite disjoint union of its active internal window copies, as defined below. We also retain the ranges projected into a fixed compact coordinate chart by the continuum curve in the contour coupling. The raw time ranges and their spatial projections remain separate coordinates.

Deleting an interior mark unions the adjacent ranges and adds their costs. Both operations are continuous in Equation (15.1), with addition in $\overline{\mathbb{R}}_+$. Repeated marks may always be inserted. Hence the deletion identities survive Hausdorff limits and retain traversal order, which a compact range alone would not record.

There are three related collections of observations. The raw array $\mathcal{H}_n$ records paths of the whole walk. A finite chart output records paths using only the internal windows and cross-window links declared by that chart; it is calculated from the same walk. Independent reference experiments will instead sample the chart inputs afresh to identify the conditional laws of these outputs. We retain their joint *laws* as auxiliary coordinates, while keeping $\mathcal{H}_n$ and its spatial projection throughout.

Extract jointly over bounded rational windows, finite rational cut lists, all numbers of marks, and the finite instruction graphs defined below. The Poisson cuts are independent of $\mathcal{H}_n$, the walk, and the coupled background. The input marginals of the reference experiments in Lemma 15.4 converge by the invariance principle and lattice approximation; their continuum laws stay fixed during later extractions. Include beforehand the images of the instruction lists under every quantum-unit transform used in the proof. These are the countably many fixed dyadic transforms. Their band endpoints may be nonrational, but remain a countable set of constants; rational bands give the same exhausted collection by the reconstruction with open margins below. Retain internal endpoint distances and time diameter bounds as well. Lemma 14.2 makes these finite-valued variables tight, while the hyperspaces are compact and the contours converge locally uniformly. A diagonal extraction in their countable product gives the *certificate topology*.

The continuum background has its fixed joint law with the limiting contour: fields, curve, reference metric, and residual embedding or imaginary-field randomness are all retained. To test this coupling, approximate the conditional expectation of a bounded background test given the contour by bounded continuous contour functions in L^1 of the fixed contour marginal. Use this approximation against bounded *raw* certificate tests, before spatial projection. The closure lemma below and Equation (15.20) give the details. This step needs no continuity of the map from contours to fields or metrics.

Proposition 15.1 (Certificate limit). *Suppose a sequence of bilateral sewing models satisfies the interval modulus and moment bounds of Lemmas 14.2 and 14.3, with $B_n \rightarrow \infty$. There is a joint subsequential certificate limit with the following properties, almost surely.*

1. *Every finite-cost certificate has approximating paths along the selected subsequence. Conversely, every sequence of paths with bounded costs and with ranges in a fixed compact subset of an open coordinate chart has a certificate subsequence. The same assertions hold with any fixed finite number of ordered subpaths marked.*
2. *If $K \subseteq U$, endpoints converging to the same point of K can be joined at costs tending uniformly to zero by paths in U . This holds for discrete endpoints, limiting endpoints, and mixtures of the two in the contour coupling.*
3. *Certificates concatenate finitely, with the sum of their costs and with arbitrarily small additional cost in an arbitrary open enlargement of their ranges. They can be cut by marking approximating paths and extracting further.*
4. *Let $F_U(x, y)$ be the infimum of the costs of certificates from x to y with compact range in U . On each connected open U this is a finite continuous pseudometric. For $V \subset U$ one has $F_U \leq F_V$ wherever both are defined. All these assertions are compatible with further certificate extractions.*

The assertion about continuity is local in $U \times U$; it does not assert convergence of distances constrained to the boundary of a fixed closed set.

Proof. The main point is local joining; compactness alone does not give it. On a fixed window let $Z_n \rightarrow Z = (L, R)$ uniformly. A first-coordinate identification in the discrete sewing has equal endpoint heights and no smaller height between them. A second-coordinate match has this property up to one height step. Consequently, every limit of a sequence of zero-cost identifications or single edges is a Brownian identification. Ordinary tree edges have adjacent contour times and satisfy the same conclusion. By compactness, the largest projected diameter of a discrete link tends to zero.

By the Brownian fiber description in Section 13, each nonsingleton class belongs to one coordinate and has two or three times. At a triple $a < b < c$, the middle time is a strict local minimum, with no further contact in $(a, b) \cup (b, c)$. We show that each such identification can be approximated by a discrete link and short internal paths.

Here is the required converse approximation. Suppose $f(a) = f(b) = H$ and $f > H$ on (a, b) , for $f = L$ or R . Fix a point in (a, b) . At levels $H + \delta$, take the last crossing before that point and the first crossing after it. For sufficiently small $\delta > 0$ the crossings exist, the contour stays above that level between them, and their times converge to a, b as $\delta \downarrow 0$. To see the last statement, for each $\epsilon > 0$ the minimum on $[a + \epsilon, b - \epsilon]$ is strictly above H . First fix a level strictly below that minimum, and then use uniform contour convergence. The corresponding discrete excursion at a nearest lattice level supplies a discrete identification, or a matched edge in the second coordinate. Its cost is at most $1/B_n$.

At $a < b < c$ with common height H , this construction above H supplies the pairs (a, b) and (b, c) . Below H it supplies (a, c) : the two outer times have values below H arbitrarily close on their exterior sides, since otherwise there would be another local minimum of height H . Taking the excursion above $H - \delta$ that contains b gives the claim. Thus all three contacts, including the outer pair, are approximable. At each stage nearby representatives are joined by short-time internal paths. Their costs tend to zero by the interval modulus, and their ranges stay in any prescribed neighborhood of the contact by continuity of η . The absence of mixed-coordinate classes shows that this treats every surface identification.

Uniform joining. The pointwise statement is uniform on compact sets. Indeed, otherwise there would be x_n, y_n with projected separation tending to zero in a compact $K \Subset U$, for which the least joining cost in U stays above some $e > 0$. Properness of the bilateral curve bounds all their time representatives in a common window. After subselection these representatives tend to a, b with $\eta(a) = \eta(b)$. The preceding chord construction, preceded and followed by short-time paths, gives a joining path of cost tending to zero. Its range tends to that common point and is eventually in U , a contradiction.

Extraction and concatenation. Hausdorff convergence of the closed certificate sets gives approximation and subextraction in both directions. A finite-cost limiting element can be approximated by elements with finite costs converging to its costs. Elements in the discrete closure are themselves approximable by actual paths. If its range is compactly in U , both approximations are eventually in U . Marking a sequence of paths first and then using the corresponding compact space proves the statement about subpaths. The continuous deletion-of-marks maps ensure consistency with an already retained unmarked certificate.

For concatenation, approximate a finite list of certificates. Endpoints that agree in the limit are connected by the uniform joining statement; the connecting paths have total cost tending to zero. The concatenated paths have a further certificate limit with the required ranges and cost. Reversal is immediate. For local finiteness, small time intervals and the finite chord description give finite-cost local connections; a finite chain of these connections along an ordinary continuous path in U connects any two prescribed points. The same joining modulus, the triangle inequality, and reversal show local continuity of F_U . Monotonicity follows from its definition. Every argument used compact spaces, strict spatial margins, and the retained interval modulus, so applies without change to a sequence of already-constructed limits. $\square$

15.2 Finite graphs determined by local cells

We now describe finite graph tests that capture every compactly confined path. Their inputs will be interval increments and relative frontier heights, all readable from the fields in the confinement region.

Add a nested family of independent Poisson cuts in area time, with intensities 2^q , $q \geq 0$, by independent superposition. Their images form the corresponding nested Poisson point processes of intensities $2^q \mu_h$. We call the image of a consecutive cut interval a cell. All cells used below have compact containment in the stated open set. Their orientation, area parametrization, and incoming and outgoing boundary arcs are part of the local data of Proposition 13.4; see also [24, Lemma 2.4] and [8, Lemma 3.7]. The abstract quantum surface of a run of consecutive cells is determined by its two contour increments [18, Lemma 4.9]. The embedded run is read from the local cells.

Reading the interval increments. For an oriented subsegment $C_{[a,t]} = \eta([a,t])$ in the retained cells and $X = L$ or R , put $m_X = \min_{[a,t]} X$. Its tagged incoming and outgoing X -frontiers have respective quantum lengths $X(a) - m_X$ and $X(t) - m_X$. These are lengths on the abstract interval surface, with frontier multiplicity; overlapping geometric arcs are not replaced by their union. Consequently

$$X(t) - X(a) = \nu_h(\partial_X^{\text{out}} C_{[a,t]}) - \nu_h(\partial_X^{\text{in}} C_{[a,t]}). \quad (15.2)$$

The increment identity is the boundary-length definition in [18, Equation (4.3), p. 35, and Figure 8, p. 36]; the four boundary segments are described in [13, discussion following the proof of Corollary 9.3, p. 152]. The locally determined oriented curve and its stopped left and right flow

lines distinguish the two sides and the incoming and outgoing frontiers of this segment. No order of other disconnected segments is needed. Relative time is local quantum area. To read the lengths, use the fact that the quantum length of an SLE₂ frontier is the fixed-normalization $(\gamma/2)$ -Gaussian multiplicative chaos of its 5/4-dimensional Minkowski-content measure; see [3, Definition 2.11 and Proposition 3.3] and the local absolute-continuity formulation in [18, paragraph following Theorem 4.1, p. 33]. On a compact frontier subarc in U , Minkowski content is determined by the arc and sufficiently small field mollifiers use only $h|_U$. The chaos limit, followed by exhaustion, is therefore locally measurable. Construct these measures simultaneously on the countable imaginary-geometry branches generating the curve *before* selecting subarcs by Poisson area cuts or other real-field data. Then restrict the measures to the selected subarcs, keeping each tagged frontier occurrence separately. This preserves multiplicity when a geometric arc appears on more than one frontier, and requires no independence between a selected subarc and the real field. Equation (15.2) therefore recovers the increments at every rational relative area time; continuity recovers the whole two increment paths. Absolute contour height and absolute global time cancel in the difference.

A chart on whole runs. We first describe the graph when each retained run is used as one whole block. A *reference block* is an oriented interval $I_b = [0, \ell_b]$ together with its two increment functions z_b^L, z_b^R , starting at zero, and a finite set of internal cuts. There are finitely many blocks in a package. For each coordinate $a \in \{L, R\}$ choose a forest on the blocks and give its edges offsets $o_{bd}^a = s_d^a - s_b^a$. A choice of zero at one block in each forest component determines all s_b^a in that component. No expression below uses heights from different components. A component-wide translation has no effect.

Give each block its own copy of the contour vertices. Inside that copy, enable the ordinary sewing witnessed in the block. Between different copies, enable only the band instructions defined next. Thus the forest offsets are the only independently supplied relative heights; the internal strings and declared bands determine every available link.

Put $m_b^a = \min z_b^a$ and $e_b^a = z_b^a(\ell_b)$. The incoming arc of block b in coordinate a is parametrized by the levels of the left-record set, and its outgoing arc by those of the right-record set:

$$\mathcal{I}_b^a = \{t : z_b^a(t) = \min_{0 \leq u \leq t} z_b^a(u)\}, \quad \mathcal{O}_b^a = \{t : z_b^a(t) = \min_{t \leq u \leq \ell_b} z_b^a(u)\}. \quad (15.3)$$

For this *continuum boundary parametrization*, repeated scalar representatives at a level refer to one boundary point. This statement does not identify primal vertices in the discrete second coordinate. The height ranges of these arcs are respectively $[m_b^a, 0]$ and $[m_b^a, e_b^a]$.

A band instruction consists of an ordered pair (b, d) , a coordinate a , two rational numbers $\alpha < \beta$, and a finite list W of blocks in the same offset component. It permits outgoing-to-incoming links at all levels H such that

$$\alpha \leq H - s_b^a \leq \beta, \quad H \leq s_w^a + m_w^a \quad (w \in W). \quad (15.4)$$

The outgoing and incoming arc conditions in Equation (15.3) are always imposed. The empty list W is allowed. Ordinary sewing is enabled only inside each block of this test. Cross-block first-coordinate links identify vertices and have cost zero. The second-coordinate rule is instead a rule for *oriented step incidences*. In unscaled integer heights an outgoing incidence is an up-step with terminal index u , lower height $h = R_{u-1}$, upper height $h+1 = R_u$, and $\min_{[u, \text{end of block}]} R \geq h+1$. An incoming incidence is a down-step with terminal index v , with $R_{v-1} = h+1$, $R_v = h$, and $\min_{[\text{start of block}, v-1]} R \geq h+1$. Test the entire strip $[h, h+1]$ against the band and minimum comparisons. A permitted pair adds an edge from the primal vertex $[u-1]_L$ to $[v-1]_L$, of cost

$1/B_n$. There is no zero-cost second-coordinate quotient, even for repeated visits to the same record level. This retains every edge cost on paths with arbitrarily many matches. Scaled heights give the identical rule with strip width $n^{-1/2}$. If a comparison is attained at a selected block minimum, use the exact discrete minimum, rather than a rounded continuum value.

Retaining old tests when runs are merged. The whole-block graphs suffice for capturing paths. Joint tests and refinements require one additional distinction. Suppose two old runs I_1 and I_2 , separated by an omitted gap, are included in a larger retained interval. The larger string determines their relative starts from its increments. But the old test still permits internal edges only when witnessed in I_1 or I_2 , together with its declared bands; a new graph on the larger interval also permits edges witnessed in the gap. Restricting the ranges of paths in that larger graph does not recover the old graph. We therefore keep both graphs as separate outputs and record the old intervals as windows of the larger string.

In the general package, call each supplied string a *master block*. A test selects a finite list of *active internal windows*, each an interval $[u, w]$ between recorded cuts of a master block M , with its own copy of the contour vertices. Its starting height is $s_M^a + z_M^a(u)$ and its increment path is $z_M^a(u + \cdot) - z_M^a(u)$. These starts are derived data, not new independent forest offsets. Enable ordinary sewing only as witnessed inside that window, and use the preceding band rules for cross-window links, with the windows' derived heights and increments. The simple case above has one whole active window per master block.

To preserve every old test under refinement, an arc or minimum may be taken on any subinterval whose endpoints are recorded cuts of its block. A band may use any recorded cut as its height anchor. If the anchor is time u in block b_0 , replace s_b^a in the band inequality by $q_A^a = s_{b_0}^a + z_{b_0}^a(u)$; thus its two endpoints are $q_A^a + \alpha$ and $q_A^a + \beta$. A minimum on a recorded window $[u, w]$ in block b_1 means exactly $s_{b_1}^a + \min_{[u, w]} z_{b_1}^a$. This adds only finite selectors from the internal cut lists to the instruction template. The anchor heights and restricted minima are evaluated from the same local increments; they are not additional independently sampled offsets. Equation (15.4) is the case of a start anchor and whole-block windows. This closure is needed because, after merging runs, the start of an old run is an internal cut of the merged run. It also lets a band retain its exact height endpoints upon refinement, instead of rounding an increment-dependent endpoint anew.

The formal graph uses only these instructions, its active-window list, the master block strings, and the forest offsets. In particular it does not query any omitted portion of the chronological walk. We retain its complete marked certificate sets, including confinement data in the constituent blocks. There are only countably many instruction lists: durations, increments, and offsets are input variables, not names in the list. Global time locations and the chronological order of blocks that have no declared seam are not inputs to an instruction graph.

For a fixed instruction list, place each active window on its own tagged copy of $[0, 1]$, using relative time divided by that window's duration. Retain the durations and cuts among the inputs. The marked output hyperspaces are then fixed compact spaces even when the input durations vary. Actual time-labelled ranges are recovered from these inputs; this is only a coordinate convention for the finite chart outputs.

Admissible bands. We next specify which instructions can be recognized from embedded cells. A directed shared seam is a common subarc of positive boundary length of an outgoing arc and an incoming arc in the same coordinate. On this arc the two height parameters differ by a constant. That constant is the offset; it is read at any two interior points, so choices of an endpoint are unnecessary. In testing admissibility, first clip the rational band to the two natural record-arc

ranges. A band with no cap is admissible if this clipped interval lies in the shared seam and has positive margin from its upper end. At its lower end it must either have positive margin or end at a natural record-arc minimum of a selected block, with a positive shared interval immediately above that minimum. In the latter case the rational lower endpoint is chosen strictly below the minimum: the exact discrete record-arc condition performs the clipping, at every resolution. A band may extend across the upper end H of a seam provided that its cap is the exact minimum of a selected block w and the three incident directed seams are visible: the outer pair overlaps below H , and each outer block overlaps w above H . All other effective band endpoints must have positive margin. We allow several such caps; a cap that is strictly larger than the whole band is harmless. Bands can also be restricted to a compact subarc of their constituent cells when confinement is tested. Call these *buffered admissible charts*. All parts of this definition are measurable from the embedded oriented cells and their boundary length parametrizations. Choosing a rational interior level, an integer lower bound on the margins, and a spanning forest can be done from a countable list.

Lemma 15.2 (Finite-band reconstruction). *Almost surely the following holds simultaneously for all compact $K \subseteq U$ in a countable basis of bounded coordinate patches and, by exhaustion, for all compact $K \subseteq U$.*

1. *Sufficiently fine selected cells in U form finite buffered admissible charts that capture every sequence of discrete paths whose projected ranges approach K . A single finite collection of whole bands suffices for all the links of all these paths, regardless of their number of seam crossings.*
2. *Conversely, a bounded-cost chart path whose limiting range has compact containment in U is realized by an actual sewn-graph path in any open enlargement of that range. Its cost is unchanged, apart from vanishing endpoint joining costs.*
3. *Taking common refinements, unions of finitely many charts, and closure with arbitrary open margins recovers exactly the certificates of Proposition 15.1 that are compactly in U . The statement includes all finite marked-subpath data. All forest offsets and chart selections required for this reconstruction are measurable from the fields and Poisson marks in U .*

The same conclusions hold inside the interior of a fixed interval cell, using only links witnessed within that interval.

Proof. The proof has three steps: identify which gap minima obstruct a link, read every necessary cutoff from local seams, and retain finitely many whole bands. The last step retains every crossing edge without an approximation error for each crossing.

The scalar test and omitted gaps. Let $I_b = [a_b, b_b]$ and $I_d = [a_d, b_d]$ be disjoint chronologically ordered runs, with $b_b < a_d$. A chord from a time in I_b to a time in I_d at height H exists precisely when the first time is on the outgoing record arc, the second is on the incoming record arc, their heights are H , and

$$H \leq \min_{b_b \leq t \leq a_d} f(t). \quad (15.5)$$

Indeed these conditions say exactly that $f \geq H$ between the two times. Splitting the intervening interval into selected runs and omitted gaps splits its minimum into the minima of those finitely many pieces. This is also an exact discrete statement, with the full-step convention for an edge. Thus a whole overlap, in height coordinates, is an interval with upper endpoint given by one of these finitely many minima, further intersected with the two record-arc ranges. This calculation proves both the sufficiency and the necessity of the band language.

Fix a positive r with the closed $3r$ -neighborhood of K compactly in U . Properness bounds its time preimage in a finite window. Choose a cut refinement for which every cell intersecting that neighborhood has diameter less than r and belongs to a slightly larger bounded time window. This is possible by uniform continuity of η and the fact that the largest Poisson gap in a fixed window tends to zero. Select all cells meeting the closed $2r$ -neighborhood of K . They are finitely many and compactly in U . Merge consecutive selected cells into maximal runs. Every visit to a point in the r -neighborhood of K lies in a selected cell. The endpoints of a maximal run stay away from K : if such an endpoint were close to K , the adjacent cell would also have been selected. Independent cut times are not nontrivial Brownian chord times, by conditioning on the cuts and using the Brownian facts in the proof of Proposition 15.1.

Consider all possible limits of links whose projections approach K . Their endpoint time pairs form a compact set. If such a pair determines a chord at level H , then *every* intervening time attaining H represents the same surface point. An omitted gap cannot have minimum H : a cell containing a time attaining that minimum would then have been selected. Hence every relevant omitted-gap minimum is strictly larger than H . There is a uniform positive difference after fixing the finite runs and restricting to the compact set of relevant link pairs. Otherwise a sequence of pairs and one of the finitely many omitted gaps would converge to a chord with its minimum attained in that omitted gap. The preceding observation rules this out. This argument also covers minima at gap endpoints, since their two adjacent cells would have been selected.

It follows that omitted gaps can be removed from Equation (15.5) for all the relevant levels, provided bands are chosen with strictly positive slack below their minima. A cutoff that remains is the exact minimum of a selected intervening run. Retaining its comparison in Equation (15.4) implements that cutoff exactly at every discrete resolution. This exact comparison matters: replacing it by an approximate real cutoff could add a spurious edge at a branch.

Reading offsets and exact cutoffs. Away from an endpoint of an overlap, a compact band in a directed shared seam reads the difference $s_d^a - s_b^a$ from the two boundary length coordinates. Suppose a cutoff H is attained in a selected run w . There are three times $t_1 < t_2 < t_3$ representing the chord point, with $t_2 \in I_w$. The scalar class description shows that t_2 is the sole intervening contact and a strict local minimum. The cut times lie away from all three visits. Choose small neighborhoods of the visits inside their respective runs. On compact subintervals of (t_1, t_2) and (t_2, t_3) the contour is strictly above H . The chord approximation in Proposition 15.1 then gives positive intervals of levels above H shared by the first and middle runs and by the middle and last runs. Immediately below H , the two outer runs share an interval of levels. The directions of these three overlaps distinguish the middle run. They recover its offset to the outer runs, and its recorded internal minimum supplies the cap. This proves local recognizability even though the branch point itself has zero boundary length. See Figure 5.

The same configuration explains a lower endpoint that must not be discarded. A seam between an outer run and the middle run begins at $H = m_w^a + s_w^a$ and has positive length above H . Choose its rational band to extend below H . The exact incoming or outgoing record condition in the middle run clips it at the discrete minimum $s_{w,n}^a + m_{w,n}^a$. Hence links at that minimum, as well as links at lattice heights converging to it from above, are retained. No new comparison with an omitted gap is introduced. Positivity of the shared interval above H already provides slack from every upper cutoff. More generally the lower endpoint of a shared seam is the larger of its two natural record-arc minima, by Equation (15.5); these exact natural conditions handle it in precisely the same way. Thus neither lower endpoints nor upper branch cutoffs require an additional joining cost for each crossing.

For each coordinate, join run vertices whenever such a positive overlap is used and retain a

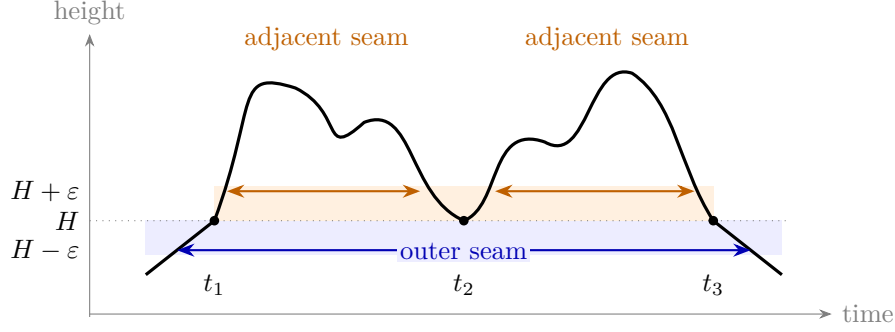


Figure 5: A triple contact in one contour coordinate. The visits $t_1 < t_2 < t_3$ at height H represent the same surface point. Overlaps below H join the outer visits; overlaps above H join adjacent visits. Their directions determine the relative offsets and identify the middle run, whose exact minimum supplies the cutoff H . Omitted gaps stay separated from the retained bands by a positive height margin. This diagram shows the incidences schematically.

spanning forest of this graph. Differences along the forest recover all offsets needed by its bands and caps. Cycles require no additional random parameters: their sums are zero because they are differences of the same height coordinates. An isolated component requires no absolute height and no comparison with another component. In particular there is no need to infer the chronological ordering of two remote runs without a shared seam. The direction of a declared seam specifies the order of its two ends; the three directed overlaps specify an intermediate cap. The omitted chronological ordering is used only in the proof that the local instructions are valid, and is not an input to them.

A finite family of whole bands. There are finitely many run pairs and gap pieces. For each relevant pair, its height set is compact. Cover it by bands with rational endpoints lying in positive seam overlaps, allowing rational lower endpoints below a natural minimum and clipping by the exact record-arc condition as just explained. Use the exact selected-minimum cap when an upper overlap endpoint belongs to the height set. The preceding positive-overlap and omitted-gap arguments supply a neighborhood of every height, so a finite subcover exists. Include the finitely many additional overlaps used to recover the offset forest. This produces one finite instruction list. Each instruction retains *all* discrete links in its band; it does not prescribe one crossing of that band. Consequently the same list captures paths with an arbitrary, even diverging, number of crossings.

Uniform contour convergence now proves capture. If a discrete link near K failed to be captured infinitely often, a limiting endpoint pair would belong to the compact set just covered. In its selected band, outgoing and incoming record conditions are the original discrete conditions, all selected minima are tested exactly, and omitted gaps have a fixed positive slack. The link therefore satisfies the instruction for all sufficiently large resolutions, a contradiction. Internal tree steps and internal matches are already included.

Realization and refinement. For realization, consider an admissible chart and a compactly confined limiting path. On a band strictly interior to a genuine shared seam, Equation (15.5) has positive margin from every omitted obstruction. At a cap the local three-overlap test identifies the selected intermediate minimum; its exact discrete comparison supplies the only possibly non-strict condition. The remaining omitted obstructions have positive slack on the confined compact set, by the same minimum argument. Thus each permitted link of the discrete approximants is an

actual link once the resolution is sufficiently large. This conclusion is uniform over all their links, because there are finitely many bands and the slack is uniform. If endpoint representatives differ, the joining conclusion of Proposition 15.1 corrects them in an arbitrary open margin at vanishing cost.

A refinement retains all old cut anchors and active windows and adds cuts. An old active window remains the same whole cut-to-cut interval; its internally enabled edge set is not changed by the extra cuts. When master strings are merged, the old windows remain selectors in the merged string, their starting heights are calculated from its increments, and their band anchors retain their exact values. Each old test graph is therefore *calculated directly* from the merged input using its old edge-witness windows and band list. It is retained as its own output component, alongside any new graph enabled on the larger union. It is not recovered by restricting ranges of paths in the larger graph, which could use newly enabled gap edges. Additional arc or minimum windows may be subdivided using the finite selector language without changing the old active internal windows. Thus every old path and every ordered marking remains exactly an output of its old test. Countably many such operations were included in the extraction. Applying capture and realization with margins decreasing to zero proves the third assertion. Measurability follows by using the countable instruction lists and rational positive margins in the local tests above.

Finally, if K is compactly in the interior of an interval cell $\eta([a, b])$, all its curve-time preimages lie in (a, b) . Indeed interval interiors are disjoint from the images of disjoint time intervals; this is also a consequence of the area parametrization and the disk-cell property of Theorem 13.2. Compactness makes the containment uniform. All cells selected near K are then internal to $[a, b]$, so the entire construction uses only links witnessed in that interval. $\square$

15.3 Conditional transfer from independent intervals

The finite graph tests use local data, but their limiting conditional laws still need to be identified. We compare independent input blocks with the same blocks in the full walk. Poissonizing the jump clock removes lattice parity; the following lemma controls the interval endpoints when we return to the original clock.

Lemma 15.3 (Stability at nonidentified cuts). *Suppose two pairs of discrete time endpoints converge to the same $a < b$ in a common bounded contour window, and the Brownian fibers of a and b are singletons. Under the uniform internal time modulus of Lemma 14.2, their internal endpoint distances, in the common cost units, differ by $o_{\mathbb{P}}(1)$. If their k th powers are uniformly integrable, their k th moments have the same limit. This applies simultaneously to every fixed finite set of deterministic or independent Poisson cuts.*

Proof. It suffices to compare the smaller and larger intervals formed from the two pairs of endpoints. The added portions are shrinking time fringes near a and b . Every edge available in the larger internal graph but not the smaller one has a witnessing step or match endpoint in one of these fringes. Likewise an old vertex represented in an added fringe is linked by a first-coordinate identification to that representative. Uniform contour convergence and singleton fibers imply that all these incidences and their other representatives lie in time neighborhoods of the same cut whose radii tend to zero in probability. Indeed, otherwise a subsequence in the containing compact window would give a link with one limiting endpoint a or b and its other endpoint a different time, contrary to the singleton assumption. The two neighborhoods are disjoint eventually.

Take a path in the larger graph between its tips. Every newly available edge is confined to one of these two neighborhoods. Keep the part after the last visit to the left neighborhood before

the first visit to the right neighborhood. Its middle uses only edges of the smaller graph. Its first and last old vertices have smaller interval representatives in the respective shrinking cut neighborhoods. Connect them internally to the smaller interval's tips using the time modulus. This adds $o_{\mathbb{P}}(1)$ once at each end. It does not pay separately for the possibly many discarded fringe visits. Taking infima bounds the smaller interval's endpoint cost by the larger one's plus this error. Graph inclusion and short endpoint pieces in the larger interval give the reverse inequality. The moment assertion follows from uniform integrability. Fixed and independent Poisson times have singleton fibers almost surely by the Brownian facts used in Proposition 15.1, so conditioning on their countable collection proves the last claim. $\square$

Put the bilateral axial walk on an independent Poisson jump clock of rate n in macroscopic time. The coordinate walks are then independent continuous-time simple walks, each of variance rate $\sigma^2 = 1/2$ before diffusive rescaling. We write $\epsilon_n = n^{-1/2}$ and use rescaled heights in $\epsilon_n \mathbb{Z}$. This removes parity constraints. The clock satisfies the uniform law of large numbers on each bounded time interval. Replacing every jump time by its jump index is an *exact* relabeling of the full sewn graph. A formal test cut at clock time T becomes a cut at index time $J_n(T)/n$; its active edge-witness windows, oriented step incidences, height anchors, and costs are identical. The law of large numbers says only that these relabeled cuts have the same limiting marks. We do not infer continuity of arbitrary formal test outputs under independently moved cuts. For internal endpoint norms with fixed macroscopic endpoints, Lemma 15.3 supplies the additional comparison. The moment argument of Lemma 14.3 also applies in Poisson time: disjoint clock intervals have independent strings, their step counts concentrate, and the probability of a count outside a fixed multiple of its mean decays exponentially. The trivial cost bound by the count controls this last event. Thus the same mixed-monomial argument gives uniform integrability, and the cut-stability lemma transfers the internal k th moments as well. This justifies returning to fixed walk lengths.

Lemma 15.4 (Finite reference experiment and conditional transfer). *Fix finitely many chart packages. A package may contain several overlapping instruction tests; tests in different packages use disjoint blocks and disjoint offset forests. Give the block durations and relative internal cuts a smooth positive density on their ordered parameter domain. Sample independent Poissonized increment strings on the blocks. Independently, sample the forest offsets with smooth positive probability densities, assigning each point of $\epsilon_n \mathbb{Z}$ the density's mass in its centered half-open lattice cell. Retain the complete marked certificate data of the package's canonical instruction graphs.*

There is a common subsequence on which these reference experiments, for all finite packages in the countable instruction language, have conditional limiting kernels

$$\nu_r(dx_r) K_r(x_r, dy_r). \quad (15.6)$$

Here x_r consists exactly of the relative block durations, internal cuts, increment paths, and forest offsets, and y_r is the package's additional certificate data. The following statements hold.

1. *Insert the selected blocks, in any chronological order consistent with the desired actual word, with positive-duration omitted gaps. Then in the limiting actual-walk experiment, conditional on the entire limiting contour and all cut marks, the law of the additional package data is $\bigotimes_r K_r(x_r, \cdot)$.*
2. *The kernel is independent of absolute time locations, integer cut labels, and the chronological ordering of pairs of blocks not used by the instruction list. It is unchanged by replacing the smooth positive reference densities. Relabeling blocks or replacing an offset forest by*

another basis for the same differences transforms the kernel by the corresponding deterministic relabeling of its inputs and outputs.

3. These assertions hold jointly for finite collections of tests, with their common refinements, and remain true after measurable selection from the jointly retained countable family of instruction templates and finite cut-mark lists. The selection may depend on the limiting contour and marks. Selection is performed after conditioning; no claim of this kind is made for selection based on the additional passage data.

Proof. We compute the density of the actual block offsets, insert the omitted intervals as bridges, and then condition on the whole contour. We finish by checking relabeling and refinement, which are needed for local chart selection.

Independent reference experiments. Under the reference law, distinct packages are independent. Their input marginals converge by the invariance principle and lattice approximation of the offsets: sample a real offset from its density and replace it by the center of its lattice cell, moving it by at most $\epsilon_n/2$. Their output coordinates belong to compact hyperspaces, so after extraction the joint laws converge to Equation (15.6). The joint limit of independently sampled packages is their product; this statement concerns joint weak convergence, and imposes no regularity on the conditional kernels K_r .

The exact change of density. First treat one coordinate and fix the continuous cut parameters. List all selected blocks in their actual chronological order $1, \dots, m$. Let s_i be their starting heights, e_i their net increments, and $g_i > 0$ the gap duration from the end of block i to the beginning of block $i + 1$. Set $s_1 = 0$, since all graphs are invariant under a common height translation. Complete the union of the package forests to a tree T on the m blocks. Add independent offset variables on its extra edges, with positive smooth densities. For an edge e with smooth positive probability density ρ_e , its discretized mass is

$$r_{n,e}(a) = \int_{a-\epsilon_n/2}^{a+\epsilon_n/2} \rho_e(u) du, \quad a \in \epsilon_n \mathbb{Z}. \quad (15.7)$$

The half-open cells partition $\mathbb{R}$, so these positive masses sum exactly to one. Uniform continuity on compact sets gives $\epsilon_n^{-1} r_{n,e}(a) \rightarrow \rho_e(a)$ uniformly for a in a compact set. Gaussian densities are one possible fixed choice; the construction applies to every smooth positive probability density.

The map from the $m - 1$ tree differences to $(s_2 - s_1, \dots, s_m - s_{m-1})$ is unimodular. To prove this, recover each s_i by summing signed edge differences along the unique path in T from 1 to i . This is an integer linear map. Its inverse is obtained by reconstructing the s_i from the consecutive differences and subtracting the appropriate two values for every tree edge. Both maps have integer entries, so their determinants are integers whose product is one. Thus their determinants are $+1$ or -1 , and no extra lattice multiplicity occurs.

Let $p_t(k)$ be the transition mass of the unscaled coordinate walk in time t . Conditional on the selected block paths, the required gap displacement is $\epsilon_n^{-1}(s_{i+1} - s_i - e_i)$. The exact density of the actual relative starts with respect to the reference offset law is

$$W_n^a = \frac{\prod_{i=1}^{m-1} p_{ng_i}(\epsilon_n^{-1}(s_{i+1} - s_i - e_i))}{\prod_{e \in T} r_{n,e}(a_e)}. \quad (15.8)$$

Indeed, summing its numerator over all consecutive gap displacements gives one, and unimodularity identifies this sum with the sum over tree offsets. Multiply Equation (15.8) for the two coordinates

and multiply by the ratio of cut-parameter densities. This is the full density W_n . If an absolute starting value is needed for reconstructing the whole contour, include an anchored additional block, or record that value separately; no chart graph uses it. External past and future portions are added conditionally on the extreme endpoints in the same way as the gaps.

Uniform density and bridge limits. The Fourier formula for a continuous-time coordinate walk is

$$p_{nt}(k) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \exp\{-n\sigma^2 t(1 - \cos \theta) - ik\theta\} d\theta. \quad (15.9)$$

After $\theta = \epsilon_n u$, the integrand converges to $\exp\{-\sigma^2 t u^2/2 - i x u\}$ when $\epsilon_n k \rightarrow x$. The bound $1 - \cos \theta \geq 2\theta^2/\pi^2$ for $|\theta| \leq \pi$ gives an integrable Gaussian majorant uniformly when $t \in [t_-, t_+] \subset (0, \infty)$. Truncating the u integral first also gives uniformity in x on compact sets. Therefore

$$\sup_{\substack{t \in [t_-, t_+] \\ |\epsilon_n k| \leq M}} \left| \epsilon_n^{-1} p_{nt}(k) - \frac{e^{-(\epsilon_n k)^2/(2\sigma^2 t)}}{\sqrt{2\pi\sigma^2 t}} \right| \rightarrow 0. \quad (15.10)$$

There are $m - 1$ numerator and denominator lattice factors in Equation (15.8); they cancel. On compact parameter sets with positive gap durations, W_n thus converges uniformly to a finite positive continuous function W .

Conditional on the selected block paths and their starts, the omitted gaps are independent bridges and are independent of every y_r . Their rescaled laws converge uniformly, on the same compact parameter sets, to the corresponding Brownian bridges. Here is a proof of the uniform assertion. Their finite-dimensional densities are products of the transition masses divided by the whole-gap mass; applying Equation (15.10) proves convergence at times bounded away from each other and the endpoints. On the first half of a gap of duration t , the bridge density with respect to the unconditioned walk started at its first endpoint is

$$\frac{p_{n(t-u)}(k - X_{nu})}{p_{nt}(k)}, \quad u \leq t/2.$$

The denominator is at least $c\epsilon_n$ for compact endpoint displacements and $t \in [t_-, t_+]$. The Fourier bound gives a numerator at most $C\epsilon_n$ uniformly in its integer argument. Thus the density is uniformly bounded. The free-walk invariance principle gives tightness on this half. Time reversal proves it on the second half. Together they prove bridge tightness; the finite-dimensional limits identify every subsequential limit, proving uniform convergence. The same argument applies to a finite vector of gaps and to external contour pieces on any bounded window.

Conditioning on the complete contour. Let a denote the added completion offsets and let z denote all omitted bridge paths. Test the joint law against products of bounded continuous functions of the inputs, outputs, and bridges. Integrate the bridge variables first. On a compact parameter set, uniform bridge convergence replaces the bridge conditional expectation by its Brownian counterpart, and uniform convergence of W_n replaces the density by W . Product convergence of the independent reference packages then gives the limiting law

$$W(x, a) \nu(dx, da) \mathcal{B}(x, a; dz) \prod_r K_r(x_r, dy_r). \quad (15.11)$$

The factor W need not factor across packages. Both it and the bridge law $\mathcal{B}$ depend only on the contour inputs, not on the additional passage variables. The complete contour and its cut marks recover (x, a, z) , including the unused completion offsets. Disintegration of Equation (15.11) therefore leaves exactly $\prod_r K_r(x_r, dy_r)$.

We remove the compact restrictions explicitly. Exhaust the finite cut-parameter domain by compact sets on which gap durations are positive, then bound endpoints and offsets, and finally bound the selected block paths in compact sets of the continuous-path topology. Under the actual law these exclusions have probability tending to zero, uniformly along the sequence, by the finite-block invariance principle, tightness, and positivity of the finitely many limiting cut gaps. If Q_n is the reference law and P_n the actual law,

$$\mathbb{E}_{Q_n}[W_n \mathbf{1}_{A^c}] = P_n(A^c).$$

Thus the same exhaustion controls the discarded density mass. On each compact set the densities were uniformly bounded. This justifies the limiting density calculation and its normalization; it does not assume a global bound for W_n . A monotone-class argument now gives the conditional identity for bounded measurable tests. Increasing bounded contour windows gives conditioning on the entire bilateral contour.

Labels, reference densities, and overlapping tests. Before any limiting operation, the canonical graph map restarts each block's clock at zero and its increments at zero. Its only noninternal numerical inputs are the specified forest differences. It never uses the original integer cut index, an absolute starting height, the starting time of a block, or any omitted jump count. The global chronological permutation appears only in Equation (15.8). Since this weight disappears from the conditional output law in Equation (15.11), that permutation is not an additional parameter of K_r .

Changing a smooth positive reference density multiplies the joint reference law by a function of the input variables alone. On a compact parameter set the ratio of the lattice cell masses converges uniformly to the continuous density ratio. The preceding test-function argument, without bridge variables, therefore leaves the limiting kernel K_r unchanged. Exhaustion is justified under the replacement law by tightness of the lattice approximations. The same argument allows any smooth positive joint density for the finite forest-offset vector within a package, independent of its strings: integrate over the product lattice cells and use their common volume. Distinct packages remain independent.

A permutation of block names is an exact relabeling of the finite graph map. A change of forest basis is a unimodular integer linear map U of the offset vector, and recovering the same height differences gives exactly the same graph. To compare the discretized input laws, write $C = [-1/2, 1/2]^d$ for the unit cell in this vector space. The pushed-forward masses integrate the transformed smooth joint density over cells $b + \epsilon_n UC$; its direct discretization uses $b + \epsilon_n C$. Both cells have volume ϵ_n^d and diameter tending to zero. Their masses divided by this volume converge uniformly on compact sets to the same positive density. Thus the ratios of the two masses tend uniformly to one there. Both lattice laws are tight, as each is a vanishing displacement of the same continuous input law, so their total variation distance tends to zero. Applying the graph map preserves this bound even when its output is discontinuous. Density invariance now identifies the new-basis kernel with the transformed old kernel. This also allows different positive reference densities for different finite tests. Consequently a locally measurable enumeration of blocks leaves no global label in the kernel.

For overlapping tests retain their joint output in one package and apply the preceding argument to the union of their finite graphs. The independently sampled blocks in this common experiment are always the maximal consecutive runs of the union. Adjacent pieces inside one such run are recorded subwindows of a single block, with their relative starts determined by that block's increments. They are never treated as separate independent blocks with a zero-duration gap or a smooth density on their forced offset. Thus all gaps to which Equation (15.8) is applied remain

positive. If two forest choices are required, express their union of differences in a single spanning forest, as above. For disjoint spatial packages, the block sets and their forests are disjoint, so their union is a forest and the same completion calculation applies, even if their chronological runs interleave.

Merging master blocks. If previously omitted gaps are included, old separate strings can become windows in a new master string. Their offsets are then functions of that master string; it would be invalid to restrict an arbitrary version of an old kernel to this forced-offset relation. Instead, start with the old independent strings, smooth forest offsets, and their old test outputs. Insert the positive-duration old gaps as conditional bridge strings and multiply by the exact density Equation (15.8). The resulting concatenated string has exactly the law of the new master string. The old output is unchanged, since its active edge-witness windows and band list are unchanged. Calculate any new union graph as an additional output component on this same master string. Applying the already-proved density and bridge convergence to the old output gives precisely Equation (15.11); hence its conditional law given the entire new master contour is its original K_7 . This identity is proved before conditioning the master path, when the gap endpoint differences have their ordinary densities. There is no conditioning of a preexisting almost-everywhere formula on a singular slice. If an old gap has duration zero, it was already within a single old master block and the active-window identity is exact without a density change. For several old tests retain their whole vector of outputs, together with the new test, before the same operation. Thus joint outputs, rather than filtered paths of a larger graph, prove common-refinement compatibility. We do not assert that a new union output is independent of the newly observed bridge strings. The calculation of the old conditional marginal first integrates out that new output; the full output vector is retained separately in the master experiment.

Extracting the countably many finite packages jointly preserves these identities by the same compact density and bridge argument. After conditioning on the full contour and cuts they hold simultaneously outside one null set, and hence also for a countably valued contour-measurable selection. This proves the lemma. $\square$

Proposition 15.5 (Spatial product kernels). *For every open coordinate domain U , the collection of all certificates with compact range in U has a regular conditional law $K_U(h|_U, \hat{h}|_U; \cdot)$ depending only on the restrictions of the two fields to U . If $U_1, \dots, U_m$ are disjoint open sets, these collections are conditionally independent given the complete fields, with respective conditional laws K_{U_i} . These laws are consistent under restriction and exhaustion. They also exist on the free-field patches obtained by local absolute continuity in Proposition 13.4. Here the local collection retains projected endpoints and ranges, costs, and ordered-subpath incidences; the original global contour-time labels have been forgotten. Relative parametrizations inside locally retained cells may be used during the construction.*

In particular one can preserve a sampled collection on each of finitely many disjoint patches and complete it to a sample of the whole certificate law, provided that the preserved collections have the specified product conditional law for the new fields on those patches. This statement concerns conditional sampling; it makes no assertion that the competitor satisfies a Weyl rule.

Proof. Finite charts with marks retained. Fix finitely many buffered charts in each of the disjoint sets. Refine and merge their instructions within each set. Different sets use disjoint cells. Maximal runs from different sets cannot share a cut: the shared point would belong to both compactly contained cells. Thus their distinct runs have positive intervening time gaps. Lemma 15.4 applies to these packages.

The reference kernel in that lemma has four kinds of input: relative area durations, increment functions, internal relative cuts, and forest height differences. Lemma 15.2 reads all four from the oriented cells and boundary lengths in the given patch. The same local data give the embedding map that sends the time-labelled ranges in the reference output to spatial ranges. The conditional output law, after that map, is therefore a function of the fields and marks *in that patch*. Lemma 15.4 also proved invariance under global block labels, reference densities, and forest bases. Thus replacing global cut indices by a local enumeration of the cells does not discard a parameter on which the kernel might still depend. This explicit invariance is necessary; conditional factorization given extra global labels would not by itself prove the present assertion.

Conditioning on the fields. Realize the continuum fields by sampling their regular conditional law given the peanosphere contour, independently of the additional raw, time-labelled reference outputs and all time cuts conditionally on that contour. This is simply the prescribed mating-of-trees coupling, with any residual embedding or imaginary-field randomness supplied at this stage. Consequently the product identity from Equation (15.11) remains true on conditioning additionally on these fields. Its right-hand side has just been shown measurable in the local fields and local marks. Taking conditional expectation with respect to the complete fields and marks gives the same product identity. No determination of the additional passage data by the continuum fields was used.

Exhaustion. Enumerate the countable instruction templates and their rational positive-margin tests. Within a patch, enumerate the locally retained cells and all finite selections from them, using any fixed measurable ordering of their embedded marked points. Every finite set of these tests has the conditional law just proved, with the consistency supplied by the common-refinement part of Lemma 15.4. Hence they define a conditional law on the countable product of compact certificate spaces. Cylinder events form a determining algebra. The capture and realization parts of Lemma 15.2, followed by an exhaustion $K_j \in U$, identify the measurable collection generated by this algebra with the local projected part of the original retained array $\mathcal{H}$. More explicitly, $\mathcal{H}_n$ and all the charts were calculated from the same word and were extracted jointly. Capture puts every compactly confined element of its limit in the chart exhaustion; realization gives the reverse inclusion. Both statements include every finite number of ordered marks. Thus no new coupling between a metric and Poisson marks is chosen at the limit. The time cuts remain independent of the original array and background, since their finite-resolution joint law was a product and they were retained jointly. They also identify the restriction maps for nested domains. The finite product identities on cylinder events extend by a monotone class to the full local collections and then to countably many subdomains when needed.

Removing the marks. Conditional on $(h, \hat{h}, \eta, \mathcal{H})$, the time cuts are still the independent Poisson processes used at the start. Since $\eta_*(dt) = \mu_h$, the Poisson mapping theorem shows that the spatial marks have their prescribed intensities proportional to μ_h . Their law consequently has that same form conditional on both fields, and their restrictions to disjoint sets are independent under this conditioning. Nested refinements are obtained by independent local superposition. Integrating the marks out of the preceding conditional product law therefore gives

$$\bigotimes_i \left(\int \mathsf{K}_{U_i}^{\text{marks}}(h|_{U_i}, \hat{h}|_{U_i}, \pi_i; \cdot) \text{Pois}_{h|_{U_i}}(d\pi_i) \right).$$

The factor in parentheses defines K_{U_i} . This proves fields-only dependence and factorization, rather than merely a mark-conditioned assertion.

Field changes and completion. On a relatively compact patch, change the field marginal by a locally absolutely continuous density and retain the same conditional kernel. The identities are

preserved because the density is a function of the conditioning variables. Two such constructions agree on their overlap by mutual absolute continuity and the established restriction consistency. An exhaustion yields the stated free-field versions. All spaces used are standard Borel, so regular conditional laws exist. Disintegrate the whole law over its restrictions to the preserved patches to obtain the completion assertion. If a recipe is subsequently selected using the full sampled configuration, restriction to the selection event gives a submeasure of this unrestricted construction; it does not require a new conditional-independence claim. $\square$

15.4 Open margins, nondegeneracy, and changes of units

The local kernels are now constructed. We next record the properties needed when the laws themselves vary: inequalities pass through open margins, the augmented construction survives further limits, and a positive internal norm prevents all local costs from vanishing.

Lemma 15.6 (Buffered tests). *Consider a convergent sequence in the certificate topology with a uniform interval modulus. Fix $K \Subset U$, a finite cost cutoff, and compact endpoint sets in K . The following implications hold.*

1. *An upper connection bound in the limit with a strict cost margin and with certificates compactly in U implies the same bound, with a smaller margin, for the approximants. Uniform bounds over compact endpoint sets follow from finitely many witnesses and the joining modulus.*
2. *A sequence of paths in K violating a fixed lower traversal bound, with bounded costs and endpoints in prescribed closed sets, has a limiting certificate violating the corresponding weak bound in U . Any finite list of ordered crossings or subpaths can be retained with it.*
3. *Suppose reference metrics D_n are retained in the joint topology and converge locally uniformly to a continuous length metric D inducing the ordinary topology. Local inequalities $F_n \leq C_n D_n$ along reference geodesics in arbitrary open margins, and lower bounds on all certificates by $c_n D_n$, pass to the limit when $C_n \rightarrow C < \infty$ and $c_n \rightarrow c \geq 0$.*

These implications also hold under any fixed absolutely continuous change of the retained continuum marginal.

Proof. For the first assertion, start with one limiting certificate. Its range has a positive distance from U^c , and its cost is strictly below the bound, so its approximants retain both margins. For a compact endpoint family, continuity and uniform joining provide neighborhoods of the endpoints in which one may use this certificate with a prescribed small additional cost. Finitely many such neighborhoods suffice. This gives the uniform finite-net assertion, with no claim at ∂U .

For the second assertion, mark the specified crossing endpoints before extraction. Proposition 15.1 gives a converging marked certificate subsequence. Closed endpoint constraints and non-strict cost upper bounds persist in this subsequence; its range is in $K \Subset U$. This proves the required contrapositive of the lower-bound transfer.

For the upper part of the third assertion, take a D -geodesic with range compactly in an open margin. Partition it into sufficiently short segments. The partition is finite. For each segment, continuity of D and positive D -distance from its small surrounding neighborhood to that neighborhood's outer boundary imply that a D_n -geodesic between its endpoints stays in this neighborhood for all sufficiently large n . To justify existence and containment one may work in a larger compact length-space chart: a competitor of sufficiently small D_n -length beats every path reaching the outer boundary, whose endpoint distance is bounded below by local uniform convergence. The sum of these D_n -distances converges to the sum of the corresponding D -distances,

which is the length of the original geodesic. Apply the assumed upper comparison on each piece, concatenate, and extract. The lower comparison follows directly by applying it to approximants of a limiting certificate. For a fixed change of marginal density w , approximate w in L^1 by bounded continuous functions of the retained continuum data. The convergence assertions pass for each such approximation, and the L^1 error bounds the discarded probabilities uniformly. $\square$

Lemma 15.7 (Closure under further limits). *Augment a continuum certificate law by its original raw, time-labelled array $\mathcal{H}$ and its projected local arrays as separate coordinates, the common fixed continuum background, the independent time cuts, and the laws of all finite independent reference packages in Lemma 15.4. The packages retain all active-window tests, old and new outputs under master-block merging, and the countably many fixed unit transforms. Their continuum input marginals are the fixed reference marginals ν_r in that lemma. Suppose a sequence of such augmented laws has a common interval modulus and the internal moment uniform integrability above. Every joint weak subsequential limit of these augmented data has the same certificate, reference-transfer, reconstruction, and spatial product-kernel properties. In particular this class is closed under the varying-law and varying-zoom extractions used below.*

Proof. The input marginals of the reference experiments are fixed. This lets us pass their density identities to a new limit, then recover the local spatial collections from the retained raw arrays.

Write the r th reference law at stage j as $\nu_r(dx_r)K_{j,r}(x_r, dy_r)$. The output spaces are compact, the variables y_r are the raw time-labelled outputs, and the input marginals are fixed. After joint extraction these laws converge to $\nu_r(dx_r)K_r(x_r, dy_r)$. For independent packages, their product laws converge to the product of these limits. This is a statement about independent full experiments, not a claim that arbitrary conditional independence is weakly closed.

For a fixed finite family, first restrict to compact input parameters with positive gap durations. The density W and Brownian bridge law in Equation (15.11) are now the same for every j . On this compact set W is bounded and continuous; the bridge law is weakly continuous in the endpoints and positive durations, by the bridge argument in Lemma 15.4. Integrating bounded continuous bridge tests therefore preserves joint weak convergence of the reference packages. Hence that equation holds with the new K_r . The error in removing the compact restrictions is the probability of the excluded parameters under the actual contour law. This law is fixed and does not depend on j or on the additional passage kernels. Its exhaustion error tends uniformly to zero. Thus the full-contour factorization, label removal, and forest-basis invariance all pass to the limit.

For merging, retain the entire vector of old and new test outputs in the master experiment. The old component is obtained at finite resolution using the same active edge-witness windows. The positive-gap density-and-bridge construction near the end of Lemma 15.4 proves its conditional marginal identity under observation of the new master string. That construction uses the same continuous W and bridge law, so its identity passes by the preceding argument. The output coordinates and their deletion-of-marks maps are continuous maps on the retained compact arrays. Joint consistency of every finite group of tests is consequently preserved. No conclusion is obtained by restricting a merged distance or by evaluating an old kernel on a forced-offset diagonal.

First pass the fixed conditional coupling of the additional continuum background with the contour, using only raw output tests. For a bounded background test, approximate its conditional expectation given the contour by bounded continuous functions in L^1 of the fixed contour marginal. Multiplication by a bounded test of the raw $\mathcal{H}$ or raw reference outputs keeps the error uniform over j . Joint weak convergence then preserves this conditional sampling identity. This argument is applied before projection; projected outputs also depend on the background and are not claimed

conditionally independent of it. It explains why the common full background and its unit transforms must be retained, as in Section 15.5, rather than sampled separately at successive scales.

Now project raw endpoints and ranges through the retained curve. On each bounded time window, $(\eta, K) \mapsto \eta(K)$ is continuous for uniform convergence of the curve and Hausdorff convergence of the compact range; the same holds for every finite list of marks. Thus the raw arrays and background determine their joint projected limits. The projected local part of the original $\mathcal{H}$ agrees with the chart exhaustion, with open margins: on each compact $K \subseteq U$, the proof of Lemma 15.2 selects finitely many bands and exact minimum tests. Positive omitted-gap and spatial margins persist in the contour/curve coupling after a slight enlargement of confinement. Selected-minimum caps, which need not be strict, remain exact comparisons inside the same retained finite test outputs. Thus their treatment does not require continuity of the graph map at an equality. Upper inclusion uses approximating certificates, finite endpoint nets, and the common joining modulus; lower inclusion uses bounded-cost marked subpath extraction, as in Lemma 15.6. A countable exhaustion in the compact, cost cutoff, number of marks, and positive margins therefore preserves both inclusions. This proves reconstruction for the projected part of the original limiting array, rather than for a newly coupled output.

Independence of the raw original array, together with its background, from the time cuts passes as a product-law identity. The locally measurable projection and the Poisson mapping theorem now give Proposition 15.5 by its same disintegration argument. Finally all fixed unit transforms are exact transforms of the retained canonical experiments; their compatibility passes by the same density calculation. This proves every assertion. $\square$

Lemma 15.8 (A nonzero internal norm gives a nonzero local cost). *Let a sequence of discrete or already-continuum certificate laws have the uniform interval modulus and uniform integrability of internal endpoint k th powers from Lemmas 14.2 and 14.3. Suppose its internal endpoint costs on $[0, 1]$ have a limiting k th moment strictly greater than zero. Then its limiting local certificate costs cannot all vanish almost surely.*

In particular, suppose reference metrics have a fixed continuum law and local upper comparison constants $C_j \downarrow 0$ along such a sequence. A joint subsequential limit cannot have a positive internal unit-interval endpoint k th moment. The assertion allows different laws and different deterministic quantum-scale zooms at each step of the sequence.

Proof. Let $Q = \eta([0, 1])$. Almost surely this is a disk with connected interior and accessible endpoint tips, and the only preimages of its interior are in $(0, 1)$; see Theorem 13.2 and Equation (13.7). There are times $a_j \downarrow 0$ and $b_j \uparrow 1$ for which $\eta(a_j)$ and $\eta(b_j)$ are interior points. Indeed every initial and terminal positive time interval contains points of the interior: its image has positive area, while the boundary of the cell has zero area. The times can be chosen from a countable dense set, since the preimage of the interior is open.

Assume every compactly local cost vanishes. Join $\eta(a_j)$ to $\eta(b_j)$ by an ordinary continuous path in the cell interior. Its compact range is covered by finitely many open sets compactly in that interior. Subdivide the path so each successive pair is in one of these sets. A zero local cost gives certificates of arbitrarily small positive cost for these finitely many pairs. Concatenation and the interior clause of Lemma 15.2 lift them to paths using only the internal links of $[0, 1]$. Prepend and append the internal short-time paths on $[0, a_j]$ and $[b_j, 1]$. The interval modulus makes their costs tend to zero as $j \rightarrow \infty$.

To avoid any interchange of these choices with a limiting infimum, first fix j and a positive middle-path cost tolerance. There are finitely many certificates, all compactly in the cell interior. They have actual approximating internal paths for all sufficiently large resolutions. The triangle

inequality for the internal endpoint distance therefore bounds its limit by the two endpoint modulus terms plus this tolerance. Send the tolerance to zero and then j to infinity. The limiting internal endpoint cost is zero almost surely. Uniform integrability now makes its limiting k th moment zero, contradicting the hypothesis.

For the last assertion, use local compactness of the reference metric law and Lemma 15.6. An upper comparison constant tending to zero supplies zero-cost limiting certificates along every sufficiently short reference geodesic. A finite chain inside an open set gives zero local costs between any two of its points; a countable dense set and the joining modulus extend this to all pairs. The first assertion is a contradiction. Its proof used only the common compactness, moment, and interior-lifting properties, so it also applies to sequences of continuum laws and changing zooms. $\square$

It remains to identify how the construction changes under a coordinate map or a change of quantum units. The latter may change the conditional kernel and will be tracked explicitly.

Write ϕ^* for the transport of endpoints and ranges under a conformal coordinate map $\phi : U \rightarrow V$; costs are unchanged. For $a > 0$, write S_a for multiplication of every cost, including every marked-subpath cost, by a . The imaginary-field pullback is the usual imaginary-geometry transform $\widehat{h} \circ \phi - \chi \arg \phi'$, taken modulo its angular period. Only its action on the directed curve and cells is used here.

Proposition 15.9 (Intrinsic transport and fixed quantum units). *The local kernels of Proposition 15.5 can be chosen consistently under restriction so that the following law identities hold.*

1. *For every fixed conformal coordinate map $\phi : U \rightarrow V$, the pullback of the conditional certificate law on V is the conditional law on U at the pulled-back fields*

$$\widetilde{h} = h \circ \phi + Q \log |\phi'|, \quad \widehat{\widetilde{h}} = \widehat{h} \circ \phi - \chi \arg \phi'.$$

This is intrinsic coordinate covariance in law.

2. *For fixed $v > 0$, put $c_v = \gamma^{-1} \log v$ and define the new normalized law*

$$\mathbf{K}_U^{(v)}(g, \widehat{g}; \cdot) := (S_{v^{-p}})_\# \mathbf{K}_U(g + c_v, \widehat{g}; \cdot). \quad (15.12)$$

As a transformation of the joint law, this gives the real field its translated marginal, namely the law of $h - c_v$, and preserves the restriction and product-kernel properties on every U . On the buffered reference patches of Proposition 13.4, local equivalence also defines it under the unchanged Gaussian reference marginal. It need not equal $\mathbf{K}_U$. For the established metric one has

$$v^{-p} D_{g+c_v} = D_g. \quad (15.13)$$

3. *All fixed dyadic v can be retained in one joint extraction. In the time- 2^j , cost- A_j experiment, the law in Equation (15.12) uses time resolution 2^{j-i} and cost denominator $A_j 2^{-ip}$ when $v = 2^{-i}$. Consequently, if the level $j - i$ is also expressed in its own A_{j-i} units, its deterministic conversion factor is*

$$b_i = 2^{ip} \lim_j \frac{A_{j-i}}{A_j}, \quad (15.14)$$

whenever this positive finite limit is part of the extraction.

4. For a fixed physical affine chart $\phi(z) = z_0 + rz$, write $h \circ \phi + Q \log r = g + c_v$. The conditional law of the physical certificate costs pulled back to this chart and divided by v^p is exactly $\mathcal{K}_U^{(v)}(g, \tilde{h}; \cdot)$. This identity remains valid on conditioning on exterior coarse variables and then choosing a dyadic index measurably from those variables, whenever the conditional interior field marginals are absolutely continuous with respect to the local reference marginals.

These are identities at each fixed index, before any limit as $v \rightarrow 0$. Changes of density under a fixed reference marginal always use patches whose full dependency region avoids its pinning circle. No uniform absolute-continuity bound for unbounded constant shifts is asserted.

Proof. An intrinsic chart records quantum areas, boundary length increments, directed arc roles, and shared seams. These quantities are unchanged by the quantum and imaginary-geometry coordinate transformations. The instruction graph consequently stays exactly the same, while the map taking its time-labelled ranges to the coordinate plane is composed with ϕ^{-1} . Reference-kernel independence of labels and parameter densities, followed by local reconstruction, therefore gives the first assertion on every finite chart. Exhaustion gives the full local collection. This is a coordinate identity for the constructed intrinsic experiment, rather than an application of metric uniqueness. Coordinate covariance of D_h is separately the known theorem [22, Theorem 1.3].

At the level of joint laws, transform the real field by $h \mapsto h - c_v$ and every cost by $S_{v^{-p}}$. Disintegration gives Equation (15.12) under the translated field marginal, on any domain U . Conditional factorization and restriction consistency pass through this deterministic transformation.

To compare with an unchanged Gaussian reference marginal, fix the pinning circle outside the entire bounded region used by the tests, their dependency buffers, and the shift supports. Extend the constant c_v on a test patch to a smooth function supported away from that circle. The extension has zero average on the pinning circle and is a Cameron–Martin direction of this same normalized field law. Hence the shifted restriction is locally equivalent to the reference restriction, so the right side of Equation (15.12) is defined almost surely there. This argument is used only on those patches; a domain containing the entire pinning circle would observe its fixed average. The established Weyl rule gives $D_{g+c_v} = e^{\xi c_v} D_g = v^p D_g$, proving Equation (15.13).

We prove that this definition is the required time zoom, since a definition of unrelated kernels would not suffice. Adding c_v multiplies the quantum area of every fixed cell by v and its quantum boundary lengths by $\sqrt{v}$. Accordingly a canonical block with duration ℓ , increment path $z(t)$, and offsets o is transformed into

$$\ell \mapsto v\ell, \quad z(t) \mapsto \sqrt{v} z(t/v), \quad o \mapsto \sqrt{v} o. \quad (15.15)$$

Internal cuts transform with the duration. Anchored band endpoints α, β are multiplied by $\sqrt{v}$, as are their anchor heights and selected minima. These are the transformed instruction templates retained at the start of this section. Lattice-cell discretization of offsets commutes exactly with this scalar change of variables. At finite resolution, a rate- n Poissonized string in a block of duration $v\ell$ has exactly the same distribution, after the time change, as a rate- nv string in a block of duration ℓ . Its height lattice also agrees: the latter lattice, multiplied by $\sqrt{v}$, is $\sqrt{v}(nv)^{-1/2}\mathbb{Z} = n^{-1/2}\mathbb{Z}$. The matching tests, full height strips, and selected-minimum comparisons are the same integer tests on the same string. Thus this is an exact finite-experiment identity, including every path and every marked subpath. Changing the positive duration and offset sampling densities under Equation (15.15) does not alter the conditional kernel, by Lemma 15.4. Taking its joint limit proves the asserted identity with time resolution nv and denominator $B_n v^p$. Substituting $n = 2^j$, $B_n = A_j$, and $v = 2^{-i}$ gives Equation (15.14).

The continuum background is transported at the same time. If $(h, \hat{h}, \eta, Z)$ is the original realization, normalizing the physical area- v patch on the same abstract surface means

$$h^{[v]} = h - c_v, \quad \eta^{[v]}(t) = \eta(vt), \quad Z^{[v]}(t) = v^{-1/2}Z(vt), \quad F^{[v]} = v^{-p}F. \quad (15.16)$$

In particular $D_{h^{[v]}} = v^{-p}D_h$. If a canonical cone embedding is desired, use its measurable conformal re-embedding and the first assertion. The Brownian scaling and cone embedding laws are the standard ones in Theorem 13.2. All fixed v use these transformations of a single background and the same microscopic experiment. Recording countably many fixed v before any further subsequence retains their compatibility. Equivalently, one may use a common Brownian/walk coupling with uniformly negligible rescaled contour error and apply exactly the transformations in Equation (15.16).

For the final assertion, the first part says that the pulled-back physical law is $K_U(g + c_v, \tilde{h}; \cdot)$. Multiplying its costs by v^{-p} is precisely Equation (15.12). One common null set suffices for the countable fixed dyadic indices and fixed coordinate charts used in the proof. Given exterior coarse data, apply these identities on the conditional interior field space. Absolute continuity of its marginal makes the same versions valid there; selection of a dyadic index is a countable measurable partition. For random harmonic residual data, the argument applies to the whole conditional field distribution. It does not substitute uncountably many deterministic field values into arbitrarily chosen versions. This proves the last assertion and the stated order of operations. $\square$

15.5 Coherent backgrounds and the law topology

We next put all scales in one space of observations. Its purpose is to retain the same continuum background under time changes and to make the later success tests stable under extraction.

Retained observations. Enumerate the bounded rational time windows, the finite Poisson/rational chart instructions of Section 15, their rational open margins and cost cutoffs, and all finite lists of subpath marks. For each instruction use its compact closed-certificate hyperspace from Proposition 15.1. Record also the internal endpoint variables in $[0, \infty]$, the continuous contour on compact windows, and the continuum background used in Proposition 15.9. The background includes the curve and reference metric in local uniform topology and the fields in their distribution topologies. It is given its usual standard Polish realization. For every bounded connected rational coordinate domain W and every compact rational exhaustion set $K \subseteq W$, retain also $D_h(\cdot, \cdot; W)|_{K \times K}$ in uniform topology. These internal-reference-metric observations have a fixed joint marginal; retaining them does not assert continuity of the map from a field or an ambient metric to its internal metric. We also retain, as deterministic coordinates, the joint laws of every auxiliary finite reference experiment in Lemma 15.4: the canonical input and complete marked output law for each package, all finite lists of independent packages, and their common refinements. Use the same prescribed smooth input densities at every scale. Include the countably many fixed quantum-unit transforms of these templates. At finite resolution the input marginals depend on the random-walk resolution and offset lattice; the invariance principle and the lattice-cell approximation of forest offsets make them converge to the prescribed continuum input marginals. These limiting input marginals are the same for every continuum law in a subsequent varying-law extraction. Each output space is a compact hyperspace, so input tightness gives precompactness of the finite-resolution weak-law coordinates as well. They are auxiliary reference laws, not extra samples from the actual contour. Their compatibility with the actual atlas, including the density-and-bridge identities, is retained by Lemma 15.7.

A bounded product metric, with coordinate weights 2^{-r} , makes this a separable metric data space $\mathcal{E}$; use the closed compatible-data subspace when a refinement identity is imposed. We equip its laws with a metric $\mathbf{d}$ for weak convergence, for example the bounded Lipschitz metric for this bounded product metric. Equivalently, this topology is specified by integrals against a fixed countable convergence-determining family of bounded continuous functions. Thus all tests below refer to one specified countable law topology.

The raw certificate coordinates are the time-labelled compact spaces in Equation (15.1). Discrete grid times are points of the same compact time window, so a finite-resolution law belongs to this ambient space without satisfying the Brownian quotient identities. All choices of representatives are retained. Spatial projection is applied only after adjoining the continuum background; the vanishing projected diameter of individual links supplies its limiting compatibility.

One Brownian coupling for all scales. Take independent two-sided Brownian motions W_1, W_2 and an independent sequence of fair coordinate choices. For each coordinate, read successive simple-walk increments from the signs of Brownian exits from the interval of radius one about its current position. Use the coordinate choices to interleave the two resulting walks; apply the same construction independently on the negative half-line. This gives exactly the bilateral axial walk. The successive Brownian exit durations are independent, have mean one, and satisfy the law of large numbers. The numbers of each coordinate choice likewise have asymptotic frequency $1/2$. Hence the Brownian clock used by coordinate a at walk time Nt , divided by N , converges uniformly on bounded t -intervals to $t/2$, in probability. The Brownian modulus of continuity after diffusive rescaling implies

$$\sup_{|t| \leq T} N^{-1/2} |Z^{\text{disc}}(\lfloor Nt \rfloor) - B(Nt)| \longrightarrow 0 \quad \text{in probability,} \quad (15.17)$$

$$B(t) = (W_1(t/2), W_2(t/2)),$$

for every fixed $T < \infty$. The interpolation error is at most one step. This argument only uses the strong Markov property, the exit-time mean, and the law of large numbers.

Attach and transport the continuum background. Let ν be the law of the full bilateral Brownian contour. Write $\nu(db)Q(b, dc)$ for its standard joint law with the full continuum background $\mathcal{C}$, including any residual imaginary-field and embedding data. This background excludes the microscopic coordinate schedule, time cuts, and additional passage data. If $S_v b(t) = v^{-1/2}b(vt)$, its intrinsic quantum-unit transform is

$$T_v \mathcal{C} = (h - c_v, \hat{h}, \eta(v \cdot), v^{-p} D_h), \quad c_v = \gamma^{-1} \log v,$$

followed by the prescribed common reembedding of its components. The decorated-cone law is invariant under (S_v, T_v) by Equation (13.12). Here the imaginary field has the uniform phase modulo its angular period specified in Proposition 13.4. Under a positive circle-average dilation, its GFF modulo constants is invariant, and the change of anchoring constant is absorbed by that uniform phase. The dilation is determined by the real field, which is independent of the imaginary field before the curve is drawn. The curve then transforms by imaginary-geometry covariance, and the reference metric by its established covariance and constant-shift rule. Thus the invariance is for the full background, including the extra imaginary randomness. A measurable representative is chosen from the intrinsic decorated surface and never from passage data.

Disintegration of this joint invariance gives

$$\mathbf{Q}(S_v b, \cdot) = (T_v)_\# \mathbf{Q}(b, \cdot) \quad \text{for } \nu\text{-almost every } b. \quad (15.18)$$

For the countable dyadic group, take one full-measure set on which these identities and their compositions all hold. Draw $\mathcal{C}$ once from $\mathbf{Q}(B, \cdot)$, independently of the coordinate schedule and all time-cut processes conditionally on B . Put $B_j = S_{2^j} B$ and $\mathcal{C}_j = T_{2^j} \mathcal{C}$. The background at $j - i$ is exactly the time- 2^{-i} transport of the background at j , intrinsically. The discrete instruction data obey the same identity, with the extra deterministic cost conversion A_{j-i}/A_j .

Let Y_j denote the raw, unprojected time-labelled passage data and time cuts at scale j . Since S_{2^j} is invertible on the full bilateral contour, B_j determines B . The conditional sampling and (15.18) give the exact finite-resolution identity

$$\text{Law}(\mathcal{C}_j \mid B_j, Y_j) = \mathbf{Q}(B_j, \cdot). \quad (15.19)$$

One must use raw data here: their projected spatial ranges also depend on $\mathcal{C}_j$ and are adjoined afterward.

Preserve the conditional law under extraction. We verify that (15.19) passes through extraction despite possible discontinuity of $\mathbf{Q}$. Suppose $(B_j, Y_j) \Rightarrow (B_\infty, Y_\infty)$. The B_j marginal is always ν , and the $\mathcal{C}_j$ marginal is always $\nu\mathbf{Q}$, so the joint triples are tight. For bounded continuous f and g , write $\mathbf{Q}g(b) = \int g(c)\mathbf{Q}(b, dc)$. Then

$$\mathbb{E}[f(B_j, Y_j)g(\mathcal{C}_j)] = \mathbb{E}[f(B_j, Y_j)\mathbf{Q}g(B_j)].$$

Approximate $\mathbf{Q}g$ in $L^1(\nu)$ by bounded continuous functions with a common bound. The replacement error is at most $\|f\|_\infty$ times this L^1 error, uniformly in j and also in the limit. Weak convergence applies to each continuous approximation. Removing the approximation yields

$$\text{Law}(\mathcal{C}_\infty \mid B_\infty, Y_\infty) = \mathbf{Q}(B_\infty, \cdot). \quad (15.20)$$

Products of continuous tests determine the joint law, giving the stated conditional identity. Thus conditioning the limiting reference experiments additionally on the continuum fields does not change their passage kernels. This is a fixed-kernel argument; closure when the passage kernels vary uses the retained reference-law coordinates and Lemma 15.7 instead.

The law at scale j . Write μ_j for the law of these complete data in time units 2^j and cost units A_j . Equation (15.17) and Proposition 15.1 give the required joint extractions. Recording the continuum background avoids asserting that an arbitrary measurable functional of a contour is continuous. Its joint marginal is fixed, and its contour equals the discrete contour limit. The exact transport identities just established identify all fixed-offset limits with the corresponding zooms in a common extraction.

Fixed local changes of density can be included in this topology without a new compactness assumption. Truncate each density and approximate it in L^1 of the fixed background marginal by bounded continuous functions. Weak convergence applies to the approximations, and their L^1 errors are uniform over the coupled laws. Use a countable list of the buffered domains and approximations required for the shell tests. In particular, the local free-field tests below are tests of the same kernels identified in Proposition 15.5.

16 Applying comparison and rigidity to the tree passage law

The independent-walk construction supplies the local passage laws needed by the common comparison theorem. We verify that interface with the actual additive normalization of the field, and then give the additional moment argument needed to initialize the tree model. Throughout this section $\gamma = \sqrt{2}$ and the geometric volume base is $\lambda = 2$.

16.1 An actual-field reference with an exterior pin

Lemma 16.1 (Ordinary cone patches and exterior Gaussian pins). *On every buffered ordinary patch compactly inside the cone's normalization disk, and on each fixed affine image of such a patch, the actual field and its passage observations can be transferred to a whole-plane Gaussian reference whose fixed pinning circle is outside all the observation buffers. The density depends only on the field, and is the same for all kernels or quantum indices attached to that field marginal. The transfer retains the actual additive constant and the reference metric's quantum units.*

Proof. Let G_1 be the whole-plane GFF with zero average on the unit circle. The circle-average cone field restricted to $B_1(0)$ has the law $G_1 - \gamma \log |z|$, by [13, Definition 4.10 and the following discussion]. Under the deterministic coordinate map $z \mapsto z/R$, its representation in $B_R(0)$ is

$$h^{\text{cone}, R}(z) = G_R(z) - \gamma \log |z| + (\gamma - Q) \log R, \quad G_R(z) = G_1(z/R). \quad (16.1)$$

Here G_R has zero average on $\partial B_R(0)$. If $U \Subset B_R(0) \setminus \{0\}$, the deterministic function added to G_R in (16.1) agrees near $\bar{U}$ with a smooth function compactly supported away from 0 and the pinning circle. This function is a Cameron–Martin direction for G_R . Thus the two actual field restrictions to U have mutually absolutely continuous laws.

For the shell tests, choose a fixed ordinary patch inside $B_1(0)$ and move it by a deterministic affine chart to the prescribed shell. For example, $\phi(z) = 1/2 + z/128$ maps $B_{16}(0)$ compactly into $B_1(0) \setminus \{0\}$. The Gaussian pullback is pinned on $\phi^{-1}(\partial B_1(0))$, outside this entire buffer, and the actual field is

$$G_1 \circ \phi - \gamma \log |\phi| + Q \log(1/128).$$

The smooth deterministic drift again has a Cameron–Martin extension supported inside the pinning circle. This is an allowed outside-pinned reference for the common comparison argument; its circle need not be centered at the shell.

Different fixed exterior pins also give mutually absolutely continuous local field laws. To see this directly, take $K \Subset W$ with both pinning circles outside $\bar{W}$. For either pinned field, its domain Markov decomposition on W is an independent zero-boundary GFF plus an exterior harmonic function. The pin imposes no constraint on the zero-boundary part, whose extension vanishes on the pinning circle. On K , each harmonic function admits a finite-energy cutoff inside W . Conditional Cameron–Martin equivalence with the zero-boundary restriction, followed by integration, proves mutual absolute continuity. This argument compares actual field laws and does not reset a random circle average.

The imaginary field is independent and has its stationary-phase law, which is preserved by these positive affine maps. Intrinsic transport in Proposition 15.9 carries the full reference experiment, its passage kernel, and the established metric in the same chart. In particular the deterministic constants in (16.1) and in the affine pullback are retained. On a fixed field restriction, if $Z = d\nu/d\mu$ is the density of one reference relative to another, adjoining any of these same conditional kernels gives, for every event E and $T > 0$,

$$(\nu \otimes K_i)(E) \leq T(\mu \otimes K_i)(E) + \mathbb{E}_\mu[Z \mathbf{1}_{\{Z > T\}}]. \quad (16.2)$$

The tail is independent of i . Hence uniform high-probability tests transfer by first truncating this one density. Constant-shift comparisons are used only on such buffered restrictions. No absolute continuity is asserted on a domain observing a complete pinning circle. Moment identities and uniform integrability are proved under the original cone and contour laws; this density argument transfers probability bounds, not moments. $\square$

16.2 The common passage-law interface

Lemma 16.2 (Tree input to the common comparison argument). *Every moment and certificate extraction satisfies the path, local-kernel, indexed-law, and joining conditions of Definition 8.2 at $\gamma = \sqrt{2}$, $\lambda = 2$, with a singleton order space. These structural conditions give the annulus-transfer framework. The varying-law closure condition holds for sequences with a common interval modulus and uniformly integrable internal endpoint k th powers. Proposition 16.4 supplies these uniform bounds for the backward quantum families and each fixed finite forward enlargement used below, together with a separate finite comparison uniform over all fixed indices.*

Proof. We check the hypotheses in their order of use.

Reference metric and path observations. Theorem 13.3, Proposition 13.4, and Proposition 13.5 give the continuous local reference metric, its established covariance and Weyl rules, fixed-pair uniqueness, and the strictly interior confluence input. These are statements at $\gamma = \sqrt{2}$. In particular $Q > 2$ leaves the strict margin in the inward box-connection estimate used by the common proof. Proposition 15.1 retains every finite ordered marking, ranges and nonnegative subpath costs, with actual open-margin realization and extraction of bounded-cost paths. Edge costs vanish because the deterministic units tend to infinity. Its second through fourth assertions give uniform local joining, finite concatenation, and continuity of the infimal internal cost F_U . The proof uses the independent Brownian contour coordinates and the SLE_8 cell topology; these conclusions are available before any comparison or rigidity conclusion. Thus the Euclidean-circle version of Lemma 8.5 applies with this joining property.

Conditional kernels and quantum units. Lemma 15.4, Lemma 15.2, and Proposition 15.5 give compatible spatial kernels from the actual real and imaginary fields. The directed local seams supply the needed ordering information; after integrating the local Poisson marks, no additional global order is an input. The order space of Definition 8.2 can therefore be chosen as $\{*\}$ with weight one. Disjoint real patches have product kernels conditional on the complete fields, and independent trial experiments use disjoint complete dependency disks. The countable closed observation spaces are standard Borel, so the prescribed product restrictions admit the stochastic completions used in the common forcing proof.

Proposition 15.9 gives the fixed-index law

$$\mathbf{K}^{(v)}(g, \widehat{g}) = (S_{v^{-p}})_{\#} \mathbf{K}(g + c_v, \widehat{g}), \quad c_v = \gamma^{-1} \log v.$$

It is the law of the actual transformed discrete experiment: durations are multiplied by v , heights and offsets by $\sqrt{v}$, and costs are read with denominator $B_n v^p$. The matched height strips and selected-minimum tests transform with the same lattice. Together with intrinsic affine transport, this is exactly (8.2)–(8.4). Lemma 16.1 supplies each field-law change under the unchanged local kernel, in its actual units. Thus exterior-measurable index selection and the same-index likelihood argument apply without a Weyl assertion for F_U .

Closure and annulus transfer. Under its common interval-modulus and uniform-integrability hypotheses, Lemma 15.7 retains the raw passage array, independent reference experiments, and fixed time offsets under further extraction, including a varying sequence of laws. The moment bridge below verifies these bounds for the backward quantum families and each fixed finite forward enlargement. The varying sequences used in amplification and rigidity have indices tending to $+\infty$ and retain only fixed offsets, so they lie in these controlled families. The continuum background is retained in its full joint coupling. Its conditional law is independent of the entire raw array given the contour and cuts; the fixed-marginal approximation in that lemma and the coherent background construction of Section 15.5 retain this property under changing base units. Each fixed retained time scale diverges in lattice steps. In particular the kernel identity is re-established from the finite reference experiments in the new law, rather than inferred from weak convergence of conditional independence. The joining estimate passes through the same marked extraction. The common Gaussian profile, annulus-transfer and local-field modification arguments now apply. Their quantitative reference is the stated zero-boundary Gaussian law; transfer to ordinary cone patches uses (16.2) only after the desired probability threshold is fixed. $\square$

Lemma 16.3 (From indexed Gaussian charts to the whole cone). *If a fixed finite upper or lower comparison holds on the ordinary Gaussian reference charts of every fixed dyadic indexed law, it holds with the same numerical constant for the original whole cone law. Conversely, a whole-cone comparison implies these indexed local comparisons. Lower chart comparisons are intrinsic, as justified for a positive constant by Lemma 8.7; the zero lower comparison is automatic. No positive lower comparison is assumed here.*

Proof. For every fixed dyadic $v > 0$, retain the joint cone zoom of (13.12). On that realization the passage costs and reference metric are both multiplied by v^{-p} , and the spatial map is $z \mapsto r_v^{-1}z$. Apply (15.16) to the raw array and the full background before spatial projection or disintegration. The coherent construction in (15.18)–(15.19) then uses the prescribed measurable reembedding of that same background. Thus Proposition 15.9 identifies the resulting passage kernel as the corresponding indexed kernel; it need not be the original kernel. This is a joint experiment identity, not a substitution of the field-dependent r_v into versions known only for each deterministic chart separately. Ordinary patches in the normalized unit disk have the mutually equivalent actual-field laws of Lemma 16.1. Thus an almost-sure comparison for all these indexed Gaussian restrictions gives the same numerical comparison on $B_{r_v}(0) \setminus \{0\}$ in the original realization. Take the countable volumes $v = 2^m$, $m \geq 0$. The levels $c_v = \gamma^{-1} \log v$ tend to infinity, whereas $r \mapsto h_r(0) + Q \log r$ is bounded above on every compact range of radii and tends to $-\infty$ as $r \downarrow 0$. Hence $r_{2^m} \rightarrow \infty$. The preceding disks exhaust the plane. Each fixed forward volume is allowed by the finite-window moment bounds and the exact transport construction. Intersect the probability-one identities for all these fixed volumes and all ordinary rational charts before choosing a covering volume for a particular realization.

Use a countable ordinary-chart atlas in these disks to make the local comparisons simultaneous. A compact reference path has a finite partition in a buffered chart cover away from zero. For a bounded-cost certificate and a positive proposed lower constant, stop approximating paths on successive exits from smaller members of such a cover. Each completed exit costs at least that constant times a positive reference separation, uniformly over the finite cover. Marked extraction therefore bounds the number of completed exits and supplies finitely many local pieces. This argument uses the approximating paths, without assuming a continuous cost parametrization of every limiting certificate. If it meets a shrinking neighborhood of zero, omit the portion from its first to its last visit. The omitted endpoints approach zero; Lemma 8.1 and the tree

joining estimate make their joining errors vanish. For a lower bound the omitted passage cost is nonnegative. Thus the same comparisons hold across zero and on every open confinement. Conversely, a comparison for the entire cone law transports under each exact joint zoom and then under the inverse ordinary-patch density. This proves both implications. The argument does not assert absolute continuity under a random repinning. $\square$

16.3 The moment bridge to a finite upper comparison

For a fixed extracted sequence put

$$q_i = \lim_j \frac{A_{j-i}}{A_j}, \quad b_i = 2^{ip} q_i,$$

whenever these finite limits have been extracted. For every fixed backward index $i \geq 0$, (14.2) gives $q_i \geq 2^{-i}$. At any index with $q_i > 0$, denote the rule in its own level- i units by G_i ; its costs are b_i^{-1} times the rule $K^{(2^{-i})}$. Internal endpoint norms are the actual limiting norms by Lemma 14.3.

We make the coarse tests precise. After enlarging M and decreasing $m > 0$, the test $\mathcal{T}(m, M)$ says, on a fixed buffered annular band, that (i) its outer circle can be connected internally at cost at most M , (ii) every crossing from its central hole to its outer boundary has cost at least m , and (iii) the same upper and lower statements hold for D . Also include one connection across the band of D -length at most M , and one passage connection across it of cost at most M . These last connections have endpoints on two fixed separated circles, so the respective lower crossing bounds apply to them. The bands are kept compactly inside the reference shell $S = \{1/16 < |z| < 4\}$. All upper assertions permit an arbitrarily small additional cost in a fixed open margin. These are measurable certificate tests by Proposition 15.1.

Proposition 16.4 (Comparison families). *Consider a limit with the certificate and local-kernel properties of Section 15, obtained with the moment compactness bound of Lemma 14.2. Either of the following assumptions is sufficient for exact-power upper endpoint moment control, compactness of the normalized backward quantum family ($v \leq 1$) and each fixed finite forward enlargement, and a finite deterministic upper comparison with D .*

1. *For some $K < \infty$, $q_i \leq K2^{-ip}$ for every $i \geq 0$.*
2. *There is an integer $i_0 \geq 0$ such that $0 < q_i < \infty$ for every $i \geq i_0$, and constants $0 < m < M < \infty$ such that every own-unit law G_i , $i \geq i_0$, satisfies $\mathcal{T}(m, M)$ with probability at least $1 - \varepsilon_0$. Here $\varepsilon_0 > 0$ is chosen sufficiently small, once, for the annulus-coverage threshold with favorable fraction $1/4$ and this fixed geometry. The finitely many scales preceding i_0 need only have finite endpoint norms and satisfy the given compactness bound.*

In the second case there is a deterministic $T_0 < \infty$, depending only on m, M and the geometry, such that

$$T_0^{-2} \leq b_i/b_j \leq T_0^2 \quad (i, j \geq i_0). \quad (16.3)$$

In either case, for a finite K' and every $i \geq 0$,

$$\|X_{2^{-i}}\|_k = q_i \leq K'2^{-ip}. \quad (16.4)$$

Each fixed finite forward enlargement has the corresponding bounded-window controls in the retained base units. A change to its own units is allowed once its limiting internal endpoint norm is positive.

Proof. First consider the second case and fix a base level $i \geq i_0$. At offset ℓ , the physical rule in the base's own units, divided by the reference length unit $2^{-\ell p}$, has law

$$\frac{b_{i+\ell}}{b_i} G_{i+\ell}. \quad (16.5)$$

This is the fixed-index transport of Proposition 15.9; it precedes any limiting choice of ℓ .

All coefficients being divided by are positive by the second hypothesis. The tests $\mathcal{T}(m, M)$ are precisely the two-sided shell tests of Proposition 8.31: outer-circle connections supply the surrounding upper bound, and the two retained crossings have endpoints subject to the opposite lower bound. The kernel, affine, and joining hypotheses needed for annulus transfer are the structural properties verified in Lemma 16.2; varying-family compactness is proved below after the norm bounds. Its volume base is $\lambda = 2$. Proposition 8.31, including the one-sided overlap argument of Lemma 8.32, therefore proves (16.3).

Equation (16.3) yields $q_i \leq T_0^2 b_{i_0} 2^{-ip}$ for $i \geq i_0$. Increasing the constant covers the finite earlier set. This proves (16.4); it is the hypothesis in the first case. Decompose an internal interval of arbitrary length $t \leq 1$ into dyadic intervals lying inside it, with at most two intervals at each depth and all lengths at most t . Stationarity, finite concatenation, and then the endpoint modulus for the two remaining tails give

$$\|X_t\|_k \leq \frac{2K'}{1 - 2^{-p}} t^p. \quad (16.6)$$

The dyadic maximum bound and the mixed-monomial argument of Lemmas 14.2 and 14.3 apply uniformly to the backward quantum family $v \leq 1$, since division by v^p cancels the factor v^p in (16.6). On each bounded time window, a fixed number of stationary pieces gives a common time modulus, tight internal observations, and uniform integrability of their endpoint k th powers. The certificate construction, with its countable charts and local joining, then gives compactness of this backward family. Each fixed finite forward enlargement has the same properties, with constants allowed to depend on that enlargement. We do not assert uniform tightness or uniform integrability as $v \rightarrow \infty$.

We spell out why this gives the needed uniform upper annular test. In every limit of the compact backward family, two nearby points have arbitrarily cheap joins in an open margin. In a connected buffered band, join finitely many points of a sufficiently fine Euclidean net by a finite tree of compact paths in the band. Subdivide its edges into finitely many of these local joining neighborhoods. Finite concatenation gives a finite random bound for all outer-circle connections. Compactness makes these random bounds tight uniformly over the family: otherwise extract laws with failure probability bounded below at successively diverging bounds; in their limit a finite set of certificates with strict cost and spatial slack gives a finite bound, and the joining modulus transfers it back, a contradiction. The common reference metric likewise has, with arbitrarily high probability, a positive crossing lower bound and a finite band upper bound, by its continuity and local length property. Choose the band in the ordinary cone patch of Lemma 16.1. Its actual-field density and (16.2) transfer these bounds to the fixed annulus reference. Their constants can be chosen uniformly over the backward zoom family.

Apply Lemma 8.12 with all sufficiently late indices favorable and then Corollary 8.6. This gives a finite deterministic upper comparison on the Gaussian reference, with a constant determined only by the common shell-test bounds and geometry.

The same numerical bound holds at every fixed dyadic volume, including forward volumes. Indeed, for $v_0 = 2^m$, $m \in \mathbb{Z}$, the exact indexed-kernel identity gives

$$(\mathbf{K}^{(2^m)})^{(2^{-\ell})} = \mathbf{K}^{(2^{m-\ell})}. \quad (16.7)$$

For $\ell \geq \max\{m, 0\}$, the right side is a member of the backward indexed family whose shell tests have the common bounds just proved. Deleting the finite prefix $\ell < m$ changes only how fine the annulus construction begins. The coverage lemma uses arbitrarily late label windows, so this deletion changes neither their favorable fraction nor the deterministic comparison ratio in Corollary 8.6. Thus one finite upper factor works for every fixed m . No convergence rate uniform in m is needed. A finite enlargement of a time window supplies the base moment modulus and joining by a fixed number of stationary pieces; it is not used to select a new upper factor depending on m .

Intersect the comparison events over the countable forward volumes. Lemma 16.3 then gives the same finite upper comparison on the original whole cone. This proves initial finiteness before any argument involving optimal constants. Finite forward enlargements also preserve the preceding endpoint and compactness conclusions, with the corresponding fixed-window constants, and do not affect the relative-unit tail argument beginning at i_0 . $\square$

16.4 A deterministic metric multiplier

An upper comparison is understood in the open-port sense of Definition 8.3; the tree joining estimate makes it an upper bound for F_U between the exact limiting endpoints. Define the deterministic optimal constants for the whole joint law by

$$\begin{aligned} C &= \inf\{q : q \text{ is an almost-sure upper comparison}\}, \\ c &= \sup\{q : q \text{ is an almost-sure lower comparison}\}. \end{aligned} \tag{16.8}$$

The countable marked observations and arbitrary open losses make the optimal constants themselves valid. Proposition 16.4 gives $C < \infty$. For a law normalized by A_j , uniform integrability makes the limiting unit internal endpoint norm equal to one. Lemma 15.8 therefore excludes $C = 0$: zero upper comparison would give zero local costs and then a zero internal endpoint norm by its disk-cell lifting argument. Consequently

$$0 \leq c \leq C < \infty, \quad C > 0. \tag{16.9}$$

The same conclusion holds for an extraction, after a change of units, of one of the controlled families furnished by Proposition 16.4, provided its limiting internal endpoint norm is positive.

The local density statement of Lemma 8.13 and the two implications of Lemma 16.3 identify the optimal constants of the whole law with those of its countable indexed Gaussian tests. The initial finite upper comparison has already been proved, with one common constant, in Proposition 16.4.

Theorem 16.5 (Law-level rigidity for the tree model). *For every normalized tree law furnished by Proposition 16.4, and every extraction of one of its controlled families after a change of units with positive limiting internal endpoint norm, the optimal constants in (16.8) satisfy $c = C > 0$. Almost surely, simultaneously on every open set U ,*

$$F_U = CD_U.$$

In particular the metric multiplier is deterministic.

Proof. We verify the remaining hypotheses of the common rigidity implication. Lemma 16.2 gives the path calculus, product kernels, exact indexed laws and closure. Proposition 16.4 gives compactness of the backward quantum family and each fixed finite forward enlargement. These

cover the sequences with indices tending to $+\infty$ and fixed offsets used below. Its separate forward-index-tail argument gives one finite upper comparison for all fixed indices. The preceding paragraph transports its optimal numerical constants to every fixed index and to the outside-pinned reference. The comparison bounds pass through further marked extraction, by the open-margin realization and local joining in Proposition 15.1. Thus Lemma 8.12 and Proposition 8.14 apply with $\lambda = 2$.

If $c < C$, apply Proposition 8.14 and retain its high and low witnesses in the full tree observation array. For each low witness retain its closed rectangular envelope, its two opposite faces and endpoint positions, and its larger denominator collar together with the actual internal reference metric there. These are legitimate additional marks: the certificate construction retains arbitrary finite subpaths and compact ranges, and the common rectangle extraction selects a first crossing between opposite faces. The range remains in the closed rectangle before the open realization margin is added. Hence Proposition 8.15 applies to this augmented extraction with its required closed envelope. Its two witness events have separate positive probabilities; their intersection is not assumed.

The conditional catalogue construction of Proposition 8.26 now uses the same tree kernels. Its order-determination condition is vacuous because the order space is a singleton. Disjoint complete dependency disks supply the independent interior Gaussian fields and fresh passage draws. The exact two-unit identity from Proposition 15.9 supplies the auxiliary law at the prescribed shifted index. One outside pin is fixed before all candidate supports, so every smooth catalogue modification is a Cameron–Martin direction of that actual Gaussian reference. The finite-energy cutoff and the density truncation in Lemma 16.1 justify all fixed reference changes. Lemma 8.29 supplies the catalogue with its closed preserved rectangle, thin preservation patch, reference connection bounds and observable outer shell crossing, including the deterministic construction required by Lemma 8.19 and the counts in Remark 8.18. Product-kernel completion retains the shortcut patch and exterior before the input is restricted to a chosen hit.

These checks verify the hypotheses of Theorem 8.30. Its proof applies the weighted Cameron–Martin argument of Proposition 8.28 and gives $c = C$, a contradiction. Therefore $c = C > 0$ by (16.9). Lemma 8.7 upgrades the positive lower comparison to $F_U \geq CD_U$; the upper port comparison and tree joining give the reverse inequality. Countable exhaustion yields the simultaneous statement on open sets. Finally, C was defined from the entire probability law in (16.8), so it is a deterministic constant before any diameter normalization is made. $\square$

17 From record scales to all time scales

The preceding comparison and rigidity arguments apply to subsequences with suitable control of their normalization. We now prove that this control holds at every sufficiently large scale. The proof has three steps: compare the internal endpoint norm with the metric multiplier in a rigid limit; fix strict tests that hold near every such limit; and rule out a first scale where those tests fail.

Throughout this section, an internal interval cost uses only the instructions witnessed in that interval. It need not equal the unrestricted distance between the tips of the limiting cell. Recall

$$X_I = d_I(\text{first tip, last tip}), \quad A_j = \|X_{[0,2^j]}\|_k, \quad s = p/2.$$

The normalizations and the constants attached to a probability law are deterministic. Stationarity and concatenation give

$$0 < A_j < \infty, \quad A_{j+1} \leq 2A_j. \quad (17.1)$$

Every extraction retains the complete interval and certificate data, including marked subpaths. This preserves the internal normalization while the ambient passage rule is being identified.

17.1 Comparing the internal norm with the metric multiplier

Call a certificate law *rigid* if its local passages agree almost surely with CD_h for one deterministic $C \in (0, \infty)$, as in Theorem 16.5. Our first task is to compare this multiplier with the internal endpoint norm used to normalize the discrete model. For such a law let q_i be the L^k norm of its internal endpoint cost on $[0, 2^{-i}]$, in its base cost units. These are the actual limiting moments supplied by Lemma 14.3, not merely upper bounds from Fatou's lemma. We consider rigid laws for which

$$H := \sup_{i \geq 0} 2^{ip} q_i < \infty. \quad (17.2)$$

Both alternatives of Proposition 16.4, followed by Theorem 16.5, give laws of this kind.

Proposition 17.1 (Universal comparison of internal norms). *There are constants $0 < \alpha \leq \beta < \infty$, depending only on the fixed Brownian and metric units and on k , with the following property. For every rigid certificate law satisfying (17.2), and every fixed integer i , its internal endpoint norm satisfies*

$$\alpha C 2^{-ip} \leq q_i \leq \beta C 2^{-ip}. \quad (17.3)$$

Negative i mean a finite enlargement of the time interval. The assertion applies also to laws obtained by joint extraction of zooms of already limiting laws. Consequently, if $q_0 = 1$, then

$$C_- := \beta^{-1} \leq C \leq \alpha^{-1} =: C_+. \quad (17.4)$$

Proof. A lower bound from the two tips. Let $W = D_h(\eta(0), \eta(1))$ in the fixed unit cone law. Its tips are distinct almost surely, so $W > 0$ almost surely. Choose $a > 0$ and $b > 0$ with $\mathbb{P}[W \geq a] \geq b$. The lower comparison for certificates, applied in successively larger open neighborhoods of the unit cell, gives $X_{[0,1]} \geq CW$. Hence

$$q_0 \geq Cab^{1/k}. \quad (17.5)$$

This argument requires no integrability statement about W at a cell tip.

An upper bound by a maximizing zoom. It suffices to bound H/C over the laws under consideration. If no such bound exists, choose laws indexed by r with $H_r/C_r \rightarrow \infty$. For each r choose an integer $i_r \geq 0$ such that

$$2^{i_r p} q_{r, i_r} > H_r/2, \quad v_r = 2^{-i_r}.$$

Change time units to v_r and cost units to $H_r v_r^p$. This is the quantum-unit transport of Proposition 15.9; its kernel is the transported law, and need not be the original kernel. In the new units the internal endpoint norms $\tilde{q}_{r,l}$ obey

$$\frac{1}{2} < \tilde{q}_{r,0} \leq 1, \quad \tilde{q}_{r,l} \leq 2^{-lp} \quad (l \geq 0). \quad (17.6)$$

Stationarity and addition extend these estimates, with constants depending only on the enlargement, to each bounded time window. Lemmas 14.2 and 14.3, and Proposition 15.1, therefore give a further joint extraction with a limiting unit endpoint variable $\tilde{X}$ satisfying

$$\mathbb{E}[\tilde{X}^k] \geq 2^{-k}.$$

The uniform integrability argument remains applicable here: internal costs in disjoint intervals have the product interval law, and their mixed monomials are controlled exactly as in Lemma 14.3. Indeed every finite vector of costs on deterministic disjoint intervals has an unconditional product law at discrete resolution, and hence after each fixed zoom extraction. Products of weakly convergent measures converge to the product, so the same assertion survives a sequence of different continuum laws and maximizing zooms. Thus this extraction does not lose the positive endpoint moment; no closure of conditional independence is being inferred in this moment argument.

In the same units, the ambient upper comparison is $(C_r/H_r)D_h$. The underlying continuum data have the fixed cone law after quantum-unit transport. The upper realization statement with open margins in Proposition 15.1 sends this comparison to zero on every compact local route. More explicitly, work first on a countable exhaustion by buffered charts and with bounded reference metric costs. Finite upper certificates realize the route in each chart; their costs tend to zero. Passing to the closed certificate sets and then removing the reference-cost cutoff makes every such limiting local cost zero almost surely. This is a joint statement with the reference metric; it does not use continuity of the metric as a function of the contour.

Lemma 15.8 now gives a contradiction. Its geometric content is useful here. Two points at curve times approaching 0 and 1 can be chosen in the interior of the unit disk cell. A finite chain of compact interior routes connects them. The routes lift using the cell's internal instructions, so their internal cost is zero in the limit. The two omitted tip pieces have arbitrarily small cost by the uniform time modulus. The unit internal endpoint cost must therefore vanish, contrary to $\mathbb{E}[\tilde{X}^k] \geq 2^{-k}$. We have proved $H \leq \beta C$ for some finite universal β . Together with (17.5), and decreasing α if necessary, this proves (17.3) for $i = 0$.

Every fixed dyadic interval. For any fixed dyadic $v > 0$, change time units to v and divide costs by v^p . The reference metric has again the fixed cone units and the comparison constant is again C . The transported law satisfies (17.2). For $v \leq 1$ this follows directly from that equation. For $v > 1$, all but finitely many of its dyadic norms are controlled by that equation, while the remaining norms are finite by addition on a bounded enlargement. Applying the already proved unit-interval statement to this law proves (17.3) for every integer i . This also proves the assertion about finite enlargements. At no point was a Weyl identity for the competing law assumed. $\square$

17.2 A fixed neighborhood of the rigid laws

We now choose tests that every rigid limit passes with a strict margin. The resulting neighborhood and numerical thresholds will remain fixed throughout the first-failure argument.

Compactness of the rigid class. Let $\mathcal{R}$ be the closure, in the topology of Section 15.5, of the rigid unit-internal-norm laws satisfying (17.2), including their admissible joint extractions. This is a compact set. To verify the assertion, Proposition 17.1 gives the uniform bounds

$$q_0 = 1, \quad q_i \leq (\beta/\alpha)2^{-ip} \quad (i \geq 0), \quad C \in [C_-, C_+]. \quad (17.7)$$

The moment and certificate compactness lemmas give precompactness on every bounded enlargement. Uniform integrability preserves $q_0 = 1$ and the bounds in (17.7). The retained reference-law coordinates and Lemma 15.7 preserve the admissible local kernels through this further extraction; the full background is attached by (15.20). After a further extraction, the deterministic constants C converge within $[C_-, C_+]$. Upper realization and lower subextraction with open margins preserve the two comparisons with this limiting constant. The limit is therefore again rigid. This proves

compactness and also explains why $\mathcal{R}$ retains all internal and certificate data: it is not obtained by discarding those data and keeping only the one-parameter family of ambient metrics.

Numerical thresholds and the ambient compact class. Put $R_0 = \beta/\alpha$. Choose R and then an integer $m \geq 1$ satisfying

$$R > 2 \max\{2, 2^p R_0, 2^{-p} R_0\}, \quad R_0 2^{-m(p-s)} < \frac{1}{2}. \quad (17.8)$$

Set

$$K_0 = \max\{R_0, 1\}(R 2^s)^{m-1}.$$

The number K_0 is fixed before any induction or stopping argument. It bounds the past tail along a successful run, as shown below. For choosing neighborhoods one can use a larger fixed constant, say $K_1 = 4(1 + 2^s + 2^p)K_0$, to include one finite forward enlargement in preceding units.

Here is the compactness fact needed to choose the neighborhood. Let $\mathcal{P}_{K_1}$ be the closure of the continuum tree certificate laws with the fixed background marginal, the limiting interval bounds of Lemma 14.2 with constant K_1 , and the retained reference-law extension. The internal interval coordinates carry the compatibility inherited from the walk: their laws are stationary under deterministic time translation, costs concatenate, and costs on every fixed finite list of deterministic intervals with disjoint interiors have their product law. These properties pass through joint weak convergence of the retained endpoint coordinates, as in the maximizing-zoom argument above. The contour coordinate agrees with the retained Brownian background contour.

The interval bounds imply uniform integrability within this class by the proof of Lemma 14.3. Indeed, subdivide a fixed rational interval into ℓ equal pieces. In the multinomial expansion of the sum of their costs to the k th power, the pure terms have total expectation at most $C_T K_1^k \ell^{1-ks}$. For each fixed ℓ , independence bounds every mixed term uniformly in $L^{k/(k-1)}$. First send the tail threshold to infinity and then ℓ to infinity. This proves the required uniform integrability since $ks > 1$.

Moment bounds on the countable windows and the fixed tight background marginal give precompactness. The uniform integrability just proved, Lemma 15.7, and (15.20) preserve the certificate and kernel properties in the closure. Thus $\mathcal{P}_{K_1}$ is compact. It includes the limiting laws of eventually successful runs: each fixed past offset has the K_0 bound, even if a finite initial portion of a discrete sequence has a larger constant. The set $\mathcal{R}$ is contained in this compact class. A finite-resolution μ_j lives in the ambient law space and need not belong to $\mathcal{P}_{K_1}$.

Stable shell tests. Choose the shell geometry and the high probability threshold required by the second alternative of Proposition 16.4. There are fixed finite upper connection and positive lower traversal constants for which every law in $\mathcal{R}$ passes those shell tests with strictly stronger constants and a strictly smaller failure probability. In fact its local rule is CD_h with $C \in [C_-, C_+]$; compactness, continuity and nondegeneracy of D_h on buffered shells give the choices, uniformly in C . Include the reference boundary-diameter and modulus tests used in that proposition. These choices involve only compact interiors of the shells.

For completeness, the passage from this statement to a weak-law neighborhood uses both halves of certificate convergence. All shell predicates here are predicates on continuum laws in $\mathcal{P}_{K_1}$. Finite-resolution laws will be tested only for membership in the ambient open neighborhood below. An upper connection bound with strict cost and spatial slack is realized by finitely many certificates for a finite net; the uniform joining modulus extends it to every point of the band. A violation of a lower traversal bound gives a bounded-cost certificate with separated marked

endpoints; compact subextraction preserves the violation at a slightly weaker threshold. Increasing finite nets and decreasing joining errors gives these statements at all points. The uniform modulus in $\mathcal{P}_{K_1}$ permits the nets and errors to be chosen before passing to the limit. In using the time modulus on a spatial compact $K \Subset U$, first enlarge K slightly inside U . Properness of η makes the preimage of this enlargement lie in a finite random time window. The common background marginal permits a deterministic window to be chosen with arbitrarily small failure probability, uniformly over $\mathcal{P}_{K_1}$. On that window use the interval modulus and the compact joining argument of Proposition 15.1, then remove the properness cutoff. Truncated density approximation gives the same implications for the specified local free-field tests. Consequently, any sequence in $\mathcal{P}_{K_1}$ converging to a law in $\mathcal{R}$ eventually satisfies the weaker shell tests. These are the uniform upper and marked lower implications of Lemma 15.6. Lower tests do not require a bound on all possible crossings: a violation is witnessed by one bounded-cost path with its crossing endpoints marked before extraction.

Let $\mathcal{F}$ be the closure in $\mathcal{P}_{K_1}$ of the laws failing these weaker tests. The preceding sequential statement, with the intervening strict slack, gives $\mathcal{F} \cap \mathcal{R} = \emptyset$. Both sets are compact. If $\mathcal{F}$ is nonempty, choose $\varepsilon > 0$ with $3\varepsilon < \mathbf{d}(\mathcal{R}, \mathcal{F})$; otherwise choose any $\varepsilon > 0$. The fixed open set

$$\mathcal{O} = \{\mu : \mathbf{d}(\mu, \mathcal{R}) < \varepsilon\} \quad (17.9)$$

contains $\mathcal{R}$, and every law in $\overline{\mathcal{O}} \cap \mathcal{P}_{K_1}$ satisfies the weaker shell tests. Notice that $\mathcal{O}$ is open in the ambient law space, not just relatively open in $\mathcal{P}_{K_1}$. This is needed when a failed scale is renormalized only after its limiting norm has been found.

Successful scales. For $j \geq m$, say that level j is *successful* when all three strict conditions hold:

$$\mu_j \in \mathcal{O}, \quad R^{-1} < \frac{A_j}{A_{j-1}} < R, \quad \frac{A_{j-m}}{A_j} < 2^{-ms}. \quad (17.10)$$

Success is a deterministic property of the law μ_j and the normalizing sequence, rather than an event for a sampled map. The topology, shell tests, R, m , and $\mathcal{O}$ are now fixed. Proposition 17.1 shows why these conditions are strict at every rigid limit, including at any fixed forward or backward scale. Indeed, at such a scale its adjacent ratio is between $R_0^{-1}2^p$ and R_02^p , while its m -step past ratio is at most

$$R_0 2^{-mp} < \frac{1}{2} 2^{-ms}. \quad (17.11)$$

17.3 No first failure after a late record

At record scales, the complete past tail gives the moment bounds needed for rigidity. We first show that a late record starts an arbitrarily long successful run. If this run has a first failure, its preceding scales still provide enough control to identify the failed scale in the preceding normalization; it will then satisfy all three success tests as well.

Proposition 17.2 (Eventual control of all dyadic scales). *There are an integer J and a finite deterministic K such that every $j \geq J$ satisfies (17.10), and*

$$\frac{A_{j-i}}{A_j} \leq K 2^{-si} \quad (j \geq J, 0 \leq i \leq j). \quad (17.12)$$

In particular, $A_j \rightarrow \infty$. Every sequence $j_r \rightarrow \infty$ has a subsequence whose joint normalized data converge to a law in $\mathcal{R}$.

Proof. Record scales start long successful runs. Lemma 14.4 supplies $r_l \rightarrow \infty$ and exponents $t_l \uparrow p$, with $s \leq t_l < p$, such that

$$\frac{A_{r_l-i}}{A_{r_l}} \leq 2^{-t_l i} \quad (0 \leq i \leq r_l). \quad (17.13)$$

These are record maxima of $2^{-t_l j} A_j$; their existence uses only the volume-growth consequence $\limsup j^{-1} \log_2 A_j \geq p$. In any joint extraction at these records, each fixed-offset internal norm is at most 2^{-ip} . The first alternative of Proposition 16.4 and Theorem 16.5 apply. The unit norm is one by Lemma 14.3, so the resulting law belongs to $\mathcal{R}$.

Fix an integer L . Extract simultaneously on the bounded enlargement needed for levels $r_l - L, \dots, r_l + L$ and their m preceding levels. Initially keep cost units A_{r_l} at all of these levels. The moment lemmas give compactness and moment convergence on this finite enlargement. The record limit is rigid, so Proposition 17.1 gives a strictly positive internal norm at every one of the enlarged or reduced scales. We may therefore divide by its own limiting norm. The resulting normalized laws are in $\mathcal{R}$, and (17.8)–(17.11) make their numerical success tests strict. Their law tests are strict since $\mathcal{O}$ is open and contains $\mathcal{R}$. If success failed infinitely often for this fixed finite collection of levels, extracting that sequence would contradict these conclusions. Thus every fixed finite run around a record eventually succeeds. A diagonal choice, relabelled r_l , makes all levels $r_l, \dots, r_l + l$ successful. The complete record tail (17.13) is retained.

The past tail along a successful run. Suppose $r \leq j$ and all levels from r to j are successful, while r has the record tail $A_q/A_r \leq 2^{-s(r-q)}$ for $q \leq r$. For $r \leq q \leq j$, write $j - q = am + b$ with $0 \leq b < m$. Iterating the m -step inequalities a times and then the adjacent inequalities b times gives

$$\frac{A_q}{A_j} \leq 2^{-ams} R^b \leq (R2^s)^{m-1} 2^{-s(j-q)}.$$

For $q < r$, apply the same estimate to A_r/A_j and multiply it by the record-tail estimate for A_q/A_r . Thus in both cases

$$\frac{A_q}{A_j} \leq K_0 2^{-s(j-q)} \quad (0 \leq q \leq j). \quad (17.14)$$

The constant has no dependence on the length of the successful run.

Extract immediately before a first failure. Suppose unsuccessful levels occur arbitrarily far out. For each selected record r_l let $f_l > r_l$ be the first unsuccessful level after it, and put $j_l = f_l - 1$. The arbitrarily long initial successful runs imply $j_l - r_l \rightarrow \infty$. Use time units 2^{j_l} and cost units A_{j_l} . Equation (17.14) gives compactness, the time modulus, and moment convergence. For every fixed $i \geq 0$, the normalized law at level $j_l - i$ is successful for large l . Its norm in the base units has a positive finite subsequential limit: the adjacent inequalities give

$$R^{-i} \leq A_{j_l-i}/A_{j_l} \leq R^i,$$

and the sharper upper estimate is (17.14). Take these limits jointly for all fixed i . Their laws lie in $\overline{\mathcal{O}}$ and their moment bounds put them in $\mathcal{P}_{K_1}$. They therefore pass the coarse shell tests. The second alternative of Proposition 16.4 applies to this scale family. It gives exact-power upper bounds and compact zoom laws; Theorem 16.5 then makes the base law rigid. Its unit internal norm is one, and its deterministic comparison constant lies in $[C_-, C_+]$.

Recover the failed scale's normalization. Keep the preceding cost units A_{j_l} . The unsuccessful level is then the interval $[0, 2]$, so it can be examined before any lower bound on its own normalization is known. Equation (17.1) gives

$$0 < A_{f_l}/A_{j_l} \leq 2, \quad (17.15)$$

but no lower bound has been used. Addition and stationarity on this finite enlargement give the same compactness and uniform integrability as before. Thus, after extraction, its internal endpoint norm is exactly

$$q_{-1} = \lim_l A_{f_l}/A_{j_l}.$$

It is now positive by Proposition 17.1, applied to the rigid law on the enlarged window:

$$\alpha C 2^p \leq q_{-1} \leq \beta C 2^p. \quad (17.16)$$

Equivalently, one may transport the enlarged interval to unit time and divide costs by $2^p A_{j_l}$; its half-interval norm is then exactly 2^{-p} , so nondegeneracy is already present before the unit interval norm is recovered. Both descriptions use preceding units first.

We may now normalize the failed-level data by A_{f_l} . Their limit is the time-2 transport of the preceding rigid law divided by its positive unit endpoint norm, hence belongs to $\mathcal{R}$. Thus their law test passes eventually. Furthermore, moment convergence on the same enlarged window and (17.3) give

$$\lim_l \frac{A_{f_l}}{A_{f_l-1}} \in [R_0^{-1} 2^p, R_0 2^p], \quad \lim_l \frac{A_{f_l-m}}{A_{f_l}} \leq R_0 2^{-mp}.$$

These limits lie strictly within the permitted adjacent and m -step ranges. All three tests at f_l therefore pass for large l , contradicting its definition. This proves eventual success.

The global bound and all subsequences. Iterating the eventual m -step inequalities and using adjacent ratios for the remainder gives (17.12) for $j - i \geq J$. The finitely many values below J are absorbed into a single larger deterministic constant K by concatenating at J . On each residue class modulo m we also have, for all sufficiently large j ,

$$A_{j+m} > 2^{ms} A_j.$$

The finitely many residue classes have positive starting values, proving $A_j \rightarrow \infty$.

Finally take any $j_r \rightarrow \infty$. Equation (17.12) gives a joint extraction with preserved unit norm. At every fixed past offset the success tests hold eventually. The limiting fixed-offset bounds use K_0 , since their entire finite runs lie beyond J . Thus the closure property of $\mathcal{O}$ again supplies the second alternative of Proposition 16.4. Rigidity and (17.4) put the resulting law in $\mathcal{R}$, as claimed. $\square$

17.4 Arbitrary time units and the interval estimate

The dyadic conclusion now extends to any integer time unit. The remaining input is the already proved time modulus, which allows a converging multiplicative change of time to be replaced by its limit. We also record the uniform interval estimate needed for the finite-sphere argument.

Theorem 17.3 (Bilateral subsequential limit at all time scales). *For a positive integer N set*

$$j(N) = \lfloor \log_2 N \rfloor, \quad B_N = A_{j(N)}.$$

Then $B_N \rightarrow \infty$. From every sequence $N_r \rightarrow \infty$ one can extract a subsequence such that the bilateral contours and all their local certificate data, in time units N_r and cost units B_{N_r} , converge jointly. The continuum background is the fixed-unit $\sqrt{2}$ -quantum cone decorated by its area-parametrized SLE₈. For one deterministic

$$\lambda \in [C_-, 2^p C_+], \quad (17.17)$$

its limiting local passage rule is

$$F_U(x, y) = \lambda D_h(x, y; U) \quad (17.18)$$

for every open U and every x, y joined by a path compactly in U . The equality uses the internal length metric on U , equivalently exhaustion by compactly confined paths, and includes the upper realization and lower subpath-extraction assertions with open margins.

There is also the following uniform discrete estimate. For each fixed $T < \infty$ there is $K_T < \infty$ such that, for every integer $N \geq 1$ and every translated integer interval I of length $|I| \leq TN$,

$$\left\| \frac{\text{diam}(V(G_I), d_I)}{B_N} \right\|_k \leq K_T \left(\frac{|I| + 1}{N} \right)^s. \quad (17.19)$$

The corresponding curve-time modulus on each bounded window is uniform, with any exponent smaller than $s - 1/k$. The fixed-interval endpoint k th moments are uniformly integrable along these extractions.

Proof. Identification in arbitrary time units. Divergence and the dyadic conclusion follow from Proposition 17.2. For arbitrary N_r , put $j_r = j(N_r)$ and pass to a subsequence for which

$$u_r = N_r/2^{j_r} \longrightarrow u \in [1, 2].$$

Extract the dyadic laws jointly on every bounded enlargement. Their limiting local rule is CD_h for a deterministic $C \in [C_-, C_+]$.

Proposition 15.9 gives transport for every fixed positive time factor. Having fixed the deterministic u , adjoin to the retained countable family the u -transforms of its templates and all their fixed dyadic offsets, and take a further joint extraction. Retain also

$$(\mathcal{C}_{j_r}, T_u \mathcal{C}_{j_r}, r_u(\mathcal{C}_{j_r})),$$

including the transformed internal reference metrics. Here $\mathcal{C}_j$ is the coherent background of Section 15.5, and r_u is the circle-average dilation in (13.12). This triple has the same joint marginal for every r . Its limiting components therefore have that same marginal and satisfy the same measurable transport identity. No continuity of the chosen reembedding is required.

The raw paths in N_r time units are the same paths with their dyadic time labels divided by u_r ; their costs remain in $A_{j_r} = B_{N_r}$ units. These time changes converge uniformly on bounded windows to division by u . For internal endpoint distances, the uniform time modulus and Lemma 15.3 identify the moving endpoints $u_r t$ with ut , since the fixed limiting cuts have singleton Brownian fibers. The same identification for local arrays follows from finite-band reconstruction with open margins and both directions of marked-certificate convergence.

To keep the spatial background in this comparison explicit, let η_j be the curve component of $\mathcal{C}_j$ and write $r_{u,j} = r_u(\mathcal{C}_j)$ and $\eta_j^{(u)}(t) = r_{u,j}^{-1} \eta_j(ut)$. In the fixed- u spatial coordinates, an N_r time label t is projected to

$$r_{u,j_r}^{-1} \eta_{j_r}(u_r t) = \eta_{j_r}^{(u)}((u_r/u)t).$$

Joint local uniform convergence of the retained curves makes its difference from $\eta_{j_r}^{(u)}(t)$ tend uniformly to zero on bounded time windows, in probability. Thus the limiting N_r arrays are the dyadic limiting arrays transported to the fixed- u background, with the original costs. The fixed conditional background law is retained as in (15.20).

Write r_u for the limiting dilation and $h^{(u)}$ for the transformed field. The local rule in these coordinates is therefore

$$F_U^{\text{new}}(x, y) = CD_h(r_u x, r_u y; r_u U) = Cu^p D_{h^{(u)}}(x, y; U).$$

The transformed background has the fixed unit cone law by (13.12). Consequently $\lambda = Cu^p$ is deterministic and satisfies (17.17). The factor u^p comes from the established reference metric's quantum-unit rule; the passage law is transported by Proposition 15.9. Equality on all open sets follows first on the countable buffered charts, then for compactly confined paths by finite covering and concatenation, and finally by exhaustion. These operations preserve upper realization and lower marked subpath extraction with open margins, as asserted.

Uniform interval bounds. Apply Lemma 14.2 to (17.12). It gives, uniformly in translated intervals of length at most $T2^j$, the internal diameter bound $K'_T((|I| + 1)/2^j)^s$ in units A_j . Since $N/2 \leq 2^{j(N)} \leq N$, enlarging the constant gives (17.19). Finitely many small N are absorbed into that constant. The same lemma gives the positive time-modulus exponent, and Lemma 14.3 gives uniform integrability, also on the finite enlargements used here. Neither the argument nor the estimate requires regular variation of A_j or convergence of its adjacent ratios. $\square$

Part III

Finite spheres and quantum area

18 The FK sphere and its quantum area

Throughout this Section, $q \in (0, 4)$ is fixed, with $\gamma \in (\sqrt{2}, 2)$ determined by Equation (1.3). Write

$$\delta_n = \frac{d_n}{B(2n)},$$

where B is the deterministic distance denominator constructed in Lemma 9.10. In particular, $B(2n) \rightarrow \infty$. Distances in this Section always use all the primal edges of M_n . Equip its vertices with

$$\mu_n(v) = \frac{\deg(v)}{2n},$$

where a loop contributes twice to the degree. Our goal is convergence of this metric probability space to the ordinary unit-area sphere.

We first specify the inputs from the preceding Sections. This also separates the local comparison theorem from the compactness argument which remains to be proved here.

(F1) For every sequence of time units tending to infinity, there is a subsequence along which the bilateral passage rule has the comparisons of Corollary 9.13, with a *deterministic* constant

$$c \in [c_-, c_+] \subset (0, \infty).$$

The same c applies to all the intrinsic local charts in that extraction. These comparisons include lower bounds for every retained path and upper bounds for following a prescribed rectifiable continuum path with endpoint slack. On subpaths compactly contained in a buffered chart, the comparison is with the intrinsic LQG metric of that chart, so that it is determined by the real field there.

- (F2) The endpoint joining conclusion of Corollary 12.8 holds in these units. It applies to arbitrary discrete endpoints whose projected locations approach one another in a compact chart, with any prescribed open clearance.
- (F3) The finite-conditioned transfer Theorem 7.10, in its spatial form Theorem 7.13, preserves the conditional local kernels on strict interior charts, with the same time and distance units. Its observables include the universal endpoint tests in (F2). We use the actual-field absolute continuity of Lemma 3.3 to transfer their metric support.

None of these inputs asserts tightness of the whole finite metric space. In particular, (F1) concerns passages with endpoint slack, whereas (F2) is needed to start paths at specified lattice vertices.

18.1 The conditioned surface and its local metric

Lemma 3.1 identifies the inventory word conditioned to reduce to the empty word with the required FK-weighted map. We now identify the surface encoded by its limit and transfer the local metric statements to that surface.

By [27, Theorem 1.8], the resolved contour pair, with time divided by $2n$ and heights by $\sqrt{2n}$, in the fixed boundary-length convention of Section 3, converges to the duration-one correlated quadrant excursion. Its correlation is

$$\rho = -\cos(\pi\gamma^2/4).$$

By [34, Theorem 1.1], this excursion determines the ordinary unit-area DMS sphere and its space-filling curve

$$(\mathcal{S}, h, \eta), \quad \eta : [0, 1] \rightarrow \mathcal{S}.$$

Here $\eta(0) = \eta(1) = p_*$, and

$$\mu_h(\eta([s, t])) = t - s, \quad 0 \leq s \leq t \leq 1.$$

We forget all distinguished points except the point p_* used temporarily in the proof. The quantum-surface law is the ordinary unit-area law. Additional area marks below serve only to choose coordinate representations; they do not change this law or reroot the exploration.

Consider an arbitrary subsequence of positive integers n . Use (F1) to select a further subsequence and its deterministic constant c , and extract the finite chart data jointly along it. We work with a coupling in which the contours converge uniformly and the countable chart tests converge. Whenever a joint coupling is refined below, we retain these tests and a countable family of their local modulus and comparison error variables. Errors converging to zero in probability can first be made almost surely convergent by a diagonal subsequence. Thus subsequent joint extractions preserve the local conclusions. All subsequent subsequences in this Section are taken from this selected sequence.

For each vertex v , choose one of its contour representatives $t_n(v) \in [0, 1]$, and set

$$\phi_n(v) = \eta(t_n(v)).$$

For the root vertex choose its representative at time 0. The choices of representatives are immaterial asymptotically: all the representatives of a primal vertex have projected diameter tending uniformly to zero. The same is true of the two endpoints of a primal edge. Here the number of encoding relations needed is bounded independently of the edge and of n . Two representatives of a primal vertex satisfy the first-coordinate contour-tree identity. A tree edge is represented by adjacent

four times. For a remaining primal edge, the two triangle steps of its quadrangle are matched in the second coordinate: use their horizontal matching relation and, if necessary, one adjacent-time move at each end to reach the primal endpoint visits. Finally one may change to the chosen representatives of those two primal vertices by two first-coordinate identities. This uses at most five relations.

Consequently a violating sequence would give, after extraction of these boundedly many times, a limiting horizontal chain in which each adjacent-time move has disappeared. The horizontal-identification rule identifies the endpoints, so its image under η is a single point, a contradiction. This argument uses the resolved tree/cotree sewing, not a graph path of an uncontrolled number of edges. Thus, for any fixed auxiliary metric σ inducing the topology of $\mathcal{S}$,

$$\max_{\{v,w\} \in E(M_n)} \sigma(\phi_n(v), \phi_n(w)) \longrightarrow 0. \quad (18.1)$$

The original root vertex projects to p_* . We can interpolate the projected edges by arcs of vanishing σ -diameter. Confinement below allows these vanishing interpolation errors. For open $U \subset \mathcal{S}$, the notation $\delta_n(v, w; U)$ denotes the infimum of normalized lengths of graph paths from v to w whose vertex projections lie in U . All uses have open clearance, so the edge interpolation convention does not change the limiting statements.

Lemma 18.1 (Spatial transfer to the ordinary sphere). *In the preceding joint extraction, the bilateral intrinsic local comparisons and endpoint tests in (F1)–(F2) hold in the finite sphere on compact patches away from p_* , with the same deterministic constant c .*

Proof. Adjoin two auxiliary points Q_1, Q_2 , sampled independently from μ_h conditional on the surface and independently of the exploration and chart observations given that surface and its contour. They may be realized as $\eta(T_1), \eta(T_2)$ for independent uniform times adjoined to the joint coupling. By the area-marking property [13, Proposition A.13] and the three-mark representation in [7, Definition 2.2 and Lemma 2.3], $(\mathcal{S}, Q_1, Q_2, p_*)$ has the ordinary three-marked unit-area sphere law. Represent it in $\mathbb{C}$ with the marks at $0, 1, \infty$. The identification with the DMS sphere includes the field, as explained in Lemma 3.3.

Before applying the quantum-area clock, the space-filling curve in [34, Theorem 1.1] has the whole-plane space-filling SLE law from infinity to infinity, independent of the real field. Construct it from an imaginary field modulo its period with stationary phase. This is the same directed imaginary-field and curve law as in the cone experiment. Conformal invariance, including the imaginary-field phase transformation, preserves independence under the affine coordinate normalizations used here.

Fix $K \Subset U$, where U is a relatively compact disk in $\mathbb{C} \setminus \{0, 1\}$, with the buffers needed by Theorem 7.13. Let μ_{sph} and μ_{cone} be the laws of the actual real-field restrictions on U , using the cone representation in Lemma 3.3. Let ν be the law of the entire directed imaginary-field and space-filling-curve object, and let Π_h be the common conditional law of the area marks in U . The field absolute continuity and the common mark kernel give

$$\mu_{\text{sph}}(dh) \nu(d\vartheta) \Pi_h(d\pi) \ll \mu_{\text{cone}}(dh) \nu(d\vartheta) \Pi_h(d\pi). \quad (18.2)$$

The entire directed imaginary-field object retains the global order of its local traversals. The finite order needed by a selected chart, its locally clocked inputs, and the relevant mark data are measurable functions of the displayed variables. Thus the order variable in Theorem 7.13 is retained in this comparison.

That theorem gives the same conditional local observation kernel in the two experiments. The intrinsic metric on U is determined by $h|_U$. Hence failure of a retained intrinsic local comparison or endpoint test is a measurable event of the local input and observation. On the cone its conditional probability is zero almost everywhere for the right-hand input law in Equation (18.2). Absolute continuity makes it zero for the sphere input law, and the common-kernel identity proves the test there. The transported metric is the internal metric on U ; an unrestricted metric depending on the exterior is not used in this step. Intersect over a countable disk cover, the protected-chart atlas, endpoint tests and rational open losses. We obtain the local statements away from p_* , Q_1 , Q_2 . Their finite kernels apply because all contour preimages of a compact set avoiding p_* lie in a strict interior time interval.

Adjoin a second independent pair Q_3, Q_4 of area points, retaining the same sphere, exploration, discrete data and reference extraction. Apply the argument to each three-marked marginal separately, in the representations $(Q_1, Q_2, p_*) = (0, 1, \infty)$ and $(Q_3, Q_4, p_*) = (0, 1, \infty)$; no field absolute continuity conditional on the other pair is asserted. The four auxiliary points are distinct and avoid p_* almost surely, since μ_h is diffuse. The two coordinate changes differ by a complex affine map. Quantum-area clocks, quantum lengths and the chart observations have the intrinsic covariance of Theorem 7.13; the internal LQG metrics have conformal covariance. Therefore the two conclusions are statements about the same intrinsic comparisons, with the same chosen c . The exploration is not rerooted, the quantum units are unchanged, and no further bilateral normalization is selected.

The union of the two off-mark covers is $\mathcal{S} \setminus \{p_*\}$, because the auxiliary mark sets are disjoint. Compact confinements admit finite subcovers, proving the lemma. $\square$

Lemma 18.2 (Local finite-volume metric statements). *In the coupling above, the following statements hold on $\mathcal{S} \setminus \{p_*\}$.*

- (i) *Let P be a D_h -rectifiable path whose image is a compact subset of an open set $U \subset \mathcal{S} \setminus \{p_*\}$. If $\phi_n(v_n) \rightarrow P(0)$ and $\phi_n(w_n) \rightarrow P(1)$, then*

$$\limsup_{n \rightarrow \infty} \delta_n(v_n, w_n; U) \leq c \operatorname{len}_{D_h}(P).$$

An arbitrarily small open enlargement of the image of P may be used in place of U .

- (ii) *For paths confined to a compact subset of one buffered chart, every limiting endpoint pair x, y satisfies*

$$\liminf_{n \rightarrow \infty} \operatorname{len}_{\delta_n}(P_n) \geq c D_h(x, y).$$

This assertion holds for all choices of the paths and endpoints.

- (iii) *The local endpoint joining modulus is uniform over all discrete endpoints in a compact chart, with paths confined to any specified open enlargement.*

Proof. A compact confinement away from p_* has all its contour preimages in a strict interior time interval: otherwise a sequence of preimages approaching 0 or 1, together with continuity of η , would put p_* in that confinement. We may therefore use the finite transfer (F3) on a countable collection of buffered charts covering the confinement.

Lemma 18.1 transfers the comparisons in (F1) and the endpoint tests in (F2), using the common spatial kernels and absolute continuity of the actual local fields. The transferred constant is the same deterministic c on every chart. An intrinsic lower bound implies the corresponding

ambient lower bound, since an internal metric dominates the ambient metric. The length of a fixed path compactly contained in a chart is unchanged when computed using its internal metric.

The lower comparison and the universal endpoint tests give (ii) and (iii) directly. For (i), first divide P into finitely many subpaths lying with clearance in buffered charts. Follow these subpaths using (F1). Their endpoints initially have only spatial slack. Part (iii) joins these endpoints and joins the first and last ones to v_n, w_n , at a total cost tending to zero. One may keep every such join inside the prescribed enlargement. Sum the subpath lengths, and then remove the finite partition and cost slack. A countable atlas and the universal endpoint observables make all three conclusions simultaneous. In particular, the last step does not infer a universal endpoint assertion from existence of a limiting path certificate. $\square$

Lemma 18.3 (Lower bounds on compact punctured routes). *Let $K \subset \mathcal{S} \setminus \{p_*\}$ be compact. Suppose that P_n has projected image in K , up to vanishing errors, and that its endpoint projections converge to x, y . Then*

$$\liminf_{n \rightarrow \infty} \text{len}_{\delta_n}(P_n) \geq cD_h(x, y).$$

Proof. It suffices to consider a subsequence on which the left side is finite and the path costs are bounded by some $L < \infty$. Take a compact neighborhood K' of K avoiding p_* . Choose finitely many buffered charts with compact cores whose interiors cover K' . For sufficiently small $\rho > 0$, every σ -ball of radius 3ρ centered in K' lies in one of these cores.

There is $a_\rho > 0$ such that, for all sufficiently large n , a retained path in the fixed compact core of one of these buffered charts, with endpoints in K' making a σ -advance of at least $\rho/2$, has cost at least a_ρ . Otherwise one could select violating paths with costs tending below half of

$$c \min\{D_h(u, v) : u, v \in K', \sigma(u, v) \geq \rho/2\} > 0,$$

extract their endpoints, and contradict Lemma 18.2(ii). The minimum is positive because D_h is a metric inducing the compact topology on K' .

Starting at the initial vertex of P_n , stop at the first vertex whose projection is at least ρ away from the preceding stop, and repeat. Equation (18.1) absorbs the overshoots. Every complete segment lies in a buffered chart and costs at least a_ρ . There are therefore at most L/a_ρ complete segments. After a further extraction their endpoints converge. Apply the local lower bound to the finitely many complete segments and sum. The terminal incomplete segment has endpoint D_h -distance at most

$$\omega_{K'}(2\rho) := \sup\{D_h(u, v) : u, v \in K', \sigma(u, v) \leq 2\rho\}.$$

The triangle inequality gives

$$cD_h(x, y) \leq \liminf_n \text{len}_{\delta_n}(P_n) + c\omega_{K'}(2\rho).$$

Uniform continuity gives $\omega_{K'}(2\rho) \rightarrow 0$, proving the claim. $\square$

18.2 Small surrounding bands

We record the continuum fact used to bypass the point p_* . An internal diameter is always taken in the indicated open neighborhood of the set.

Lemma 18.4 (Vanishing surrounding bands). *Almost surely there are compact annular bands K_m and open neighborhoods U_m , with*

$$K_m \subset U_m \subset \mathcal{S} \setminus \{p_*\}, \quad \sup_{z \in U_m} D_h(z, p_*) \longrightarrow 0,$$

such that each K_m contains, with positive topological clearance, a separator between p_ and the complement of a neighborhood shrinking to p_* , and*

$$\text{diam}_{D_h(\cdot, \cdot; U_m)} K_m \longrightarrow 0.$$

The bands can be chosen with an inner and an outer collar, so that paths crossing them are stable under vanishing perturbations.

Proof. First work in a conformal chart on a compact set avoiding the field marks. The internal small-box connection estimate in Theorem 3.5 gives a random upper bound tending to zero for the internal diameter of a small box in a fixed bounded enlargement. The bound is uniform over boxes in this compact set. Cover a square annulus by a fixed number of overlapping boxes, using enlargements which avoid its central point. Chaining through the overlaps gives an internal diameter tending to zero for the annulus. Thinner concentric square annuli provide both collars. Since the box estimate holds simultaneously on compacts, this construction applies to every point of the chart. A countable chart cover gives the property at every point except the finitely many field marks.

The property just established says that the point has shrinking surrounding bands with vanishing internal joining diameter. It is a property of the metric surface at the point, and is independent of which auxiliary marks were used to embed the field. The quantum area measure is diffuse, so the property holds at μ_h -almost every point of the unmarked surface. The area-rerooting property of the ordinary sphere says that the distinguished point p_* , conditionally on that surface, is sampled from its area measure; see [34, Remark 1.3], citing [13, Proposition A.13]. It follows that the property holds at p_* . Finally, the metric induces the surface topology, so the bands and their sufficiently small collars have $\sup_{z \in U_m} D_h(z, p_*) \rightarrow 0$. By taking strict bounds and collars, all choices can be made measurably from countable chart and rational-box configurations. $\square$

Remark 18.5. The internal connection estimate in this proof is essential. The conclusion of Lemma 18.4 is not being deduced solely from geodesicity and the topology of the sphere. The rerooting argument transfers a property already proved on ordinary local field charts; it does not require a uniform Radon–Nikodym bound at a shrinking neighborhood of a field mark.

Proposition 18.6 (Uniform punctured distances). *For every compact $K \subset \mathcal{S} \setminus \{p_*\}$,*

$$\sup_{\substack{v, w \in V(M_n) \\ \phi_n(v), \phi_n(w) \in K}} |\delta_n(v, w) - cD_h(\phi_n(v), \phi_n(w))| \longrightarrow 0.$$

The distances on the left are unrestricted graph distances. An empty supremum of nonnegative errors is interpreted as zero.

Proof. Fix sequences of endpoints with projections converging to $x, y \in K$. For the upper bound take a D_h -geodesic from x to y . If it meets a sufficiently small hole around p_* , replace its portion between first entrance and last exit of a surrounding band by a path in U_m . The additional

length tends to zero by Lemma 18.4. The resulting path avoids p_* , is compactly contained in its complement, and has length at most $D_h(x, y) + o_m(1)$. Lemma 18.2(i) gives

$$\limsup_n \delta_n(v_n, w_n) \leq cD_h(x, y).$$

For the lower bound, choose paths from v_n to w_n with costs within $1/n$ of $\delta_n(v_n, w_n)$. The upper bound makes their costs bounded. Fix a small surrounding band with collars. If a path enters the inner hole, retain only its part up to its first entrance and its part after its last exit. The retained parts lie in a compact punctured region, and the entrance and exit vertices lie in a small neighborhood of p_* , with an error vanishing by Equation (18.1). Apply Lemma 18.3 separately to these two parts. The omitted cost is nonnegative. The D_h -distance between the two limiting entrance points tends to zero as the band shrinks, by the triangle inequality through p_* . If the path never enters the inner hole, apply Lemma 18.3 to the whole path. In both cases,

$$\liminf_n \delta_n(v_n, w_n) \geq cD_h(x, y).$$

The collars ensure that no edge can skip the band.

This proves convergence for arbitrary sequences of endpoints with limiting locations in K . If the asserted uniform convergence failed, compactness of K would supply violating endpoint sequences with convergent projections. The result just proved would contradict that violation. $\square$

18.3 Sample arrays and the two roots

Punctured convergence leaves uncontrolled only those vertices whose projections approach the exploration root. We use two conditionally independent uniform corner roots of the same map, together with one common sequence of corner samples. The sample arrays identify the two continuum completions, and the roots become distinct points in that completion. A vertex escaping every finite sample net would have to approach both roots. Separating bands determine its distance differences to the samples and force the two root points to coincide.

Conditional on the unrooted decorated map, let R_n^0 be its original root corner, let R_n^1 be a second independent uniform corner, and let

$$X_n^1, X_n^2, \dots$$

be further independent uniform corners. Each corner is represented in the metric by its incident primal vertex. Reroot at R_n^1 and form the second inventory exploration.

A corner corresponds to its outgoing dart, so an n -edge map has $2n$ corners. Exactly $\deg(v)$ are incident to a vertex v , including two for each loop at v . Thus each sampled corner, represented by its vertex, has the probability law μ_n defined above. This remains true for a one-vertex map with loops. Conditional on the unrooted decorated map, the original root and these additional corners are exchangeable. Indeed the map weight does not depend on its root. Choosing a uniform corner in a fixed representative gives the uniform rooted isomorphism class: the automorphism group acts freely on oriented corners, so all its corner orbits have the same size. The subsequent independent choices preserve exchangeability of every metric distance array.

Both explorations have the same marginal law. Extract their contours and local tests jointly, using the same normalization $B(2n)$ and the same bilateral reference extraction. They give

$$(\mathcal{S}^j, cD^j, \mu^j, \eta^j, p_j), \quad j = 0, 1.$$

The discrete samples retain their common labels in the two explorations.

Lemma 18.7 (Common dense sample array). *After a joint extraction, for each j the locations X_n^i converge in exploration j to points $X^{j,i}$ which are iid from μ^j , conditionally on that surface and its curve. For every $k < \infty$,*

$$\left(\delta_n(X_n^i, X_n^\ell)\right)_{i,\ell \leq k} \longrightarrow \left(cD^j(X^{j,i}, X^{j,\ell})\right)_{i,\ell \leq k}.$$

The labeled arrays identify the two limiting compact metric probability spaces by a unique measure-preserving isometry sending $X^{0,i}$ to $X^{1,i}$ for every i .

Proof. The triangle tour passes through every Tutte edge once, and Tutte edges correspond to corners of the primal map. Consequently an independent uniform corner is a uniform tour time, with only the fixed endpoint/triangle convention of the encoding. Its normalized index is uniform on the discrete time grid. Conditional on an exploration, the indices of independently sampled corners are independent uniform indices. They converge jointly with the contours to iid uniform times U_i^j , independent of the limiting contour data. Thus

$$X^{j,i} = \eta^j(U_i^j).$$

Area parametrization gives the asserted conditional area law.

Every fixed sample avoids p_j almost surely. Apply Proposition 18.6 to the finitely many locations in a fixed array, and then use a diagonal extraction over k . The numerical arrays obtained from the two explorations are identical, since their discrete entries were identical.

Each area measure has full support. The marginal sample sequences are therefore dense almost surely, and Lemma 2.2 extends the common labeled array to a unique measure-preserving isometry of the two compact spaces. This uses the two marginal sampling laws; no conditional iid assertion given both explorations, or prior convergence of metric-measure spaces, is needed. $\square$

Use this isometry to regard the limits as one compact metric probability space (S, d, μ_h) , where $d = cD_h$, and write X^i for the common labeled samples. We next include the root labels in the limiting arrays. Their finite arrays are tight by exchangeability with sample arrays, so they can be included in the same diagonal extraction.

Lemma 18.8 (Root completion points). *Each root label determines a point $Z_j \in S$ such that*

$$\delta_n(R_n^j, X_n^i) \longrightarrow d(Z_j, X^i) \quad \text{for every } i.$$

Their mutual distance also converges to $d(Z_0, Z_1)$. Moreover $Z_0 \neq Z_1$ almost surely.

Proof. Let $l_{j,i}$ denote the limiting distance from root j to sample i . Exchangeability gives the same law to every finite array including the root labels as to a sample array with the corresponding number of additional labels. It follows that

$$\inf_{i \geq 1} l_{j,i} = 0 \quad \text{almost surely.} \tag{18.3}$$

One can see this without reconstructing a limiting measure: for fixed k , the law of $\min_{i \leq k} l_{j,i}$ is the law of the distance from one iid area sample to the first k other samples. These distances decrease to zero by their almost sure density.

Choose indices i_m with $l_{j,i_m} \rightarrow 0$. The triangle inequalities in the finite arrays pass to the limit, giving

$$d(X^{i_m}, X^{i_\ell}) \leq l_{j,i_m} + l_{j,i_\ell}.$$

Thus X^{i_m} is Cauchy and has a limit $Z_j \in S$. The same inequalities imply

$$|l_{j,i} - d(X^{i_m}, X^i)| \leq l_{j,i_m},$$

so $l_{j,i} = d(Z_j, X^i)$ for all i . This also proves uniqueness. If l_{01} is the limiting root-to-root distance, then

$$|l_{01} - l_{1,i}| \leq l_{0,i}.$$

Taking samples approaching Z_0 gives $l_{01} = d(Z_0, Z_1)$.

Finally, by exchangeability l_{01} has the law of the distance between two independent area samples in (S, d, μ_h) . The area measure is diffuse, so this distance is strictly positive almost surely. $\square$

We need a simple deterministic observation about a separating set in a graph.

Lemma 18.9 (Separation and distance profiles). *Let (V, δ) be a finite graph with its length metric. Let $C \subset V$ meet every path from v to each of the vertices $x_1, \dots, x_k$, and put $e = \delta(v, C)$. Then*

$$e + \inf_{z \in C} \delta(z, x_i) \leq \delta(v, x_i) \leq e + \sup_{z \in C} \delta(z, x_i).$$

In particular, if $\sup_{z \in C} |\delta(z, x_i) - a_i| \leq t$ for each i , then

$$|(\delta(v, x_i) - \delta(v, x_1)) - (a_i - a_1)| \leq 2t.$$

The last bound is independent of e .

Proof. A shortest path from v to x_i first meets C at some vertex z . Its two portions have costs at least e and $\delta(z, x_i)$, giving the lower bound. For the upper bound, join v to a point minimizing its distance to C , then continue to x_i . Subtract the two resulting intervals to obtain the last assertion. $\square$

For each exploration j , the bands of Lemma 18.4, their collars, and Equation (18.1) produce discrete vertex sets $C_{n,m}^j$ with the following properties. For fixed m , vertices projecting sufficiently close to p_j are separated by $C_{n,m}^j$ from every fixed sample outside the band. For all large m , every fixed finite list of samples is outside. Moreover, by Proposition 18.6,

$$\lim_{m \rightarrow \infty} \limsup_{n \rightarrow \infty} \sup_{z \in C_{n,m}^j} |\delta_n(z, X_n^i) - d(p_j, X^i)| = 0 \quad \text{for every fixed } i. \quad (18.4)$$

Here and below p_j is viewed in S by the label-preserving isometry. To construct these sets, take the vertices whose projections lie in a closed thick band around a separator compactly inside K_m . A projected path from its inner side to its outer side has a vertex in that thick band once the edge mesh is smaller than its collars. The band remains a compact punctured set for fixed m , so Proposition 18.6 applies uniformly there. Its continuum distance profile tends to that of p_j because its points converge uniformly to p_j .

Lemma 18.10 (Identification of the root with the missing point). *In the common completion, $Z_j = p_j$, for $j = 0, 1$. Consequently $p_0 \neq p_1$ almost surely.*

Proof. Set $e_{n,m}^j = \delta_n(R_n^j, C_{n,m}^j)$. For any fixed sample i , all sufficiently small bands separate the root from that sample. Therefore

$$e_{n,m}^j \leq \delta_n(R_n^j, X_n^i).$$

First take a limsup in n , then in m , and finally the infimum over i . Equation (18.3) gives

$$\lim_{m \rightarrow \infty} \limsup_n e_{n,m}^j = 0.$$

Apply Lemma 18.9 with $v = R_n^j$, and use Equation (18.4). For every i ,

$$d(Z_j, X^i) = d(p_j, X^i).$$

The sample set is dense, so $Z_j = p_j$. The last assertion follows from Lemma 18.8. $\square$

Proposition 18.11 (Tight finite metric nets). *For every $\varepsilon > 0$,*

$$\lim_{k \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P} \left[\max_{v \in V(M_n)} \min_{1 \leq i \leq k} \delta_n(v, X_n^i) > \varepsilon \right] = 0$$

along the selected subsequence.

Proof. Suppose otherwise. There are $\varepsilon, \theta > 0$, integers $k_n \rightarrow \infty$, and a further subsequence on which the displayed event with $k = k_n$ has probability at least θ . On this event choose a violating vertex v_n by a fixed finite ordering; elsewhere choose any vertex. Include the event indicator and contour representatives of v_n in both explorations in the joint extraction. These additional variables are tight: the indicator lies in $\{0, 1\}$ and the representative times lie in $[0, 1]$. We do not condition the laws of either limiting sphere on the event. In the resulting coupling the indicators converge almost surely to a variable $I \in \{0, 1\}$. Put $E = \{I = 1\}$. Then $\mathbb{P}(E) \geq \theta$, and on E the selected vertices violate the net condition for all sufficiently large n .

On E , the projected vertex must converge to p_j in each exploration. Indeed, a limiting location different from p_j lies in a compact punctured region. A fixed sample can be chosen arbitrarily close to that location. Proposition 18.6 would then give $\delta_n(v_n, X_n^i) < \varepsilon$ for some fixed i and all sufficiently large n , contrary to $k_n \rightarrow \infty$ and the choice of v_n .

Fix i . In either exploration, for each sufficiently large fixed m , all paths from v_n to X_n^i and X_n^1 meet $C_{n,m}^j$, eventually on E . Equations (18.4) and Lemma 18.9 imply

$$\delta_n(v_n, X_n^i) - \delta_n(v_n, X_n^1) \longrightarrow d(p_j, X^i) - d(p_j, X^1).$$

The possibly unbounded escape cost from v_n cancels. The numerical expression on the left is identical in both explorations, so on E

$$d(p_0, X^i) - d(p_0, X^1) = d(p_1, X^i) - d(p_1, X^1) \quad \text{for every } i.$$

Density and continuity imply that $d(p_0, x) - d(p_1, x)$ is constant for $x \in S$. At $x = p_0$ its value is $-d(p_0, p_1)$, whereas at $x = p_1$ its value is $d(p_0, p_1)$. Thus $p_0 = p_1$ on E , contradicting Lemma 18.10. $\square$

Corollary 18.12 (The subsequential metric and area limit). *Along the selected subsequence,*

$$(V(M_n), \delta_n, \mu_n) \implies (S, cD_h, \mu_h)$$

in Gromov–Hausdorff–Prokhorov topology. Every original subsequence has a further subsequence of this form, with c deterministic and in $[c_-, c_+]$.

Proof. The samples X_n^i , represented by their incident vertices, are conditionally iid with law μ_n . Lemma 18.7 gives convergence of every fixed finite distance array to the array of conditionally iid area samples in (S, cD_h, μ_h) . The limiting area measure has full support. Proposition 18.11 gives precisely the covering condition in Lemma 2.1. Applying that lemma proves the GHP assertion with the identified quantum-area measure.

The constant c was selected in (F1) before either finite exploration was extracted. It is deterministic and common to all charts. Neither sampling, completion nor the measured convergence criterion changes this multiplier. The construction began with an arbitrary subsequence, which proves the last assertion. $\square$

18.4 A deterministic normalization for all integers

We apply the common normalization lemma to the measured limit just proved. Let

$$H_n = \text{diam}(V(M_n), d_n), \quad H = \text{diam}(S, D_h),$$

and let m be the center of Lemma 2.4. The variable H is finite and strictly positive almost surely. Define the deterministic numbers

$$b_n = \exp(m(\log(H_n \vee 1)) - m(\log H)), \quad a_n(q) = b_n^{-1}. \quad (18.5)$$

The truncation includes maps with a single vertex and loops, whose unscaled graph diameter is zero.

Proposition 18.13 (The full-sequence FK limit). *The numbers in Equation (18.5) satisfy $a_n(q) > 0$, $a_n(q) \rightarrow 0$, and*

$$(V(M_n), a_n(q)d_n, \mu_n) \implies (S, D_h, \mu_h)$$

in Gromov–Hausdorff–Prokhorov topology as $n \rightarrow \infty$ through all positive integers.

Proof. Set $s_n = B(2n)$, which tends to infinity by Lemma 9.10. Corollary 18.12 supplies every hypothesis of Lemma 2.5, including the deterministic nature of the subsequential multiplier. The normalization prescribed by that lemma is exactly Equation (18.5). Its conclusion proves the proposition, with the quantum-area measure unchanged. This proves the critical FK clause of Theorem 1.1. $\square$

The normalization uses unconditional diameter laws and is deterministic. The proof uses the deterministic multiplier assertion in (F1); it does not replace a random subsequential multiplier by a deterministic one. All constants and all subsequence choices in this Section may depend on the fixed value of q . No endpoint statement or uniformity as q approaches 0 or 4 is used.

19 From tree excursions to the measured quantum sphere

Throughout this section, $\gamma = \sqrt{2}$.

The local tree passage law has now been identified up to a deterministic constant. We transfer this identification to a finite excursion, control the two ends of its contour, and recover all unrestricted distances. The contour parametrization will identify the limiting probability measure at the same time.

Write $N = 2n$ and let Z^{disc} be the walk whose independent increments choose each of the four axial unit steps with probability $1/4$. Put $\mathcal{Q} = \mathbb{Z}_{\geq 0}^2$ and

$$q_m(x, y) = \mathbb{P}_x[Z_m^{\text{disc}} = y, Z_j^{\text{disc}} \in \mathcal{Q} \text{ for } 0 \leq j \leq m].$$

Thus q_m includes the killing condition. We denote the law conditioned on $Z_N^{\text{disc}} = 0$ and survival until N by $\mathbb{P}_N^{\text{exc}}$. Theorem 13.1 identifies this law with the uniform pair (M_n, T_n) defined in Section 1. In particular, every graph distance below uses all primal edges of that pair. The internal interval distances used as upper bounds retain the tree edges and the matched non-tree edges witnessed in that interval; they do not change the definition of the ambient distance.

We use the preliminary deterministic units

$$B_N = A_{\lfloor \log_2 N \rfloor}. \quad (19.1)$$

Theorem 17.3 says that $B_N \rightarrow \infty$, that every sequence $N \rightarrow \infty$ has a further joint extraction of the bilateral contours and passage data, and that on this extraction the local rule is ΛD_h , for a deterministic Λ in a fixed compact subset of $(0, \infty)$. Brownian time is normalized to have total duration one; the deterministic factor from $N/2^{\lfloor \log_2 N \rfloor}$ is included in Λ . We also use its uniform interval bound: if I is an integer interval with $1 \leq |I| \leq N$, and $\mathcal{D}(I)$ is its internal contour-vertex diameter, then

$$\|B_N^{-1} \mathcal{D}(I)\|_{L^k(\mathbb{P})} \leq C \left(\frac{|I|}{N} \right)^s. \quad (19.2)$$

A bound with $|I| + 1$ is equivalent here. These statements hold also on any fixed finite enlargement of the time window. No limit of adjacent normalizing ratios will be assumed.

19.1 Conditioning and the additional passage data

Lemma 19.1 (Excursion densities). *There are constants $c, C > 0$ such that, for all $m \geq 0$ and $x \in \mathcal{Q}$,*

$$q_m(0, x) \leq C(x_1 + 1)(x_2 + 1)(m + 1)^{-3} \exp\left(-\frac{c|x|^2}{m + 1}\right). \quad (19.3)$$

For positive even N ,

$$q_N(0, 0) \asymp (N + 1)^{-3}, \quad \lim_{N \rightarrow \infty, N \text{ even}} N^3 q_N(0, 0) = \frac{32}{\pi}. \quad (19.4)$$

For $1 \leq l \leq N/4$, the density of the increment path on $[l, 2l]$ under $\mathbb{P}_N^{\text{exc}}$, relative to the law of l free increments, is

$$f_{N,l}(\omega) = \frac{1}{q_N(0, 0)} \sum_{x \in \mathcal{Q}} q_l(0, x) \mathbf{1}_{\{x + \omega_j \in \mathcal{Q}, 0 \leq j \leq l\}} q_{N-2l}(x + \Delta, 0), \quad \Delta = \omega_l. \quad (19.5)$$

It satisfies

$$f_{N,l}(\omega) \leq C \prod_{i=1}^2 \left(1 + \frac{|\Delta_i|}{\sqrt{l}}\right) \leq C \left(1 + \frac{|\Delta|}{\sqrt{l}}\right)^2, \quad \sup_{N,l} \mathbb{E}[f_{N,l}^2] < \infty. \quad (19.6)$$

Fix $0 < a < b < 1$. The densities for the increment block $[[aN], [bN]]$ converge, under the free invariance principle, to a continuous function of the limiting block's coordinate minima and net increment. They are uniformly integrable, indeed uniformly bounded in L^2 . Consequently, if arbitrary additional block-measurable passage data converge jointly with the free rescaled increments along a subsequence, their excursion-conditioned joint laws converge along that subsequence by the same change of density. Conditional on the limiting block increments, this change preserves their conditional kernel.

Proof. Let $p_r(h)$ be the simple symmetric one-dimensional walk transition probability from 0 to h , and let $q_r^{(1)}$ denote its kernel killed on entering -1 . Reflection gives, on the admissible parity lattice,

$$q_r^{(1)}(0, h) = p_r(h) - p_r(h+2) = \frac{2(h+1)}{r+h+2} p_r(h), \quad h \geq 0. \quad (19.7)$$

The formula is interpreted directly at $r = 0$ and at the ends of the walk's support. The binomial formula for p_r , or Stirling's inequalities, gives

$$q_r^{(1)}(0, h) \leq C(h+1)(r+1)^{-3/2} e^{-ch^2/(r+1)}.$$

Given the number J of first-coordinate moves in m steps, the two coordinate walks are independent and J has the binomial law $\text{Bin}(m, 1/2)$. Thus

$$q_m(0, x) = 2^{-m} \sum_{j=0}^m \binom{m}{j} q_j^{(1)}(0, x_1) q_{m-j}^{(1)}(0, x_2). \quad (19.8)$$

For $m/4 \leq j \leq 3m/4$, the preceding bound proves (19.3) term by term. The complementary binomial probability is at most Ce^{-cm} . On the support of $q_m(0, x)$, $|x| \leq m$; after decreasing c , this error is at most $C(m+1)^{-3} e^{-c|x|^2/(m+1)}$ and is absorbed in the same bound. The kernel is symmetric, so the bound also applies to $q_m(x, 0)$.

Set $e_r = q_r^{(1)}(0, 0)$. The exact formula is $e_{2j} = \binom{2j}{j} / (4^j(j+1))$, with $e_0 = 1$ and $e_{2j+1} = 0$. Hence $e_r \sim 2^{3/2} \pi^{-1/2} r^{-3/2}$ through even r . Apply this to (19.8) with $x = 0$ and $m = N$. On $|J - N/2| \leq N^{3/4}$, the product of the two nonzero Catalan probabilities is $(64/\pi + o(1))N^{-3}$, uniformly. Exactly half the binomial mass is on even J ; the mass outside this window is exponentially small compared with N^{-3} . This proves the limit in (19.4); increasing the constants covers all positive even N .

The Markov property at times l and $2l$ proves (19.5). Since $N - 2l \geq N/2$, use (19.3) twice and (19.4). Drop the survival indicator and the exit Gaussian. Terms with $x + \Delta \notin \mathcal{Q}$ have zero weight, and for the other terms $x_i + \Delta_i + 1 \leq x_i + |\Delta_i| + 1$. We obtain

$$f_{N,l}(\omega) \leq Cl^{-3} \prod_{i=1}^2 \sum_{h \geq 0} (h+1)(h+|\Delta_i|+1) e^{-ch^2/l}.$$

For $u \geq 0$ and $l \geq 1$, comparison with Gaussian integrals gives

$$\sum_{h \geq 0} (h+1)(h+u+1) e^{-ch^2/l} \leq C(l^{3/2} + ul).$$

This proves the pointwise bound in (19.6). Free bounded increments have uniformly bounded moments of every fixed order after division by $\sqrt{l}$: expanding an even moment, every nonzero term pairs each independent centered increment with another occurrence of that increment, giving a bound $C_r l^r$ for the $2r$ th moment. In particular the fourth moment proves the asserted L^2 bound. The same argument gives an L^r bound for every fixed finite r .

Here are the local limits and the precise change-of-measure argument. They also fix all parity factors. Set $c_* = 32/\pi$ and

$$K_t(u) = c_* t^{-3} u_1 u_2 e^{-|u|^2/t}, \quad t > 0, \quad u \in \mathbb{R}_{\geq 0}^2.$$

Extend K_t by zero outside the quadrant, and use the same zero convention for killed kernels evaluated outside their state space. Uniformly when t stays in a compact subset of $(0, \infty)$ and $x/\sqrt{N}$ stays in a fixed compact subset of the quadrant, Stirling's formula in (19.7) and binomial concentration in (19.8) imply

$$N^2 q_{\lfloor tN \rfloor}(0, x) - K_t(x/\sqrt{N}) \longrightarrow 0 \quad (19.9)$$

on the admissible lattice $x_1 + x_2 \equiv \lfloor tN \rfloor \pmod{2}$. To check its coefficient, the one-dimensional local limit on its parity lattice is $p_j(h) \sim \sqrt{2/(\pi j)} e^{-h^2/(2j)}$. The product in (19.8), with $j \sim tN/2$, therefore has coefficient $64/\pi$; its admissible binomial schedules have asymptotic mass $1/2$. Boundary points cause no problem for uniform absolute convergence: if one rescaled coordinate tends to zero, both sides of (19.9) tend to zero by (19.3). This proves the stated uniform version.

Let $a_N = \lfloor aN \rfloor$, $b_N = \lfloor bN \rfloor$, and linearly interpolate a free block after dividing heights by $\sqrt{N}$ and times by N . For its limiting path w of duration $b - a$, write $d = w(b - a)$ and $m_i = \min_t w_i(t)$. The limiting density is

$$f_{a,b}(w) = \frac{1}{2c_*} \int_{\mathbb{R}_{\geq 0}^2} K_a(u) K_{1-b}(u + d) \mathbf{1}_{\{u_i + m_i \geq 0, i=1,2\}} du. \quad (19.10)$$

The discrete formula is (19.5) with entrance length a_N , block length $b_N - a_N$, and exit length $N - b_N$. Each summation lattice has asymptotic density $N/2$ after rescaling; (19.4) and (19.9) give (19.10). More explicitly, if rescaled paths w_N converge uniformly to w , their minima and increments converge. On bounded u -sets the Riemann sums converge, since the boundaries $u_i = -m_i$ have zero Lebesgue measure. On their complements the Gaussian bound (19.3), with bounded d , gives an integrable common majorant. It follows that $f_N(w_N) \rightarrow f_{a,b}(w)$. The same argument proves continuity of $f_{a,b}$ in the uniform path topology. Dropping the exit Gaussian as above also gives

$$f_N(w) \leq C_{a,b} \left(1 + \frac{|\Delta|}{\sqrt{N}} \right)^2.$$

Thus the densities are uniformly bounded in L^2 under the free law.

For completeness, suppose that (W_N, H_N) denotes the rescaled free block together with any additional block data, taking values in the Polish data spaces of Proposition 15.1, and that $(W_N, H_N) \Rightarrow (W, H)$. The preceding sequential convergence and uniform integrability imply, for every bounded continuous Φ ,

$$\mathbb{E}[f_N(W_N) \Phi(W_N, H_N)] \longrightarrow \mathbb{E}[f_{a,b}(W) \Phi(W, H)]. \quad (19.11)$$

Indeed, realize the unweighted convergence almost surely, truncate the densities at a fixed height, and then remove the truncation using their uniform L^2 bound. Taking $\Phi = 1$ shows that the

limiting density integrates to one. If the unweighted limit disintegrates as $\mu(dw)\mathbf{K}(w,dz)$, its weighted law is exactly $f_{a,b}(w)\mu(dw)\mathbf{K}(w,dz)$. This proves the assertion about additional data and their conditional kernel, not just convergence of the contour marginal. $\square$

The corresponding full-contour invariance principle has a useful direct interpretation. Given a schedule with J first-coordinate moves, excursion conditioning makes the two coordinate strings independent nonnegative bridges of lengths J and $N - J$. The conditional mass of J is

$$\frac{2^{-N} \binom{N}{J} e_J e_{N-J}}{q_N(0,0)}. \quad (19.12)$$

It is concentrated at $J/N = 1/2$. For each $\epsilon > 0$, the probability of $|J - N/2| > \epsilon N$ is at most $CN^3 e^{-c_\epsilon N}$, by the ordinary binomial bound and (19.4). Conditional on J , all schedules are uniform and independent of the two coordinate bridges. Their partial counts satisfy $\sup_{r \leq N} |J_r - rJ/N|/N \rightarrow 0$ in probability. One way to see the last assertion is to use

$$\mathbb{P}[J_r = u \mid J] = \binom{r}{u} \binom{N-r}{J-u} / \binom{N}{J}$$

and Stirling's inequalities: for J/N in a compact subset of $(0,1)$, the mass with $|u - rJ/N| > \epsilon N$ is at most $CN^C e^{-c_\epsilon N}$, uniformly in r ; summing over r suffices. Consequently the schedule time changes converge uniformly to half speed. The reflection formula gives the nonnegative-bridge finite-dimensional limits as the Brownian excursion transition densities. It also gives tightness: on a fixed interior time interval use the bounded L^2 density with respect to free increments; close to each endpoint use the one-dimensional version of (19.6) and sum dyadic free-walk maximum bounds. The latter sum is bounded by $C \sum_{j \geq 0} \sqrt{\delta 2^{-j}} = C' \sqrt{\delta}$. This proves the excursion invariance principle, including the endpoints. Thus $N^{-1/2} Z_{[Nt]}^{\text{disc}}$ converges to two independent Brownian excursions with the variance convention inherited from the axial walk.

By Theorem 13.2, equivalently [34, Theorem 1.1], this pair encodes the area-one quantum sphere, with its area-parametrized space-filling SLE_8 . This is the ordinary sphere of [13, Definition 4.21(ii)]. Its distinguished points can be forgotten, by [13, Proposition A.13]; we write q for the common point $\eta(0) = \eta(1)$ used by the encoding. The strict positivity of both excursions on $(0,1)$ gives

$$\eta^{-1}(\{q\}) = \{0,1\}. \quad (19.13)$$

All deterministic Brownian boundary-length conventions here are the same ones as in the bilateral construction and Theorem 13.2.

19.2 The metric on interior sphere charts

Lemma 19.2 (Interior sphere metric). *In the joint extraction with distance units B_N and deterministic multiplier Λ selected above, the limiting finite-excursion local passage rule satisfies*

$$F_U(x,y) = \Lambda D_h(x,y;U)$$

for every buffered chart U compactly contained in $S \setminus \{q\}$ and every x,y joined by a path compactly in U . This identification retains the upper realization and lower subpath-extraction conclusions of Theorem 17.3.

Proof. Fix rational $0 < \delta < 1/2$, and write $a = \delta$, $b = 1 - \delta$. Retain the complete collection of internal interval-chart passage data constructed solely from the increments on $[a_N, b_N]$. It includes every certificate whose range has positive clearance from the boundary of the interval surface, by Lemma 15.2. Apply (19.11) to this collection, first in finitely many compact charts and then by the countable extraction of Proposition 15.1. In the bilateral reference law its compact-interior local passage rule equals ΛD , by Theorem 17.3; the equality event has probability one and is therefore preserved by the limiting density.

The restricted increments determine an abstract decorated quantum surface, including its quantum boundary-length units. For the cone this is [18, Lemma 4.9]. The sphere statement needed here is supplied separately by the final paragraph of the proof of [34, Theorem 1.1, the case $\gamma \leq \sqrt{2}$, p. 49]: conditionally on the contour values at δ and $1 - \delta$, the quantum surface complementary to $\eta([0, \delta]) \cup \eta([1 - \delta, 1])$ has the corresponding cone conditional law and is determined, together with its curve, by the restricted contour. This is the symmetric interval statement recorded in Theorem 13.2. Its quantum units agree with those in the density calculation. Metric locality and conformal covariance therefore identify its internal D with the internal metric on its image in the area-one sphere. The transported equality is an equality with this sphere's D_h in the fixed normalization.

This equality is an event of the restricted contour and its passage data, since those contour increments determine the abstract interval surface. Adjoining the outside contour therefore preserves the same probability-one equality.

Finally, if $K \subset S \setminus \{q\}$ is compact, continuity of η and (19.13) imply that $\eta^{-1}(K)$ is contained in $(\delta, 1 - \delta)$ for some rational $0 < \delta < 1/2$. Choose a smaller positive rational δ if necessary. The image of the omitted compact time intervals is disjoint from K , so K lies in the interior of the interval surface and has positive spatial clearance from its boundary. Its local certificates are therefore among the ones just transferred. By a countable exhaustion, the local passage rule off q equals ΛD_h simultaneously in these charts, with the same deterministic Λ . $\square$

We also record how the transfer behaves when the complete outside contour is retained. The following conditional-kernel statement allows the inside passage observations to be used jointly with outside contour observations.

Lemma 19.3 (The kernel after adjoining the outside contour). *Fix $0 < a < b < 1$. In the setup of Equation (19.11), suppose the free joint limit of the block increments W_N and block-measurable passage data H_N disintegrates as $\mu(dw)K(w, dz)$. Write W, H for the corresponding interior increment path and passage datum in their joint excursion-conditioned limit with the complete contour. The conditional law of H given that contour is $K(W, \cdot)$.*

Proof. The assertion follows from the Markov factorization: given the entrance height x and the inside increments, the outside consists of a killed entrance bridge and a killed exit bridge, independent of the inside data. Their weights are the two factors in (19.5). On compact positive-height ranges, the reflection formula for $q_m(x, y)$ (the four coordinate reflections of the free two-dimensional kernel) gives uniform bridge finite-dimensional limits. For tightness at the zero endpoint, the one-dimensional reflection formula gives $q_r^{(1)}(u, v) \leq C(u + 1)(v + 1)(r + 1)^{-3/2}$; telescope step-two differences of p_r in the smaller of u, v , using symmetry of the kernel and the binomial Gaussian bound. Averaging over schedules as in (19.8) consequently gives $q_m(y, z) \leq C_R(y_1 + 1)(y_2 + 1)N^{-2}$ when $m \asymp N$ and $|z| \leq R\sqrt{N}$. On a fixed positive-height compact set, $q_{a_N}(0, z) \geq c_R N^{-2}$, so the proof of (19.6) applies to the small dyadic blocks of this entrance

bridge. At its other endpoint, reverse the bridge: the increment density on an initial short block is bounded by $C_R(1 + |\Delta|/\sqrt{N})^2$, using (19.3). Free-walk maximum bounds therefore give tightness there too. This proves the uniform bridge convergence used here.

Both entrance and exit heights can be restricted to positive compact sets before this calculation. Indeed their marginal law at time j is

$$\mathbb{P}_N^{\text{exc}}[Z_j^{\text{disc}} = x] = \frac{q_j(0, x)q_{N-j}(x, 0)}{q_N(0, 0)}.$$

Uniformly for j/N in a compact subinterval of $(0, 1)$, (19.3) bounds this by $CN^{-3}(x_1 + 1)^2(x_2 + 1)^2e^{-c|x|^2/N}$. Summing shows that the probability of either coordinate below $\epsilon\sqrt{N}$ has limsup at most $C\epsilon^3$, whereas the probability of $|x| > R\sqrt{N}$ tends to zero uniformly as $R \rightarrow \infty$. Apply these bounds at both a_N and b_N .

Retain the rescaled entrance height $u = x/\sqrt{N}$ jointly in the Riemann sum proving (19.11). On the preceding compact sets, the two bridge kernels converge uniformly and are independent of the additional inside data given (u, w) . The limiting joint law consequently has the form

$$\mu(dw)\mathbf{K}(w, dz) \frac{K_a(u)K_{1-b}(u+d)}{2c_*} \mathbf{1}_{\{u_i+m_i \geq 0, i=1,2\}} du \mathbf{B}_a(0, u; d\omega^-) \mathbf{B}_{1-b}(u+d, 0; d\omega^+),$$

where $\mathbf{B}$ are the limiting killed bridge laws. The marginal tail bounds just proved remove the truncations. Conditioning on the complete contour (ω^-, w, ω^+) therefore leaves the inside kernel $\mathbf{K}(w, dz)$ unchanged, exactly as in Lemma 15.4. $\square$

19.3 Endpoint diameters

The density estimate controls the metric information which an interior change of measure cannot see. Let v_i be the primal vertex at contour time i , with $v_0 = v_N$, and set

$$T_N(\delta) = \{v_i : 0 \leq i \leq \lfloor \delta N \rfloor \text{ or } N - \lfloor \delta N \rfloor \leq i \leq N\}.$$

For $1 \leq l \leq N/4$, Cauchy–Schwarz, (19.6), and (19.2) give

$$\mathbb{E}_N^{\text{exc}} \left[B_N^{-1} \mathcal{D}([l, 2l]) \right] \leq \|f_{N,l}\|_2 \left\| B_N^{-1} \mathcal{D}([0, l]) \right\|_2 \leq C(l/N)^s. \quad (19.14)$$

The internal distances in this display are genuine functions of the block increments, so the density calculation applies to them exactly. If only endpoint norms are used in Theorem 17.3, the diameter bound follows by balanced binary subdivision: at depth r , the maximum of at most 2^r endpoint costs has L^k norm at most $C(l/N)^s 2^{-r(s-1/k)}$ in B_N units; summing these bounds is finite since $s > 1/k$. Rounded subdivision lengths change the constant only.

For $0 < \delta \leq 1/8$ and $\delta N \geq 1$, choose J with $2^J \leq \delta N < 2^{J+1}$. The chain

$$[0, 1], [1, 2], [2, 4], \dots, [2^J, 2^{J+1}]$$

covers $[0, \delta N]$ and its consecutive intervals share endpoints. Every block except $[0, 1]$ in this chain satisfies the hypothesis of (19.14). Its first interval has graph diameter at most one. Connect points through the common endpoints using the interval paths and sum the resulting diameter bounds. Time reversal gives the same bound at N , and the two chains meet at $v_0 = v_N$. If $\delta N < 1$, the endpoint set consists only of this vertex. We have proved

$$\mathbb{E}_N^{\text{exc}} \left[B_N^{-1} \text{diam}_{d_n} T_N(\delta) \right] \leq C\delta^s + 2B_N^{-1}, \quad 0 < \delta \leq 1/8. \quad (19.15)$$

In particular,

$$\lim_{\delta \downarrow 0} \limsup_{N \rightarrow \infty} \mathbb{P}_N^{\text{exc}} \left[B_N^{-1} \text{diam}_{d_n} T_N(\delta) > \epsilon \right] = 0 \quad \text{for every } \epsilon > 0. \quad (19.16)$$

This is an estimate for the actual ambient graph metric; the internal metrics appeared only to construct admissible paths.

On a closed interior time interval, the uniform L^2 density in Lemma 19.1 transfers the free time-modulus estimate of Lemma 14.2. Explicitly, if an event has free probability r , its conditioned probability is at most $C_{a,b}\sqrt{r}$. Together with (19.16), this gives tightness of the cyclic time modulus of the rescaled ambient distances on $[0, 1]$. Indeed first cover $[\delta, 1 - \delta]$ by finitely many interior blocks, and then join its ends to the two endpoint chains just constructed. The same argument bounds the full diameter in probability.

19.4 Identification of unrestricted distances

Lemma 19.4 (The measure counted by the tree tour). *Let v_i be the primal vertex represented by contour time i , with $v_N = v_0$. Then, for every $v \in V(M_n)$,*

$$\#\{0 \leq i < N : v_i = v\} = \deg_{M_n}(v). \quad (19.17)$$

Consequently, if

$$\mu_n = \frac{1}{2n} \sum_{v \in V(M_n)} \deg_{M_n}(v) \delta_v, \quad f_n(t) = v_{\lfloor Nt \rfloor}, \quad 0 \leq t \leq 1,$$

then $(f_n)_* \text{Leb}_{[0,1]} = \mu_n$. Conditional on the map, independent uniform times give independent samples from μ_n .

Proof. Associate time slot $i \in \{0, \dots, N-1\}$ with the step departing at i . The two horizontal traversals of a tree edge depart once from each endpoint, so these slots count its two incidences. Every vertical step belongs to a unique matched pair with terminal times $u < v$. By (13.3), this pair adds a non-tree edge between v_{u-1} and v_{v-1} ; its two departure slots count exactly those two incidences. This also counts a loop twice. Every slot belongs to exactly one such pair, proving (19.17). Each slot has time mass $1/N = 1/(2n)$, which proves the pushforward identity and the sampling assertion. $\square$

Proposition 19.5 (Sphere identification). *For every sequence of positive integers $n \rightarrow \infty$, there is a further subsequence and a deterministic $\Lambda \in (0, \infty)$ such that*

$$(V(M_n), B_{2n}^{-1} d_n, \mu_n) \Rightarrow (S, \Lambda D_h, \mu_h) \quad (19.18)$$

in the Gromov–Hausdorff–Prokhorov topology. The constants Λ lie in a fixed deterministic compact subset of $(0, \infty)$.

Proof. Choose a simultaneous bilateral extraction as in Theorem 17.3, with its deterministic multiplier Λ . Lemma 19.2 gives convergent sphere certificates on every compact subset of $S \setminus \{q\}$, with local passage rule ΛD_h . The local certificate assertions include realization with open margins, subpath extraction, and uniformly negligible costs for joining nearby endpoints.

We give the compactness and path argument explicitly. Put

$$\rho_N(t, u) = B_N^{-1} d_n(v_{\lfloor Nt \rfloor}, v_{\lfloor Nu \rfloor}), \quad 0 \leq t, u \leq 1.$$

Consecutive contour vertices are equal or joined by a tree edge, so replacing ρ_N by its bilinear interpolation changes it by at most $2/B_N$. The cyclic time modulus and diameter bound proved above imply tightness of these interpolations in $C([0, 1]^2)$. Extract jointly with the contours and all certificate charts, and use an almost sure representation. Then ρ_N converges uniformly to a continuous pseudometric ρ on $[0, 1]$, with $\rho(0, 1) = 0$. The limiting contour determines the sphere and its continuous surjection η . In particular the endpoint time sets have limiting ρ -diameters tending to zero. We will show that

$$\rho(t, u) = \Lambda D_h(\eta(t), \eta(u)) \quad (0 \leq t, u \leq 1). \quad (19.19)$$

The projection part of Proposition 15.1 applies to the finite excursion sewing as well: its proof uses uniform contour convergence and the defining link relations. Thus all representatives of a single vertex have D_h -close images, uniformly over vertices, and the two endpoints of every graph edge have D_h -distance tending to zero, uniformly over edges. To see the uniform assertion directly, a contrary sequence has convergent contour times; the limiting tree identification or matched second-coordinate chord identifies the two images. This also covers links involving time 0 or 1, which project to q . Choose any representative of each vertex and write $\pi_N(v)$ for its image under η at the rescaled time. Let $\varepsilon_N \rightarrow 0$ bound both representative errors and edge-image jumps. These maps are only used to locate paths in the limiting surface; all costs remain discrete all-edge costs.

We first record a uniform local crossing fact. If $K \subset S \setminus \{q\}$ is compact, take finitely many pairs of open sets $V_i \Subset U_i$ covering K , all contained in interval-surface interiors. Choose $r > 0$ so small that every D_h -ball of radius $3r$ about a point of K is contained in one of the U_i with positive clearance from its boundary. A discrete path beginning at such a point, stopped on first reaching D_h -displacement r , stays in that compact chart after an ε_N enlargement. For all sufficiently large N , it has scaled cost at least $\Lambda r/2$. Otherwise a subsequence of these paths has bounded cost and compact confinement; subpath extraction gives a certificate with endpoints at distance r and cost at most $\Lambda r/2$, contradicting the local lower comparison. Compactness of K and the finite cover make this assertion uniform in its starting point and in the path. Paths of larger cost already satisfy it. It follows that a path of scaled cost at most L can be divided into at most $2L/(\Lambda r) + 2$ such steps, with a final shorter step, as long as it stays in K .

For the upper bound in (19.19), fix t, u and a D_h -geodesic P from $x = \eta(t)$ to $y = \eta(u)$. A compact subarc disjoint from q has a finite subdivision into pieces lying with open margins in the sets U_i of a finite local cover. For each piece the local upper comparison and realization supply approximating discrete paths with cost at most Λ times that piece's D_h -length, plus an arbitrarily prescribed positive error. Uniform joining connects their endpoints, also with vanishing cost. Consequently a geodesic whose range avoids q gives $\rho(t, u) \leq \Lambda D_h(x, y)$.

Here is the endpoint modification when P meets q , also valid if one or both endpoints are q . Given $\delta > 0$, compactness and (19.13) give a radius $r_\delta > 0$ such that

$$\eta^{-1}(\overline{B}_{D_h}(q, 2r_\delta)) \subset [0, \delta] \cup [1 - \delta, 1]. \quad (19.20)$$

If P enters $B_{D_h}(q, r_\delta)$, retain its portion before the first entry and its portion after the last exit. Both are compactly away from q except for their retained endpoints, which are still at positive distance from q unless the whole portion is empty. Apply the finite local subdivision to these portions. The two cut endpoints have contour representatives in the endpoint time set of (19.20). Their approximating vertices can be connected at cost at most the scaled diameter of $T_N(\delta)$, with a vanishing joining error; the same applies if an original endpoint lies in this set. The sum of the retained geodesic lengths is at most $D_h(x, y)$. Uniform convergence of ρ_N therefore gives

$$\rho(t, u) \leq \Lambda D_h(x, y) + \text{diam}_\rho([0, \delta] \cup [1 - \delta, 1]).$$

The last term tends to zero by continuity of ρ at 0 and 1. This proves the upper bound in all cases.

For the lower bound, choose shortest discrete paths between the two contour vertices. Their scaled lengths converge to $\rho(t, u)$ and are bounded. Fix $r > 0$. If a path has vertices whose images lie in $B_{D_h}(q, r)$, remove the portion between its first and last such vertices, retaining its prefix and suffix. The removed portion has nonnegative cost. Its two cut endpoints have D_h -distance at most $2r$. Apart from these endpoints, the retained pieces stay outside $B_{D_h}(q, r)$; since edge-image jumps are at most ε_N , they are contained in $S \setminus B_{D_h}(q, r/2)$ for large N . If there is no visit, retain the entire path. Cases with an endpoint inside the ball simply have an empty retained piece.

Cover $S \setminus B_{D_h}(q, r/2)$ by the finite buffered local charts above and choose a fixed step size $r' > 0$ for that cover. Divide each retained piece at successive first D_h -displacements r' . The uniform crossing fact bounds the number of pieces independently of N . Pass to a further extraction recording their endpoints, costs, and marked subpaths. Each piece has a limiting certificate compactly in one chart. Its limiting cost is at least Λ times its endpoints' D_h -distance. Add these inequalities, including the final shorter pieces. The triangle inequality in D_h then gives

$$\rho(t, u) \geq \Lambda(D_h(x, y) - 2r).$$

No cost is assigned to the removed segment in this bound. Letting $r \downarrow 0$ proves the required lower inequality. In particular, it rules out a short global path escaping a local chart and returning at a different point; such a path has paid for its successive chart crossings. We have proved (19.19).

The construction applies first on a countable dense set of contour pairs. Continuity of ρ , D_h , and η extends it to every pair. As $\rho_N \rightarrow \rho$ uniformly, the correspondence

$$\mathcal{R}_N = \{(v_{\lfloor Nt \rfloor}, \eta(t)) : 0 \leq t \leq 1\}$$

has distortion tending to zero when the discrete metric is $B_N^{-1}d_n$ and the sphere metric is ΛD_h . It covers both spaces: every primal vertex has a contour visit and η is surjective. Moreover, Lemma 19.4 gives $(f_n)_*\text{Leb} = \mu_n$, and quantum-area parametrization gives $\eta_*\text{Leb} = \mu_h$. Lemma 2.3 therefore bounds the GHP distance by half the same distortion. This proves (19.18). The bounds and deterministic nature of Λ come from Theorem 17.3; a change of density cannot change a fixed almost sure equality constant. $\square$

19.5 A deterministic normalization for every integer

Proof of the spanning-tree case of Theorem 1.1. Let $X = \text{diam}_{D_h} S$. The ordinary area-one quantum sphere is compact and nontrivial, so $0 < X < \infty$ almost surely. With the center m from Lemma 2.4, define, for every positive integer n ,

$$X_n = \text{diam}_{d_n} V(M_n), \quad a_n = \exp \{m(\log X) - m(\log(X_n \vee 1))\}. \quad (19.21)$$

These are positive deterministic constants; the truncation includes one-vertex maps. Proposition 19.5 supplies every hypothesis of Lemma 2.5, with $s_n = B_{2n} \rightarrow \infty$ and a deterministic positive subsequential multiplier Λ . Hence $a_n \rightarrow 0$ and

$$(V(M_n), a_n d_n, \mu_n) \Longrightarrow (S, D_h, \mu_h)$$

in GHP topology along all positive integers. All distances use every primal edge, and forgetting the root and spanning tree gives the unmarked metric probability spaces in the theorem. $\square$

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